Southern Nuclear

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Docket Nos.: 52-025
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NL-26-0017
10 CFR 50.90

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Southern Nuclear Operating Company
Vogtle Electric Generating Plant Units 3 and 4
License Amendment Request to Revise Technical Specifications
to Include a Surveillance Frequency Control Program

Ladies and Gentlemen:

In accordance with the provisions of Section 50.90 of Title 10 of the *Code of Federal Regulations* (10 CFR), Southern Nuclear Operating Company (SNC) requests an amendment to revise the combined licenses (COLs) for Vogtle Electric Generating Plant (VEGP) Unit 3 (License No. NPF-91) and Unit 4 (License No. NPF-92).

The proposed amendment would modify TS requirements to permit the use of a Surveillance Frequency Control Program consistent with the methodology described in Technical Specifications Task Force (TSTF) traveler TSTF-425, Revision 3, "Relocate Surveillance Frequencies to Licensee Control - RITSTF Initiative 5b" [ADAMS Accession No. ML090850642], TSTF-563, Revision 0, "Revise Instrument Testing Definitions to Incorporate the Surveillance Frequency Control Program," dated December 6, 2018, and TSTF-596, Revision 2, "Expand the Applicability of the Surveillance Frequency Control Program (SFCP)," [ML25251A241]. Safety evaluations were provided by the NRC for TSTF-425 in a Federal Register posting on July 6, 2009 [74 *FR* 31996 - 32006], and in letters to the TSTF for TSTF-563 [ML18333A144] and for TSTF-596 [ML24362A056].

- Attachment 1 provides a description and assessment of the proposed change, the requested confirmation of applicability, and plant-specific verifications.
- Attachment 2 provides documentation of PRA technical adequacy.
- Attachment 3 identifies the changes and the associated differences and variations from the NUREG-1431 Westinghouse Standard Technical Specifications markups provided in each of the subject TSTF travelers
- Attachment 4 provides the Technical Specifications (TS) pages marked up to show the proposed changes.

- Attachment 5 provides existing TS Bases pages marked up to show the proposed changes and is provided for information only.
- Attachment 6 provides the Proposed No Significant Hazards Consideration.

SNC requests approval of the proposed license amendment request no later than nine months after acceptance. SNC expects to implement the proposed changes within 180 days of issuance of the amendment.

SNC has concluded that the proposed change presents no significant hazards consideration under the standards set forth in 10 CFR 50.92, “Issuance of amendment.”

In accordance with 10 CFR 50.91, SNC is notifying the State of Georgia of this license amendment request by transmitting a copy of this letter, with attachments and enclosures, to the designated State Official.

This letter contains no regulatory commitments. This letter has been reviewed and determined not to contain security-related information.

If you have any questions, please contact Ryan Joyce at 205.992.6468.

I declare under penalty of perjury that the foregoing is true and correct.
Executed on the 26th day of March 2026.

Respectfully submitted,

Jamie Coleman

Digitally signed by JAMIEMCO
Date: 2026.02.26 08:56:12 -06'00'

Jamie M. Coleman
Regulatory Affairs Director

JMC/RMJ

Attachments:
1. Description and Assessment
2. Documentation of PRA Adequacy
3. Identification of Changes, Differences and Variations from Standard Technical Specifications Mark-ups
4. Proposed Technical Specification Changes (Mark-up)
5. Proposed Technical Specification Bases Changes (Mark-up) – For Information Only
6. Proposed No Significant Hazards Consideration

cc: NRC Regional Administrator, Region II
NRR Project Manager – Vogtle 3&4
Senior Resident Inspector – Vogtle 3&4
Director, Environmental Protection Division - State of Georgia
Document Services RTYPE: VND.LI.L00

Attachment 1 to NL-26-0017
Description and Assessment

License Amendment Request to Revise Technical Specifications to Include a Surveillance Frequency Control Program

1. DESCRIPTION

2. ASSESSMENT
   - 2.1. Applicability of Published Safety Evaluations
   - 2.2. Optional Changes and Variations
   - 2.3. Additional Change Requests

3. REGULATORY SAFETY ANALYSIS
   - 3.1. No Significant Hazards Consideration Analysis
   - 3.2. Conclusion

4. ENVIRONMENTAL CONSIDERATION

ATTACHMENT 1

DESCRIPTION AND ASSESSMENT

1.0 DESCRIPTION

The proposed amendment would modify the Vogtle Electric Generating Plant (VEGP) Units 3 and 4 Technical Specification (TS) requirements by relocating the periodic Surveillance Frequencies from the TS and placing the Frequencies under licensee control in accordance with a new program, the Surveillance Frequency Control Program (SFCP). The SFCP, which is added to TS Section 5.5, “Programs and Manuals,” describes the requirements for a program to control changes to the relocated Surveillance Frequencies.

The process for making changes to a Surveillance Frequency in accordance with the SFCP is described in NEI 04-10, "Risk-Informed Technical Specifications Initiative 5b, Risk-Informed Method for Control of Surveillance Frequencies, Industry Guidance Document,” Revision 1 [ADAMS Accession No. ML071360456], as approved by the NRC [ML072570267]. Adherence to NEI 04-10-A is required by the SFCP.

The proposed amendment is consistent with the methodology described in Technical Specifications Task Force (TSTF) traveler TSTF-425, "Provide Risk-Informed Extended Completion Times - RITSTF Initiative 4b," Revision 3 [ML090850642], and includes incorporation of TSTF-563, “Revise Instrument Testing Definitions to Incorporate the Surveillance Frequency Control Program,” Revision 0, dated December 6, 2018, (TSTF-563), and TSTF-596-A, “Expand the Applicability of the Surveillance Frequency Control Program (SFCP),” Revision 2 [ML25251A241] (TSTF-596). However, since neither TSTF-425 nor TSTF-596 provided markups of the NUREG-2194, Standard Technical Specifications for Westinghouse Advanced Passive 1000 (AP1000) Plants, the proposed changes to the VEGP Units 3 and 4 TS differ from those provided in these two travelers.

Variations to the changes described in TSTF-425 and TSTF-596 are identified in Section 2.2, Optional Changes and Variations. There are no variations proposed to the TSTF-563 markups. Although TSTF-563 and TSTF-596 describe and evaluate changes that are not consistent with the criteria for exceptions of TSTF-425, changes that are proposed in accordance with TSTF-563 and TSTF-596 are not identified as variations to TSTF-425, as they have previously been reviewed and approved by the NRC.

Markups showing the TS and Updated Final Safety Analysis Report (UFSAR) changes are provided in Attachment 4. Conforming TS Bases changes are provided in Attachment 5 for information only. The existing Bases information describing the basis for the Surveillance Frequency will be relocated to the licensee-controlled Surveillance Frequency Control Program. Various editorial changes are made to the ISTS Bases as needed to facilitate the incorporation of TSTF-425 and TSTF-596. These changes are made consistent with the Technical Specifications (TS) Bases Control Program and determined to not require NRC approval.

In addition, SNC proposes changes to remove completed or otherwise outdated TS language that is no longer applicable.

2.0 ASSESSMENT

2.1 Applicability of Published Safety Evaluations

Southern Nuclear Operating Company (SNC) has reviewed TSTF-425, Revision 3, and the associated model Safety Evaluation published in the July 6, 2009 Federal Register (74 FR 31996 – 32006). This review included the supporting information provided to support TSTF-425 and the safety evaluation for NEI 04-10, Revision 1. SNC has concluded that the technical basis is applicable to VEGP Units 3 and 4 and supports incorporation of this amendment in the VEGP TS.

SNC has also reviewed the safety evaluations for TSTF-563, Revision 0, and TSTF-596, Revision 2, provided to the Technical Specifications Task Force in letters dated December 4, 2018 and December 30, 2024, respectively. This review included the NRC staff's evaluation, as well as the information provided in the respective TSTF travelers. SNC has concluded that the justifications presented in each of these TSTF travelers and their NRC safety evaluations are applicable to VEGP Units 3 and 4 and justify this amendment for incorporation of the changes to the VEGP Units 3 and 4 TS.

SNC confirms that the SFCP shall contain a list of the frequencies controlled by the program, including Frequencies of Surveillances that implement the requirements of 10 CFR 50.55a(f).

SNC recognizes that TSTF-425 and TSTF-596 were not identified as applicable to NUREG-2194, Standard Technical Specifications for Westinghouse Advanced Passive 1000 (AP1000) Plants. However, it should be noted that the VEGP Units 3 and 4 TS were originally developed from NUREG-1431, Standard Technical Specifications (STS) for Westinghouse Plants. Thus, TSTF-425 and TSTF-596 can be readily adapted to the VEGP Units 3 and 4 TS through comparison to the SFCP markups identified for NUREG-1431. Furthermore, the specific criteria provided in TSTF-425 for determining exceptions to the Surveillance Frequencies to be relocated to an SFCP, as applied to NUREG-1431, can equally be applied to identify the Frequencies that cannot be relocated from the VEGP Units 3 and 4 TS to the SFCP. TSTF-563 was identified as applicable to NUREG-2194 and accordingly can be readily adapted to the VEGP Units 3 and 4 TS.

2.2 Optional Changes and Variations

The proposed amendment is consistent with the intent of the changes described in TSTF-425, TSTF-563, and TSTF-596; however, SNC proposes variations from these travelers, including variations from the NUREG-1431, Westinghouse STS markups in TSTF-425 and TSTF-596, as identified the Attachment 3. Attachment 3 also identifies the changes to TS Surveillances with Surveillance Requirement (SR) numbers that differ from those in NUREG-1431.

SNC is proposing the following variations from the TS changes described in TSTF-425, and TSTF-596 or the applicable parts of the NRC staff's proposed model safety evaluation published in the Federal Register on July 6, 2009 (74 FR 31996 – 32006) and provided in a letter to the TSTF [ML25251A241], respectively. These variations are identified and justified below, as applicable to each traveler. SNC is proposing no variations from the TS changes described in TSTF-563.

General Variations

The following general variations are proposed to the changes addressed by TSTF-425, TSTF-563, and TSTF-596:

1. VEGP Surveillance Requirements that have identical numbers to the corresponding NUREG-1431 Surveillance Requirements are not variations from TSTF-425 or TSTF-596, except for administrative variations (if any) such as formatting. These variations are administrative with no impact on the NRC proposed model safety evaluation. The VEGP TS are compared to NUREG-1431 since this was the original NUREG from which the AP1000 TS were developed.

2. VEGP TS Surveillances that are not applicable to Surveillances identified in the NUREG-1431 markups in TSTF-425 or TSTF-596 are included in the scope of this LAR. These variations are administrative, because exclusion of the proposed changes is based on applying the relocation exception criteria in Section 2.0 of TSTF-425 (except as the criterion regarding Frequencies that reference other approved programs, which is the subject of the TSTF-596 based changes), rather than the content of the Surveillance (not the Frequency). This is an administrative variation from TSTF-425 or TSTF-596 with no impact on the NRC model safety evaluation.

3. VEGP TS Surveillances that are applicable to similar Surveillances identified in the NUREG-1431 markups in TSTF-425 or TSTF-596 are included in the scope of this LAR. These variations are administrative, because exclusion of the proposed changes is based on applying the relocation exception criteria in Section 2.0 of TSTF-425 (except as the criterion regarding Frequencies that reference other approved programs, which is the subject of the TSTF-596 based changes), rather than the content of the Surveillance (not the Frequency). This is an administrative variation from TSTF-425 or TSTF-596 with no impact on the NRC model safety evaluation.

4. TSTF-425 or TSTF-596 Surveillances applicable to NUREG-1431 that are not applicable to VEGP are not included in the proposed VEGP TS. This is an administrative variation from TSTF-425 or TSTF-596 with no impact on the NRC model safety evaluation.

5. The model applications provided in TSTF-425 (in 74 *FR* 31996), TSTF-563, and TSTF-596 include an attachment for revised, clean typed TS pages reflecting the proposed changes. This request does not include such an attachment due to the number of TS pages included in this submittal and the straightforward nature of the proposed changes. Providing only mark-ups of the proposed TS changes satisfies the requirements of 10 CFR 50.90, "Application for amendment of license, construction permit, or early site permit," in that the mark-ups fully describe the changes desired. This is an administrative deviation from TSTF-425, TSTF-563, and TSTF-596 with no impact on the NRC's model safety evaluations. Because of this deviation, the contents and numbering of the attachments for this amendment request differ from the attachments specified in the model applications in TSTF-425, TSTF-563, and TSTF-596.

6. The term “Surveillance Frequency Control Program” is replaced with the commonly accepted abbreviation for this program, “SFCP,” throughout the TS and TS Bases. For example, the model applications provided in TSTF-425 and TSTF-596 include mark-ups to Surveillance Frequencies using the text "In accordance with the Surveillance

Frequency Control Program." This application proposes more succinct Frequencies to read, "In accordance with the SFCP." Additionally, this variation applies to the proposed changes to the defined terms in TS Section 1.1 associated with the adoption of TSTF-563. The abbreviation SFCP is defined in the title of the new TS Section 5.5.15, "Surveillance Frequency Control Program (SFCP)." Therefore, this variation is applicable to each of the three travelers (i.e., TSTF-425, TSTF-563, and TSTF-596) to be adopted by this amendment request.

This is an administrative deviation from TSTF-425, TSTF-563, and TSTF-596, with no impact on the NRC's model safety evaluation. The abbreviation SFCP is defined in the title of the new TS Section 5.5.15, "Surveillance Frequency Control Program (SFCP)." VEGP Units 3 and 4 are the last of SNC's eight operating units to develop an SFCP; thus, SNC has been using this terminology for many years consistent with industry use. TSTF-GG-05-01, "Writer's Guide for Plant Specific Improved Technical Specifications," (June 2005) states that "Some acronyms and initialisms are commonly used and understood. In many instances they are more commonly understood than the complete spelled out phrase. These may be used without initially defining them with the full phrase." Therefore, the use of the acronym SFCP within the proposed TS meets the TSTF-GG-05-01 guidance.

TSTF-425 Variations

The following variations are applicable specifically to TSTF-425, Rev. 3:

1. The definition of STAGGERED TEST BASIS is being retained in VEGP TS Definition Section 1.1 since this terminology is used in the Completion Time for TS 3.3.19, "DAS Manual Controls," Condition B.1, to "Perform SR 3.3.7.1." Changes to Completion Times are not the subject of this amendment request and this application of STAGGERED TEST BASIS is not proposed to be changed. This is an administrative deviation from TSTF-425 with no impact on the NRC staff's model safety evaluation dated July 6, 2009 (74 FR 31996).

2. As noted above, SNC is also incorporating changes to TSTF-425 identified in TSTF-596. TSTF-596 proposes changes that relocate the Frequency and Bases for surveillances that refer to the Inservice Testing (IST) Program and other approved programs to the SFCP. The changes addressed in TSTF-596 supersede the changes excluded by the TSTF-425 exception criterion regarding frequencies that reference other approved programs for the specific interval. Therefore, the variation from TSTF-425 to relocate Surveillance Frequencies and the associated Bases to the licensee-controlled SFCP for Frequencies that reference the IST Program is justified by discussion in TSTF-596. Similarly, the revision to the SFCP description in TS 5.5.16 is consistent with the SFCP description in TSTF-596 and is justified by the discussion in TSTF-596. This variation from TSTF-425 has no impact on the NRC staff's final safety evaluation for TSTF-596, Revision 2, dated December 30, 2024 [ADAMS Accession No. ML24362A056].

TSTF-596 Variations

The following variations are applicable specifically to TSTF-596, Rev. 2:

1. TSTF-596 includes a change to delete the definition of “Inservice Testing Program” from STS 1.1. The VEGP 3&4 TS do not currently define Inservice Testing Program; therefore, an administrative variation is made to disregard the provision to delete this term. The definition of Inservice Testing Program in Section 1.1 identifies it as the licensee program that fulfills the requirements of 10 CFR 50.55a(f). Regardless of whether the IST Program is defined in the TS or not, the IST Program is, and will be, required to be performed in accordance with 10 CFR 50.55a(f), which incorporates by reference the ASME Code and the ASME Operations and Maintenance (OM) Code. Therefore, this is an administrative variation, which has no impact on the changes described and justified in TSTF-596 or in the NRC’s Safety Evaluation for this traveler.

2. TSTF-596 provided changes to four STS Section 5.5 programs containing specific testing frequencies by replacing the periodic testing frequencies to state that they are tested at “a Frequency in accordance with the [SFCP].” Two of these programs (i.e., Primary Coolant Sources Outside Containment Program and Diesel Oil Testing Program), are not currently included in the VEGP TS. Accordingly, an administrative variation is made to disregard the provision to revise Section 5.5 for these two programs.

   Relocating frequencies for the periodic test requirements of TS Section 5.5 programs only allows for changes to the current surveillance frequencies and does not add, delete, or modify the scope or methods for performance of any of these surveillances. Accordingly, as the current VEGP TS do not include Section 5.5 programs for primary coolant sources outside containment or diesel oil testing, the changes proposed by TSTF-594 for these two regulatory programs do not infer the addition of these programs to the VEGP 3&4 TS. Furthermore, as these programs are independent of the other VEGP 3&4 Section 5.5 programs (as well as other VEGP 3&4 SRs), omission of the TSTF-596 changes to the STS Primary Coolant Sources Outside Containment Program and Diesel Oil Testing Program has no impact on the TSTF-596 justifications or the NRC Safety Evaluation for this traveler.

3. TSTF-596 provided changes to NUREG-1431, Section 1.1, Definitions, to remove the defined term INSERVICE TESTING PROGRAM. The Vogtle 3&4 TS do not include a definition of the INSERVICE TESTING PROGRAM, but instead include TS 5.5.3 identifies the Inservice Testing Program as “the licensee program that fulfills the requirements of 10 CFR 50.55a(f).” The Vogtle TS 5.5.3 IST Program description is identical to the NUREG-1431 definition of INSERVICE TESTING PROGRAM, which is proposed for deletion by TSTF-596. Therefore, the proposed change to delete the Inservice Testing Program from Vogtle TS 5.5.3 is consistent with the intent of the TSTF-596 change to delete the definition of INSERVICE TESTING PROGRAM from the NUREG-1431 Westinghouse STS.

4. TSTF-596 provided changes to the NUREG-1431, Section 5.5.10, Ventilation Filter Testing Program (VFTP), which is addressed in Section 5.5.13 of the VEGP TS. In addition to the administrative numbering difference between the NUREG-1431 STS and the VEGP TS, the difference between the descriptions of the VFTP in the STS versus that in the VEGP TS necessitates a variation from the markup of the VEGP in TSTF-425.

Whereas the brief introduction to the VFTP in NUREG-1431 only requires a single change to identify the SFCP Frequencies as an option to the event-based frequencies specified in the regulatory guidance, the organization of the VEGP TS requires separate changes for the frequencies of the high efficiency particulate air (HEPA) filter and charcoal adsorber inplace testing, the charcoal adsorber laboratory testing, and the Main Control Room Emergency Habitability System (VES) pressure drop testing. Therefore, the TSTF-596 change to allow the option of performing periodic ventilation filter testing only needs to be made once, while the corresponding change to the VEGP 3&4 program requires similar changes to be made to each of the three introductory paragraphs.

This is considered an administrative variation, as the end result is that the proposed option allows the performance of HEPA filter and charcoal adsorber inplace testing, charcoal adsorber laboratory testing, and VES pressure drop testing at a frequency determined in accordance with the SFCP, as described and justified in TSTF-596 and in the NRC's Safety Evaluation for this traveler. Therefore, this administrative variation does not change the conclusion of the NRC Safety Evaluation for TSTF-596, that the SR frequencies for the VFTP do not meet any of the exclusion criteria in the approved TSTF-425, the frequencies are fixed periodic frequencies, and the proposed changes are consistent with the intent of TSTF-425.

5. TSTF-596 provided changes to the NUREG-1431, Section 5.5.17, Control Room Envelope (CRE) Habitability Program, which is referred to as the Main Control Room Envelope (MCRE) Habitability Program in Section 5.5.12 of the VEGP TS. In addition to the administrative numbering and Program naming differences between the NUREG-1431 STS and the VEGP TS, there are minor differences between the descriptions of the air inleakage testing and the CRE/MCRE boundary testing. The differences in the current content of NUREG-1431, Section 5.5.17, CRE Habitability Program, and the VEGP 3&4 TS, Section 5.5.12, MCRE Habitability Program, include the TS number, program title, ventilation system designation [i.e., the STS CRE ventilation system is the Control Room Emergency Filtration System (CREFS) and the VEGP MCRE ventilation system is referred to as the Main Control Room Emergency Habitability System (VES)].

   These differences are administrative and do not impact adoption of the proposed changes as described and justified in TSTF-596 and in the NRC's Safety Evaluation for this traveler. Therefore, this administrative variation does not change the conclusion of the NRC Safety Evaluation for TSTF-596, that the SR frequencies for the CRE Habitability Program do not meet any of the exclusion criteria in the approved TSTF-425, the frequencies are fixed periodic frequencies, and the proposed changes are consistent with the intent of TSTF-425.

6. TSTF-596 includes a provision that the SRs being revised to remove references to the IST in the Surveillance (not the Frequency) should also include a description of the test, similar to other SRs. The VEGP 3&4 TS include six SR in which the Surveillance refers to the IST Program, four of which are revised to describe the test, similar to other SRs. These four SRs are identified below, and do not constitute a variation, as the change to describe the test is consistent with the TSTF provision:
   - SR 3.4.6.1, "Pressurizer Safety Valves,"
   - SR 3.4.14.5, "Low Temperature Overpressure Protection (LTOP),"
   - SR 3.6.9.3, "Vacuum Relief Valves," and
   - SR 3.7.1.1, "Main Steam Safety Valves (MSSVs)."

The two SRs that specify requirements to test squib valves in accordance with the IST Program are:

- SR 3.4.11.3, “Automatic Depressurization System (ADS) – Operating,” for testing of the ADS stage 4 valves, and
- SR 3.5.6.8, “In-containment Refueling Water Storage Tank (IRWST) – Operating,” for testing of the IRWST injection and containment recirculation squib valves.

To maintain the level of detail currently provided for these two SRs and remove the reference to the IST Program while providing reference to the source of the testing requirements, a variation to the TSTF provision is proposed that changes the reference to the IST Program to refer to 10 CFR 50.55a(f). Reference to §50.55a(f) is an appropriate alternate to providing a description of the Code-required testing, as 10 CFR 50.55a(f) incorporates by reference the IST requirements of the ASME OM Code, including ISTC-5260, Explosively Actuated Valves, which specifies the requirements for testing these valves. Therefore, this proposed change is consistent with the current TS and provides an acceptable variation to the TSTF-596 provision to replace the reference to the IST Program with a description of the surveillance, while maintaining the level of detail currently provided for these two SRs and removing the reference to the IST Program.

2.3 Additional Change Requests

In addition to the changes to adopt TSTF-425, TSTF-563, and TSTF-596, the LAR proposes changes to remove completed and obsolete TS requirements and TS language that is no longer applicable. These changes are to (1) delete the “At first refueling” from the Frequency of SR 3.6.6.6, for performance of the PCS flow and water coverage performance System Level OPERABILITY Test, (2) delete the “initial” inplace testing of the VES high efficiency particulate air (HEPA) filter and charcoal adsorber per TS 5.5.13.a.1 and 5.5.13.1.2, and (3) delete the exception for Unit 3 cycle 1 for VES pressure drop testing per TS 5.5.13.a.1, 5.5.13.a.2, and TS 5.5.13.a.4.

VEGP Unit 3 and Unit 4 are both operating in their second fuel cycle, so the stated Frequencies referring to the “first refueling,” “initial” testing, and “Unit 3 cycle 1” are obsolete and can be omitted.

3.0 REGULATORY SAFETY ANALYSIS

3.1 No Significant Hazards Consideration Analysis

SNC has reviewed the proposed no significant hazards consideration determinations (NSHC) published in the **Federal Register** on July 6, 2009 (74 FR 31996 – 32006) for TSTF-425, and the model applications attached to TSTF-563-A [ML18333A146] and TSTF-596-A, Revision 2 [ML25251A241]. SNC has concluded that the proposed NSHC presented in the **Federal Register** notice for TSTF-425 and the model applications for TSTF-563 and TSTF-596 are applicable to VEGP Units 3 and 4 and are used as the basis for the NSHC provided as Attachment 6 to this amendment request, which satisfies the requirements of 10 CFR 50.91(a).

3.2 Conclusion

In conclusion, based on the considerations discussed above, (1) there is reasonable assurance that the health and safety of the public will not be endangered by operation in the proposed manner, (2) such activities will be conducted in compliance with the Commission's regulations, and (3) the issuance of the amendment will not be inimical to the common defense and security or to the health and safety of the public.

4.0 ENVIRONMENTAL CONSIDERATION

SNC has reviewed the environmental evaluation included in the model safety evaluations associated with TSTF-425, TSTF-563, and TSTF-596. SNC has concluded that the NRC staff findings presented in the evaluations are applicable to VEGP Units 3 and 4.

The proposed change would change a requirement with respect to installation or use of a facility component located within the restricted area, as defined in 10 CFR 20, or would change an inspection or surveillance requirement. However, the proposed change does not involve (i) a significant hazards consideration, (ii) a significant change in the types or significant increase in the amounts of any effluents that may be released offsite, or (iii) a significant increase in individual or cumulative occupational radiation exposure. Accordingly, the proposed change meets the eligibility criterion for categorical exclusion set forth in 10 CFR 51.22(c)(9).

Therefore, pursuant to 10 CFR 51.22(b), no environmental impact statement or environmental assessment need be prepared in connection with the proposed change.

Attachment 2 to NL-26-0017
Documentation of PRA Technical Adequacy

License Amendment Request to Revise Technical Specifications to Include a Surveillance Frequency Control Program

1.0 Overview

This attachment addresses the assessment of the technical adequacy of the probabilistic risk assessment (PRA) that will be used to support the implementation of the Surveillance Frequency Control Program at Vogtle Units 3&4. The implementation of the Surveillance Frequency Control Program (also referred to as Technical Specification Initiative 5b) will follow the methodology provided in NEI 04-10, Revision 1 (Reference 2) in evaluating proposed surveillance test interval (STI; also referred to as "surveillance frequency") changes. NEI 04-10 provides guidance for implementation of a generic TS improvement that establishes licensee control of surveillance test frequencies for the majority of the TS surveillances. Since the scope of potential Initiative 5b applications (i.e., individual Surveillance Test Interval change evaluations) is very broad and the impact of various assumptions differs from application to application, the NEI 04-10 methodology provides that PRA attributes be addressed with each STI interval change and documented in the supporting evaluation. The analysis process is described in detail in NEI 04-10 and implemented by plant-specific procedures.

The NEI 04-10 methodology requires plants to evaluate the adequacy of the PRA models used for this application following the guidance provided in Regulatory Guide 1.200, (References 3 and 4). When an STI cannot be modeled in the plant PRA, a qualitative or bounding analysis is performed to provide justification for the acceptability of the proposed STI revision.

In support of adoption of Technical Specification Initiative 5b (Reference 1), this attachment demonstrates technical adequacy of the VEGP 3 &4 PRA Model to be used as the basis for the VEGP SFCP. The following sections describe the scope of risk contributors addressed by the PRA model, and aspects of model development, including technical acceptability and relevant impacts of assumptions and approximations.

2.0 PRA Model Scope

The proposed application utilizes PRA models for internal events, internal flooding, internal fire and seismic hazards, developed with capability for quantitative risk assessment to evaluate risk contributors (hazard groups and modes of operation). The scope of the integrated VEGP Units 3&4 PRA Model consists of full-scope Level 1 and Level 2 “at-power” PRA that includes these hazard groups to support decision-making for the SFCP. Other external hazards were screened from development of a quantifiable PRA model in accordance with Part 6 of Reference 5; screening analysis insights will be used to qualitatively evaluate risk impacts of these hazards with respect to STI changes.

All parts of the PRA models identified in the following subsections are applicable to the surveillance frequency control process. The methodology for using the PRA model to support risk-informed Technical Specifications frequencies is described in NEI 04-10 (Reference 2).

2.1 Internal Events

The PRA model supporting the application for a SFCP for VEGP 3 & 4 is the 2025 PRA model of record as documented in calculation SV0-PRA-GSC-322 (Reference 6) and SV0-PRA-GSC-376 (Reference 7), and the related PRA supporting documents.

Quantification of the VEGP Units 3&4 internal events PRA generated the following CDF/LERF results:

CDF (/reactor-year)	LERF (/reactor-year)
3.59E-07	3.06E-08

2.2 Internal Flooding

An internal flooding PRA was performed based on VEGP Units 3&4 site-specific information.
Fluid systems that are potential flood sources were identified for the internal flooding PRA. For these systems, information on pipe length, pipe diameter and associated flood area was collected. The piping information obtained was used in addition to industry pipe rupture frequency data to quantify the initiating event frequencies for a specific flood source system in a specific room.
Flood source capacity along with door and penetration information were used to identify propagation pathways for each flood scenario to determine impacted PRA-related structures, systems, and components (SSCs).

Flood scenarios were screened and identified for further evaluation. Each of the flood scenarios were characterized by the flood area, flood source, failure mechanism, potential failure modes of the flood source, potential impact on components in the flood area and areas along the propagation path and plant impact. Four distinct failure modes (spray, flood, major flood, high-energy line break (HELB)) were considered in characterizing flood scenarios.
Quantification of the VEGP Units 3&4 internal flooding PRA generated the following CDF/LERF results (Reference 8):

CDF (/reactor-year)	LERF (/reactor-year)
1.98E-07	5.05E-08

2.3 Internal Fire

The at-power fire PRA was developed following NUREG/CR-6850 and supplemental methodology as endorsed by the NRC. The fire PRA analysis was based on the Level 1 and Level 2 release categories from the internal events modified to account for fire specific failures and multiple spurious actuations. The plant fire PRA model includes: 1) Plant Boundary Definition and Partitioning, 2) Component Selection, 3) Cable selection, 4) Qualitative Screening, 5) Fire Risk Model, 6) Ignition Frequency Development, 7) Quantitative Screening, 8) Scoping Fire Modeling, 9) Detailed Circuit Failure Analysis, 10) Circuit Failure Likelihood Analysis, 11) Detailed Fire Modeling, 12) Fire Human Reliability Analysis, 13) Fire Risk Quantification, 14) Uncertainty and Sensitivity Analysis, and 15) detailed documentation of the quantification process and results
Quantification of the VEGP Units 3&4 fire PRA generated the following CDF/LERF results (Reference 9):

Unit	CDF (/reactor-year)	LERF (/reactor-year)
3	7.37E-07	2.33E-07
4	7.38E-07	2.32E-07

2.4 Seismic

The VEGP Units 3&4 Seismic PRA (SPRA) follows industry recognized guidance to estimate the frequencies of core damage and Level 2 release categories. The SPRA assembles the results of the seismic hazard analysis, fragility analysis, and the seismic accident sequence model into a large fault tree including seismic initiating events (ground motion intervals) with separate top events for CDF and Level 2 release categories. The quantification is a convolution of the seismic hazard fragility for each screened in SSCs including contributions of random hardware and human failures to contribute to the CDF and Level 2 results.

Quantification of the VEGP Units 3&4 Seismic PRA generated the following CDF/LERF results (Reference 10):

CDF (/reactor-year)	LERF (/reactor-year)
3.60E-07	3.35E-07

2.5 Summary of Baseline CDF & LERF

CDF/LERF (/reactor-year) at VEGP 3&4 from all PRA-modeled hazard groups used for the SFCP application are summarized in the table below:

Unit 3		
Source	CDF	LERF
Internal Events	3.59E-07	3.06E-08
Flood	1.98E-07	5.05E-08
Fire	7.37E-07	2.33E-07
Seismic	3.60E-07	3.35E-07
Total Unit 3	1.65E-06	6.49E-07

Unit 4		
Source	CDF	LERF
Internal Events	3.59E-07	3.06E-08
Flood	1.98E-07	5.05E-08
Fire	7.38E-07	2.32E-07
Seismic	3.60E-07	3.35E-07
Total Unit 4	1.65E-06	6.48E-07

2.6 Non-Modeled Hazards

2.6.1 Other External Events

The at-power External Hazards PRA Notebook (Reference 11) concluded that all identified external hazards (excluding Internal Flooding, Seismic, and Internal Fire) were individually screened out with CDFs between 1E-09 per year to approximately 1E-14 per year. Qualitative and quantitative screening of external hazards was performed in accordance with the requirements of the ASME/ANS PRA Standard (ASME/ANS RA-Sa-2009) for this external events analysis.

The full-power, integrated PRA model used to support the SFCP does not include explicit consideration of external events such as severe winds, and external flooding. In the absence of quantifiable PRA models for all external hazards, NEI 04-10 allows STI change evaluations to be performed with review of qualitative impacts.

2.6.2 Shutdown Risk Considerations

SNC does not maintain a shutdown PRA model for VEGP 3&4. Consistent with the NEI 04-10, Revision 1 (Reference 2) guidance, qualitative information must be developed that supports the acceptability of the STI change with respect to the shutdown risk or it must be screened as not having an impact on the CDF and large early release frequency (LERF) metrics.

Shutdown risk is evaluated using a Shutdown Defense-In-Depth (DID) model (Reference 12) developed to support implementation of NUMARC 91-06 (Reference 22). Plant procedures establish shutdown DID risk assessment to support the requirements of Article (a)(4) of 10CFR50.65. This model defines key safety functions applicable to low power and shutdown conditions and associated status tree decision criteria for the identified plant operating states.

3.0 Technical Adequacy of Vogtle 3&4 PRA Models

A structured approach to establishing and maintaining the acceptability, adequacy and fidelity of the PRA models for is used for VEGP 3&4. This approach includes both a proceduralized PRA maintenance and update process, supplemented with independent peer reviews. The following information describes this approach as it applies to the VEGP 3&4 PRA.

3.1 PRA Maintenance and Update

Southern Nuclear (SNC) employs a multi-faceted approach to establishing and maintaining the technical adequacy and fidelity of PRA models. This approach includes proceduralized PRA maintenance (Reference 13 and Reference 14) and an established process for the use of self-assessments and independent peer reviews. PRA model changes are evaluated with respect to Capability Category (CC) II of the ASME/ANS PRA Standard (ASME/ANS RA-Sa-2009) as endorsed by Regulatory Guide 1.200, Revision 3 (Reference 4) to the extent possible for an early operational plant. Any model maintenance activity that constitutes a model upgrade will be identified through this process (Reference 13 and Reference 14.)

PRA maintenance is performed on a regularly scheduled basis. This process involves a periodic review and update cycle to model any changes in the plant design or operation. Plant modifications and procedure changes are routinely reviewed to determine if they impact the PRA. Changes that warrant a prompt PRA update are incorporated as soon as practical consistent with the significance in the base model and expected impact on PRA applications. These processes help assure that the VEGP 3&4 PRA reflects the as-built, as-operated plant within the limitations of the PRA methodology.

As part of the PRA evaluation for each STI change request, a review of open items in the maintenance database will be performed and an assessment of the impact on the results of the application will be made prior to presenting the results of the risk analysis to the Integrated Decision-making Panel (IDP). If a non-trivial impact is expected, then the analysis will include performance of additional sensitivity studies or PRA model changes to confirm the risk impact is acceptable.

3.2 Key PRA Assumptions and Sources of Uncertainty

The current VEGP 3&4 PRA model of record includes documentation of a wide range of generic and plant specific contributors to modeling uncertainties (Reference 23) and supporting notebooks References 6 through 11); these contributors were examined to aid in the decision-making process of potential applications. With respect to model completeness, evaluation of proposed STI changes will reference bounding analyses or update the model to incorporate the missing scope elements, as appropriate. Lack of detail in the model generally tends toward a conservative bias which may be removed (by adding detail) if so, required by a particular application. The completeness consideration is also recognized and addressed in the NEI 04-10 methodology.

The Technical Specifications Initiative 5b process is a risk-informed process with the PRA model results providing one of the inputs to determine if an STI change is acceptable. The NEI 04-10 methodology recognizes that a key area of uncertainty for this application is the failure rate utilized in the determination of the STI extension impact. Therefore, the methodology requires the performance of selected sensitivity studies on the standby failure rate of the component(s) of interest for the STI assessment. Other relevant key assumptions will depend on the SSCs being considered for an STI revision. For each proposed STI change, PRA assumptions will be reviewed and documented to identify potential impacts.

3.3 Requirements Related to PRA Scope and Quality

The VEGP 3 & 4 PRA model has been revised substantially since its origination and the various model revisions subjected to careful evaluation and review to assure the scope and quality of the model are adequate for planned risk informed applications. The PRA model supporting the application reflect the as-built, as-operated plant including Structures, Systems and Components (SSCs) at VEGP 3&4 (e.g., front-line, support safety-related systems) modeled in the plant specific PRA models.

The VEGP 3 & 4 PRA models of record are maintained as controlled documents and are updated on a periodic basis to represent the as-built, as-operated plant (Reference 14, and Reference 13). Initial peer reviews performed for each of the PRA models documented assessment against Regulatory Guide (RG) 1.200, Revision 2 (Reference 3) Follow on peer reviews assessed VEGP PRA models against RG 1.200, Revision 3 (Reference 4). Subsequent peer reviews to provide independent assessment for closure of peer review findings in accordance with NEI guidance (Appendix E, Reference 15). The VEGP 3 & 4 PRA models are sufficiently robust and suitable for use in risk-informed processes such as regulatory decision-making. The peer reviews that have been conducted and the resolution of findings from those reviews demonstrate that the internal events, internal flooding, fire, and seismic PRA models have been performed in a technically sound manner. The assumptions and approximations used in development of the PRA have also been reviewed and are appropriate for their application.

The PRA analysis for VEGP Units 3&4 consists of full-scope Level 1 and Level 2 “at-power” PRA. As described in the Updated Final Safety Analysis (UFSAR) (Reference 24), detailed PRA models have been developed for internal events, internal floods, internal fire, and seismic hazards. Other external hazards were screened from detailed evaluation in accordance with the qualitative and quantitative screening requirements of Part 6 of Reference 5.

3.4 Technical Adequacy - Internal Events PRA Model

The VEGP 3 & 4 internal events PRA model (References 6 and 7) is an at-power model (i.e., it directly addresses plant configurations during plant modes 1 and 2 of reactor operation). The model includes both at-power internal events core damage frequency (CDF) and large early release frequency (LERF).

SNC considers the Vogtle 3&4 Internal Events PRA as adequate to support the requested application. A Peer Review was conducted in 2013 evaluating technical elements against Capability Category II (CC-II) of ASME RA-Sa-2009 (Reference 5). Peer review reported results summarized in the following sections can be used to reliably evaluate the technical adequacy of the PRA against PRA standard and regulatory guidance (RG 1.200, Rev. 3, Reference 4).

3.4.1 RG 1.200 Previous and Current Peer Reviews - Internal Events PRA

The At-Power Internal Events PRA was initially peer reviewed against Part 2 of ASME/ANS RA-Sa-2009 (Reference 5) in 2013. Since the 2013 peer review (Reference 16), the Vogtle Units 3&4 plant At-Power Internal Events PRA has been updated to address several peer review Findings & Observations (F&Os), to the extent possible for an early operational plant. Follow on work addressed further model improvement focused on as-built, as-operated plant information.

Subsequently, the VEGP 3 & 4 Internal Events PRA Model was subject to a closure and focused-scope industry peer review (Reference 25) conducted by the Pressurized Water Reactor Owners' Group (PWROG) following self-assessment in 2025. The peer review and subsequent independent assessments demonstrated that the VEGP 3 & 4 Internal Events PRA substantially met the ASME/ANS PRA standard RA-Sa-2009.)

3.4.2 Resolution of F&Os from RG 1.200 Internal Events Peer Review

The VEGP 3 & 4 PRA Internal Events Model References 6 and 7 was approved and incorporated changes, as applicable, to support the resolutions of the 2025 peer review findings. The findings from 2025 were closed by an independent assessment conducted (Reference 25), with a number of open findings remaining either out of scope (not applicable) or not fully resolved to meet Category II or better. The findings that remain open are summarized in Table E2.1 documenting CC II and associated justifications relative to SFCP. The closure assessment was conducted in accordance with Appendix E to NEI 17-07 (Reference 15), the NRC endorsed PRA peer review guidance consolidated for different hazard groups. The independent assessment is documented in the closure report (Reference 25).

The open findings summarized in Table E2-1 have been reviewed in support of the requested application.

Table E2-1 Summary of Peer Review Findings - Internal Events

F&O Number	F&O Details	Associated Supporting Requirement (SR)	Peer Review Team Assessment	Impact on Application
IE-A3-01	A number of SRs were not achievable in the IE element because insufficient plant-specific data are available for update.	IE-A3	SR MET	Generic Initiating event data includes operational data from Vogtle Units 1 & 2. Update to include plant specific operating experience is not expected to significantly alter the risk profile. Without an update, results are potentially non-conservative, but STI evaluation should not be impacted.
		IE-A7	SR MET	
		IE-A9	SR Met: (CC II)	
		IE-C2	Not Met	
		IE-C4	Not Met	
DA-C2-01	Site specific guidance establishes methodologies for performing a data update. Due to the early operational state of the plant, sufficient operating experience is not yet available to support this task.	DA-C2	Not Met	Generic availability and reliability data utilized. Update to include plant specific operating experience is not expected to significantly alter the risk profile. Without an update, results are potentially non-conservative, but STI evaluations should not be impacted.
		DA-C3	Not Met	
		DA-C4	Not Met	
		DA-C5	Not Met	
		DA-C6	Not Met	
		DA-C7	Not Met	
		DA-C8	Not Met	
		DA-C9	Not Met	
		DA-C11	Not Met	
		DA-C13	Not Met	
		DA-C14	Not Met	
		DA-D1	SR MET: (CC I)	
		DA-D4	SR MET: (CC I)	
QU-D4-01	No similar plants exist for comparison.	QU-D4	SR MET: (CC I)	Review of PRAs from similar plants, if available, could reveal information that could alter the risk profile. Such information is not expected to impact STI evaluations.

Table E2-1 Summary of Peer Review Findings - Internal Events				
F&O Number	F&O Details	Associated Supporting Requirement (SR)	Peer Review Team Assessment	Impact on Application
LE-E2-01	Model results show that LERF is dominated by the two phenomena of 1) hydrogen Deflagration-to-Detonation Transition, and 2) diffusion flame. This SR is characterized as CC I because of the conservative nature of the assumptions leading to early containment failure.	LE-D1	SR MET: (CC I)	This is a conservative approach that impacts LERF. STI evaluations should not be impacted.
		LE-E2	SR MET: (CC I)	

3.5 Technical Adequacy - Internal Flooding PRA Model

The current internal flood model (Reference 8) was integrated into the PRA model of record to quantify flood-induced scenarios to determine their importance to plant risk. As part of quantification, the effects of flood-induced failures were combined with the effects of other types of equipment failures or unavailability. This includes the effects caused by flooding and failures coincident with the flooding due to maintenance unavailability, independent equipment faults and other credible causes. The direct and indirect effects of the flood were included in the quantification.

SNC considers the Vogtle 3&4 Internal Flooding PRA as adequate to support the requested application. A Peer Review was conducted in 2013 evaluating technical elements against Capability Category II (CC-II) of ASME RA-Sa-2009 (Reference 5). Peer review reported results summarized in the following sections can be used to reliably evaluate the technical adequacy of the PRA against PRA standard and regulatory guidance (RG 1.200, Rev. 3, Reference 4).

3.5.1 RG 1.200 Previous and Current Peer Reviews - Internal Flooding PRA

The At-Power Internal Flooding PRA was initially peer reviewed against Part 3 of ASME/ANS RA-Sa-2009 in 2013. Since the 2013 peer review, the Vogtle Units 3 & 4 plant At-Power Internal Flooding PRA was updated to address peer review F&Os, to the extent possible for an early operational plant. Follow on work addressed further model improvement focused on as-built, as-operated plant information. It should be noted that certain supporting requirements were deemed not applicable because the AP1000 plant design is a modular single unit plant; for example, there is no need to include shared areas, systems or structures in the plant partitioning task. An additional focused scope review was performed in 2020 (Reference 20).

Following further F&O closure efforts, model maintenance, and self-assessment in September 2025, the VEGP 3 & 4 Internal Flooding PRA Model was subject to a focused-scope industry peer review conducted by the Pressurized Water Reactor Owners' Group (PWROG) that included a closure review to address previous findings. The peer review and subsequent independent assessments demonstrated that the VEGP 3 & 4 Internal Flooding PRA met the ASME/ANS PRA standard RA-Sa-2009 (Reference 5) at Capability Category (CC) II or better.

3.5.2 Resolution of F&Os from RG 1.200 Internal Flooding Peer Review

The VEGP 3 & 4 PRA internal flooding model (Reference 8) was approved and incorporated changes, as applicable, to support the resolutions of the 2025 peer review findings. The findings from 2025 were closed by an independent assessment conducted (Reference 25.) No findings remain open. The closure assessment was conducted in accordance with Appendix E to NEI 17-07 (Reference 15). The independent assessment is documented in the closure report (Reference 25).

3.6 Technical Adequacy - Fire PRA Model

The fire PRA model (Reference 9) is built to integrate with the internal events PRA model using fire PRA logic models (plant response model for CDF and LERF) and database inputs comprising identified fire scenarios, frequencies, and target sets. CDF and LERF results include contribution from all credible fire sources and unscreened fire compartments within the fire PRA global plant analysis boundary. Multi-unit impacts for Vogtle 3 & 4 Fire PRA are limited.

- The AP1000 plant is a single independent plant. Unit 3 shares a switchyard with Vogtle Units 1 and 2. The Unit 4 switchyard is separate. The transformers for Unit 3 and 4 are separate but in close proximity.
- The plant does not share systems or components with another plant. Unit 3 and Unit 4 are located within close proximity to one another; however, the units do not share any components or cables. The separation of the standard plant units is a fundamental design criterion of the AP1000 plant units.
- The other operating units at the site are separate from the AP1000 plant units aside from the switchyard.

Elements from the ASME standard were included in the scope of self-assessments and peer reviews: Plant Partitioning (PP), Equipment Selection (ES), Cable Selection and Location (CS), Qualitative Screening (QLS), FPRA Plant Response Model (PRM), Fire Scenario Selection and Analysis (FSS), Ignition Frequency (IGN), Circuit Failures (CF), Human Reliability Analysis (HRA), Fire Risk Quantification (FQ), and Uncertainty and Sensitivity Analysis (UNC). Reference to these elements leads to a consistent application of industry consensus methods and data sources in FPRA development and model reviews.

SNC considers the Vogtle 3&4 Fire PRA as adequate to support the requested application. A Peer Review was conducted in 2017 evaluating technical elements against Capability Category II (CC-II) of ASME RA-Sa-2009 (Reference 5). Peer review reported results summarized in the following sections can be used to reliably evaluate the technical adequacy of the PRA against PRA standard and regulatory guidance (RG 1.200, Rev. 3, Reference 4).

3.6.1 RG 1.200 Previous and Current Peer Reviews - Fire PRA

The At-Power Fire PRA was initially peer reviewed against Part 4 of ASME/ANS RA-Sa-2009 (Reference 5) in 2017. Since the 2017 peer review (Reference 17), the Vogtle Units 3&4 plant At-Power Fire PRA has been updated to address several peer review F&Os, to the extent possible for an early operational plant. Follow on work addressed further model improvement focused on as-built, as-operated plant information. Subsequently, the VEGP 3 & 4 Fire PRA Model was subject to a full-scope industry peer review in 2020 (Reference 21).

Following further F&O closure efforts, model maintenance, and self-assessment in September 2025, the VEGP 3 & 4 Fire PRA Model was subject to a focused-scope industry peer review conducted by the Pressurized Water Reactor Owners' Group (PWROG) which included a closure review to address previous findings. The peer review and subsequent independent assessments demonstrated that the VEGP Fire PRA substantially met the ASME/ANS PRA standard RA-Sa-2009 (Reference 5) at Capability Category (CC) II or better.

3.6.2 Resolution of F&Os from RG 1.200 Internal Fire Peer Review

The VEGP PRA Fire Model (Reference 9) was approved and incorporated changes, as applicable, to support the resolutions of peer review findings. The findings were closed by an independent assessment conducted in 2025 (Reference 25.) The one finding that remains open is summarized in Table E2-2 and associated justifications relative to the SFCP. The closure assessment was conducted in accordance with Appendix E to NEI 17-07 (Reference 15). The independent assessment is documented in the closure report (Reference 25.)

Table E2-2 Summary of Peer Review Findings - Internal Fire				
F&O Number	F&O Details	Associated Supporting Requirement (SR)	Peer Review Team Assessment	Impact on Application
06-001	Corrections are indicated for an event tree that performs calculations for the non-suppression probability to ensure consistency with data sources.	FSS-C7	SR MET	The values affected are occupancy and maintenance factors, reliability values supporting fire suppression systems, and a typographical error. The associated Fire Scenario Selection notebook attachment was updated to address discrepancies identified. No impact for the SFCP application.

3.7 Technical Adequacy - Seismic

The Seismic PRA model (Reference 10) uses the internal events PRA model as the starting point to develop a seismic-PRA systems model, adding screened in SSCs while maintaining consistency with the internal events systems model regarding cause and effect relationships of the failures. The resulting plant response analysis and quantification encompasses the major seismic PRA technical elements for hazard treatment, fragility treatment and model logic. These elements from the ASME standard were included in the scope of self-assessments and peer reviews to establish technical adequacy of methodologies and their application.

SNC considers the Vogtle 3&4 Seismic PRA as adequate to support the requested application. A Peer Review was conducted in 2017 evaluating technical elements against Capability Category II (CC-II) of ASME RA-Sa-2009 (Reference 5). Peer review reported results summarized in the following sections can be used to reliably evaluate the technical adequacy of the PRA against PRA standard and regulatory guidance (RG 1.200, Rev. 3, Reference 4).

3.7.1 RG 1.200 Previous and Current Peer Reviews - Seismic

The At-Power Seismic PRA (SPRA) was initially peer reviewed against Part 4 of ASME/ANS RA-Sa-2009 (Reference 5) in 2017. Since the 2017 peer review follow on work focused on resolution of peer review F&Os and incorporation of as-built, as-operated plant information followed by an independent assessment was performed to close SPRA findings and reassess supporting requirements not met to Capability Category (CC) II or better (Reference 18).

Following further F&O closure efforts, model maintenance, and self-assessment in September 2025, the VEGP 3 & 4 SPRA was subject to a focused-scope industry peer review conducted by the Pressurized Water Reactor Owners' Group (PWROG) which included a closure review to address previous findings. The peer review and subsequent independent assessments demonstrated that the VEGP PRA substantially met the ASME/ANS PRA standard RA-Sa-2009 (Reference 5) at CC II or better.

3.7.2 Resolution of F&Os from RG 1.200 Seismic Peer Review

The VEGP PRA Seismic Model (Reference 10) was approved and incorporated changes, as applicable, to support the resolutions of the 2025 peer review findings. The findings from 2025 were closed by an independent assessment conducted in 2025 (Reference 26), with no open findings remaining. The findings that remain open are summarized in Table E2-3 and associated justifications relative to the SFCP.

The closure assessment was conducted in accordance with Appendix E to NEI 17-07 (Reference 15).

3.8 Technical Adequacy - Other External Hazard Screening

The information provided in this section demonstrates that the VEGP 3 & 4 other external events screening meets expectations for PRA scope and technical adequacy as presented in NRC RG 1.200, Revision 3 (Reference 4). Conservative analysis has been demonstrated through

progressive screening and application of qualitative and quantitative criteria, with results documented in Reference 11.

The ASME/ANS PRA standard (Reference 5) contains a total of 17 numbered supporting requirements for High Level Requirements for Other External Hazards: Requirements for Screening and Conservative Analysis. During development and self-assessment, two additional SRs supporting high winds (WHA-1 and WHA-3) were determined to meet the ASME standard with respect to tornado analysis and frequency of high straight winds. The hazards were screened via this analysis, with the exception of two hazards, Internal Flooding and Seismic, documented as "Not Screenable," and addressed separately from this analysis via a full scope at-power PRA

3.8.1 RG 1.200 Previous and Current Peer Reviews - Other External Hazard Screening

The At-Power External Events analysis was peer reviewed against Part 6 of ASME/ANS RA-Sa-2009 in 2016. Since the 2016 peer review (Reference 19), the Vogtle Units 3 & 4 At-Power external events PRA has been updated to address peer review F&Os. Other significant external events with potential impact on risk are evaluated in accordance with the PRA standard to support the application.

The VEGP 3 & 4 Other External Hazard Screening analysis was subject to a focused-scope industry peer review conducted by the Pressurized Water Reactor Owners' Group (PWROG) following self-assessment in 2025. The peer review and subsequent independent assessments demonstrated that the VEGP 3 & 4 Other External Hazard Screening substantially met the ASME/ANS PRA standard RA-Sa-2009 (Reference 5) at Capability Category (CC) II or better.

3.8.2 Resolution of F&Os from RG 1.200 Other External Hazard Screening Peer Review

The VEGP 3 & 4 Other External Hazard Screening (Reference 11) was approved and incorporated changes, as applicable, to support the resolutions of the 2025 peer review findings. The findings from 2025 were closed by an independent assessment conducted (Reference 25), with no open findings remaining. The closure assessment was conducted in accordance with Appendix E to NEI 17-07 (Reference 15). The independent assessment is documented in the closure report (Reference 25.)

3.9 General Conclusions Regarding PRA Capability

The VEGP 3&4 PRA technical capability evaluations and the maintenance and update processes described above provide a robust basis for concluding that the VEGP 3&4 PRA is suitable for use in risk-informed processes such as that proposed for the implementation of the Surveillance Frequency Control Program.

4.0 References

1. Technical Specification Task Force Improved Standard Technical Specifications Change Traveler TSTF-425, Revision 3, "Relocate Surveillance Frequencies to Licensee Control - RITSTF Initiative 5b," dated March 18, 2009.
2. Nuclear Energy Institute (NEI) 04-10, "Risk-Informed Technical Specifications Initiative Sb, Risk-Informed Method for Control of Surveillance Frequencies" (ML071360456), Revision 1, April 2007
3. Regulatory Guide (RG) 1.200, "An Approach for Determining the Technical Adequacy of Probabilistic Risk Assessment Results for Risk-Informed Activities," Revision 2, March 2009.
4. RG 1.200, Revision 3, "An Approach for Determining the Technical Adequacy of Probabilistic Risk Assessment Results for Risk-Informed Activities."
5. ASME/ANS RA-Sa-2009, Addenda to ASME/ANS RA-S-2008 Standard for Level 1/Large Early Release Frequency Probabilistic Risk Assessment for Nuclear Power Plant Applications
6. SV0-PRA-GSC-322, Revision 3, “AP1000® Plant At-Power Internal Events PRA, Quantification Notebook.”
7. SV0-PRA-GSC-376, Revision 2, “AP1000® Plant At-Power Internal Events PRA, Level 2 Quantification Results Notebook”
8. SV0-PRA-GSC-326, Revision 1, “AP1000® Plant At-Power Internal Flooding PRA, Quantification and Results Notebook.”
9. SV0-PRA-GSC-370, Revision 2, “AP1000® Plant At-Power Fire PRA, Quantification and Results Notebook.”
10. SV0-PRA-GSC-334, Revision 1, “AP1000® Plant At-Power Seismic PRA, Quantification Notebook.”
11. SV0-PRA-GSC-335, Revision 2, “AP1000® Plant At-Power External Hazards PRA Notebook.”]
12. SV0-PRA-GSC-383, Revision A, “AP1000® Plant PRA, Shutdown Defense-In-Depth Risk Monitor Model Notebook.”
13. RIE-001, Revision 17.0, “Generation and Maintenance of Probabilistic Risk Assessment Models and Associated Updates.”
14. APP-GW-GMP-025, Revision B, AP1000® Plant Probabilistic Risk Assessment Maintenance and Update Procedure
15. Nuclear Energy Institute (NEI) Topical Report (TR) NEI 17-07, " Performance of PRA Peer Reviews Using the ASME/ANS PRA Standard," Revision 2, August 2019.
16. AP1000 PRA Peer Review Report Using ASME PRA Standard Requirements, November 2013
17. Peer Review – Fire Peer Review of the AP1000 Internal Fire Probabilistic Risk Assessment for V.C Summer & Vogtle, June 2017
18. Report #05214-RPT-01, Seismic PRA Fact and Observation Independent Assessment & Focused-scope Peer Review, Vogtle Electric Generating Plant (Units 3&4), August 2020.
19. Vogtle 3 and 4 And V. C. Summer 2 and 3 Other External Hazards PRA Peer Review Report Using ASME/ANS PRA Standard Requirements, October 2016
20. V34-RIE-IF-U00-006, Version 1.0, “2020 Appendix X F&O Closure and Focused Scope Peer Review for Plant Vogtle 2&4 Internal Flooding PRA Model”, September 2020
21. P3512-1000-01, Rev. 0, “Peer Review of the Vogtle 3&4 Internal Fire Probabilistic Risk Assessment, July 2020

22. NUMARC 91-06, "Guidelines for Industry Actions to Address Shutdown Management," December 1991.
23. SV0-PRA-GSC-357, Revision 1, “AP1000® Plant At-Power Internal Events PRA, Uncertainties Notebook.”
24. VEGP 3&4 UFSAR,” Chapter 19, Revision 12.1, July 2023
25. PWROG-25030-P, Revision 0, “Vogtle Unit 3&4 F&O Closure and Focused Scope Peer Review Report,” November 2025
26. PWROG-25031-P, Revision 0, “Vogtle 3&4 SPRA F&O Closure and FSPR,” December 2025.

Attachment 3 to NL-26-0017
Identification of Changes, Differences and Variations
from Standard Technical Specifications Mark-ups

License Amendment Request to Revise Technical Specifications
to Include a Surveillance Frequency Control Program

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
Shutdown Margin (SDM)	3.1.1.1	3.1.1.1	X		None.
Core Reactivity	3.1.2.1	3.1.2.1	X		None.
Rod Group Alignment Limits	3.1.4.1	3.1.4.1	X		None.
Rod Group Alignment Limits	3.1.4.2	3.1.4.2	X		None.
Shutdown Bank Insertion Limits	3.1.5.1	3.1.5.1	X		None.
Control Bank Insertion Limits	3.1.6.2	3.1.6.2	X		None.
Control Bank Insertion Limits	3.1.6.3	3.1.6.3	X		None.
PHYSICS TESTS Exceptions – MODE 2	3.1.8.1	3.1.8.2	X		Numbering difference only.
PHYSICS TESTS Exceptions – MODE 2	3.1.8.2	3.1.8.3	X		Numbering difference only.
PHYSICS TESTS Exceptions – MODE 2	3.1.8.3	3.1.8.4	X		Numbering difference only.
Chemical and Volume Control System (CVS) Demineralized Water Isolation Valves and Makeup Line Isolation Valves	3.1.9.1	n/a 50.55a(f)		X	Variation #2 (administrative): Vogtle 3&4 SR 3.1.9.1 is to verify two CVS demineralized water isolation valves and two CVS makeup line isolation valves stroke closed. This Surveillance is to be performed at a Frequency specified by the Inservice Testing (IST) Program. NUREG-1431 does not have a similar SR to this Vogtle 3&4 SR, which would be tested in accordance with 10 CFR 50.55a(f). The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
CVS Demineralized Water Isolation Valves and Makeup Line Isolation Valves	3.1.9.2	3.6.3.5		X	Variation #3 (administrative): Vogtle 3&4 SR 3.1.9.2 is to verify closure time of each CVS makeup isolation valve is within limits on an actual or simulated actuation signal. This Surveillance is to be performed at a Frequency specified by the Inservice Testing (IST)

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					Program. This SR is functionally analogous to NUREG-1431, SR 3.6.3.5, to verify the isolation time of each automatic power operated containment isolation valve is within limits. The proposed change to relocate the Frequency to the SFCP is consistent with the markup of NUREG-1431, SR 3.6.3.8, in TSTF-596.
CVS Demineralized Water Isolation Valves and Makeup Line Isolation Valves	3.1.9.3	3.6.3.8	X		Variation #3 (administrative): Vogtle 3&4 SR 3.1.9.3 is to verify each CVS demineralized water isolation valve actuates to the isolation position on an actual or simulated actuation signal. This Surveillance is to be performed at a 24-month Frequency. NUREG-1431 has a similar SR to verify each automatic containment isolation valve that is not locked, sealed or otherwise secured in position, actuates to the isolation position on an actual or simulated actuation signal. The proposed change to relocate the Frequency to the SFCP is consistent with the markup of NUREG-1431, SR 3.6.3.8, in TSTF-425.
Rod Withdrawal Test Exception – MODE 5	3.1.10.1	LCO 3.4.7 Note 1.a	X		Variation #2 (administrative): Vogtle 3&4 SR 3.1.10.1 is to verify boron concentration of the reactor coolant system is greater than the all rods out (ARO) boron concentration providing $k_{eff} < 0.99$. This Surveillance is to be performed at a 12-hour Frequency. This SR is functionally analogous to NUREG-1431, LCO 3.4.7 Note 1.a, which prohibits removal from service of an RHR pump of the loop in operation for ≤ 1 hour per 8 hour period provided, “No operations are permitted that would cause introduction of coolant into the RCS with boron concentration less than required to meet the SDM of LCO 3.1.1. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
Heat Flux Hot Channel Factor ($F_Q(Z)$) (Constant Axial Offset Control (CAOC) W(Z) Methodology)	3.2.1.3	3.2.1.1	X		Numbering difference only.
$F_Q(Z)$ (CAOC W(Z) Methodology)	3.2.1.4	3.2.1.2	X		Numbering difference only.
Nuclear Enthalpy Rise Hot Channel Factor ($F^N_{\Delta H}$)	3.2.2.2	3.2.2.1	X		Numbering difference only.
AXIAL FLUX DIFFERENCE (AFD) (Constant Axial Offset Control (CAOC) Methodology)	3.2.3.1	3.2.3.1	X		None.
AXIAL FLUX DIFFERENCE (AFD) (Constant Axial Offset Control (CAOC) Methodology)	3.2.3.2	3.2.3.2	X		None.
AXIAL FLUX DIFFERENCE (AFD) (Constant Axial Offset Control (CAOC) Methodology)	3.2.3.3	3.2.3.3	X		None.
QUADRANT POWER TILT RATIO (QPTR)	3.2.4.1	3.2.4.1	X		None.
QPTR	3.2.4.2	3.2.4.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.2.4.2 is to verify QPTR is within limit using a minimum of 4 symmetric pairs of fixed incore detectors. This SR is analogous to NUREG-1431 SR 3.2.4.2 with the addition of a clarification that the QPTR limit is verified “using a minimum of 4 symmetric pairs” of fixed incore detectors. However, this clarification does not affect the Frequency of this SR. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
On-Line Power Distribution Monitoring System (OPDMS)-Monitored Parameters	3.2.5.1	a. n/a b. 3.2.2.1 c. 2.1.1.1 d. 3.1.1.1	X		Variation #2 (administrative): Vogtle 3&4 SR 3.2.5.1 is to verify the identified OPDMS parameters are within limits. This Surveillance is to be performed at a 24-hour Frequency. Parameters b, c, and d are verified by NUREG-1431, SRs 3.2.2.1, 2.1.1.1, and 3.1.1.1, respectively. However, NUREG-1431 does not have a similar SR to verify Parameter "a," Peak Linear Heat Rate. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Reactor Trip System (RTS) Instrumentation	3.3.1.1	3.3.1.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.1.1 is to compare results of calorimetric heat balance to nuclear instrument channel output. This Surveillance is to be performed at a 24-hour Frequency. This SR is similar to NUREG-1431 SR 3.3.1.2 to compare results of calorimetric heat balance calculation to power range channel output and adjust power range channel output if the calorimetric heat balance calculation results exceed power range channel output by more than +2% RTP. The additional requirement in the STS SR to adjust power range channel output if required is an administrative variation in that it does not affect the Frequency for this Surveillance. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RTS Instrumentation	3.3.1.2	3.3.1.2	X		None.
RTS Instrumentation	3.3.1.3	3.3.1.3	X		None.
RTS Instrumentation	3.3.1.4	3.3.1.6	X		Numbering difference only.
RTS Instrumentation	3.3.1.5	3.3.1.10	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.1.5 is to perform CHANNEL CALIBRATION in accordance with Setpoint Program. This Surveillance is performed at a 24-month Frequency. This SR is similar to NUREG-1431 SR 3.3.1.10 to perform CHANNEL CALIBRATION.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					The additional requirement in the Vogtle 3&4 SR referring to the Setpoint Program for the CHANNEL CALIBRATION is an administrative variation in that it does not affect the Frequency for this Surveillance. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RTS Instrumentation	3.3.1.6	3.3.1.11	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.1.6 is to perform CHANNEL CALIBRATION in accordance with Setpoint Program. This Surveillance is performed at a 24-month Frequency. This SR is similar to NUREG-1431 SR 3.3.1.11 to perform CHANNEL CALIBRATION. The additional requirement in the Vogtle 3&4 SR referring to the Setpoint Program for the CHANNEL CALIBRATION is an administrative variation in that it does not affect the Frequency for this Surveillance. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RTS Instrumentation	3.3.1.7	3.3.1.14	X		Numbering difference only.
RTS Instrumentation	3.3.1.8	3.3.1.16	X		Numbering difference only.
RTS Source Range Instrumentation	3.3.2.1	3.3.1.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.2.1 is to perform CHANNEL CHECK. This Surveillance is performed at a 12-hour Frequency. This SR is similar to NUREG-1431 SR 3.3.1.1 to perform a CHANNEL CHECK. The NUREG-1431 STS includes source range instrumentation within the scope of TS 3.3.1, whereas the Vogtle 3&4 TS that addresses the source range instrumentation is in a separate TS (3.3.2), but this does not affect the Frequency of this Surveillance. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
RTS Source Range Instrumentation	3.3.2.2	3.3.1.11	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.2.2 perform CHANNEL CALIBRATION in accordance with Setpoint Program. This Surveillance is performed at a 24-month Frequency. This SR is similar to NUREG-1431 SR 3.3.1.11 to perform a CHANNEL CALIBRATION. The NUREG-1431 STS includes source range instrumentation within the scope of TS 3.3.1, whereas the Vogtle 3&4 TS addresses the source range instrumentation in a separate TS (3.3.2), but this does not affect the Frequency of this Surveillance. The Vogtle 3&4 Surveillance reference to the Setpoint Program does not affect the Frequency for this Surveillance. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RTS Source Range Instrumentation	3.3.2.3	3.3.1.16	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.2.3 is to verify RTS RESPONSE TIME is within limits. This Surveillance is performed at a 24-month on a STAGGERED TEST BASIS Frequency. This SR is similar to NUREG-1431 SR 3.3.1.16 to verify RTS RESPONSE TIME is within limits. The NUREG-1431 STS includes source range instrumentation within the scope of TS 3.3.1, whereas the Vogtle 3&4 TS addresses the source range instrumentation in a separate TS (3.3.2), but this does not affect the Frequency of this Surveillance. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RTS Intermediate Range Instrumentation	3.3.3.1	3.3.1.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.3.1 is to perform a CHANNEL CHECK. This Surveillance is performed at a 12-hour Frequency. This SR is similar to NUREG-1431 SR 3.3.1.1 to perform a CHANNEL CHECK. The NUREG-1431 STS includes intermediate range instrumentation within the scope of TS 3.3.1,

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					whereas the Vogtle 3&4 TS addresses the intermediate range instrumentation is in a separate TS (3.3.3), but this does not affect the Frequency of this Surveillance. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RTS Intermediate Range Instrumentation	3.3.3.2	3.3.1.11	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.3.2 is to perform CHANNEL CALIBRATION in accordance with Setpoint Program. This Surveillance is performed at a 24-month Frequency. This SR is similar to NUREG-1431 SR 3.3.1.11 to perform CHANNEL CALIBRATION. The NUREG-1431 STS includes intermediate range instrumentation within the scope of TS 3.3.1, whereas the Vogtle 3&4 TS addresses the intermediate range instrumentation is in a separate TS (3.3.3), but this does not affect the Frequency of this Surveillance. The Vogtle 3&4 Surveillance reference to the Setpoint Program does not affect the Frequency for this Surveillance. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RTS Intermediate Range Instrumentation	3.3.3.3	3.3.1.16	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.3.3 is to verify RTS RESPONSE TIME is within limits. This Surveillance is performed at a 24-month on a STAGGERED TEST BASIS Frequency. This SR is similar to NUREG-1431 SR 3.3.1.16 to verify RTS RESPONSE TIME is within limits. The NUREG-1431 STS includes intermediate range instrumentation within the scope of TS 3.3.1, whereas the Vogtle 3&4 TS addresses the intermediate range instrumentation is in a separate TS (3.3.3), but this does not affect the Frequency of this Surveillance. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
RTS Engineered Safety Feature Actuation System (ESFAS) Instrumentation	3.3.4.1	3.3.1.16	X		Numbering difference only. (The STS Note excluding neutron detectors from response time testing does not affect the Frequency of this Surveillance.)
RTS Manual Actuation	3.3.5.1	3.3.1.14	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.5.1 is to perform TADOT. This Surveillance is performed at a 24-month Frequency. This SR is similar to NUREG-1431 SR 3.3.1.14 to perform TADOT. The NUREG-1431 STS includes RTS manual reactor trip within the scope of SR 3.3.1.14, whereas the Vogtle 3&4 TS surveillance for the performance of a Trip Actuating Device Operational Test (TADOT) of the RTS manual actuation trip actuating device is in a separate TS (3.3.5), but this does not affect the Frequency of this Surveillance. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RTS Trip Actuation Devices	3.3.7.1	3.3.1.14	X		Variation #2 (administrative): Vogtle 3&4 SR 3.3.7.1 is to perform TADOT on both reactor trip breakers in one division. This Surveillance is to be performed at a Frequency of 92 days on a STAGGERED TEST BASIS. This SR is functionally analogous to NUREG-1431, SR 3.3.1.14, to perform TADOT on identified Functions, including Function 19, Reactor Trip Breakers. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
ESFAS Instrumentation	3.3.8.1	3.3.2.1	X		Numbering difference only.
ESFAS Instrumentation	3.3.8.2	3.3.2.9	X		Numbering difference only. (The STS Note excluding neutron detectors from CHANNEL CALIBRATION does not affect the Frequency of this Surveillance.)
ESFAS Instrumentation	3.3.8.3	3.3.2.10	X		Numbering difference only. (The Vogtle TS Notes and the STS Note excluding certain Functions from this SR do not affect the Frequency of this Surveillance.)

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
ESFAS Manual Initiation	3.3.9.1	3.3.2.8	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.9.1 is to perform TADOT of the Functions identified in Table 3.3.9-1. This Surveillance is to be performed at a 24-month Frequency. This SR is similar to NUREG-1431 SR 3.3.2.8 to perform TADOT of the Functions identified in Table 3.3.2-1. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
ESFAS Reactor Coolant System (RCS) Hot Leg Level Instrumentation	3.3.10.1	3.3.2.9	X		Numbering difference only.
ESFAS RCS Hot Leg Level Instrumentation	3.3.10.2	3.3.2.10	X		Numbering difference only.
ESFAS Startup Feedwater Flow Instrumentation	3.3.11.1	3.3.2.9	X		Numbering difference only.
ESFAS Startup Feedwater Flow Instrumentation	3.3.11.2	3.3.2.10	X		Numbering difference only.
ESFAS Reactor Trip Initiation	3.3.12.1	3.3.1.15	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.12.1 is to perform TADOT of the blocks for the reactor trip (P-4) input from the integrated protection cabinets. This Surveillance is to be performed at a 24-month Frequency. This SR is similar to NUREG-1431 SR 3.3.1.14 to perform a TADOT of the Manual Reactor Trip, RCP Breaker Position, and the SI Input from ESFAS. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
ESFAS Main Control Room Isolation, Air Supply Initiation, and Electrical Load De-energization	3.3.13.1	3.3.7.9	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.13.1 is to perform CHANNEL CALIBRATION of the ESFAS Main Control Room Isolation, Air Supply Initiation, and Electrical Load De-energization instrumentation in accordance with Setpoint Program. This Surveillance is to be performed at a 24-month Frequency. This SR is similar to NUREG-1431 SR 3.3.7.9 to perform CHANNEL CALIBRATION of the Control Room Radiation instrumentation (Control Room Atmosphere and Control Room Air Intakes) within the scope of SR 3.3.7.9. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
ESFAS Main Control Room Isolation, Air Supply Initiation, and Electrical Load De-energization	3.3.13.2	3.3.2.10	X		Numbering difference only.
ESFAS In-containment Refueling Water Storage Tank (IRWST) and Spent Fuel Pool Level Instrumentation	3.3.14.1	3.3.2.9 (partial)	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.14.1 is to perform CHANNEL CALIBRATION of the ESFAS IRWST and spent fuel pool level instrumentation in accordance with the Setpoint Program. This Surveillance is to be performed at a 24-month Frequency. This aspect of this SR is analogous to NUREG-1431 SR 3.3.7.9 to perform CHANNEL CALIBRATION of the RWST level instrumentation, with a Surveillance Frequency which is relocated by TSTF-425 to the SFCP. Variation #2 (administrative): NUREG-1431 does not include a TS for spent fuel pool level instrumentation. The proposed change to relocate the 24-month Surveillance Frequency to refer to the SFCP is consistent with the intent of TSTF-425.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
ESFAS IRWST and Spent Fuel Pool Level Instrumentation	3.3.14.2	3.3.2.10 (partial)	X		Variation #3 (administrative): The NUREG-1431 SR to verify the ESF RESPONSE TIME of the RWST level instrumentation is addressed by SR 3.3.2.10, with a Surveillance Frequency which is relocated by TSTF-425 to the SFCP. Variation #2 (administrative): NUREG-1431 does not include a TS for spent fuel pool level instrumentation. The proposed change to relocate the 24-month STAG-GERED TEST BASIS Surveillance Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Post Accident Monitoring (PAM) Instrumentation	3.3.17.1	3.3.3.1	X		Numbering difference only.
PAM Instrumentation	3.3.17.2	3.3.3.2	X		Numbering difference only.
Remote Shutdown Workstation (RSW)	3.3.18.1	3.3.4.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.18.1 is to verify each required transfer switch is capable of performing the required function. This Surveillance is to be performed at a 24-month Frequency. This SR is similar to NUREG-1431 SR 3.3.3.4.2 to verify each required control circuit and transfer switch is capable of performing the intended function. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RSW	3.3.18.2	3.3.4.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.18.1 is to verify the RSW communicates indication and controls with Division A, B, C and D of the Protection and Safety Monitoring System (PMS). This SR is similar to NUREG-1431 SR 3.3.4.1, to perform a CHANNEL CHECK for each required instrumentation channel of the Remote Shutdown System that is normally energized. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
RSW	3.3.18.3	3.3.4.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.18.3 is to verify the OPERABILITY of the RSW hardware and software. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.3.4.3 to perform a CHANNEL CALIBRATION for each instrumentation channel of the Remote Shutdown System. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RSW	3.3.18.4	3.3.4.4	X		Numbering difference only.
Diverse Actuation System (DAS) Manual Controls	3.3.19.1	n/a	X		Variation #2 (administrative): Vogtle 3&4 SR 3.3.19.1 is to perform a TADOT of the DAS manual trip and actuation controls for the specified safety-related equipment. This Surveillance is to be performed at a 24-month Frequency. NUREG-1431 does not have a similar SR to this Vogtle 3&4 SR. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Automatic Depressurization System (ADS) and IRWST Injection Blocking Device	3.3.20.1	3.3.1.5	X		Variation #2 (administrative): Vogtle 3&4 SR 3.3.20.1 is to verify each ADS and IRWST Injection Block switch is in the “unblock” position. This Surveillance is to be performed at a 7-day Frequency. This SR is functionally analogous to NUREG-1431, SR 3.3.1.5, to perform an ACTUATION LOGIC TEST of the Reactor Trip System (RTS) instrumentation. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
ADS and IRWST Injection Blocking Device	3.3.20.2	3.3.1.10 3.3.1.11	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.20.2 is to perform CHANNEL CALIBRATION in accordance with Setpoint Program. This Surveillance is to be

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.3.1.10 and SR 3.3.1.11 to perform CHANNEL CALIBRATION of RTS instrumentation. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
ADS and IRWST Injection Blocking Device	3.3.20.3	3.3.1.5	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.20.3 is to perform ACTUATION LOGIC TEST of ADS and IRWST Injection Blocking Devices. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.3.1.5 to perform ACTUATION LOGIC TEST of RTS instrumentation. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
ADS and IRWST Injection Blocking Device	3.3.20.4	3.3.1.15	X		Variation #3 (administrative): Vogtle 3&4 SR 3.3.20.4 is to perform TADOT of ADS and IRWST Injection Block manual switches. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.3.1.15 to perform TADOT of RTS instrumentation. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RCS Pressure, Temperature, and Flow Departure from Nucleate Boiling (DNB) Limits	3.4.1.1	3.4.1.1	X		None.
RCS Pressure, Temperature, and Flow DNB Limits	3.4.1.2	3.4.1.2	X		None.
RCS Pressure, Temperature, and Flow DNB Limits	3.4.1.3	3.4.1.3	X		None.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
RCS Pressure, Temperature, and Flow DNB Limits	3.4.1.4	3.3.1.12	X		Variation #3 (administrative): Vogtle 3&4 SR 3.4.1.4 is to perform a CHANNEL CALIBRATION of differential pressure RCS total flow rate indication channels. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.3.1.12, to CHANNEL CALIBRATION of RTS instrumentation. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RCS Pressure, Temperature, and Flow DNB Limits	3.4.1.5	3.4.1.4	X		This is a numbering difference only. The specific means of determining RCS total flow rate, while specified in the Vogtle 3&4 and NUREG-1431 Surveillances, have no impact on the Surveillance Frequency. Therefore, the proposed change to relocate the 24-month Surveillance Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RCS Minimum Temperature for Criticality	3.4.2.1	3.4.2.1	X		None.
RCS Pressure and Temperature (P/T) Limits	3.4.3.1	3.4.3.1	X		None.
RCS Loops	3.4.4.1 (M1, M2 and M3, M4, M5 with PCS capable of rod withdrawal)	3.4.4.1 (M1/M2), 3.4.5.1 (M3), 3.4.6.1 (M4)	X		Variation #3 (administrative): Vogtle 3&4 SR 3.4.4.1 is to verify each RCS loop is in operation with variable speed control bypassed. This Surveillance is to be performed at a 12-hour Frequency. This SR is similar to NUREG-1431 SR 3.4.4.1 to verify each RCS loop is in operation (Modes 1 and 2), SR 3.4.5.1 to verify required RCS loops are in operation (Mode 3), and SR 3.4.6.1 to verify required RHR or RCS loop is in operation (Mode 4). The Vogtle 3&4 TS and NUREG-1431 STS both verify each RCS loop is in operation; however, the Vogtle 3&4 SR also specifies that this surveillance is performed "with variable speed control bypassed." This additional specificity is unique to the AP1000 design and does not impact the Frequency of this surveillance. Therefore,

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					the proposed change to relocate the 24-month Surveillance Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Pressurizer	3.4.5.1	3.4.9.1	X		Numbering difference only.
Pressurizer Safety Valves	3.4.6.1	3.4.10.1		X	The Vogtle 3&4 TSs refer to the IST Program within the SR 3.4.6.1 Surveillance to verify each pressurizer safety valve OPERABLE, as well as in the Surveillance Frequency. Consistent with direction in TSTF-596, this Surveillance is revised to include a description of the test, including the lift setting range provided in the current Limiting Condition for Operation (LCO). The proposed change is consistent with the NUREG-1431 markup in TSTF-596. In addition, the sentence requiring the lift settings to be within ± 1% is deleted, as the PSVs are required to be within the specified range (≥2460 psig and ≤2510 psig), which equates to the nominal setting of 2485 psig ± 1%. Deletion of the ambiguous use of the 1% tolerance without specifying the nominal value to which this tolerance is applied, is an administrative enhancement to eliminate the potential for misinterpreta-tion of the SR. Therefore, the proposed change to revise the Surveillance to include only a description of the test and to revise the Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
RCS Operational LEAKAGE	3.4.7.1	3.4.13.1	X		Numbering difference only.
RCS Operational LEAKAGE	3.4.7.2	3.4.13.2	X		Numbering difference only.
Minimum RCS Flow	3.4.8.1	3.4.5.1 3.4.6.1 3.4.7.1 3.4.8.1	X		Variation #2 (administrative): Vogtle 3&4 SR 3.4.8.1 is to verify at least one RCP is in operation with total flow through the core ≥ 3,000 gpm. This Surveillance applicable in Modes 3, 4, and 5 with unborated water sources not isolated from the RCS and is to be performed at a 12-hour Frequency. This SR is functionally analogous to NUREG-1431, SRs 3.4.5.1,

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					3.4.6.1, 3.4.7.1, and 3.4.8.1, to verify required RCS or RHR loop(s) are in operation in Modes 3, 4, and 5 with loops filled and loops not filled, respectively. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RCS Leakage Detection Instrumentation	3.4.9.1	3.4.15.1	X		Variation #3 (administrative): The Vogtle 3&4 SR 3.4.9.1 is for a CHANNEL CHECK. The TS Bases clarifies that this SR is applicable to both the containment atmosphere F18 particulate monitor and the required containment sump level channels. The CHANNEL CHECK required by the NUREG-1431 SR 3.4.15.1 is only for the containment atmospheric radioactivity monitor. However, this difference does not affect the Frequency of this Surveillance. The proposed change to relocate the Surveillance Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RCS Leakage Detection Instrumentation	3.4.9.2	3.4.15.2	X		Numbering difference only.
RCS Leakage Detection Instrumentation	3.4.9.3	3.4.15.3, 3.4.15.4	X		Variation #3 (administrative): The Vogtle 3&4 SR 3.4.9.3 is for a CHANNEL CHECK. The TS Bases clarifies that this SR is applicable to both the containment atmosphere F18 particulate monitor and the required containment sump level channels. The CHANNEL CHECK required by the NUREG-1431 SR 3.4.15.1 is only for the containment atmospheric radioactivity monitor. However, this difference does not affect the Frequency of this Surveillance. The proposed change to relocate the Surveillance Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
RCS Specific Activity	3.4.10.1	3.4.16.1	X		Numbering difference only.
RCS Specific Activity	3.4.10.2	3.4.16.2	X		Numbering difference only.
Automatic Depressurization System (ADS) – Operating	3.4.11.1	3.5.2.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.4.11.1 is to verify the motor operated valve in series with each 4th

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					stage ADS valve is open. This Surveillance is to be performed at a 12-hour Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.2.2, to verify each ECCS manual, power operated, and automatic valve in the flow path, that is not locked, sealed, or otherwise secured in position, is in the correct position. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
ADS - Operating	3.4.11.2	3.5.2.5		X	Variation #3 (administrative): Vogtle 3&4 SR 3.4.11.2 is to verify each stage 1, 2, and 3 ADS valve strokes open. This Surveillance is to be performed at a Frequency in accordance with the IST Program. This SR is functionally analogous to NUREG-1431, SR 3.5.2.5, to verify each ECCS automatic valve in the flow path that is not locked, sealed, or otherwise secured in position, actuates to the correct position on an actual or simulated actuation signal. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
ADS - Operating	3.4.11.3	n/a		X	TSTF-596 Variation #6: The Vogtle 3&4 TSs reference the IST Program in both the SR 3.4.11.3 Surveillance to verify each stage 4 ADS valve OPERABLE and in the Surveillance Frequency. A detailed description of the surveillance requirements for squib valves, such as these stage 4 ADS valves, is provided in the ASME OM Code, which is incorporated by reference in 10 CFR 50.55a(f). Therefore, to minimize the potential for misrepresenting the detailed testing requirements specified in the Code, the reference to the IST Program in the Surveillance is replaced with a reference to 10 CFR 50.55a(f), and the reference to the IST Program in the Frequency is replaced with a reference to the SFCP. Based on the detail provided in the ASME OM Code,

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					with its regulatory basis in 10 CFR 50.55a(f), the proposed change to revise the Surveillance to refer to 10 CFR 50.55a(f) and the Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
ADS - Operating	3.4.11.4	3.5.2.5	X		Variation #3 (administrative): Vogtle 3&4 SR 3.4.11.4 is to verify each stage 1, 2, and 3 ADS valve actuates to the open position on an actual or simulated actuation signal. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.2.5 to verify each ECCS automatic valve in the flow path that is not locked, sealed, or otherwise secured in position, actuates to the correct position on an actual or simulated actuation signal. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
ADS - Operating	3.4.11.5	3.3.2.2		X	Variation #3 (administrative): Vogtle 3&4 SR 3.4.11.5 is to verify continuity of the circuit from the Protection Logic Cabinets to each stage 4 ADS valve. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.3.2.2, to perform ACTUATION LOGIC TEST to verify that the ESFAS logic modules are OPERABLE and that there is an intact voltage signal to the master relay coils. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
ADS – Shutdown, RCS Open	3.4.13.1	3.5.2.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.4.13.1 is to verify each required ADS stage 1, 2, and 3 valve is in the open position. This Surveillance is to be performed at a 12-hour Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.2.2 to verify each ECCS manual, power operated, and automatic valve in the flow path, that is not locked, sealed, or otherwise secured in position, is in the correct position. The

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Low Temperature Overpressure Protection (LTOP)	3.4.14.1	3.4.12.3	X		Numbering difference only.
LTOP	3.4.14.2	3.4.12.4	X		Variation #3 (administrative): Vogtle 3&4 SR 3.4.14.2 is to verify both RNS suction isolation valves in one RNS suction flow path are open. This SR is similar to NUREG-1431 SR 3.4.12.4 to verify residual heat removal (RHR) suction valve is open for each required RHR suction relief valve. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
LTOP	3.4.14.3	3.4.12.3	X		Numbering difference only.
LTOP	3.4.14.4	3.4.12.5	X		Variation #3 (administrative): Vogtle 3&4 SR 3.4.14.4 is to verify RCS vent ≥ 4.15 square inches is open. This SR is to be performed at a Frequency of 12 hours for unlocked open vents and 31 days for locked open vents. This SR is similar to NUREG-1431 SR 3.4.12.5 to verify required RCS vent ≥ [2.07] square inches open. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
LTOP	3.4.14.5	n/a 50.55a(f)		X	The Vogtle 3&4 TSs reference the IST Program in the SR 3.4.14.5 Specification to verify the lift setting of each normal residual heat removal system (RNS) suction relief valve, as well as in the Surveillance Frequency. The proposed change removes the reference to the IST Program in the Surveillance while retaining the description of the surveillance to verify the lift setting of each of these valves, consistent with direction in TSTF-596. Additionally, the Frequency is revised to refer to

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					the SFCP. Therefore, the proposed change to revise the Surveillance to include only a description of the test and to revise the Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
RCS Pressure Isolation Valve (PIV) Integrity	3.4.15.1	3.4.14.1	X		Numbering difference only.
Reactor Vessel Head Vent (RVHV)	3.4.16.1	n/a 50.55a(f)		X	Variation #2 (administrative): Vogtle 3&4 SR 3.4.1.6.1 is to verify each RVHV valve strokes open. This Surveillance is to be performed at a Frequency in accordance with the IST Program. NUREG-1431 does not have a similar SR to this Vogtle 3&4 SR. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
Pressurizer Heater Circuit Breakers	3.4.18.1	3.3.15.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.4.18.1 is to verify each pressurizer heater circuit breaker trips open on an actual or simulated actuation signal, except for circuit breaker maintained in the tripped condition. This Surveillance is performed on a 24-month Frequency. This SR is similar to NUREG-1431 SR 3.3.15.2 to verify pressurizer heater circuit breakers trip open on an actual or simulated actuation signal. The exception in the Vogtle SR has no impact on the Frequency of this SR. Therefore, the proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Auxiliary Spray and Purification Line Isolation Valves	3.4.19.1	3.3.15.6	X		Numbering difference only.
CVS Letdown Isolation Valves	3.4.20.1	3.3.16.3	X		Numbering difference only.
Accumulators	3.5.1.1	3.5.1.1	X		None.
Accumulators	3.5.1.2	3.5.1.2	X		None.

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Accumulators	3.5.1.3	3.5.1.3	X		None.
Accumulators	3.5.1.4	3.5.1.4	X		None.
Accumulators	3.5.1.5	3.5.1.5	X		None.
Accumulators	3.5.1.6	3.8.1.20	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.1.6 is to verify system flow performance of each accumulator in accordance with the System Level OPERABILITY Testing Program. This Surveillance is to be performed at a 10-year Frequency. This SR is functionally analogous to NUREG-1431, SR 3.8.1.20, which demonstrates OPERABILITY of the diesel generators on a 10-year frequency. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Core Makeup Tanks (CMTs) – Operating	3.5.2.1	3.5.4.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.2.1 is to verify the temperature of the borated water in each CMT is < 120°F. This Surveillance is to be performed at a 24-hour Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.4.1 to verify the RWST borated water temperature is within limits. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
CMTs - Operating	3.5.2.2	3.5.1.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.2.2 is to verify each CMT inlet isolation valve is fully open. This Surveillance is to be performed at a 12-hour Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.1.1, to verify each accumulator isolation valve is fully open. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
CMTs - Operating	3.5.2.3	3.5.2.3	X		None.

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CMTs - Operating	3.5.2.4	3.5.4.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.2.4 is to verify the boron concentration in each CMT is ≥ 3400 ppm and ≤ 4500 ppm. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.4.3 to verify RWST boron concentration is within limits. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
CMTs - Operating	3.5.2.5	3.6.14.4 50.55a(f)		X	Variation #3 (administrative): Vogtle 3&4 SR 3.5.2.5 is to verify each CMT outlet isolation valve strokes open. This Surveillance is to be performed at a Frequency in accordance with the IST Program. This SR is functionally analogous to NUREG-1431, SR 3.6.14.4, to verify each motor operated valve in the hydrogen collection header that is not locked, sealed, or otherwise secured in position, opens on an actual or simulated actuation signal after a specified delay time. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
CMTs - Operating	3.5.2.6	n/a 3.5.2.5	X		Variation #2 (administrative): Vogtle 3&4 SR 3.5.2.6 is to verify each CMT outlet isolation valve actuates to the open position on an actual or simulated actuation signal. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.2.5, to verify each ECCS automatic valve in the flow path that is not locked, sealed, or otherwise secured in position, actuates to the correct position on an actual or simulated actuation signal. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
CMTs - Operating	3.5.2.7	3.8.1.20	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.2.7 is to verify system flow performance of each CMT in

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					accordance with the System Level OPERABILITY Testing Program. This Surveillance is to be performed at a 10-year Frequency. This SR is functionally analogous to NUREG-1431, SR 3.8.1.20, which demonstrates OPERABILITY of the diesel generators on a 10-year frequency. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
CMTs – RCS Intact	3.5.3.3	3.5.4.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.3.3 is to verify the CMT borated water volume is ≥ 2450 cu. ft. This Surveillance is to be performed at a 7-day Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.4.2, to verify RWST borated water volume is ≥ a specified minimum limit. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Passive Residual Heat Removal Heat Exchanger (PRHR HX) – Operating	3.5.4.1	3.5.2.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.4.1 is to verify the PRHR HX outlet manual isolation valve is fully open. This Surveillance is to be performed at a 12-hour Frequency. This SR is similar to NUREG-1431 SR 3.5.2.2 to verify each ECCS manual, power operated, and automatic valve in the flow path, that is not locked, sealed, or otherwise secured in position, is in the correct position. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
PRHR HX – Operating	3.5.4.2	3.5.2.1, 3.5.2.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.4.2 is to verify the PRHR HX inlet motor operated isolation valve is open. This Surveillance is to be performed at a 12-hour Frequency. This SR is similar to NUREG-1431 SR 3.5.2.2 to verify each ECCS manual, power operated, and automatic valve in the flow path, that is not locked, sealed, or otherwise secured in position, is in the correct

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					position. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
PRHR HX – Operating	3.5.4.3	3.5.2.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.4.3 is to verify the volume of noncondensible gases in the PRHR HX inlet line has not caused the high-point water level to drop below the sensor. This Surveillance is to be performed at a 24-hour Frequency. This SR is similar to NUREG-1431 SR 3.5.2.3 to verify ECCS locations susceptible to gas accumulation are sufficiently filled with water. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
PRHR HX – Operating	3.5.4.4	3.4.4.1 3.4.5.1 3.4.6.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.4.4 is to verify one Loop 1 RCP is in operation. This Surveillance is to be performed at a 12-hour Frequency. This SR is functionally analogous to NUREG-1431, SRs 3.4.4.1, 3.4.5.1, and 3.4.6.1, which verify required RCS loops are in operation. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
PRHR HX – Operating	3.5.4.5	3.5.2.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.4.5 is to verify power is removed from the PRHR HX inlet motor operated isolation valve. This Surveillance is to be performed on a 31-day frequency. This SR is similar to NUREG-1431 SR 3.5.2.1 to verify various identified ECCS valves are in the specified position “with power to the valve operator removed.” The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.

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PRHR HX – Operating	3.5.4.6	3.5.2.5		X	Variation #3 (administrative): Vogtle 3&4 SR 3.5.4.6 is to verify both PRHR HX air operated outlet isolation valves stroke open and both IRWST gutter isolation valves stroke closed. This Surveillance is to be performed on a Frequency in accordance with the IST Program. This SR is similar to NUREG-1431 SR 3.5.2.5 to verify each ECCS automatic valve in the flow path that is not locked, sealed, or otherwise secured in position, actuates to the correct position on an actual or simulated actuation signal, which is performed at an 18-month Frequency. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
PRHR HX – Operating	3.5.4.7	3.6.19.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.4.7 is to verify by visual inspection that the IRWST gutter and downspout screens are not restricted by debris. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.6.19.1, to verify, by visual inspection, the containment sump(s) does not show structural damage, abnormal corrosion, or debris blockage. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
PRHR HX – Operating	3.5.4.8	3.5.2.5	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.4.8 is to verify both PRHR HX air operated outlet isolation valves actuate to the open position and both IRWST gutter isolation valves actuate to the isolation position on an actual or simulated actuation signal. This Surveillance is to be performed on a 24-month frequency. This SR is similar to NUREG-1431 SR 3.5.2.5 to verify each ECCS automatic valve in the flow path that is not locked, sealed, or otherwise secured in position, actuates to the correct position on an actual or simulated actuation

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					signal. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
PRHR HX – Operating	3.5.4.9	3.8.1.20	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.4.9 is to verify PRHR HX heat transfer performance in accordance with the System Level OPERABILITY Testing Program. This Surveillance is to be performed at a 10-year Frequency. This SR is functionally analogous to NUREG-1431, SR 3.8.1.20, which demonstrates OPERABILITY of the diesel generators on a 10-year frequency. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
IRWST – Operating	3.5.6.1	3.5.4.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.6.1 is to verify the IRWST water temperature is < 120°F. This Surveillance is to be performed at a 24-hour Frequency. This SR is similar to NUREG-1431 SR 3.8.1.11 to verify RWST borated water temperature is within a specified range. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
IRWST – Operating	3.5.6.2	3.5.4.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.6.2 is to verify the IRWST borated water volume is ≥ 73,100 cu. ft. This Surveillance is to be performed at a 24-hour Frequency. This SR is similar to NUREG-1431 SR 3.8.1.11 to verify RWST borated water volume is ≥ a specified volume. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
IRWST – Operating	3.5.6.3	3.5.2.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.6.3 is to verify the volume of noncondensible gases in each of the four IRWST injection squib valve outlet line pipe stubs has not caused the high-point water level to drop

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					below the sensor. This Surveillance is to be performed at a 24-hour Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.2.3, to verify ECCS locations susceptible to gas accumulation are sufficiently filled with water. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
IRWST – Operating	3.5.6.4	3.5.4.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.6.4 is to verify the IRWST boron concentration is ≥ 2600 ppm and ≤ 2900 ppm. This Surveillance is to be performed at a 31-day Frequency. This SR is similar to NUREG-1431 SR 3.5.4.3 to verify RWST boron concentration is within a specified range. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
IRWST – Operating	3.5.6.5	3.5.2.1 3.5.2.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.6.5 is to verify each motor operated IRWST isolation valve is fully open. This Surveillance is to be performed at a 12-hour Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.2.1 and SR 3.5.2.2, to verify identified ECCS valves are in the listed or correct position. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
IRWST – Operating	3.5.6.6	3.5.2.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.6.6 is to verify power is removed from each motor operated IRWST isolation valve. This Surveillance is to be performed at a 31-day Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.2.1, to verify identified ECCS valves are in the listed position with power to the valve operator removed. The proposed change to relocate the Frequency to the

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
IRWST – Operating	3.5.6.7	3.5.2.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.6.7 is to verify each motor operated containment recirculation isolation valve is fully open. This Surveillance is to be performed at a 31-day Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.2.2, to verify each ECCS manual, power operated, and automatic valve in the flow path, that is not locked, sealed, or otherwise secured in position, is in the correct position. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
IRWST – Operating	3.5.6.8	n/a 50.55a(f)		X	TSTF-596 Variation #6: The Vogtle 3&4 TSs reference the IST Program in both the SR 3.4.11.3 Surveillance to verify each verify each IRWST injection and containment recirculation squib valve OPERABLE, as well in as the Surveillance Frequency. A detailed description of the surveillance requirements for squib valves, such as the IRWST injection and containment recirculation squib valves, is provided in the ASME OM Code, which is incorporated by reference in 10 CFR 50.55a(f). Therefore, to minimize the potential for misrepresenting the detailed testing requirements specified in the Code, the reference to the IST Program in the Surveillance is replaced with a reference to 10 CFR 50.55a(f), and the reference to the IST Program in the Frequency is replaced with a reference to the SFCP. Based on the detail provided in the ASME OM Code, with its regulatory basis in being in 10 CFR 50.55a(f), the proposed change to revise the Surveillance to refer to 10 CFR 50.55a(f) and the Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
IRWST – Operating	3.5.6.9	3.5.2.5	X		Variation #2 (administrative): Vogtle 3&4 SR [3.] is to verify continuity of the circuit from the Protection Logic

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					Cabinets to each IRWST injection and containment recirculation squib valve on an actual or simulated actuation signal. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.2.5, to verify each ECCS automatic valve in the flow path that is not locked, sealed, or otherwise secured in position, actuates to the correct position on an actual or simulated actuation signal. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
IRWST – Operating	3.5.6.10	3.6.19.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.6.10 is to verify by visual inspection that the IRWST screens and the containment recirculation screens are not restricted by debris. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.6.19.1, to verify, by visual inspection, the containment sump(s) does not show structural damage, abnormal corrosion, or debris blockage. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
IRWST – Operating	3.5.6.11	3.8.1.20	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.6.11 is to verify IRWST injection and recirculation system flow performance in accordance with the System Level OPERABILITY Testing Program. This SR is functionally analogous to NUREG-1431, SR 3.8.1.20, which demonstrates OPERABILITY of the diesel generators on a 10-year frequency. NUREG-1431 does not have a similar SR to this Vogtle 3&4 SR. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
IRWST – Shutdown, MODE 5	3.5.7.1	3.6.3.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.7.1 is to verify Spent Fuel Pool Cooling System containment isolation valves are closed. This Surveillance is to be performed at a 31-day Frequency. This SR is functionally analogous to NUREG-1431, SR 3.6.3.3, to verify each containment isolation manual valve and blind flange that is located outside containment and not locked, sealed, or otherwise secured and required to be closed during accident conditions is closed. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
IRWST – Shutdown, MODE 6	3.5.8.1	3.5.4.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.8.1 is to verify the IRWST and refueling cavity water temperature is < 120°F. This Surveillance is to be performed at a 24-hour Frequency and is applicable in MODE 6 only. This SR is functionally analogous to NUREG-1431, SR 3.5.4.1, to verify RWST borated water temperature is within limits. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
IRWST – Shutdown, MODE 6	3.5.8.2	3.5.4.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.8.2 is to verify the IRWST and refueling cavity water total borated water volume is ≥ 73,100 cu. ft. This Surveillance is to be performed at a 24-hour Frequency and is applicable in MODE 6 only. This SR is functionally analogous to NUREG-1431, SR 3.5.4.2, to verify RWST borated water volume is ≥ a specified limit. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
IRWST – Shutdown, MODE 6	3.5.8.3	3.6.3.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.8.3 is to verify Spent Fuel Pool Cooling System containment isolation valves are closed. This Surveillance is to be performed at a 31-day Frequency and is applicable in MODE 6 only. This SR is functionally analogous to NUREG-1431, SR 3.6.3.3, to verify each containment isolation manual valve and blind flange that is located outside containment and not locked, sealed, or otherwise secured and required to be closed during accident conditions is closed. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
IRWST – Shutdown, MODE 6	3.5.8.4	3.5.4.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.8.4 is to verify the IRWST and refueling cavity boron concentration is ≥ 2600 ppm and ≤ 2900 ppm. This Surveillance is to be performed at a 31-day Frequency and is applicable in MODE 6 only. This SR is functionally analogous to NUREG-1431, SR 3.5.4.3, to verify RWST boron concentration is within specified limits. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Reactor Coolant Pump (RCP) Breakers	3.5.9.1	3.3.1.4	X		Variation #3 (administrative): Vogtle 3&4 SR 3.5.9.1 is to verify each RCP breaker trips open on an actual or simulated actuation signal, except for breaker maintained in the tripped condition. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.3.1.4, to perform TADOT of the reactor trip breakers. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Containment Air Locks	3.6.2.2	3.6.2.2	X		None.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
Containment Isolation Valves	3.6.3.1	3.6.3.2	X		None.
Containment Isolation Valves	3.6.3.2	3.6.3.3	X		None.
Containment Isolation Valves	3.6.3.4	3.6.3.5		X	Numbering difference only.
Containment Isolation Valves	3.6.3.5	3.6.3.8	X		None.
Containment Pressure	3.6.4.1	3.6.4A.1	X		None.
Containment Air Temperature	3.6.5.1	3.6.5A.1	X		None.
Passive Containment Cooling System (PCS)	3.6.6.1	3.5.4.1	X		Variation #3 (administrative): Vogtle SR 3.6.6.1 verifies the water storage tank temperature is within specified limits. This Surveillance is to be performed at a 24-hour Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.4.1, to verify RWST borated water temperature is within specified limits. The proposed change to relocate the Surveillance Frequency to the SFCP is consistent with the intent of TSTF-425.
PCS	3.6.6.2	3.5.4.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.6.6.2 verifies the water storage tank volume is within a specified limit. This Surveillance is to be performed at a 7-day Frequency. This SR is functionally analogous to NUREG-1431, SR 3.5.4.2, to verify RWST borated water volume is ≥ a specified limit. The proposed change to relocate the Surveillance Frequency to the SFCP is consistent with the intent of TSTF-425.
PCS	3.6.6.3	3.6.6.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.6.6.3 verifies each passive containment cooling system manual, power operated, and automatic valve in each flow path that is not locked, sealed, or otherwise secured in position, is in the correct position. This Surveillance is to be performed at a 31-day Frequency. This SR is similar to NUREG-1431 SR 3.6.6.1, which verifies the status of the containment spray manual, power operated and automatic valves. The proposed

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
PCS	3.6.6.4	3.6.6.6	X		Variation #3 (administrative): Vogtle 3&4 SR 3.6.6.4 is to verify each passive containment cooling system automatic valve in each flow path that is not locked, sealed, or otherwise secured in position, actuates to the correct position on an actual or simulated actuation signal. This Surveillance is to be performed at a 24-month Frequency. This SR is similar to NUREG-1431 SR 3.8.1.11 to verify each automatic containment spray valve in the flow path that is not locked, sealed, or otherwise secured in position, actuates to the correct position on an actual or simulated actuation signal. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
PCS	3.6.6.5	3.6.6A.9 3.6.15.4	X		Variation #3 (administrative): Vogtle 3&4 SR 3.6.6.5 is to verify the air flow path from the shield building annulus inlet to the exit is unobstructed and, that all air baffle sections are in place. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.6.6A.9 and 3.6.15.4, which demonstrate the OPERABILITY of containment systems by testing and visual inspections. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
PCS	3.6.6.6	3.8.1.20	X		Variation #3 (administrative): Vogtle 3&4 SR 3.6.6.6 is to verify passive containment cooling system flow and water coverage performance in accordance with the System Level OPERABILITY Testing Program. This Surveillance is to be performed at the first refueling outage and subsequently on a 10-year Frequency. This SR is functionally analogous to NUREG-1431, SR 3.8.1.20, which demonstrates OPERABILITY of the

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					diesel generators on a 10-year frequency. The proposed change to relocate the periodic (10-year) Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425. Additionally, a change is proposed to revise the TS by deleting the initial test, which was performed at the first refueling outage. This change is discussed separately in this amendment request.
Containment Penetrations	3.6.7.1 (Modes 5 & 6)	3.6.3.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.6.7.1 is to verify each required containment penetration is in the required status. This Surveillance is to be performed at a 7-day Frequency. This SR is functionally analogous to NUREG-1431, SR 3.6.3.3, to verify each containment isolation manual valve and blind flange that is located outside containment and not locked, sealed, or otherwise secured and required to be closed during accident conditions is closed, The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Containment Penetrations	3.6.7.2	n/a	X		Variation #2 (administrative): Vogtle 3&4 SR 3.6.7.2 is to verify the hardware, tools, equipment and power source necessary to close the equipment hatch are available. This Surveillance is to be performed prior to hatch removal and at a 7-day Frequency. NUREG-1431 does not have a similar SR to this Vogtle 3&4 SR. The proposed change to relocate the recurring 7-day Frequency to the SFCP and revise the SR 7-day Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
pH Adjustment	3.6.8.1	3.6.15.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.6.8.1 is to verify the pH adjustment baskets contain ≥ 26,460 lbs. of trisodium phosphate (TSP). This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.6.15.3, to

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					verify that the ice mass of each basket sampled is > a specified minimum weight. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
pH Adjustment	3.6.8.2	3.6.15.5 3.6.15.7	X		Variation #3 (administrative): Vogtle 3&4 SR 3.6.8.2 is to verify sample from the pH adjustment baskets provides adequate pH adjustment of the post-accident water. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.6.15.5 and SR 3.6.15.7, to verify the ice baskets contain sufficient boron to meet the boron concentration and pH requirements for the ECCS recirculation mode of operation. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Vacuum Relief Valves	3.6.9.1	3.6.5.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.6.9.1 is to verify containment inside to outside differential air temperature is ≤ 90°F. This Surveillance is to be performed at a 12-hour Frequency. This SR is functionally analogous to NUREG-1431, SR 3.6.5.1, to verify containment average air temperature. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Vacuum Relief Valves	3.6.9.2	3.6.3.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.6.9.2 is to verify each vacuum relief isolation valve is closed. This Surveillance is to be performed at a 31-day Frequency. This SR is functionally analogous to NUREG-1431, SR 3.6.3.3, to verify each containment isolation manual valve and blind flange that is located outside containment and … required to be closed during accident conditions is closed. The proposed change to relocate the Frequency to the SFCP and revise the SR

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Vacuum Relief Valves	3.6.9.3	3.6.12.1		X	The Vogtle 3&4 TSs reference the IST Program in the SR 3.4.6.1 Specification to verify each vacuum relief check valve and each vacuum relief isolation valve is OPERABLE, as well as in the Surveillance Frequency. Consistent with direction in TSTF-596, this Surveillance is revised to include a description of the test by specifying "(a) each vacuum relief check valve opens and closes, with set pressure within limits, and (b) each vacuum relief isolation valve opening time is within limits." Additionally, the Frequency is revised to refer to the SFCP. Therefore, the proposed change to revise the Surveillance to include a description of the test and to revise the Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
Vacuum Relief Valves	3.6.9.4	3.6.3.8	X		Variation #3 (administrative): Vogtle 3&4 SR 3.6.9.4 is to verify each vacuum relief isolation valve actuates on actual or simulated signals. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.6.3.8, to verify each automatic containment isolation valve that is not locked, sealed or otherwise secured in position, actuates to the isolation position on an actual or simulated actuation signal. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Main Steam Safety Valves (MSSVs)	3.7.1.1	3.7.1.1		X	The Vogtle 3&4 TSs reference the IST Program within the SR 3.7.1.1 Specification to verify each MSSV setpoint per Table 3.7.1-2, as well as in the Surveillance Frequency. Consistent with direction in TSTF-596, this Surveillance is revised by deleting the reference to the IST Program, while retaining the reference to Table 3.7.1-2 for the lift settings. Additionally, the Frequency is revised to refer to the SFCP. Therefore, the proposed

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					change to revise the Surveillance to include only a description of the test and to revise the Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
Main Steam Line Flow Path Isolation Valves	3.7.2.1	3.7.2.1		X	Variation #3 (administrative): Vogtle 3&4 SR 3.7.2.1 verifies MSIV closure time is within limits on an actual or simulated actuation signal. This SR is performed on a Frequency in accordance with the IST Program. This SR is analogous to NUREG-1431 SR 3.7.2.1 to verify the isolation time of each MSIV is within a specified limit. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
Main Steam Line Flow Path Isolation Valves	3.7.2.2	3.7.2.1		X	Variation #3 (administrative): Vogtle 3&4 SR 3.7.2.2 verifies required turbine stop, turbine control, turbine bypass, and moisture separator reheater 2nd stage steam isolation valves' closure time is within limits on an actual or simulated actuation signal. This SR is performed on a Frequency in accordance with the IST Program. This SR is functionally analogous to NUREG-1431, SR 3.7.2.1, to verify the isolation time of each MSIV is within limits. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
Main Steam Line Flow Path Isolation Valves	3.7.2.3	3.7.2.1		X	Variation #3 (administrative): Vogtle 3&4 SR 3.7.2.3 verifies the isolation time of each MSIV bypass valve and main steam line drain isolation valve is within limits. This Surveillance is to be performed at a Frequency in accordance with the IST Program. This SR is functionally analogous to NUREG-1431, SR 3.7.2.1, to verify the isolation time of each MSIV is within limits. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-596.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
Main Steam Line Flow Path Isolation Valves	3.7.2.4	3.7.2.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.7.2.4 verifies each MSIV bypass valve and main steam line drain isolation valve actuates to the isolation position on an actual or simulated actuation signal. This SR is performed on a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.7.2.2, to verify each MSIV actuates to the isolation position on an actual or simulated actuation signal. The proposed change to relocate the (24-month) Surveillance Frequency to the SFCP is consistent with the intent of TSTF-425.
Main Feedwater Isolation Valves (MFIVs) and Main Feedwater Control Valves (MFCVs)	3.7.3.1	3.7.3.1		X	None.
MFIVs and MFCVs	3.7.3.2	n/a 3.3.2.2	X		Variation #2 (administrative): Vogtle 3&4 SR 3.7.3.2 verifies “each MFW pump breaker trips open on an actual or simulated actuation signal, except for pump breaker maintained in the tripped condition.” This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.3.2.2, to perform ACTUATION LOGIC TEST of Function 5.a, Turbine Trip and Feedwater Isolation, Automatic Actuation Logic and Actuation Relays. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Secondary Specific Activity	3.7.4.1	3.7.18.1	X		Numbering difference only.
Spent Fuel Pool Water Level	3.7.5.1	3.7.15.1	X		Numbering difference only.
Main Control Room Emergency Habitability System (VES)	3.7.6.1	3.8.3.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.7.6.1 verifies “the compressed air storage tanks contain > 327,574 scf of compressed air.” This Surveillance is to be performed at a 24-hour Frequency. This SR is

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					functionally analogous to NUREG-1431, SR 3.8.3.1, to verify each fuel oil storage tank contains an adequate inventory of fuel oil to support each DG's operation for [7 days] at full load. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the TSTF-425 intent.
VES	3.7.6.2	n/a	X		Variation #2 (administrative): Vogtle 3&4 SR 3.7.6.2 verifies thermal mass for the main control room envelope and adjacent locations is within limit. This Surveillance is to be performed at a 24-hour Frequency. NUREG-1431 does not have a similar SR to this Vogtle 3&4 SR. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the TSTF-425 intent.
VES	3.7.6.3	3.7.10.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.7.6.3 is to operate VES for ≥ 15 minutes. This SR is analogous to NUREG-1431 SR 3.7.10.1 to operate each control room emergency filtration system for [≥ 10 continuous hours with the heaters operating or (for systems without heaters) ≥ 15 minutes]. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
VES	3.7.6.4	3.6.3.2 3.6.3.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.7.6.4 is to verify each VES air header manual isolation valve is in an open position. This Surveillance is to be performed at a 31-day Frequency. This SR is functionally analogous to NUREG-1431, SR 3.6.3.2 and SR 3.6.3.3, to verify that each [8] inch purge valve and each containment isolation manual valve that is … required to be closed during accident conditions is closed (i.e., in its required safety position). The proposed change to relocate the Frequency to the SFCP and revise the SR

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
VES	3.7.6.5	3.8.3.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.7.6.5 is to verify the air quality of the compressed air storage tanks meets the requirements of CGA G-7.1, Commodity Specification for Air, Grade E, with a pressure dew point of ≤ 40°F at ≥ 3400 psig. This Surveillance is to be performed at a 92-day Frequency. This SR is functionally analogous to NUREG-1431, SR 3.8.3.3, to verify fuel oil properties of new and stored fuel oil are tested in accordance with, and maintained within the limits of, the Diesel Fuel Oil Testing Program. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
VES	3.7.6.6	3.6.3.8	X		Variation #3 (administrative): Vogtle 3&4 SR 3.7.6.6 is to verify each MCRE isolation valve is OPERABLE and will close upon receipt of an actual or simulated actuation signal, except for valves that are locked, sealed, or otherwise secured in the closed position. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.6.3.8, to verify each automatic containment isolation valve that is not locked, sealed or otherwise secured in position, actuates to the isolation position on an actual or simulated actuation signal. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
VES	3.7.6.7	50.55a(f) 3.6.12.1		X	Variation #3 (administrative): Vogtle 3&4 SR 3.7.6.6 is to verify each VES pressure relief isolation valve within the MCRE pressure boundary is OPERABLE. This Surveillance is to be performed at a Frequency in accordance with the IST Program. This SR is functionally analogous to NUREG-1431, SR 3.6.12.1, to verify each vacuum relief line is OPERABLE in

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					accordance with the INSERVICE TESTING PROGRAM. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
VES	3.7.6.8	3.7.12.5 3.7.13.5	X		Variation #3 (administrative): Vogtle 3&4 SR 3.7.6.6 is to verify each VES pressure relief damper is OPERABLE. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.7.12.5 and SR 3.7.13.5, which verify that ventilation system filter bypass dampers can be closed, except for dampers that are locked, sealed, or otherwise secured in the closed position. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
VES	3.7.6.11	3.8.1.11.b	X		Variation #3 (administrative): Vogtle 3&4 SR 3.7.6.11 is to verify the MCR load shed function actuates upon receipt of an actual or simulated actuation signal. This Surveillance is to be performed at a 24-month Frequency. This SR is similar to NUREG-1431 SR 3.8.1.11 to verify on an actual or simulated loss of offsite power signal (b) load shedding from emergency buses. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
VES	3.7.6.12	3.7.10.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.7.6.12 is to verify each VES main air delivery isolation valve actuates to the correct position upon receipt of an actual or simulated actuation signal. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.7.10.3, to verify each CREFS train actuates on an actual or simulated actuation signal. The proposed change to relocate the Frequency to the SFCP and revise the SR

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Startup Feedwater Isolation and Control Valves	3.7.7.1	50.55a(f)		X	Variation #2 (administrative): Vogtle 3&4 SR 3.7.7.1 is to verify each SFW isolation and control valve is OPERABLE. This Surveillance is to be performed at a Frequency in accordance with the IST Program. This SR is functionally analogous to other NUREG-1431 Surveillances that verify valve OPERABILITY in accordance with the IST Program, such as SR 3.4.10.1 for pressurizer safety valves and SR 3.6.12.1 for vacuum relief lines. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
Startup Feedwater Isolation and Control Valves	3.7.7.2	3.7.5.3	X		Variation #3 (administrative): Vogtle 3&4 SR 3.7.7.2 is to verify each SFW isolation and control valve actuates to the isolation position on an actual or simulated actuation signal. This Surveillance is to be performed at a 24-month Frequency. This SR is similar to NUREG-1431 SR 3.7.5.3 to verify each AFW automatic valve that is not locked, sealed, or otherwise secured in position, actuates to the correct position on an actual or simulated actuation signal. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Startup Feedwater Isolation and Control Valves	3.7.7.3	3.3.1.4	X		Variation #3 (administrative): Vogtle 3&4 SR 3.7.7.3 is to verify each SFW pump breaker trips open on an actual or simulated actuation signal, except for pump breaker maintained in the tripped condition. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.3.1.4, to perform TADOT of the reactor trip breakers. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					SFCP is consistent with the intent of TSTF-425. Start vs stop but both are logic tests.
Steam Generator (SG) Isolation Valves	3.7.10.1	3.5.2.5 50.55a(f)		X	Variation #3 (administrative): Vogtle 3&4 SR 3.7.10.1 is to verify each SG PORV, PORV block valve, and SG blowdown isolation valve strokes closed. This Surveillance is to be performed at a Frequency in accordance with the IST Program. This SR is functionally analogous to other NUREG-1431 Surveillances that verify valves actuate to the correct position on an actual or simulated signal, such as SR 3.5.2.5 for ECCS automatic valves. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
SG Isolation Valves	3.7.10.2	3.7.2.1 50.55a(f)		X	Variation #3 (administrative): Vogtle 3&4 SR 3.7.10.2 is to verify the isolation time of each PORV block valve and SG blowdown isolation valve is within limits. This Surveillance is to be performed at a Frequency in accordance with the IST Program. This SR is functionally analogous to other NUREG-1431 Surveillances that verify valve isolation times are within limits in accordance with the IST Program, such as SR 3.7.2.1 for MSIVs. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-596.
SG Isolation Valves	3.7.10.3	3.7.2.2	X		Variation #3 (administrative): Vogtle 3&4 SR 3.7.10.3 is to verify each SG PORV, PORV block valve, and SG blowdown isolation valve actuates to the isolation position on an actual or simulated actuation signal. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to other NUREG-1431 Surveillances that verify valve actuation to the isolation position, such as SR 3.7.2.2 for MSIVs. The proposed change to relocate the Frequency to the

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Spent Fuel Pool Boron Concentration	3.7.11.1	3.7.16.1	X		Numbering difference only.
DC Sources – Operating	3.8.1.1	3.8.4.1	X		Numbering difference only.
DC Sources – Operating	3.8.1.2	3.8.4.2	X		Numbering difference only.
DC Sources – Operating	3.8.1.3	3.8.4.3	X		Numbering difference only.
DC Sources – Operating	3.8.1.4	n/a 3.8.1.11	X		Variation #2 (administrative): Vogtle 3&4 SR 3.8.1.4 is to main control room air supply radiation monitor sample pump deenergizes on an actual or simulated actuation signal. This Surveillance is to be performed at a 24-month Frequency. This SR is functionally analogous to NUREG-1431, SR 3.8.1.11, which demonstrates shedding of nonessential loads and energization of the emergency buses and respective loads upon an actual or simulated loss of offsite power signal. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Inverters – Operating	3.8.3.1	3.8.7.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.8.3.1 is to verify correct inverter voltage, frequency, and alignment to required AC instrument and control buses. This Surveillance is to be performed at a 7-day Frequency. This SR is similar to NUREG-1431 SR 3.8.7.1 to verify correct inverter voltage, [frequency,] and alignment to required vital buses. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Inverters – Shutdown	3.8.4.1	3.8.8.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.8.4.1 is to verify correct inverter voltage, frequency, and alignments to required AC instrument and control buses. This Surveillance is to be performed at a 7-day

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
					Frequency. This SR is similar to NUREG-1431 SR 3.8.7.1 to verify correct inverter voltage, [frequency,] and alignments to required vital buses. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Distribution Systems – Operating	3.8.5.1	3.8.9.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.8.5.1 is to verify correct breaker and switch alignments and voltage to required DC and AC instrument and control electrical power distribution subsystems. This surveillance is performed on a 7-day Frequency. This SR is similar to NUREG-1431 SR 3.8.9.1 to verify correct breaker alignments and voltage to [required] AC, DC, and AC vital bus electrical power distribution subsystems. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Distribution Systems – Shutdown	3.8.6.1	3.8.10.1	X		Variation #3 (administrative): Vogtle 3&4 SR 3.8.6.1 is to verify correct breaker and switch alignments and voltage to required DC and AC instrument and control electrical power distribution subsystems. This surveillance is performed on a 7-day Frequency. This SR is similar to NUREG-1431 SR 3.8.10.1 to verify correct breaker alignments and voltage to [required] AC, DC, and AC vital bus electrical power distribution subsystems. The proposed change to relocate the Frequency to the SFCP and revise the SR Frequency to refer to the SFCP is consistent with the intent of TSTF-425.
Battery Parameters	3.8.7.1	3.8.6.1	X		Numbering difference only.
Battery Parameters	3.8.7.2	3.8.6.2	X		Numbering difference only.
Battery Parameters	3.8.7.3	3.8.6.3	X		Numbering difference only.
Battery Parameters	3.8.7.4	3.8.6.4	X		Numbering difference only.
Battery Parameters	3.8.7.5	3.8.6.5	X		Numbering difference only.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
Battery Parameters	3.8.7.6	3.8.6.6	X		Numbering difference only. (The proposed change only relocates the periodic (60-month) Frequency to the SFCP and revises the SR Frequency to refer to the SFCP, consistent with TSTF-425. It does not change the aspect of the Frequency that is related to battery age; i.e., 25% of expected life.)
Boron Concentration	3.9.1.1	3.9.1.1	X		None.
Unborated Water Source Flow Paths	3.9.2.1	3.9.2.1	X		None.
Nuclear Instrumentation	3.9.3.1	3.9.3.1	X		None.
Nuclear Instrumentation	3.9.3.2	3.9.3.2	X		None.
Refueling Cavity Water Level	3.9.4.1	3.9.7.1	X		Numbering difference only.
Inservice Testing Program	5.5.3	n/a		X	TSTF-596 Variation #3: Vogtle 3&4 TS 5.5.3 identifies the Inservice Testing Program as “the licensee program that fulfills the requirements of 10 CFR 50.55a(f).” The Vogtle TS 5.5.3 IST Program description is identical to the NUREG-1431 definition of INSERVICE TESTING PROGRAM, which is proposed for deletion by TSTF-596. Therefore, the proposed change to delete the Inservice Testing Program from Vogtle TS 5.5.3 is consistent with the intent of the TSTF-596 change to delete the definition of INSERVICE TESTING PROGRAM from the NUREG-1431 Westinghouse STS.
Main Control Room Envelope Habitability Program	5.5.12	5.5.17		X	TSTF-596 Variation #5: In addition to the TS numbering and Program naming differences between the NUREG-1431 STS and the VEGP TS, there are minor differences between the descriptions of the air inleakage testing and the CRE/MCRE boundary testing and the ventilation system designations. These are considered administrative variations. Therefore, the proposed change to relocate the testing frequency to the SFCP and revise the Program description to refer to the SFCP is consistent with the intent of TSTF-596.

TS Title	VEGP 3&4 SR	NUREG-1431 STS SR	TSTF-425	TSTF-596	Difference or Variation
Ventilation Filter Testing Program (VFTP)	5.5.13	5.5.10		X	TSTF-596 Variation #4: In addition to the TS numbering difference between the NUREG-1431 STS and the VEGP TS, there are minor differences between how the various aspects of the VFTP are described in the VEGP TS versus the NUREG-1431 STS. These are considered administrative variations that affect how the markup is presented, without having a material impact on the TSTF described changes. Therefore, the proposed change to relocate the testing frequency to the SFCP and revise the Program description to refer to the SFCP is consistent with the intent of TSTF-596.
Surveillance Frequency Control Program (SFCP)	5.5.16	5.5.17			Numbering difference only. The proposed changes to add the SFCP to TS Section 5.5, Programs and Manuals, and to expand the applicability of the SFCP to include other periodic testing frequencies, are consistent with the changes presented in TSTF-425 and TSTF-596, respectively.

Attachment 4 to NL-26-0017
Proposed Technical Specification Changes (Mark-ups)

License Amendment Request to Revise Technical Specifications to Include a Surveillance Frequency Control Program

Insertions Denoted by underlined Blue text and Deletions by ~~Red Strikethrough~~

1.0 USE AND APPLICATION

1.1 Definitions

- NOTE -

The defined terms of this section appear in capitalized type and are applicable throughout these Technical Specifications and Bases.

Term	Definition
ACTIONS	ACTIONS shall be that part of a Specification that prescribes Required Actions to be taken under designated Conditions within specified Completion Times.
ACTUATION LOGIC TEST	An ACTUATION LOGIC TEST shall be the application of various simulated or actual input combinations in conjunction with each possible interlock logic state required for OPERABILITY of a logic circuit and the verification of the required logic output. The ACTUATION LOGIC TEST may be performed by means of any series of sequential, overlapping, or total steps.
AXIAL FLUX DIFFERENCE (AFD)	AFD shall be the difference in normalized flux signals between the top and bottom halves of a two-section excore neutron detector.
CHANNEL CALIBRATION	A CHANNEL CALIBRATION shall be the adjustment, as necessary, of the channel output such that it responds within the necessary range and accuracy to known values of the parameter that the channel monitors. The CHANNEL CALIBRATION shall encompass all devices in the channel required for OPERABILITY. Calibration of instrument channels with resistance temperature detector (RTD), thermocouple, or reactor coolant pump speed sensors may consist of an inplace qualitative assessment of sensor behavior and normal calibration of the remaining adjustable devices in the channel. The CHANNEL CALIBRATION may be performed by means of any series of sequential, overlapping, or total channel steps, and each step must be performed within the Frequency in the SFCP for the devices included in this step.

1.1 Definitions

CHANNEL CHECK	A CHANNEL CHECK shall be the qualitative assessment, by observation, of channel behavior during operation. This determination shall include, where possible, comparison of the channel indication and status to other indications or status derived from independent instrument channels measuring the same parameter.
CHANNEL OPERATIONAL TEST (COT)	A COT shall be the injection of a simulated or actual signal into the channel as close to the sensor as practicable to verify OPERABILITY of all devices in the channel required for channel OPERABILITY. The COT shall include adjustments, as necessary, of the required alarm, interlock, and trip setpoints required for channel OPERABILITY such that the setpoints are within the necessary range and accuracy. The COT may be performed by means of any series of sequential, overlapping, or total channel steps, and each step must be performed within the Frequency in the SFCP for the devices included in this step.
CORE OPERATING LIMITS REPORT (COLR)	The COLR is the unit specific document that provides cycle specific parameter limits for the current reload cycle. These cycle specific parameter limits shall be determined for each reload cycle in accordance with Specification 5.6.3. Plant operation within these limits is addressed in individual Specifications.
DOSE EQUIVALENT I-131	DOSE EQUIVALENT I-131 shall be that concentration of I-131 (microcuries/gram) that alone would produce the same committed effective dose equivalent as the quantity and isotopic mixture of I-130, I-131, I-132, I-133, I-134, and I-135 actually present. The dose conversion factors used for this calculation shall be those listed in Table 2.1 of EPA Federal Guidance Report No. 11, “Limiting Values of Radionuclide Intake and Air Concentration and Dose Conversion Factors for Inhalation, Submersion, and Ingestion,” EPA-520/ 1-88-020, September 1988.
DOSE EQUIVALENT XE-133	DOSE EQUIVALENT XE-133 shall be that concentration of Xe-133 (microcuries per gram) that alone would produce the same effective dose equivalent as the quantity and isotopic mixture of noble gases (Kr-85m, Kr-85, Kr-87, Kr-88, Xe-131m, Xe-133m, Xe-133, Xe-135m, Xe-135, and Xe-138) actually present. The dose conversion factors used for this calculation shall be those listed in Table III.1 of EPA Federal Guidance Report No. 12, “External Exposure to

1.1 Definitions

TRIP ACTUATING DEVICE OPERATIONAL TEST (TADOT)	A TADOT shall consist of operating the trip actuating device and verifying the OPERABILITY of all devices in the channel required for trip actuating device OPERABILITY. The TADOT shall include adjustment, as necessary, of the trip actuating device so that it actuates at the required setpoint within the necessary accuracy. The TADOT may be performed by means of any series of sequential, overlapping, or total channel steps, and each step must be performed within the Frequency in the SFCP for the devices included in this step.
VENTED	VENTED shall be the condition when all required flow paths in ADS stage 1, 2, and 3, or alternative flow path with equivalent area, required by LCO 3.4.13, "Automatic Depressurization System (ADS) - Shutdown, RCS Open," are open.

3.1 REACTIVITY CONTROL SYSTEMS

3.1.1 SHUTDOWN MARGIN (SDM)

LCO 3.1.1 The SDM shall be within the limits specified in the COLR.

APPLICABILITY: MODE 2 with $k_{eff} < 1.0$.
MODES 3, 4, and 5.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. SDM not within limits.	A.1 Initiate boration to restore SDM to within limits.	15 minutes

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.1.1.1	Verify SDM to be within limits.	~~24 hours~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.1.2.1	---------- **- NOTE -** The predicted reactivity values may be adjusted (normalized) to correspond to the measured core reactivity prior to exceeding a fuel burnup of 60 effective full power days (EFPD) after each fuel loading. ----------	
	Verify measured core reactivity is within ±1% Δk/k of predicted values.	Once prior to entering MODE 1 after each refueling AND ---------- **- NOTE -** Only required to be performed after 60 EFPD ---------- In accordance with the SFCP ~~31 EFPD thereafter~~

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
C. Required Action and associated Completion Time of Condition B not met.	C.1 Be in MODE 3.	6 hours
D. More than one rod not within alignment limit.	D.1.1 Verify SDM is within the limits specified in the COLR. OR D.1.2 Initiate boration to restore required SDM to within limit. AND D.2 Be in MODE 3.	1 hour 1 hour 6 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.1.4.1	---------- **- NOTE -** Not required to be performed for rods associated with inoperable digital rod position indication or demand position indication. ---------- Verify individual rod positions within alignment limit.	~~12 hours~~ In accordance with the SFCP
SR 3.1.4.2	Verify rod freedom of movement (trippability) by moving each RCCA not fully inserted in the core ≥ 10 steps in either direction.	~~92 days~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.1.5.1	Verify each shutdown bank is within the insertion limits specified in the COLR.	~~12 hours~~ In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
B. Control bank sequence or overlap limits not met.	B.1.1 Verify SDM is within the limits specified in the COLR.	1 hour
	OR	
	B.1.2 Initiate boration to restore SDM to within limits.	1 hour
	AND	
	B.2 Restore control bank sequence and overlap to within limits.	2 hours
C. Required Action and associated Completion Time not met.	C.1 Be in MODE 2 with $k_{eff} < 1.0$.	6 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.1.6.1	Verify the estimated critical control bank position is within limits specified in the COLR.	Within 4 hours prior to achieving criticality
SR 3.1.6.2	Verify each control bank insertion is within the limits specified in the COLR.	~~12 hours~~ In accordance with the SFCP
SR 3.1.6.3	Verify sequence and overlap limits, specified in the COLR, are met for control banks not fully withdrawn from the core.	~~12 hours~~ In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
D. Required Action and Associated Completion Time of Condition C not met.	D.1 Be in MODE 3.	15 minutes

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.1.8.1	Verify the RCS lowest loop average temperature is ≥ 541°F.	~~30 minutes~~ In accordance with the SFCP
SR 3.1.8.2	Verify THERMAL POWER is ≤ 5% RTP.	~~30 minutes~~ In accordance with the SFCP
SR 3.1.8.3	Verify SDM is within the limits specified in the COLR.	~~24 hours~~ In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
B. Required Action and associated Completion Time of Condition A not met. OR Two CVS demineralized water isolation valves inoperable. OR Two CVS makeup line isolation valves inoperable.	B.1 Isolate the affected flow path(s) to the Reactor Coolant System by use of at least one closed manual or one closed and de-activated automatic valve.	1 hour

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.1.9.1	Verify two CVS demineralized water isolation valves and two CVS makeup line isolation valves stroke closed.	In accordance with the ~~Inservice Testing Program~~SFCP
SR 3.1.9.2	Verify closure time of each CVS makeup isolation valve is within limits on an actual or simulated actuation signal.	In accordance with the ~~Inservice Testing Program~~SFCP
SR 3.1.9.3	Verify each CVS demineralized water isolation valve actuates to the isolation position on an actual or simulated actuation signal.	~~24 months~~ In accordance with the SFCP

3.1 REACTIVITY CONTROL SYSTEMS

3.1.10 Rod Withdrawal Test Exception – MODE 5

LCO 3.1.10 During the performance of rod movement and rod drop time testing, the requirements of LCO 3.4.4, "RCS Loops," may be suspended provided boron concentration of the reactor coolant system is greater than the all rods out (ARO) boron concentration that provides $k_{eff} < 0.99$.

APPLICABILITY: MODE 5 with LCO 3.4.4 not met.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. Requirements of the LCO not met.	A.1 Initiate action to fully insert all rods.	Immediately
	AND	
	A.2 Place the Plant Control System in a condition incapable of rod withdrawal.	1 hour

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.1.10.1	Verify boron concentration of the reactor coolant system is greater than the ARO boron concentration providing $k_{eff} < 0.99$.	~~12 hours~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.2.1.3	---------- **- NOTE -** Not required to be performed until 24 hours after OPDMS not monitoring parameters. ---------- Verify $F_Q^C(Z)$ within limit.	Once within 24 hours after achieving equilibrium conditions after exceeding, by ≥ 10% RTP, the THERMAL POWER at which $F_Q^C(Z)$ was last verified AND ~~31 effective full power days (EFPD) thereafter~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.2.1.4	---------- **- NOTE -** Not required to be performed until 24 hours after OPDMS not monitoring parameters. ---------- Verify $F_Q^W(Z)$ within limits.	Once within 24 hours after achieving equilibrium conditions after exceeding, by ≥ 10% RTP, the THERMAL POWER at which $F_Q^W(Z)$ was last verified AND ~~31 EFPD thereafter~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.2.2.1	---------- **- NOTE -** Not required to be performed if OPDMS was monitoring parameters upon exceeding 75% RTP. ---------- Verify $F^N_{\Delta H}$ within limits specified in the COLR.	Once after each refueling prior to THERMAL POWER exceeding 75% RTP
SR 3.2.2.2	---------- **- NOTE -** Not required to be performed until 24 hours after OPDMS not monitoring parameters. ---------- Verify $F^N_{\Delta H}$ within limits specified in the COLR.	~~31 effective full power days (EFPD)~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

- NOTE -

Not required to be performed until 7 days after the last verification of OPDMS parameters.

SURVEILLANCE		FREQUENCY
SR 3.2.3.1	Verify AFD within limits for each OPERABLE excore channel.	~~7 days~~ In accordance with the SFCP
SR 3.2.3.2	Update the target flux difference.	Once within 31 EFPD after each refueling AND ~~31 EFPD thereafter~~ In accordance with the SFCP
SR 3.2.3.3	--- **- NOTE -** The initial target flux difference after each refueling may be determined from design predictions. --- Determine, by measurement, the target flux difference.	Once within 31 EFPD after each refueling AND ~~92 EFPD thereafter~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

- NOTE -

Not required to be performed until 12 hours after the last verification of OPDMS parameters.

SURVEILLANCE		FREQUENCY
SR 3.2.4.1	--- **- NOTES -** 1. With one Power Range Neutron Flux channel inoperable and THERMAL POWER < 75% RTP, the remaining three Power Range Neutron Flux channels can be used for calculating QPTR. 2. SR 3.2.4.2 may be performed in lieu of this Surveillance. --- Verify QPTR within limit by calculation.	~~7 days~~ In accordance with the SFCP
SR 3.2.4.2	--- **- NOTE -** Not required to be performed until 12 hours after input from one or more Power Range Neutron Flux channels are inoperable with THERMAL POWER ≥ 75% RTP. --- Verify QPTR is within limit using a minimum of 4 symmetric pairs of fixed incore detectors.	~~12 hours~~ In accordance with the SFCP

3.2 POWER DISTRIBUTION LIMITS

3.2.5 On-Line Power Distribution Monitoring System (OPDMS)-Monitored Parameters

LCO 3.2.5 The following parameters shall not exceed their operating limits as specified in the COLR:

a. Peak Linear Heat Rate;
b. $F^N_{\Delta H}$;
c. DNBR; and
d. SDM.

APPLICABILITY: MODE 1 with THERMAL POWER ≥ 25% RTP and with OPDMS monitoring parameters a, b, and c.
MODE 1 with OPDMS monitoring parameter d.
MODE 2 with k_{eff} ≥ 1.0 and OPDMS monitoring parameter d.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. One or more of the parameters a. through c. above not within limits.	A.1 Restore all parameters to within limits.	1 hour
B. Required Action and associated Completion Time of Condition A not met.	B.1 Reduce THERMAL POWER to < 25% RTP.	4 hours
C. Parameter d above not within limits.	C.1 Initiate boration to restore SDM to within limits.	15 minutes

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.2.5.1	Verify the parameters a. through d. to be within their limits.	~~24 hours~~ In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
E. As required by Required Action C.1 and referenced in Table 3.3.1-1.	E.1 Reduce THERMAL POWER to < P-10.	6 hours

SURVEILLANCE REQUIREMENTS

- NOTE -

Refer to Table 3.3.1-1 to determine which SRs apply for each RTS Function.

SURVEILLANCE		FREQUENCY
SR 3.3.1.1	**- NOTES -** 1. Required to be met within 12 hours after reaching 15% RTP. 2. If the calorimetric heat balance is ≥ 15% RTP, and if the nuclear instrumentation channel indicated power is: a. lower than the calorimetric measurement by > 5% RTP, then adjust the nuclear instrumentation channel upward to match the calorimetric measurement. b. higher than the calorimetric measurement, then no adjustment is required. Compare results of calorimetric heat balance to nuclear instrument channel output.	~~24 hours~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.3.1.2	---------- **- NOTES -** 1. Adjust the conversion factor, ΔT°, in the ΔT power calculation ($q_{\Delta T}$) if absolute difference between $q_{\Delta T}$ and the calorimetric measurement is > 3% RTP. 2. Required to be met within 12 hours after reaching 50% RTP. 3. If the calorimetric heat balance is < 70% RTP, and if $q_{\Delta T}$ is: a. lower than the calorimetric measurement by > 5%, then adjust ΔT° to match the calorimetric measurement. b. higher than the calorimetric measurement, then no adjustment is required. ----------	
	Compare results of calorimetric heat balance to the ΔT power calculation ($q_{\Delta T}$) output.	~~24 hours~~In accordance with the SFCP
SR 3.3.1.3	---------- **- NOTES -** 1. Adjust nuclear instrument channel in PMS if absolute difference is ≥ 1.5% AFD. 2. Required to be met within 24 hours after reaching 20% RTP. ----------	
	Compare results of the incore detector measurements to nuclear instrument channel AXIAL FLUX DIFFERENCE.	~~31 effective full power days (EFPD)~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.3.1.4	---------- **- NOTE -** Required to be met within 24 hours after reaching 50% RTP. ---------- Calibrate excore channels to agree with incore detector measurements.	~~92 EFPD~~In accordance with the SFCP
SR 3.3.1.5	---------- **- NOTE -** This Surveillance shall include verification that the time constants are adjusted to within limits. ---------- Perform CHANNEL CALIBRATION in accordance with Setpoint Program.	~~24 months~~In accordance with the SFCP
SR 3.3.1.6	---------- **- NOTE -** Neutron detectors are excluded from CHANNEL CALIBRATION. ---------- Perform CHANNEL CALIBRATION in accordance with Setpoint Program.	~~24 months~~In accordance with the SFCP
SR 3.3.1.7	---------- **- NOTE -** Verification of setpoint is not required. ---------- Perform TADOT.	~~24 months~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.3.1.8	---------- **- NOTE -** Neutron detectors are excluded from response time testing. ---------- Verify RTS RESPONSE TIME is within limits.	~~24 months on a STAGGERED TEST BASIS~~In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
E. Required Action and associated Completion Time of Condition D not met.	E.1 Intiate action to fully insert all rods. AND E.2 Place the Plant Control System in a condition incapable of rod withdrawal.	1 hour 1 hour
F. Three or more channels inoperable.	F.1 Open reactor trip breakers (RTBs).	Immediately

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.3.2.1	Perform CHANNEL CHECK.	~~12 hours~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.3.2.2	---------- **- NOTE -** Neutron detectors are excluded from CHANNEL CALIBRATION. ---------- Perform CHANNEL CALIBRATION in accordance with Setpoint Program.	~~24 months~~In accordance with the SFCP
SR 3.3.2.3	---------- **- NOTE -** Neutron detectors are excluded from response time testing. ---------- Verify RTS RESPONSE TIME is within limits.	~~24 months on a STAGGERED TEST BASIS~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.3.3.1	---------- **- NOTE -** Not required to be met in MODE 1. ---------- Perform CHANNEL CHECK.	~~12 hours~~In accordance with the SFCP
SR 3.3.3.2	---------- **- NOTE -** Neutron detectors are excluded from CHANNEL CALIBRATION. ---------- Perform CHANNEL CALIBRATION in accordance with Setpoint Program.	~~24 months~~In accordance with the SFCP
SR 3.3.3.3	---------- **- NOTE -** Neutron detectors are excluded from response time testing. ---------- Verify RTS RESPONSE TIME is within limits.	~~24 months on a STAGGERED TEST BASIS~~In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
D. Required Action and associated Completion Time of Condition C not met. OR One or more Functions with three or more channels inoperable in MODE 3, 4, or 5.	D.1 Initiate action to fully insert all rods. AND D.2 Place the Plant Control System in a condition incapable of rod withdrawal.	1 hour 1 hour

SURVEILLANCE REQUIREMENTS

- NOTE -
Refer to Table 3.3.4-1 to determine to which RTS ESFAS Function the SR applies.

SURVEILLANCE		FREQUENCY
SR 3.3.4.1	Verify RTS RESPONSE TIME is within limit.	~~24 months on a STAGGERED TEST BASIS~~In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
C. Required Action and associated Completion Time of Condition A not met in MODE 3, 4, or 5. OR One or more Functions with two manual actuation channels inoperable in MODE 3, 4, or 5.	C.1 Initiate action to fully insert all rods. AND C.2 Place the Plant Control System in a condition incapable of rod withdrawal.	1 hour 1 hour

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.3.5.1	---------- **- NOTE -** Verification of setpoint is not required. ---------- Perform TADOT.	~~24 months~~In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
D. Required Action and associated Completion Time of Condition A or B not met in MODE 3, 4, or 5. OR One or both Functions within three or more divisions inoperable in MODE 3, 4, or 5.	D.1 Initiate action to fully insert all rods. AND D.2 Place the Plant Control System in a condition incapable of rod withdrawal.	6 hours 6 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.3.7.1	Perform TADOT on both reactor trip breakers in one division.	~~92 days on a STAGGERED TEST BASIS~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.3.8.1	---------- **- NOTE -** Only required to be met for Source Range Neutron Flux Doubling. ---------- Perform CHANNEL CHECK.	~~12 hours~~In accordance with the SFCP
SR 3.3.8.2	---------- **- NOTES -** 1. This surveillance shall include verification that the time constants are adjusted to within limits. 2. Neutron detectors are excluded from CHANNEL CALIBRATION. ---------- Perform CHANNEL CALIBRATION in accordance with Setpoint Program.	~~24 months~~ In accordance with the SFCP
SR 3.3.8.3	---------- **- NOTES -** 1. Not applicable to Function 1.a. 2. Neutron detectors are excluded from response time testing. ---------- Verify ESF RESPONSE TIME is within limit.	~~24 months on a STAGGERED TEST BASIS~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.3.9.1	---------- **- NOTE -** Verification of setpoint not required . ---------- Perform TRIP ACTUATING DEVICE OPERATIONAL TEST (TADOT).	~~24 months~~ In accordance with the SFCP.

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.3.10.1	---------- **- NOTE -** This surveillance shall include verification that the time constants are adjusted to within limits. ---------- Perform CHANNEL CALIBRATION in accordance with Setpoint Program.	~~24 months~~ In accordance with the SFCP
SR 3.3.10.2	Verify ESF RESPONSE TIME is within limit.	~~24 months on a STAGGERED TEST BASIS~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.3.11.1	---------- **- NOTE -** This surveillance shall include verification that the time constants are adjusted to within limits. ---------- Perform CHANNEL CALIBRATION in accordance with Setpoint Program.	~~24 months~~ In accordance with the SFCP
SR 3.3.11.2	Verify ESF RESPONSE TIME is within limit.	~~24 months on a STAGGERED TEST BASIS~~ In accordance with the SFCP

3.3 INSTRUMENTATION

3.3.12 Engineered Safety Feature Actuation System (ESFAS) Reactor Trip Initiation

LCO 3.3.12 Three ESFAS Reactor Trip (P-4) divisions shall be OPERABLE.

APPLICABILITY: MODES 1, 2, 3, and 4.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. One required division inoperable.	A.1 Restore required division to OPERABLE status.	6 hours
B. Required Action and associated Completion Time of Condition A not met. OR Two or three required divisions inoperable.	B.1 Declare affected isolation valve(s) inoperable. AND B.2 Be in MODE 3. AND B.3 Be in MODE 4.	Immediately 6 hours 12 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.3.12.1	Perform TRIP ACTUATING DEVICE OPERATIONAL TEST (TADOT).	~~24 months~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

- NOTE -

Refer to Table 3.3.13-1 to determine which SRs apply for each ESFAS Main Control Room Isolation, Air Supply Initiation, and Electrical Load De-energization Function.

SURVEILLANCE		FREQUENCY
SR 3.3.13.1	--- **- NOTE -** This surveillance shall include verification that the time constants are adjusted to within limits. --- Perform CHANNEL CALIBRATION in accordance with Setpoint Program.	~~24 months~~ In accordance with the SFCP
SR 3.3.13.2	Verify ESF RESPONSE TIME is within limit.	~~24 months on a STAGGERED TEST BASIS~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.3.14.1	---------- **- NOTE -** This surveillance shall include verification that the time constants are adjusted to within limits. ---------- Perform CHANNEL CALIBRATION in accordance with Setpoint Program.	~~24 months~~ In accordance with the SFCP
SR 3.3.14.2	Verify ESF RESPONSE TIME is within limit.	~~24 months on a STAGGERED TEST BASIS~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

- NOTE -
Refer to Table 3.3.17-1 to determine which SRs apply for each PAM Function.

SURVEILLANCE		FREQUENCY
SR 3.3.17.1	**- NOTE -** Not required to be met for Neutron Flux (Intermdiate Range) in MODE 1. Perform CHANNEL CHECK for each required instrumentation channel that is normally energized.	~~31 days~~In accordance with the SFCP
SR 3.3.17.2	**- NOTE -** Neutron detectors are excluded from CHANNEL CALIBRATION. Perform CHANNEL CALIBRATION.	~~24 months~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

- NOTE -
Refer to Table 3.3.17-1 to determine which SRs apply for each PAM Function.

SURVEILLANCE		FREQUENCY
SR 3.3.17.1	**- NOTE -** Not required to be met for Neutron Flux (Intermdiate Range) in MODE 1. Perform CHANNEL CHECK for each required instrumentation channel that is normally energized.	~~31 days~~In accordance with the SFCP
SR 3.3.17.2	**- NOTE -** Neutron detectors are excluded from CHANNEL CALIBRATION. Perform CHANNEL CALIBRATION.	~~24 months~~In accordance with the SFCP

3.3 INSTRUMENTATION

3.3.18 Remote Shutdown Workstation (RSW)

LCO 3.3.18 The RSW shall be OPERABLE.

APPLICABILITY: MODES 1, 2, and 3.
MODE 4 with Reactor Coolant System (RCS) average temperature (T_{avg}) ≥ 350°F.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. RSW inoperable.	A.1 Restore to OPERABLE status.	30 days
B. Required Action and associated Completion Time not met.	B.1 Be in MODE 3. AND B.2 Be in MODE 4 with T_{avg} < 350°F.	6 hours 12 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.3.18.1	Verify each required transfer switch is capable of performing the required function.	~~24 months~~ In accordance with the SFCP
SR 3.3.18.2	Verify the RSW communicates indication and controls with Division A, B, C and D of the Protection and Safety Monitoring System (PMS).	~~24 months~~ In accordance with the SFCP
SR 3.3.18.3	Verify the OPERABILITY of the RSW hardware and software.	~~24 months~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.3.18.4	Perform TADOT of the reactor trip breaker open/closed indication.	~~24 months~~ In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
D. Required Action and associated Completion Time of Condition B not met. OR Required Action and associated Completion Time of Condition C not met.	D.1 Be in MODE 3. AND D.2 Be in MODE 5.	6 hours 36 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.3.19.1	---------- **- NOTE -** Verification of setpoint not required. ---------- Perform TRIP ACTUATION DEVICE OPERATIONAL TEST (TADOT).	~~24 months~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

- NOTE -
Refer to Table 3.3.20-1 to determine which SRs apply for each ADS and IRWST Injection Blocking Device Function.

SURVEILLANCE		FREQUENCY
SR 3.3.20.1	Verify each ADS and IRWST Injection Block switch is in the "unblock" position.	~~7 days~~ In accordance with the SFCP
SR 3.3.20.2	Perform CHANNEL CALIBRATION in accordance with Setpoint Program.	~~24 months~~ In accordance with the SFCP
SR 3.3.20.3	Perform ACTUATION LOGIC TEST of ADS and IRWST Injection Blocking Devices.	~~24 months~~ In accordance with the SFCP
SR 3.3.20.4	**- NOTE -** Verification of setpoint not required. Perform TRIP ACTUATING DEVICE OPERATIONAL TEST (TADOT) of ADS and IRWST Injection Block manual switches.	~~24 months~~ In accordance with the SFCP
SR 3.3.20.5	The following SRs of Specification 3.5.2, "Core Makeup Tanks (CMTs) – Operating" are applicable for each CMT: SR 3.5.2.2 SR 3.5.2.5 SR 3.5.2.6	In accordance with applicable SRs

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.1.1	Verify pressurizer pressure is greater than or equal to the limit specified in the COLR.	~~12 hours~~In accordance with the SFCP
SR 3.4.1.2	Verify RCS average temperature is less than or equal to the limit specified in the COLR.	~~12 hours~~In accordance with the SFCP
SR 3.4.1.3	Verify RCS total flow rate is ≥ 301,670 gpm and greater than or equal to the limit specified in the COLR.	~~12 hours~~In accordance with the SFCP
SR 3.4.1.4	Perform a CHANNEL CALIBRATION of differential pressure RCS total flow rate indication channels.	~~24 months~~In accordance with the SFCP
SR 3.4.1.5	---------- **- NOTE -** Not required to be performed until 24 hours after ≥ 90% RTP. ---------- Verify RCS total flow rate is ≥ 301,670 gpm and greater than or equal to the limit specified in the COLR as determined by precision heat balance or differential pressure RCS total flow rate indication measurements.	~~24 months~~In accordance with the SFCP

3.4 REACTOR COOLANT SYSTEM (RCS)

3.4.2 RCS Minimum Temperature for Criticality

LCO 3.4.2 Each RCS loop average temperature (T_{avg}) shall be ≥ 551°F.

APPLICABILITY: MODE 1,
MODE 2 with k_{eff} ≥ 1.0.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. T_{avg} in one or more RCS loops not within limit.	A.1 Be in MODE 2 with k_{eff} < 1.0.	30 minutes

SURVEILLANCE REQUIREMENTS

SURVEILLANCE	FREQUENCY
SR 3.4.2.1 Verify RCS T_{avg} in each loop ≥ 551°F.	~~12 hours~~In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
C. ---------- **- NOTE -** Required Action C.2 shall be completed whenever this Condition is entered. ---------- Requirements of LCO not met any time in other than MODE 1, 2, 3, or 4.	C.1 Initiate action to restore parameter(s) to within limits. AND C.2 Determine RCS is acceptable for continued operation.	Immediately Prior to entering MODE 4

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.3.1	---------- **- NOTE -** Only required to be performed during RCS heatup and cooldown operations and RCS inservice leak and hydrostatic testing. ---------- Verify RCS pressure, RCS temperature, and RCS heatup and cooldown rates within limits specified in the PTLR.	~~30 minutes~~In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
B. ---------- **- NOTE -** Required Actions must be completed whenever Condition B is entered. ---------- Requirements of LCO not met in MODE 3, 4, or 5.	B.1 Suspend start of any RCP. AND B.2 Initiate action to fully insert all rods. AND B.3 Place the Plant Control System in a condition incapable of rod withdrawal.	Immediately 1 hour 1 hour

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.4.1	Verify each RCS loop is in operation with variable speed control bypassed.	~~12 hours~~In accordance with the SFCP

3.4 REACTOR COOLANT SYSTEM (RCS)

3.4.5 Pressurizer

LCO 3.4.5 The pressurizer water level shall be ≤ 92% of span.

APPLICABILITY: MODES 1, 2, and 3,
MODE 4 with all four cold leg temperatures > 275°F.

ACTIONS

CONDITION	REQUIRED ACTION		COMPLETION TIME
A. Pressurizer water level not within limit.	A.1	Be in MODE 3.	6 hours
	AND		
	A.2	Initiate action to fully insert all rods.	6 hours
	AND		
	A.3	Place the Plant Control System in a condition incapable of rod withdrawal.	6 hours
	AND		
	A.4	Be in MODE 4 with at least one cold leg temperature ≤ 275°F.	24 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.5.1	Verify pressurizer water level ≤ 92% of span.	~~12 hours~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.6.1	Verify each pressurizer safety valve OPERABLE ~~in accordance with the Inservice Testing Program~~ with lift settings ≥ 2460 psig and ≤ 2510 psig. Following testing, lift settings shall be within ±1%.	In accordance with the ~~Inservice Testing Program~~SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
C. Required Action and associated Completion Time not met. OR Primary to secondary LEAKAGE not within limit.	C.1 Be in MODE 3. AND C.2 Be in MODE 5.	6 hours 36 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.7.1	---------- **- NOTES -** 1. Not required to be performed until 12 hours after establishment of steady state operation. 2. Not applicable to primary to secondary LEAKAGE. ---------- Verify RCS operational LEAKAGE is within limits by performance of RCS water inventory balance.	~~72 hours~~In accordance with the SFCP
SR 3.4.7.2	---------- **- NOTE -** Not required to be performed until 12 hours after establishment of steady state operation. ---------- Verify primary to secondary LEAKAGE is ≤ 150 gallons per day through any one SG.	~~72 hours~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.8.1	Verify at least one RCP is in operation with total flow through the core ≥ 3,000 gpm.	~~12 hours~~In accordance with the SFCP

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
D. Required Action and associated Completion Time of Condition A, B, or C not met.	D.1 Be in MODE 3. AND D.2 Be in MODE 5.	6 hours 36 hours
E. All required monitors inoperable.	E.1 Enter LCO 3.0.3.	Immediately

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.9.1	Perform a CHANNEL CHECK.	~~12 hours~~In accordance with the SFCP
SR 3.4.9.2	Perform a COT of containment atmosphere F18 particulate monitor.	~~92 days~~In accordance with the SFCP
SR 3.4.9.3	Perform a CHANNEL CALIBRATION.	~~24 months~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.10.1	Verify reactor coolant DOSE EQUIVALENT XE-133 specific activity ≤ 280 μCi/gm.	~~7 days~~ In accordance with the SFCP
SR 3.4.10.2	---------- **- NOTE -** Only required to be performed in MODE 1. ---------- Verify reactor coolant DOSE EQUIVALENT I-131 specific activity ≤ 1.0 μCi/gm.	~~14 days~~ In accordance with the SFCP AND Between 2 to 6 hours after a THERMAL POWER change of ≥ 15% of RTP within a 1 hour period

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
D. Required Action and associated Completion Time of Condition A, B, or C not met. OR Condition A and Condition B entered concurrently. OR Three or more flow paths in ADS stage 1, 2, and 3 inoperable. OR LCO not met for reasons other than Condition A, B, or C.	D.1 Be in MODE 3. AND D.2 Be in MODE 5.	6 hours 36 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.11.1	Verify the motor operated valve in series with each 4th stage ADS valve is open.	~~12 hours~~In accordance with the SFCP
SR 3.4.11.2	Verify each stage 1, 2, and 3 ADS valve strokes open.	In accordance with the ~~Inservice Testing Program~~SFCP
SR 3.4.11.3	Verify each stage 4 ADS valve is OPERABLE in accordance with 10 CFR 50.55a(f). ~~the Inservice Testing Program.~~	In accordance with the ~~Inservice Testing Program~~SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.4.11.4	Verify each stage 1, 2, and 3 ADS valve actuates to the open position on an actual or simulated actuation signal.	~~24 months~~In accordance with the SFCP
SR 3.4.11.5	---------- **- NOTE -** Squib actuation may be excluded. ---------- Verify continuity of the circuit from the Protection Logic Cabinets to each stage 4 ADS valve.	 ~~24 months~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.13.1	Verify each required ADS stage 1, 2, and 3 valve is in the open position.	~~12 hours~~In accordance with the SFCP
SR 3.4.13.2	For each required flow path in ADS stage 4, the following SRs are applicable: SR 3.4.11.1 SR 3.4.11.3 SR 3.4.11.5	In accordance with applicable SRs

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.14.1	---------- **- NOTE -** Only required to be met when accumulator pressure is greater than or equal to the maximum RCS pressure for the existing RCS cold leg temperature allowed by the P/T limit curves provided in the PTLR. ---------- Verify each accumulator is isolated.	~~12 hours~~In accordance with the SFCP
SR 3.4.14.2	---------- **- NOTE -** Only required to be met when complying with LCO 3.4.14.a. ---------- Verify both RNS suction isolation valves in one RNS suction flow path are open.	~~12 hours~~In accordance with the SFCP
SR 3.4.14.3	---------- **- NOTE -** Only required to be met when complying with LCO 3.4.14.a. ---------- Verify CVS makeup line containment isolation valve, CVS-PL-V091, is closed.	~~12 hours~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.4.14.4	---------- **- NOTE -** Only required to be met when complying with LCO 3.4.14.b. ---------- Verify RCS vent ≥ 4.15 square inches is open.	~~12 hours for unlocked-open vent~~ ~~AND~~ ~~31 days for locked-open vent~~In accordance with the SFCP
SR 3.4.14.5	---------- **- NOTE -** Only required to be met when complying with LCO 3.4.14.a. ---------- Verify the lift setting of each RNS suction relief valve ~~in accordance with the Inservice Testing Program~~ is within limits.	In accordance with the ~~Inservice Testing Program~~SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
	A.2 Verify a second OPERABLE PIV can meet the leakage limits. This valve is required to be a check valve, or a closed valve, if it isolates a line that penetrates containment.	72 hours
B. Required Action and associated Completion Time not met.	B.1 Be in MODE 3. AND B.2 Be in MODE 5.	6 hours 36 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.15.1	Verify leakage of each RCS PIV is equivalent to ≤ 0.5 gpm per nominal inch valve size up to a maximum of 5 gpm at an RCS pressure ≥ 2215 and ≤ 2255 psig.	~~24 months~~In accordance with the SFCP

3.4 REACTOR COOLANT SYSTEM (RCS)

3.4.16 Reactor Vessel Head Vent (RVHV)

LCO 3.4.16 The Reactor Vessel Head Vent shall be OPERABLE.

APPLICABILITY: MODES 1, 2, and 3,
MODE 4 with all four cold leg temperatures > 275°F.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. One flow path inoperable.	A.1 Restore flow path to OPERABLE status.	72 hours
B. Two flow paths inoperable.	B.1 Restore at least one flow path to OPERABLE status.	6 hours
C. Required Action and associated Completion Time not met.	C.1 Be in MODE 3. AND C.2 Be in MODE 4 with at least one cold leg temperature ≤ 275°F.	6 hours 24 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.16.1	Verify each RVHV valve strokes open.	In accordance with the ~~Inservice Testing Program~~SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.18.1	Verify each pressurizer heater circuit breaker trips open on an actual or simulated actuation signal, except for circuit breaker maintained in the tripped condition.	~~24 months~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.19.1	Verify each auxiliary spray and purification line isolation valve actuate to the isolation position on an actual or simulated actuation signal.	~~24 months~~In accordance with the SFCP

3.4 REACTOR COOLANT SYSTEM (RCS)

3.4.20 Chemical and Volume Control System (CVS) Letdown Isolation Valves

LCO 3.4.20 Each CVS letdown isolation valve shall be OPERABLE.

APPLICABILITY: MODE 5 below P-12 (Pressurizer Level) interlock,
MODE 6 with water level < 23 feet above the top of the reactor vessel flange.

ACTIONS

- NOTE -
CVS letdown flow path may be unisolated intermittently under administrative controls.

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. One CVS letdown isolation valve inoperable.	A.1 Restore CVS letdown isolation valve to OPERABLE status.	72 hours
B. Required Action and associated Completion Time of Condition A is not met. OR Two CVS letdown isolation valves inoperable.	B.1 Initiate action to isolate CVS letdown flow path by use of at least one closed and deactivated automatic valve.	Immediately

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.4.20.1	Verify CVS letdown isolation valves actuate to the isolation position on an actual or simulated actuation signal.	~~24 months~~In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.5.1.1	Verify each accumulator isolation valve is fully open.	~~12 hours~~ In accordance with the SFCP
SR 3.5.1.2	Verify the borated water volume in each accumulator is ≥ 1667 cu. ft., and ≤ 1732 cu. ft.	~~12 hours~~ In accordance with the SFCP
SR 3.5.1.3	Verify the nitrogen cover gas pressure in each accumulator is ≥ 637 psig and ≤ 769 psig.	~~12 hours~~ In accordance with the SFCP
SR 3.5.1.4	Verify the boron concentration in each accumulator is ≥ 2600 ppm and ≤ 2900 ppm.	~~31 days~~ In accordance with the SFCP AND ---------- **- NOTE -** Only required for affected accumulators. ---------- Once within 6 hours after each solution volume increase of ≥ 51 cu. ft. that is not the result of addition from the in-containment refueling water storage tank
SR 3.5.1.5	Verify power is removed from each accumulator isolation valve operator when pressurizer pressure is ≥ 2000 psig.	~~31 days~~ In accordance with the SFCP
SR 3.5.1.6	Verify system flow performance of each accumulator in accordance with the System Level OPERABILITY Testing Program.	~~10 years~~ In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
D. One CMT inlet line with noncondensible gas volume not within limit.	D.1 Restore CMT inlet line noncondensible gas volume to within limit.	24 hours
E. One CMT inoperable for reasons other than Condition A, B, or D.	E.1 Restore CMT to OPERABLE status.	1 hour from discovery of LCO 3.5.2 Condition E entry concurrent with LCO 3.5.1 Condition B entry AND 8 hours
F. Required Action and associated Completion Time of Condition A, B, C, D, or E not met. OR Two CMTs inoperable for reasons other than Condition C.	F.1 Be in MODE 3. AND F.2 Be in MODE 5.	6 hours 36 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.5.2.1	Verify the temperature of the borated water in each CMT is < 120°F.	~~24 hours~~ In accordance with the SFCP
SR 3.5.2.2	Verify each CMT inlet isolation valve is fully open.	~~12 hours~~ In accordance with the SFCP
SR 3.5.2.3	Verify the volume of noncondensible gases in each CMT inlet line has not caused the high-point water level to drop below the sensor.	~~24 hours~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.5.2.4	---------- **- NOTE -** Not required to be performed for Unit 4 until entry into TS 3.5.3 Applicability during startup from 4R01. ---------- Verify the boron concentration in each CMT is ≥ 3400 ppm and ≤ 4500 ppm.	~~31 days~~ In accordance with the SFCP
SR 3.5.2.5	Verify each CMT outlet isolation valve strokes open.	In accordance with the ~~Inservice Testing Program~~ SFCP
SR 3.5.2.6	Verify each CMT outlet isolation valve actuates to the open position on an actual or simulated actuation signal.	~~24 months~~ In accordance with the SFCP
SR 3.5.2.7	Verify system flow performance of each CMT in accordance with the System Level OPERABILITY Testing Program.	~~10 years~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.5.3.3	---------- **- NOTE -** Only required to be met in MODE 5 with RCS not VENTED. ---------- Verify the borated water volume is ≥ 2450 cu. ft.	~~7 days~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.5.4.1	Verify the PRHR HX outlet manual isolation valve is fully open.	~~12 hours~~ In accordance with the SFCP
SR 3.5.4.2	Verify the PRHR HX inlet motor operated isolation valve is open.	~~12 hours~~ In accordance with the SFCP
SR 3.5.4.3	Verify the volume of noncondensible gases in the PRHR HX inlet line has not caused the high-point water level to drop below the sensor.	~~24 hours~~ In accordance with the SFCP
SR 3.5.4.4	---------- **- NOTE -** Only required to be met when one or more reactor coolant pumps (RCPs) are in operation. ---------- Verify one Loop 1 RCP is in operation.	~~12 hours~~ In accordance with the SFCP
SR 3.5.4.5	Verify power is removed from the PRHR HX inlet motor operated isolation valve.	~~31 days~~ In accordance with the SFCP
SR 3.5.4.6	Verify both PRHR HX air operated outlet isolation valves stroke open and both IRWST gutter isolation valves stroke closed.	In accordance with the ~~Inservice Testing Program~~ SFCP
SR 3.5.4.7	Verify by visual inspection that the IRWST gutter and downspout screens are not restricted by debris.	~~24 months~~ In accordance with the SFCP
SR 3.5.4.8	Verify both PRHR HX air operated outlet isolation valves actuate to the open position and both IRWST gutter isolation valves actuate to the isolation position on an actual or simulated actuation signal.	~~24 months~~ In accordance with the SFCP
SR 3.5.4.9	Verify PRHR HX heat transfer performance in accordance with the System Level OPERABILITY Testing Program.	~~10 years~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.5.6.1	Verify the IRWST water temperature is < 120°F.	~~24 hours~~ In accordance with the SFCP
SR 3.5.6.2	Verify the IRWST borated water volume is ≥ 73,100 cu. ft.	~~24 hours~~ In accordance with the SFCP
SR 3.5.6.3	Verify the volume of noncondensible gases in each of the four IRWST injection squib valve outlet line pipe stubs has not caused the high-point water level to drop below the sensor.	~~24 hours~~ In accordance with the SFCP
SR 3.5.6.4	Verify the IRWST boron concentration is ≥ 2600 ppm and ≤ 2900 ppm.	~~31 days~~ In accordance with the SFCP AND Once within 6 hours after each solution volume increase of ≥ 15,000 gal
SR 3.5.6.5	Verify each motor operated IRWST isolation valve is fully open.	~~12 hours~~ In accordance with the SFCP
SR 3.5.6.6	Verify power is removed from each motor operated IRWST isolation valve.	~~31 days~~ In accordance with the SFCP
SR 3.5.6.7	Verify each motor operated containment recirculation isolation valve is fully open.	~~31 days~~ In accordance with the SFCP
SR 3.5.6.8	Verify each IRWST injection and containment recirculation squib valve is OPERABLE in accordance with ~~the Inservice Testing Program~~10 CFR 50.55a(f).	In accordance with the ~~Inservice Testing Program~~ SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.5.6.9	---------- - NOTE - Squib actuation may be excluded. ---------- Verify continuity of the circuit from the Protection Logic Cabinets to each IRWST injection and containment recirculation squib valve on an actual or simulated actuation signal.	~~24 months~~ In accordance with the SFCP
SR 3.5.6.10	Verify by visual inspection that the IRWST screens and the containment recirculation screens are not restricted by debris.	~~24 months~~ In accordance with the SFCP
SR 3.5.6.11	Verify IRWST injection and recirculation system flow performance in accordance with the System Level OPERABILITY Testing Program.	~~10 years~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.5.7.1	---------- **- NOTE -** Penetration flow path(s) may be unisolated intermittently under administrative controls. ---------- Verify Spent Fuel Pool Cooling System containment isolation valves are closed.	~~31 days~~ In accordance with the SFCP
SR 3.5.7.2	For the IRWST and flow paths required to be OPERABLE, the following SRs are applicable: SR 3.5.6.1 SR 3.5.6.2 SR 3.5.6.4 SR 3.5.6.5 SR 3.5.6.6 SR 3.5.6.7 SR 3.5.6.8 SR 3.5.6.9 SR 3.5.6.10 SR 3.5.6.11	In accordance with applicable SRs
SR 3.5.7.3	---------- **- NOTE -** Not required to be met during RCS vacuum fill operations. ---------- For the IRWST and flow paths required to be OPERABLE, the following SR is applicable: SR 3.5.6.3	In accordance with applicable SR

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.5.8.1	Verify the IRWST and refueling cavity water temperature is < 120°F.	~~24 hours~~ In accordance with the SFCP
SR 3.5.8.2	Verify the IRWST and refueling cavity water total borated water volume is ≥ 73,100 cu. ft.	~~24 hours~~ In accordance with the SFCP
SR 3.5.8.3	---------- **- NOTES -** 1. Penetration flow path(s) may be unisolated intermittently under administrative controls. 2. Only required to be met with refueling cavity and spent fuel pool volumes not in communication. ---------- Verify Spent Fuel Pool Cooling System containment isolation valves are closed.	~~31 days~~ In accordance with the SFCP
SR 3.5.8.4	Verify the IRWST and refueling cavity boron concentration is ≥ 2600 ppm and ≤ 2900 ppm.	~~31 days~~ In accordance with the SFCP AND Once within 6 hours after each solution volume increase of ≥ 15,000 gal
SR 3.5.8.5	For the IRWST and flow paths required to be OPERABLE, the following SRs are applicable: SR 3.5.6.3 SR 3.5.6.6 SR 3.5.6.8 SR 3.5.6.10 SR 3.5.6.5 SR 3.5.6.7 SR 3.5.6.9 SR 3.5.6.11	In accordance with applicable SRs

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
D. Required Action and associated Completion Time of Condition C is not met. OR One or more RCPs with two breakers inoperable in MODE 5.	D.1 Initiate action to trip affected RCP breaker(s).	Immediately

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.5.9.1	Verify each RCP breaker trips open on an actual or simulated actuation signal, except for breaker maintained in the tripped condition.	~~24 months~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.6.2.1	---------- **- NOTES -** 1. An inoperable air lock door does not invalidate the previous successful performance of the overall air lock leakage test. 2. Results shall be evaluated against acceptance criteria applicable to SR 3.6.1.1. ---------- Perform required air lock leakage rate testing in accordance with the Containment Leakage Rate Testing Program.	In accordance with the Containment Leakage Rate Testing Program
SR 3.6.2.2	Verify only one door in the air lock can be opened at a time.	~~24 months~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.6.3.1	Verify each 16 inch containment purge valve is closed, except when the 16 inch containment purge valves are open for pressure control, ALARA or air quality considerations for personnel entry, or for Surveillances which require the valves to be open.	~~31 days~~ In accordance with the SFCP
SR 3.6.3.2	---------- **- NOTE -** Valves and blind flanges in high radiation areas may be verified by use of administrative controls. ---------- Verify each containment isolation manual valve and blind flange that is located outside containment and not locked, sealed, or otherwise secured and required to be closed during accident conditions is closed, except for containment isolation valves that are open under administrative controls.	~~31 days~~ In accordance with the SFCP
SR 3.6.3.3	---------- **- NOTE -** Valves and blind flanges in high radiation areas may be verified by use of administrative controls. ---------- Verify each containment isolation manual valve and blind flange that is located inside containment and not locked, sealed, or otherwise secured and required to be closed during accident conditions is closed, except for containment isolation valves that are open under administrative controls.	Prior to entering MODE 4 from MODE 5 if not performed within the previous 92 days
SR 3.6.3.4	Verify the isolation time of each automatic power operated containment isolation valve is within limits.	In accordance with the ~~Inservice Testing Program~~ SFCP
SR 3.6.3.5	Verify each automatic containment isolation valve that is not locked, sealed or otherwise secured in position, actuates to the isolation position on an actual or simulated actuation signal.	~~24 months~~ In accordance with the SFCP

3.6 CONTAINMENT SYSTEMS

3.6.4 Containment Pressure

LCO 3.6.4 Containment pressure shall be ≥ -0.2 psig and ≤ +1.0 psig.

APPLICABILITY: MODES 1, 2, 3, and 4.
MODES 5 and 6 without an open containment air flow path ≥ 6 inches in diameter.

- NOTE -

The high pressure LCO limit is not applicable in MODES 5 or 6.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. Containment pressure not within limits.	A.1 Restore containment pressure to within limits.	1 hour
B. Required Action and associated Completion Time of Condition A not met in MODE 1, 2, 3, or 4.	B.1 Be in MODE 3. AND B.2 Be in MODE 5.	6 hours 36 hours
C. Required Action and associated Completion Time of Condtion A not met in MODE 5 or 6.	C.1 Initiate action to restore containment pressure to within limits.	Immediately

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.6.4.1	Verify containment pressure is within limits.	~~12 hours~~ In accordance with the SFCP

3.6 CONTAINMENT SYSTEMS

3.6.5 Containment Air Temperature

LCO 3.6.5 Containment average air temperature shall be ≤ 120°F.

APPLICABILITY: MODES 1, 2, 3, and 4,
MODES 5 and 6 with both containment equipment hatches and both containment airlocks closed.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. Containment average air temperature not within limit.	A.1 Restore containment average air temperature to within limit.	8 hours
B. Required Action and associated Completion Time of Condition A not met in MODE 1, 2, 3, or 4.	B.1 Be in MODE 3. AND B.2 Be in MODE 5.	6 hours 36 hours
C. Required Action and associated Completion Time of Condition A not met in MODE 5 or 6.	C.1 Initiate action to restore containment average air temperature to within limit.	Immediately

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.6.5.1	Verify containment average air temperature is within limit.	~~12 hours~~ In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
E. Required Action and associated Completion time of Condition A, B, or C not met in MODE 5 or 6. OR LCO not met for reasons other than Condition A, B, or C in MODE 5 or 6.	E.1 Initiate action to restore passive containment cooling to OPERABLE status. AND E.2 Suspend positive reactivity additions.	Immediately Immediately

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.6.6.1	Verify the water storage tank temperature ≥ 40°F and ≤ 120°F.	~~24 hours~~ In accordance with the SFCP
SR 3.6.6.2	Verify the water storage tank volume ≥ 756,700 gallons.	~~7 days~~ In accordance with the SFCP
SR 3.6.6.3	Verify each passive containment cooling system manual, power operated, and automatic valve in each flow path that is not locked, sealed, or otherwise secured in position, is in the correct position.	~~31 days~~ In accordance with the SFCP
SR 3.6.6.4	Verify each passive containment cooling system automatic valve in each flow path that is not locked, sealed, or otherwise secured in position, actuates to the correct position on an actual or simulated actuation signal.	~~24 months~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.6.6.5	Verify the air flow path from the shield building annulus inlet to the exit is unobstructed and, that all air baffle sections are in place.	~~24 months~~ In accordance with the SFCP
SR 3.6.6.6	Verify passive containment cooling system flow and water coverage performance in accordance with the System Level OPERABILITY Testing Program.	~~At first refueling~~ ~~AND~~ ~~10 years~~ In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
B. Required Action and associated Completion Time of Condition A not met. OR LCO not met for reasons other than Condition A.	B.1 Initiate action to restore compliance with the LCO. AND B.2 Suspend positive reactivity additions.	Immediately Immediately

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.6.7.1	Verify each required containment penetration is in the required status.	~~7 days~~ In accordance with the SFCP
SR 3.6.7.2	---------- **- NOTE -** Only required to be met for an open equipment hatch. ---------- Verify the hardware, tools, equipment and power source necessary to close the equipment hatch are available.	 Prior to hatch removal AND ~~7 days~~ In accordance with the SFCP

3.6 CONTAINMENT SYSTEMS

3.6.8 pH Adjustment

LCO 3.6.8 The pH adjustment baskets shall contain ≥ 26,460 lbs of trisodium phosphate (TSP).

APPLICABILITY: MODES 1, 2, 3, and 4.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. The weight of TSP in the pH adjustment baskets not within limit.	A.1 Restore weight of TSP in the pH adjustment baskets to within limit.	72 hours
B. Required Action and associated Completion Time not met.	B.1 Be in MODE 3. AND B.2 Be in MODE 5.	6 hours 84 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.6.8.1	Verify the pH adjustment baskets contain ≥ 26,460 lbs of TSP.	~~24 months~~ In accordance with the SFCP
SR 3.6.8.2	Verify a sample from the pH adjustment baskets provides adequate pH adjustment of the post-accident water.	~~24 months~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.6.9.1	Verify containment inside to outside differential air temperature is ≤ 90°F.	~~12 hours~~ In accordance with the SFCP
SR 3.6.9.2	---------- **- NOTES -** 1. Not required to be met for vacuum relief valves open during Surveillances. 2. Not required to be met for vacuum relief valves open when performing their vacuum relief function. ---------- Verify each vacuum relief isolation valve is closed.	~~31 days~~ In accordance with the SFCP
SR 3.6.9.3	Verify (a) each vacuum relief check valve opens and closes with set pressure within limits, and (b) each vacuum relief isolation valve opening time is within limits~~is OPERABLE in accordance with the Inservice Testing Program~~.	In accordance with the ~~Inservice Testing Program~~ SFCP
SR 3.6.9.4	Verify each vacuum relief isolation valve actuates on actual or simulated signals.	~~24 months~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.7.1.1	---------- **- NOTE -** Only required to be performed in MODES 1 and 2. ---------- Verify each MSSV lift setpoint per Table 3.7.1-2 ~~in accordance with the Inservice Testing Program~~.	In accordance with the ~~Inservice Testing Program~~SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.7.2.1	---------- **- NOTE -** Only required to be performed prior to entry into MODE 2. ---------- Verify MSIV closure time is within limits on an actual or simulated actuation signal.	In accordance with the ~~Inservice Testing Program~~ SFCP
SR 3.7.2.2	---------- **- NOTE -** Only required to be performed prior to entry into MODE 2. ---------- Verify required turbine stop, turbine control, turbine bypass, and moisture separator reheater 2nd stage steam isolation valves' closure time is within limits on an actual or simulated actuation signal.	In accordance with the ~~Inservice Testing Program~~ SFCP
SR 3.7.2.3	Verify the isolation time of each MSIV bypass valve and main steam line drain isolation valve is within limits.	In accordance with the ~~Inservice Testing Program~~ SFCP
SR 3.7.2.4	Verify each MSIV bypass valve and main steam line drain isolation valve actuates to the isolation position on an actual or simulated actuation signal.	~~24 months~~ In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
D. Required Action and associated Completion Time not met.	D.1 Be in MODE 3.	6 hours
	AND	
	D.2 Be in MODE 4 with the Reactor Coolant System (RCS) cooling provided by the Normal Residual Heat Removal System (RNS).	24 hours
	AND	
	D.3 Be in MODE 5.	36 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.7.3.1	---------- **- NOTE -** Only required to be performed prior to entry into MODE 2. ---------- Verify the closure time of each MFIV and MFCV is within limits on an actual or simulated actuation signal.	In accordance with the ~~Inservice Testing Program~~ SFCP
SR 3.7.3.2	Verify each MFW pump breaker trips open on an actual or simulated actuation signal, except for pump breaker maintained in the tripped condition.	~~24 months~~ In accordance with the SFCP

3.7 PLANT SYSTEMS

3.7.4 Secondary Specific Activity

LCO 3.7.4 The specific activity of the secondary coolant shall be ≤ 0.01 µCi/gm DOSE EQUIVALENT I-131.

APPLICABILITY: MODES 1, 2, 3 and 4.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. Specific activity not within limit.	A.1 Be in MODE 3.	6 hours
	AND	
	A.2 Be in MODE 5.	36 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.7.4.1	Verify the specific activity of the secondary coolant ≤ 0.01 µCi/gm DOSE EQUIVALENT I-131.	~~31 days~~ In accordance with the SFCP

3.7 PLANT SYSTEMS

3.7.5 Spent Fuel Pool Water Level

LCO 3.7.5 The spent fuel pool water level shall be ≥ 23 ft above the top of irradiated fuel assemblies seated in the storage racks.

APPLICABILITY: When irradiated fuel assemblies are stored in the spent fuel pool.

ACTIONS

- NOTE -

LCO 3.0.3 is not applicable.

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. Spent fuel pool water level < 23 ft.	A.1 Suspend movement of irradiated fuel assemblies in the spent fuel pool.	Immediately
	AND	
	A.2 Initiate action to restore water level to ≥ 23 ft.	1 hour

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.7.5.1	Verify the spent fuel pool water level is ≥ 23 ft above the top of the irradiated fuel assemblies seated in the storage racks.	~~7 days~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.7.6.1	Verify the compressed air storage tanks contain > 327,574 scf of compressed air.	~~24 hours~~ In accordance with the SFCP
SR 3.7.6.2	Verify thermal mass for the following heat sink locations is within limit: a. MCRE; b. Each required individual room adjacent to and below MCRE; c. Each required room-pair adjacent to and below MCRE; and d. Room above MCRE.	~~24 hours~~ In accordance with the SFCP
SR 3.7.6.3	Operate VES for ≥ 15 minutes.	~~31 days on a STAGGERED TEST BASIS~~ In accordance with the SFCP
SR 3.7.6.4	Verify each VES air header manual isolation valve is in an open position.	~~31 days~~ In accordance with the SFCP
SR 3.7.6.5	Verify the air quality of the compressed air storage tanks meets the requirements of CGA G-7.1, Commodity Specification for Air, Grade E, with a pressure dew point of ≤ 40°F at ≥ 3400 psig.	~~92 days~~ In accordance with the SFCP
SR 3.7.6.6	Verify each MCRE isolation valve is OPERABLE and will close upon receipt of an actual or simulated actuation signal, except for valves that are locked, sealed, or otherwise secured in the closed position.	~~24 months~~ In accordance with the SFCP
SR 3.7.6.7	Verify each VES pressure relief isolation valve within the MCRE pressure boundary is OPERABLE.	In accordance with the ~~Inservice Testing Program~~ SFCP
SR 3.7.6.8	Verify each VES pressure relief damper is OPERABLE.	~~24 months~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.7.6.9	Perform required MCRE unfiltered air inleakage testing in accordance with the Main Control Room Envelope Habitability Program.	In accordance with the Main Control Room Envelope Habitability Program
SR 3.7.6.10	Perform required VES Passive Filtration system filter testing in accordance with the Ventilation Filter Testing Program (VFTP).	In accordance with the VFTP
SR 3.7.6.11	Verify the MCR load shed function actuates upon receipt of an actual or simulated actuation signal.	~~24 months~~ In accordance with the SFCP
SR 3.7.6.12	Verify each VES main air delivery isolation valve actuates to the correct position upon receipt of an actual or simulated actuation signal.	~~24 months~~ In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
D. Required Action and associated Completion Time not met.	D.1 Be in MODE 3.	6 hours
	AND	
	D.2 Be in MODE 4 with the Reactor Coolant System (RCS) cooling provided by the Normal Residual Heat Removal System (RNS).	24 hours
	AND	
	D.3 Be in MODE 5.	36 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.7.7.1	Verify each SFW isolation and control valve is OPERABLE.	In accordance with the ~~Inservice Testing Program~~ SFCP
SR 3.7.7.2	Verify each SFW isolation and control valve actuates to the isolation position on an actual or simulated actuation signal.	~~24 months~~ In accordance with the SFCP
SR 3.7.7.3	Verify each SFW pump breaker trips open on an actual or simulated actuation signal, except for pump breaker maintained in the tripped condition.	~~24 months~~ In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
E. Required Action and associated Completion Time not met.	E.1 Be in MODE 3. AND E.2 Be in MODE 4 with the RCS cooling provided by the RNS. AND ---------- **- NOTE –** Not applicable for inoperable PORV(s). ---------- E.3 Be in MODE 5.	6 hours 24 hours 36 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.7.10.1	Verify each SG PORV, PORV block valve, and SG blowdown isolation valve strokes closed.	In accordance with the ~~Inservice Testing Program~~ SFCP
SR 3.7.10.2	Verify the isolation time of each PORV block valve and SG blowdown isolation valve is within limits.	In accordance with the ~~Inservice Testing Program~~ SFCP
SR 3.7.10.3	Verify each SG PORV, PORV block valve, and SG blowdown isolation valve actuates to the isolation position on an actual or simulated actuation signal.	~~24 months~~ In accordance with the SFCP

3.7 PLANT SYSTEMS

3.7.11 Spent Fuel Pool Boron Concentration

LCO 3.7.11 The spent fuel pool boron concentration shall be ≥ 2300 ppm.

APPLICABILITY: When fuel assemblies are stored in the spent fuel pool.

ACTIONS

- NOTE -

LCO 3.0.3 is not applicable.

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. Spent fuel pool boron concentration not within limit.	A.1 Suspend movement of fuel assemblies in the spent fuel pool.	Immediately
	AND	
	A.2 Initiate action to restore spent fuel pool boron concentration to within limit.	Immediately

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.7.11.1	Verify the spent fuel pool boron concentration is within limit.	~~7 days~~ In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
C. One or more batteries in one division inoperable.	C.1 Restore batteries to OPERABLE status.	6 hours
D. One or more batteries in two divisions inoperable.	D.1 Restore batteries to OPERABLE status.	2 hours
E. One DC electrical power subsystem inoperable for reasons other than Condition A or C.	E.1 Restore DC electrical power subsystem to OPERABLE status.	6 hours
F. Two DC electrical power subsystems inoperable for reasons other than B or D.	F.1 Restore DC electrical power subsystem to OPERABLE status.	2 hours
G. Required Action and associated Completion Time not met.	G.1 Be in MODE 3. AND G.2 Be in MODE 5.	6 hours 36 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.8.1.1	Verify battery terminal voltage is greater than or equal to the minimum established float voltage.	~~7 days~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.8.1.2	Verify each battery charger supplies ≥ 150 amps at greater than or equal to the minimum established float voltage for ≥ 8 hours. OR Verify each battery charger can recharge the battery to the fully charged state within 24 hours while supplying the largest combined demands of the various continuous steady state loads, after a battery discharge to the bounding design basis event discharge state.	~~24 months~~ In accordance with the SFCP
SR 3.8.1.3	---------- **- NOTE -** The modified performance discharge test in SR 3.8.7.6 may be performed in lieu of SR 3.8.1.3. ---------- Verify battery capacity is adequate to supply, and maintain in OPERABLE status, the required emergency loads for the design duty cycle when subjected to a battery service test.	~~24 months~~ In accordance with the SFCP
SR 3.8.1.4	---------- **- NOTE -** Only required to be met when the main control room air supply radiation monitor sample pumps are energized. ---------- Verify main control room air supply radiation monitor sample pump deenergizes on an actual or simulated actuation signal.	~~24 months~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.8.3.1	Verify correct inverter voltage, frequency, and alignment to required AC instrument and control buses.	~~7 days~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.8.4.1	Verify correct inverter voltage, frequency, and alignments to required AC instrument and control buses.	~~7 days~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.8.5.1	Verify correct breaker and switch alignments and voltage to required DC and AC instrument and control electrical power distribution subsystems.	~~7 days~~ In accordance with the SFCP

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. (continued)	A.2.4 Initiate actions to restore required DC and AC instrument and control electrical power distribution subsystems to OPERABLE status.	Immediately

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.8.6.1	Verify correct breaker and switch alignments and voltage to required DC and AC instrument and control electrical power distribution subsystems.	~~7 days~~ In accordance with the SFCP

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
F. Required Action and associated Completion Time not met. OR One or more batteries in one division with one or more battery cells float voltage < 2.07 V and float current > 2 amps.	F.1 Declare associated battery inoperable.	Immediately

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.8.7.1	---------- **- NOTE -** Not required to be met when battery terminal voltage is less than the minimum established float voltage of SR 3.8.1.1. ---------- Verify each battery float current is ≤ 2 amps.	~~7 days~~ In accordance with the SFCP
SR 3.8.7.2	Verify each battery pilot cell float voltage is ≥ 2.07 V.	~~31 days~~ In accordance with the SFCP
SR 3.8.7.3	Verify each battery connected cell electrolyte level is greater than or equal to minimum established design limits.	~~31 days~~ In accordance with the SFCP
SR 3.8.7.4	Verify each battery pilot cell temperature is greater than or equal to minimum established design limits.	~~31 days~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.8.7.5	Verify each battery connected cell float voltage is ≥ 2.07 V.	~~92 days~~ In accordance with the SFCP
SR 3.8.7.6	Verify battery capacity is ≥ 80% of the manufacturer's rating when subjected to a performance discharge test or a modified performance discharge test.	~~60 months~~In accordance with the SFCP or 25% of expected life, whichever is less AND 12 months when battery shows degradation, or has reached 85% of the expected life with capacity < 100% of manufacturer's rating AND 24 months when battery has reached 85% of the expected life with capacity ≥ 100% of manufacturer's rating

3.9 REFUELING OPERATIONS

3.9.1 Boron Concentration

LCO 3.9.1 Boron concentration of the Reactor Coolant System (RCS), the fuel transfer canal, and the refueling cavity shall be maintained within the limit specified in COLR.

APPLICABILITY: MODE 6.

- NOTE -

Applicable to the fuel transfer canal and the refueling cavity only when connected to the RCS.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. Boron concentration not within limit.	A.1 Suspend positive reactivity additions.	Immediately
	AND	
	A.2 Initiate actions to restore boron concentration to within limits.	Immediately

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.9.1.1	Verify boron concentration is within the limit specified in the COLR.	~~72 hours~~ In accordance with the SFCP

3.9 REFUELING OPERATIONS

3.9.2 Unborated Water Source Flow Paths

LCO 3.9.2 One valve in each unborated water source flow path shall be secured in the closed position.

APPLICABILITY: MODE 6.

ACTIONS

- NOTE -

Separate condition entry is allowed for each unborated water source flow path.

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. ---------- **- NOTE -** Required Action A.2 must be completed whenever Condition A is entered. ---------- One or more unborated water source flow paths with no valve secured in the closed position.	A.1 Initiate actions to secure one valve in the flow path in the closed position. AND A.2 Perform SR 3.9.1.1.	Immediately 4 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.9.2.1	Verify one valve in each unborated water source flow path is secured in the closed position.	~~31 days~~ In accordance with the SFCP

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.9.3.1	Perform a CHANNEL CHECK.	~~12 hours~~ In accordance with the SFCP
SR 3.9.3.2	---------- **- NOTE-** Neutron detectors are excluded from CHANNEL CALIBRATION. ---------- Perform CHANNEL CALIBRATION.	 ~~24 months~~ In accordance with the SFCP

3.9 REFUELING OPERATIONS

3.9.4 Refueling Cavity Water Level

LCO 3.9.4 Refueling Cavity Water Level shall be maintained ≥ 23 ft above the top of the reactor vessel flange.

APPLICABILITY: During movement of irradiated fuel assemblies within containment.

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. Refueling cavity water level not within limit.	A.1 Suspend movement of irradiated fuel assemblies within containment.	Immediately

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.9.4.1	Verify refueling cavity water level is ≥ 23 ft above the top of reactor vessel flange.	~~24 hours~~ In accordance with the SFCP

5.5 Programs and Manuals

5.5.2 Radioactive Effluent Control Program (continued)

ii. For iodine-131, iodine-133, tritium, and all radionuclides in particulate form with half-lives greater than 8 days: a dose rate ≤ 1500 mrem/yr to any organ;

8. Limitations on the annual and quarterly air doses resulting from noble gases released in gaseous effluents from each unit to areas beyond the site boundary, conforming to 10 CFR 50, Appendix I;

9. Limitations on the annual and quarterly doses to a member of the public from iodine-131, iodine-133, tritium, and all radionuclides in particulate form with half lives > 8 days in gaseous effluents released from each unit to areas beyond the site boundary, conforming to 10 CFR 50, Appendix I; and

10. Limitations on the annual dose or dose commitment to any member of the public, beyond the site boundary, due to releases of radioactivity and to radiation from uranium fuel cycle sources, conforming to 40 CFR 190.

b. The provisions of SR 3.0.2 and SR 3.0.3 are applicable to the Radioactive Effluent Controls Program surveillance frequency.

5.5.3 ~~Inservice Testing Program~~Not used.

~~The Inservice Testing Program is the licensee program that fulfills the requirements of 10 CFR 50.55a(f).~~

5.5.4 Steam Generator (SG) Program

A Steam Generator Program shall be established and implemented to ensure that SG tube integrity is maintained. In addition, the Steam Generator Program shall include the following provisions:

a. Provisions for condition monitoring assessments. Condition monitoring assessment means an evaluation of the “as found” condition of the tubing with respect to the performance criteria for structural integrity and accident induced leakage. The "as found" condition refers to the condition of the tubing during an SG inspection outage, as determined from the inservice inspection results or by other means, prior to the plugging of tubes. Condition monitoring assessments shall be conducted during each outage during which the SG tubes are inspected or plugged, to confirm that the performance criteria are being met.

5.5 Programs and Manuals

5.5.12 Main Control Room Envelope Habitability Program

A Main Control Room Envelope (MCRE) Habitability Program shall be established and implemented to ensure that MCRE habitability is maintained such that, with an OPERABLE Main Control Room Emergency Habitability System (VES), MCRE occupants can control the reactor safely under normal conditions and maintain it in a safe condition following a radiological event, hazardous chemical release, or a smoke challenge. The program shall ensure that adequate radiation protection is provided to permit access and occupancy of the MCRE under design basis accident (DBA) conditions without personnel receiving radiation exposures in excess of 5 rem total effective dose equivalent (TEDE) for the duration of the accident. The program shall include the following elements:

a. The definition of the MCRE and the MCRE boundary.

b. Requirements for maintaining the MCRE boundary in its design condition, including configuration control and preventive maintenance.

c. Requirements for (i) determining the unfiltered air inleakage past the MCRE boundary into the MCRE at Frequencies in accordance with the SFCP and in accordance with the testing methods and at the event-driven frequencies ~~Frequencies~~ specified in Sections C.1 and C.2 of Regulatory Guide 1.197, "Demonstrating Control Room Envelope Integrity at Nuclear Power Reactors," Revision 0, May 2003, and (ii) assessing MCRE habitability at the Frequencies specified in Sections C.1 and C.2 of Regulatory Guide 1.197, Revision 0.

d. Measurement, at designated locations, of the MCRE pressure relative to all external areas adjacent to the MCRE boundary during the pressurization mode of operation of one VES air delivery flow path, operating at the required flow rate of 65 ± 5 scfm, at a Frequency in accordance with the SFCP~~of 24 months on a STAGGERED TEST BASIS~~. The results shall be trended and used as part of the ~~24 month~~periodic assessment of the MCRE boundary.

e. The quantitative limits on unfiltered air inleakage into the MCRE. These limits shall be stated in a manner to allow direct comparison to the unfiltered air inleakage measured by the testing described in paragraph c. The unfiltered air inleakage limit for radiological challenges is the inleakage flow rate assumed in the licensing basis analyses of DBA consequences. Unfiltered air inleakage limits for hazardous chemicals must ensure that exposure of MCRE occupants to these hazards will be within the assumptions in the licensing basis.

f. The provisions of SR 3.0.2 are applicable to the Frequencies for assessing MCRE habitability, determining MCRE unfiltered inleakage, and measuring

5.5 Programs and Manuals

5.5.13 Ventilation Filter Testing Program (VFTP)

a. A program shall be established to implement the following required testing of the VES.

Tests described in Specification 5.5.13.a.1 and 5.5.13.a.2 shall be performed: i) ~~initially~~at Frequencies in accordance with the SFCP, ii) ~~once each 24 months except for Unit 3 cycle 1 when testing shall be performed prior to startup from first refueling outage, iii)~~ after partial or complete replacement of a HEPA filter or charcoal adsorber, ~~iv~~iii) following detection of, or evidence of, penetration or intrusion of water or other material into any portion of the VES that may have an adverse effect on the functional capability of the filters, and ~~v~~iv) following painting, fire, or chemical release in any ventilation zone communicating with the VES that may have an adverse effect on the functional capability of the system.

Tests described in Specification 5.5.13.a.3 shall be performed: i) ~~after each 720 hours of system operation or at least once each 24 months, whichever comes first~~at a Frequency in accordance with the SFCP, ii) following painting, fire, or chemical release in any ventilation zone communicating with the VES that may have an adverse effect on the functional capability of the carbon media, and iii) following detection of, or evidence of, penetration or intrusion of water or other material into any portion of the VES that may have an adverse effect on the functional capability of the carbon media.

Tests described in 5.5.13.a.4 shall be performed ~~once per 24 months except for Unit 3 cycle 1 when testing shall be performed prior to startup from first refueling outage~~at a Frequency in accordance with the SFCP.

1. Demonstrate for the VES that an inplace test of the high efficiency particulate air (HEPA) filter shows a penetration and system bypass ≤ 0.05% when tested in accordance with Regulatory Guide 1.52, Revision 3, and ASME N510-1989 at a flow rate at least 600 cfm greater than the VES makeup flow rate.

Ventilation System	**Flow Rate**
VES	≥ 600 + VES makeup flow rate (cfm)

2. Demonstrate for the VES that an inplace test of the charcoal adsorber shows a penetration and system bypass ≤ 0.05% when tested in accordance with Regulatory Guide 1.52, Revision 3, and ASME N510-1989 at a flow rate at least 600 cfm greater than the VES makeup flow rate.

Ventilation System	**Flow Rate**
VES	≥ 600 + VES makeup flow rate (cfm)

5.5 Programs and Manuals

5.5.16 Surveillance Frequency Control Program (SFCP)

This program provides controls for Surveillance Frequencies and the frequencies of other periodic testing required by these Technical Specifications (herein also referred to as Surveillance Frequencies). The program shall ensure that Surveillance Requirements and other required testing specified in these Technical Specifications are performed at intervals sufficient to assure the associated Limiting Conditions for Operation and other requirements are met.

a. The SFCP shall contain a list of Frequencies controlled by the program.

b. Changes to the frequencies listed in the SFCP shall be made:

1. In accordance with the requirements of 10 CFR 50.55a for testing required by 10 CFR 50.55a(f);

2. In accordance with the requirements of 10 CFR 50.69(d)(2) for testing permitted under 10 CFR 50.69(b)(1)(v) in lieu of testing required by 10 CFR 50.55a(f);

3. Otherwise, in accordance with NEI 04-10, “Risk-Informed Method for Control of Surveillance Frequencies,” Revision 1.

c. The provisions of Surveillance Requirements 3.0.2 and 3.0.3 are applicable to the Frequencies established in the SFCP, except:

1. For testing required by 10 CFR 50.55a(f), Surveillance Requirement 3.0.2 is only applicable as permitted by that regulation; and

2. Surveillance Requirement 3.0.3 is only applicable to other required testing when invoked by a Chapter 5 Specification.

Attachment 5 to NL-26-0017
Proposed Technical Specification Bases Changes (Mark-up)
- For Information Only -

License Amendment Request to Revise Technical Specifications to Include a Surveillance Frequency Control Program

BASES

SR 3.0.2 (continued)

requirements of regulations take precedence over the TS. Examples of where SR 3.0.2 does not apply are the Containment Leakage Rate Testing Program required by 10 CFR 50, Appendix J, and the inservice testing of pumps and valves in accordance with ~~applicable American Society of Mechanical Engineers Operation and Maintenance Code, as required by~~ the requirements of 10 CFR 50.55a(f). These programs establish testing requirements and Frequencies in accordance with the requirements of regulations. The TS cannot in and of themselves extend a test interval specified in the regulations directly or by reference.

As stated in SR 3.0.2, the 25% extension also does not apply to the initial portion of a periodic Completion Time that requires performance on a "once per …" basis. The 25% extension applies to each performance after the initial performance. The initial performance of the Required Action, whether it is a particular Surveillance or some remedial action, is considered a single action with a single Completion Time. One reason for not allowing the 25% extension to this Completion Time is that such an action usually verifies that no loss of function has occurred by checking the status of redundant or diverse components or accomplishes the function of the inoperable equipment in an alternative manner.

The provisions of SR 3.0.2 are not intended to be used repeatedly to extend Surveillance intervals (other than those consistent with refueling intervals) or periodic Completion Time intervals beyond those specified.

SR 3.0.3

SR 3.0.3 establishes the flexibility to defer declaring affected equipment inoperable or an affected variable outside the specified limits when a Surveillance has not been completed within the specified Frequency. A delay period of up to 24 hours or up to the limit of the specified Frequency, whichever is greater, applies from the point in time that it is discovered that the Surveillance has not been performed, in accordance with SR 3.0.2, and not at the time that the specified Frequency was not met.

This delay period provides adequate time to complete Surveillances that have been missed. This delay period permits the completion of a Surveillance before compliance with Required Actions or other remedial measures that might preclude completion of the Surveillance.

The basis for this delay period includes consideration of unit conditions, adequate planning, availability of personnel, the time required to perform

BASES

SURVEILLANCE REQUIREMENTS (continued)

e. Xenon concentration;

f. Samarium concentration; and

g. Isothermal Temperature Coefficient (ITC).

Using the ITC accounts for Doppler reactivity in this calculation because the reactor is subcritical and the fuel temperature will be changing at the same rate as the RCS.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 hours is based on the generally slow change in required boron concentration and the low probability of an accident occurring without the required SDM. This allows time for the operator to collect the required data, which includes performing a boron concentration analysis, and complete the calculation.~~

REFERENCES

1. 10 CFR 50, Appendix A, GDC 26.

2. FSAR Chapter 15, “Accident Analysis.”

3. 10 CFR 50.34.

4. NUREG-0800, Standard Review Plan, Section 4.2, Revision 3, “Fuel System Design” (Appendix B).

5. WCAP-16943-P-A, "Enhanced GRCA (Gray Rod Cluster Assembly) Rodlet Design."

BASES

ACTIONS (continued)

The required Completion Time of 7 days is adequate for preparing whatever operating restrictions or Surveillances that may be required to allow continued reactor operation.

B.1

If the core reactivity cannot be restored to within the 1% Δk/k limit, the plant must be brought to a MODE in which the LCO does not apply. To achieve this status, the plant must be brought to at least MODE 3 within 6 hours. If the SDM for MODE 3 is not met, then the boration required by LCO 3.1.1 Required Action A.1 would occur. The allowed Completion Time is reasonable, based on operating experience, for reaching MODE 3 from full power conditions in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.1.2.1

Core reactivity is verified by periodic comparisons of measured and predicted RCS boron concentrations. The comparison is made considering that other core conditions are fixed or stable, including control rod position, moderator temperature, fuel temperature, fuel depletion, xenon concentration, and samarium concentration. The Surveillance is performed prior to entering MODE 1 as an initial check on core conditions and design calculations at BOC.

The SR is modified by a Note. The Note indicates that the normalization of predicted core reactivity to the measured value must take place within the first 60 effective full power days (EFPDs) after each fuel loading. This allows sufficient time for core conditions to reach steady state, but prevents operation for a large fraction of the fuel cycle without establishing a benchmark for the design calculations. The Surveillance Frequency is controlled under the SFCP. ~~The required subsequent Frequency of 31 EFPDs following the initial 60 EFPDs after entering MODE 1 is acceptable based on the slow rate of core changes due to fuel depletion and the presence of other indicators (QPTR, OPDMS, AFD, etc.) for prompt indication of an anomaly.~~

REFERENCES

1. 10 CFR 50, Appendix A, GDC 26, GDC 28, and GDC 29.
2. FSAR, Chapter 15, “Accident Analysis.”

BASES

ACTIONS (continued)

D.2

If more than one rod is found to be misaligned or becomes misaligned because of bank movement, the unit conditions fall outside of the accident analysis assumptions. Since automatic bank sequencing could continue to cause misalignment, the unit must be brought to a MODE or Condition in which the LCO requirements are not applicable. To achieve this status, the unit must be brought to at least MODE 3 within 6 hours.

The allowed Completion Time is reasonable, based on operating experience, for reaching MODE 3 from full power in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.1.4.1

Verification that individual rod positions are within alignment limits ~~at a Frequency of 12 hours~~ provides a history that allows the operator to detect that a rod is beginning to deviate from its expected position. The Surveillance Frequency is controlled under the SFCP. ~~The specified Frequency takes into account other rod position information that is continuously available to the operator in the main control room so that during actual rod motion, deviations can immediately be detected.~~

The SR is modified by a Note that permits it to not be performed for rods associated with an inoperable demand position indication or an inoperable DRPI. The alignment limit is based on the position indication, and would not be available if the indication is inoperable. LCO 3.1.7, “Rod Position Indication,” provides Actions to verify the rods are in alignment when one or more rod position indications are inoperable.

Verification that individual rod positions are within alignment limits may utilize any position indication data, e.g., DRPI and/or incore detector data.

SR 3.1.4.2

Verifying each RCCA is OPERABLE would require that each rod be tripped. However, in MODES 1 and 2, tripping each RCCA would result in radial or axial power tilts, or oscillations. Exercising each individual RCCA ~~every 92 days~~ provides increased confidence that all rods continue to be OPERABLE without exceeding the alignment limit, even if they are not regularly tripped. Moving each RCCA by 10 steps will not cause radial or axial power tilts, or oscillations, to occur.

BASES

SURVEILLANCE REQUIREMENTS (continued)

The Surveillance Frequency is controlled under the SFCP. ~~The 92 day Frequency takes into consideration other information available to the operator in the control room and SR 3.1.4.1, which is performed more frequently and adds to the determination of OPERABILITY of the RCCAs.~~ Between required performances of SR 3.1.4.2 (determination of RCCA OPERABILITY by movement), if a RCCA(s) is discovered to be immovable, but remains trippable and aligned, the RCCA(s) is considered to be OPERABLE. At any time, if a RCCA(s) is immovable, a determination of the trippability (OPERABILITY) of the RCCA(s) must be made, and appropriate action taken.

GRCA are excluded from this Surveillance because they are not considered in the calculation of SDM in MODES 1 and 2 with $k_{eff} \geq 1.0$. In Mode 2 with $k_{eff} < 1.0$, SR 3.1.4.2 is not applicable because GRCAs must be verified to be fully inserted to be credited in the calculation of SDM.

SR 3.1.4.3

Verification of RCCA drop times allows the operator to determine that the maximum RCCA drop time permitted is consistent with the assumed rod drop time used in the safety analysis. Measuring RCCA drop times prior to reactor criticality, after each reactor vessel head removal and each earthquake requiring plant shutdown, ensures that the reactor internals and rod drive mechanism will not interfere with RCCA motion or RCCA drop time, and that no degradation in these systems has occurred that would adversely affect control rod motion or drop time. This testing is performed with all RCPs operating and the average moderator temperature ≥ 500°F to simulate a reactor trip under conservative conditions. GRCA are excluded from this Surveillance because they are not considered in the calculation of SDM in MODES 1 and 2 with $k_{eff} \geq 1.0$. In MODE 2 with $k_{eff} < 1.0$, SR 3.1.4.3 is not applicable because GRCAs must be verified to be fully inserted to be credited in the calculation of SDM.

This Surveillance is performed during a plant outage due to the plant conditions needed to perform the SR and the potential for an unplanned plant transient if the Surveillance were performed with the reactor at power.

REFERENCES

1. 10 CFR 50, Appendix A, GDC 10 and GDC 26.
2. 10 CFR 50.46.
3. FSAR Chapter 15, “Accident Analysis.”

BASES

ACTIONS (continued)

The allowed Completion Time of 1 hour for verification of SHUTDOWN MARGIN is an acceptable time for an operator to interface with the OPDMS to update the rod position of the control rods in violation of the Shutdown Bank Insertion Limits, or for the performance of the reactivity balance calculation.

The allowed Completion Time of 2 hours provides an acceptable time for evaluating and repairing minor problems without allowing the plant to remain in an unacceptable condition for an extended period of time.

B.1

If the shutdown banks cannot be restored to within their insertion limits within 2 hours, the unit must be brought to a MODE where the LCO is not applicable. The allowed Completion Time of 6 hours is reasonable based on operating experience, for reaching the required MODE from full power conditions in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.1.5.1

Verification that the shutdown banks are within their insertion limits prior to an approach to criticality ensures that when the reactor is critical, or being taken critical, the shutdown banks will be available to shut down the reactor, and the required SDM will be maintained following a reactor trip. This SR ~~and Frequency ensure~~ ensures that the shutdown banks are withdrawn before the control banks are withdrawn during a unit startup.

The Surveillance Frequency is controlled under the SFCP. ~~Since the shutdown banks are positioned manually by the main control room operator, a verification of shutdown bank position at a Frequency of 12 hours, after the reactor is taken critical, is adequate to ensure that they are within their insertion limits. Also, the 12 hours Frequency takes into account other information available in the main control room for the purpose of monitoring the status of shutdown rods.~~

REFERENCES

1. 10 CFR 50, Appendix A, GDC 10, GDC 26, and GDC 28.
2. 10 CFR 50.46.
3. FSAR, Chapter 15, “Accident Analysis.”

BASES

ACTIONS (continued)

C.1

If Required Actions A.1 and A.2, or B.1 and B.2 cannot be completed within the associated Completion Times, the plant must be brought to MODE 2 with $k_{eff} < 1.0$, or to a lower MODE, where the LCO is not applicable. The allowed Completion Time of 6 hours is reasonable based on operating experience for reaching the required MODE from full power condition in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.1.6.1

This Surveillance is required to ensure that the reactor does not achieve criticality with the control banks below their insertion limits.

The estimated critical position (ECP) depends upon a number of factors, one of which is xenon concentration. If the ECP was verified long before criticality, xenon concentration could change sufficiently to make the ECP substantially in error. Conversely, verifying the ECP immediately before criticality could be an unnecessary burden. There are a number of unit parameters requiring operator attention at that point. Performing the ECP verification within 4 hours prior to criticality allows the operator some flexibility to schedule the ECP verification with other startup activities.

SR 3.1.6.2

The Surveillance Frequency is controlled under the SFCP. ~~Verification of the control banks insertion limits at a Frequency of 12 hours is sufficient to detect control banks that may be approaching the insertion limits since the insertion limits are monitored and alarms will occur on approach to and/or the exceeding of the limit.~~

SR 3.1.6.3

When control banks are maintained within their insertion limits as checked by SR 3.1.6.2 above, it is unlikely that their sequence and overlap will not be in accordance with requirements provided in the COLR. The sequence and overlap requirements must be met while in the MODES of APPLICABILITY except during rod trippability testing per SR 3.1.4.2, or GRCA bank sequence exchange when the LCO does not apply for up to one hour. The Surveillance Frequency is controlled under the SFCP. ~~A Frequency of 12 hours is consistent with the insertion limit check above in SR 3.1.6.2.~~

BASES

SURVEILLANCE REQUIREMENTS

SR 3.1.8.1

Verification that the RCS lowest loop T_{avg} is ≥ 541°F will ensure that the unit is not operating in a condition that could invalidate the safety analyses. The Surveillance Frequency is controlled under the SFCP. ~~Verification of the RCS temperature at a Frequency of 30 minutes during the performance of the PHYSICS TESTS will provide assurance that the initial conditions of the safety analyses are not violated.~~

SR 3.1.8.2

Verification that the THERMAL POWER is ≤ 5% RTP will ensure that the plant is not operating in a condition that could invalidate the safety analyses. The Surveillance Frequency is controlled under the SFCP. ~~Verification of the THERMAL POWER at a Frequency of 30 minutes during the performance of the PHYSICS TESTS will ensure that the initial conditions of the safety analyses are not violated.~~

SR 3.1.8.3

The SDM is verified by performing a reactivity balance calculation, considering the following reactivity effects:

a. RCS boron concentration;

b. Control bank position;

c. RCS average temperature;

d. Fuel burnup based on gross thermal energy generation;

e. Xenon concentration;

f. Samarium concentration; and

g. Isothermal temperature coefficient (ITC).

Using the ITC accounts for Doppler reactivity in this calculation because the reactor is subcritical, and the fuel temperature will be changing at the same rate as the RCS.

SDM may also be verified using the OPDMS in place of a reactivity balance calculation. The OPDMS continuously monitors the available SDM during PHYSICS TESTS and considers all of the above effects.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 hours is based on the generally slow change in required boron~~

~~concentration and on the low probability of an accident occurring without the required SDM.~~

BASES

ACTIONS (continued)

B.1

If the Required Actions and associated Completion Time of Condition A are not met, or if both CVS demineralized water isolation valves or both CVS makeup line isolation valves are not OPERABLE (i.e., not able to be closed automatically), then the affected flow path(s) must be isolated.

For inoperabilities of CVS makeup isolation valves, isolating the "affected flow path" must prevent flow from the CVS makeup pumps to the RCS in order to provide protection from overfilling. Isolation can be accomplished by manually closing the CVS makeup isolation MOVs or alternatively, manual valve(s) in the makeup line between the makeup pumps and the RCS.

For inoperabilities of CVS demineralized water isolation valves isolation from the demineralized water supply flow path to the RCS can be accomplished by manually isolating the CVS demineralized water isolation valve(s) or by positioning the 3-way blend valve to only take suction from the boric acid tank. Alternatively, the dilution path may be isolated by closing appropriate isolation valve(s) in the flow path(s) from the demineralized water storage tank to the reactor coolant system.

SURVEILLANCE REQUIREMENTS

SR 3.1.9.1

Verification that the CVS demineralized water isolation valves and makeup line isolation valves stroke closed demonstrates that the valves can perform their safety related function. The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f). ~~The Frequency is in accordance with the Inservice Testing Program.~~

SR 3.1.9.2

Verification that the closure time of each RCS makeup isolation valve is less than that assumed in the safety analysis (i.e., < 30 seconds), is performed by measuring the time required for each valve to close on an actual or simulated actuation signal. The actual or simulated actuation signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valves. The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f). ~~The Frequency is in accordance with the Inservice Testing Program.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.1.9.3

This SR verifies that each CVS demineralized water isolation valve actuates to the correct position on an actual or simulated actuation signal. The actual or simulated actuation signal is processed through the component interface module to verify the continuity between the output of component interface module and the valve. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months is based on the need to perform this surveillance during periods in which the plant is shutdown for refueling to prevent any upsets of plant operation.~~

REFERENCES

1. FSAR Chapter 15, “Accident Analysis.”

BASES

ACTIONS

A.1 and A.2

If the LCO is not met, the reactor must be brought to a MODE in which the LCO does not apply. To achieve this status, action is initiated immediately to fully insert all rods and, within 1 hour, the PLS is placed in a condition incapable of rod withdrawal. This precludes rod withdrawal events which are unanalyzed. The Completion Times for Required Actions A.1 and A.2 are reasonable, based on operating experience, to reach the specified condition in an orderly manner and without challenging plant systems

SURVEILLANCE REQUIREMENTS

SR 3.1.10.1

This SR requires verification that boron concentration of the reactor coolant system is greater than the all-rods-out boron concentration providing a $k_{eff} < 0.99$. Assuming all rods are fully withdrawn ensures that the reactor will remain subcritical for all planned tests.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 12 hours is based on the generally slow change in required boron concentration and the low probability of an accident occurring without the required boron concentration. This allows time for the operator to collect the required data, which includes performing a boron concentration analysis, and complete the calculation.~~

REFERENCES

None

BASES

SURVEILLANCE REQUIREMENTS

SR 3.2.1.1 and SR 3.2.1.2 are modified by a Note, which applies to the situation where the OPDMS is not monitoring parameters at the beginning of cycle startup, i.e., the Note applies during the first power ascension after a refueling. It states that performance of these SRs is not required if OPDMS was monitoring parameters upon exceeding 75% RTP. Because $F_Q^C(Z)$ and $F_Q^W(Z)$ could not have previously been measured in this reload core, the SR 3.2.1.1 and SR 3.2.1.2 Frequency is applicable only for reload cores, and requires determination of these parameters before exceeding 75% RTP. This ensures that some determination of $F_Q^C(Z)$ and $F_Q^W(Z)$ are made at a lower power level at which adequate margin is available before going to 100% RTP. Also, SR 3.2.1.1 and SR 3.2.1.2 Frequency, together with the SR 3.2.1.3 and SR 3.2.1.4 first Frequency requiring verification of $F_Q^C(Z)$ and $F_Q^W(Z)$ following a power increase of more than 10%, ensures that they are verified as soon as RTP (or any other level for extended operation) is achieved. In the absence of these Frequency conditions, it is possible to increase power to RTP and operate for 31 days without verification of $F_Q^C(Z)$ and $F_Q^W(Z)$. The SR 3.2.1.3 and SR 3.2.1.4 first Frequency is not intended to require verification of these parameters after every 10% increase in power level above the last verification. They only require verification after an equilibrium power level is achieved for extended operation that is 10% higher than that power at which $F_Q(Z)$ was last measured.

The SR 3.2.1.3 Note and SR 3.2.1.4 Note 1 apply to the situation where the OPDMS is no longer monitoring parameters while the plant is in MODE 1. Without the continuous monitoring capability of the OPDMS, F_Q limits must be monitored on a periodic basis. The first measurement must be made within 24 hours of the OPDMS being declared not monitoring parameters. This is expected to be performed using the most recent contingency surveillance report created automatically during OPDMS monitoring. The Frequency of subsequent surveillances is controlled under the SFCP. ~~Subsequent surveillances are required consistent with the 31 EFPD Frequency.~~

SR 3.2.1.1 and SR 3.2.1.3

Verification that $F_Q^C(Z)$ is within its specified limits involves increasing the measured values of $F_Q^C(Z)$ to allow for manufacturing tolerance and measurement uncertainties in order to obtain $F_Q^C(Z)$. Specifically, $F_Q^M(Z)$ is the measured value of $F_Q(Z)$ obtained from incore flux map results and $F_Q^C(Z) = F_Q^M(Z) * F_Q^{MU}(Z)$. $F_Q^C(Z)$ is then compared to its specified limits.

SURVEILLANCE REQUIREMENTS (continued)

The limit to which $F_Q^C(Z)$ is compared varies inversely with power above 50% RTP.

Performing the Surveillance in MODE 1 prior to exceeding 75% RTP following a refueling outage assures that some determination of $F_Q^C(Z)$ is made prior to achieving a significant power level where the peak linear heat rate could approach the limits assumed in the safety analyses.

If THERMAL POWER has been increased by ≥ 10% RTP since the most recent determination of $F_Q^C(Z)$, another evaluation of this factor is required 24 hours after achieving equilibrium conditions at this higher power level (to assure that $F_Q^C(Z)$ values are being reduced sufficiently with power increase to stay within the LCO limits). Equilibrium conditions are achieved when the core is sufficiently stable at the intended operating conditions required to perform the Surveillance.

The allowance of up to 24 hours after achieving equilibrium conditions at the increased THERMAL POWER level to complete the next $F_Q^C(Z)$ surveillance applies to situations where the $F_Q^C(Z)$ has already been measured at least once at a reduced THERMAL POWER level. The observed margin in the previous surveillance will provide assurance that increasing power up to the next plateau will not exceed the F_Q limit, and that the core is behaving as designed.

This Frequency condition is not intended to require verification of these parameters after every 10% increase in RTP above the THERMAL POWER at which the last verification was performed. It only requires verification after a power level is achieved for extended operation that is 10% higher than the THERMAL POWER at which $F_Q^C(Z)$ was last measured.

The Frequency of subsequent surveillances is controlled under the SFCP. In the absence of the~~se~~ Frequency conditions ~~(~~discussed above~~)~~ it is possible to operate for a Frequency determined under the SFCP ~~31 EFPD~~ without verification of $F_Q^C(Z)$.

~~The Frequency of 31 effective full power days (EFPDs) is adequate to monitor the change of power distribution with core burnup because such changes are slow and well controlled when the plant is operated in accordance with Technical Specifications.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

Equilibrium conditions are achieved when the core is sufficiently stable at the intended operating conditions required to perform the Surveillance.

If a previous Surveillance of $F_Q^W(Z)$ was performed at part power conditions, SR 3.2.1.4 also requires that $F_Q^W(Z)$ be verified at power levels ≥ 10% RTP above the THERMAL POWER of its last verification within 24 hours after achieving equilibrium conditions. This ensures that $F_Q^W(Z)$ is within its limit using radial peaking factors measured at the higher power level.

The allowance of up to 24 hours after achieving equilibrium conditions will provide a more accurate measurement of $F_Q^W(Z)$ by allowing sufficient time to achieve equilibrium conditions and obtain the power distribution measurement.

In the absence of these Frequency conditions (discussed above) it is possible to operate for 31 EFPD without verification of $F_Q^W(Z)$. The Frequency of subsequent surveillances is controlled under the SFCP. ~~The Surveillance Frequency of 31 EFPDs is adequate to monitor the change of power distribution because such a change is sufficiently slow, when the plant is operated in accordance with Technical Specifications, to preclude the occurrence of adverse peaking factors between 31 EFPD Surveillances. The Surveillance may be done more frequently if required by the results of $F_Q(Z)$ evaluations.~~

REFERENCES

1. 10 CFR 50.46, “Acceptance Criteria for Emergency Core Cooling Systems for Light Water Nuclear Power Reactors,” 1974.

2. NUREG-0800, Standard Review Plan, Section 4.2, Revision 3, “Fuel System Design” (Appendix B).

3. 10 CFR 50, Appendix A, GDC 26.

4. WCAP-7308-L-P-A, “Evaluation of Nuclear Hot Channel Factor Uncertainties,” June 1988 (Westinghouse Proprietary) and WCAP-7308-L-A (Non-Proprietary).

5. WCAP-10216-P-A, Revision 1A, “Relaxation of Constant Axial Offset Control F_Q Surveillance Technical Specification,” February 1994 (Westinghouse Proprietary) and WCAP-10217-A (Non-Proprietary).

BASES

SURVEILLANCE REQUIREMENTS (continued)

After each refueling, with the OPDMS not monitoring parameters, $F^N_{\Delta H}$ must be determined prior to exceeding 75% RTP. This requirement ensures that $F^N_{\Delta H}$ limits are met at the beginning of each fuel cycle.

SR 3.2.2.1 is modified by a Note, which applies to the situation where the OPDMS is not monitoring parameters at the beginning of cycle startup, i.e., applies during the first power ascension after a refueling. It states that performance is not required if OPDMS was monitoring parameters upon exceeding 75% RTP.

The Surveillance Frequency is controlled under the SFCP. ~~With the OPDMS not monitoring parameters, the 31 EFPDs Frequency is acceptable because the power distribution will change relatively slowly over this amount of fuel burnup. This Frequency is short enough so that the $F^N_{\Delta H}$ limit will not be exceeded for any significant period of operation.~~

SR 3.2.2.2 is modified by a Note, which applies to the situation where the OPDMS is no longer monitoring parameters. Without the continuous monitoring capability of the OPDMS, $F^N_{\Delta H}$ limits must be monitored on a periodic basis. The first measurement must be made within 24 hours of the OPDMS being declared not monitoring parameters. This is expected to be performed using the most recent contingency surveillance report created automatically during OPDMS monitoring. The Frequency of subsequent surveillances is controlled under the SFCP.~~Subsequent surveillances are required consistent with the 31 EFPD Frequency.~~

REFERENCES

1. NUREG-0800, Standard Review Plan, Section 4.2, Revision 3, “Fuel System Design” (Appendix B).

2. 10 CFR 50, Appendix A, GDC 26.

3. 10 CFR 50.46, “Acceptance Criteria for Emergency Core Cooling Systems for Light Water Nuclear Power Reactors.”

BASES

SURVEILLANCE REQUIREMENTS

The surveillances are modified by a Note allowing 7 days without the continuous monitoring capability of the OPDMS before AFD must be initially verified. The first performance must be made within 7 days of the most recent date where the OPDMS data has verified parameters. ~~This is consistent with the 7 day Surveillance Frequency for AFD.~~

SR 3.2.3.1

This surveillance verifies that the AFD, as indicated by the PMS excore channel, is within the target band. The Surveillance Frequency is controlled under the SFCP. ~~The Surveillance Frequency of 7 days is adequate considering that the AFD is monitored by a computer and any deviation from requirements is alarmed.~~

~~The AFD should be monitored and logged more frequently in periods of operation for which the power level or control bank positions are changing to allow corrective measures when the AFD is more likely to move outside the target band.~~

SR 3.2.3.2

The Surveillance Frequency is controlled under the SFCP. ~~This Surveillance requires that the target flux difference is updated at a Frequency of 31 effective full power days (EFPD) to account for small changes that may occur in the target flux differences in that period due to burnup by performing SR 3.2.3.3.~~

SR 3.2.3.3

Measurement of the target flux difference is determined from an incore power distribution measurement at, or very near equilibrium xenon conditions, preferably at high power levels with the control banks at a reference position. This power distribution measurement provides the reference axial power distribution from which the target value can be determined. The target flux difference varies slowly with core burnup.

The Surveillance Frequency is controlled under the SFCP. ~~A Frequency of 31 EFPD after each refueling and 92 EFPD thereafter for remeasuring the target flux differences adjusts the target flux difference for each excore channel to the value measured at steady-state conditions. This is the basis for the CAOC. Remeasurement at this Surveillance interval also establishes the AFD target flux difference values that account for changes in incore-excore calibrations that may have occurred in the interim.~~

A Note modifies this SR to allow the predicted beginning of cycle AFD from the cycle nuclear design to be used to determine the initial target flux difference after each refueling.

BASES

ACTIONS (continued)

then the peaking factor surveillances must be performed within 48 hours after increasing THERMAL POWER above the limit of Required Action A.1. These Completion Times are intended to allow adequate time to increase THERMAL POWER to above the limit of Required Action A.1, while not permitting the core to remain with unconfirmed power distributions for extended periods of time.

Required Action A.6 is modified by a Note that states that the peaking factor surveillances may only be done after the excore detectors have been normalized to restore QPTR to within limit (i.e., Required Action A.5). The intent of this Note is to have the peaking factor surveillances performed at operating power levels, which can only be accomplished after the excore detectors are calibrated to show zero tilt and the core returned to power.

B.1

If Required Actions A.1 through A.6 are not completed within their associated Completion Times, the unit must be brought to a MODE or condition in which the requirements do not apply. To achieve the status, THERMAL POWER must be reduced to < 50% RTP within 4 hours. The allowed Completion Time of 4 hours is reasonable based on operating experience regarding the amount of time required to reach the reduced power level without challenging plant systems.

SURVEILLANCE REQUIREMENTS

The Surveillances are modified by a Note allowing 12 hours without the continuous monitoring capability of the OPDMS before QPTR must be initially verified. The first verification must be made within 12 hours of the most recent date where the OPDMS data has verified parameters. ~~This is consistent with the 12 hour Surveillance Frequency for QPTR in SR 3.2.4.2.~~

SR 3.2.4.1

SR 3.2.4.1 is modified by two Notes. Note 1 allows QPTR to be calculated with three Power Range Neutron Flux channels if THERMAL POWER is < 75% RTP and the input from one Power Range Neutron Flux channel is inoperable. Note 2 allows performance of SR 3.2.4.2 in lieu of SR 3.2.4.1.

This Surveillance verifies that the QPTR as indicated by the Protection and Safety Monitoring System (PMS) excore channels is within its limits. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 7 days takes into account other information and alarms available to the operator in the control room.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

For those causes of QPTR that occur quickly (a dropped rod), there are other indications of abnormality that prompt a verification of core power tilt.

SR 3.2.4.2

This Surveillance is modified by a Note, which states that it is not required until 12 hours after the input from one or more Power Range Neutron Flux channels are inoperable and the THERMAL POWER is ≥ 75% RTP.

With a PMS Power Range Neutron Flux channel inoperable, tilt monitoring for a portion of the reactor core becomes degraded. Large tilts would likely be detected with the remaining channels, but the capability for detection of small power tilts in some quadrants is decreased. The Surveillance Frequency is controlled under the SFCP. ~~Performing SR 3.2.4.2 at a Frequency of 12 hours provides an accurate alternative means for assuring that any tilt remains within its limits.~~

For purposes of monitoring the QPTR when one Power Range Neutron Flux channel is inoperable, the incore detectors are used to confirm that the normalized symmetric power distribution is acceptable.

With the OPDMS not monitoring parameters and one PMS channel inoperable, the surveillance of the incore power distribution on a 12 hour basis is sufficient to maintain peaking factors within their normal limits, especially, considering the other LCOs and ACTIONS required when the OPDMS is out of service.

REFERENCES

1. Title 10, Code of Federal Regulations, Part 50.46, “Acceptance Criteria for Emergency Core Cooling Systems for Light Water Nuclear Power Reactors.”

2. NUREG-0800, Standard Review Plan, Section 4.2, Revision 3, “Fuel System Design” (Appendix B).

3. Title 10, Code of Federal Regulations, Part 50, Appendix A, “General Design Criteria for Nuclear Power Plants,” GDC 26, “Reactivity Control System Redundancy and Capability.”

BASES

SURVEILLANCE REQUIREMENTS

With OPDMS monitoring parameters, the power distribution parameters and SDM are continuously computed and compared against their limits, and margins to limits displayed. Two levels of alarms are provided to the operator. The first alarm provides a warning before these parameters (peak linear heat rate, $F^N_{\Delta H}$, DNBR), or SDM exceed their limits. The second alarm indicates when they actually reach their limits. A third alarm indicates trouble with the OPDMS system.

SR 3.2.5.1

This Surveillance requires the operator to verify that the power distribution parameters are within their limits. This confirmation is a verification in addition to the automated checking performed by the OPDMS system. The Surveillance Frequency is controlled under the SFCP. ~~A 24 hour Surveillance interval provides assurance that the system is functioning properly and that the core limits are met.~~

REFERENCES

1. Title 10, Code of Federal Regulations, Part 50.46, “Acceptance Criteria for Emergency Core Cooling Systems for Light Water Nuclear Power Reactors,” 1974.

2. NUREG-0800, Standard Review Plan, Section 4.2, Revision 3, “Fuel System Design” (Appendix B).

3. WCAP-12472-P-A, (Westinghouse Proprietary) and WCAP-12473-A (Non-Proprietary), “BEACON Core Monitoring and Operations Support System,” August 1994, Addendum 1, May 1996, and Addendum 2, March 2001 and WCAP-12472-P-A (Westinghouse Proprietary) and WCAP-12472-NP-A (Non-Proprietary) Addendum 4, September 2012.

BASES

ACTIONS (continued)

more channels are inoperable for a Function, thermal power must be reduced to below the P-10 interlock; a condition in which the LCO does not apply. The allowed Completion Time is reasonable, based on operating experience, to reach the specified condition from full power conditions in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

The SRs for each RTS Function are identified in the SRs column of Table 3.3.1-1 for that Function.

A Note has been added to the SR table stating that Table 3.3.1-1 determines which SRs apply to which RTS Functions.

SR 3.3.1.1

This SR compares the calorimetric heat balance to the nuclear instrumentation channel output every 24 hours. If the calorimetric measurement at ≥ 15% RTP, differs from the nuclear instrument channel output by > 5% RTP, the nuclear instrument channel is not declared inoperable, but must be adjusted. If the nuclear instrument channel output cannot be properly adjusted, the channel is declared inoperable.

Two Notes modify this SR. The first Note clarifies that this Surveillance is required only if THERMAL POWER is ≥ 15% RTP and that 12 hours is allowed for performing the first Surveillance after reaching 15% RTP. At lower power levels the calorimetric data from feedwater flow venturi measurements are less accurate. The second Note is required because, at power levels ≥ 15% RTP, calorimetric uncertainty and control rod insertion create the potential for miscalibration of the nuclear instrumentation channel. Therefore, if the calorimetric heat measurement is ≥ 15% RTP, and if the nuclear instrumentation channel indicated power is lower than the calorimetric measurement by > 5% RTP, then the nuclear instrumentation channel shall be adjusted upward to match the calorimetric measurement. No nuclear instrumentation channel adjustment is required if the nuclear instrumentation channel is higher than the calorimetric measurement.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of every 24 hours is adequate based on plant operating experience, considering instrument reliability and operating history data for instrument drift.~~

~~Together, these factors demonstrate the change in the absolute difference between nuclear instrumentation and heat balance calculated powers rarely exceeds 5% RTP in any 24 hours period.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

~~In addition, main control room operators periodically monitor redundant indications and alarms to detect deviations in channel outputs.~~

SR 3.3.1.2

This SR compares the calorimetric heat balance to the calculated ΔT power ($q_{\Delta T}$) in each Division every 24 hours. If the calorimetric measurement between 70% and 100% RTP, differs from the calculated ΔT power by > 3% RTP, the Function is not declared inoperable, but the conversion factor, ΔT°, must be adjusted. If ΔT° cannot be properly adjusted, the Function is declared inoperable in the affected Division(s).

Three Notes modify this SR. The first Note indicates that ΔT° shall be adjusted consistent with the calorimetric results if the absolute difference between the calculated ΔT power and the calorimetric measurement between 70% and 100% RTP is > 3% RTP.

The second Note clarifies that this Surveillance is required only if THERMAL POWER is ≥ 50% RTP and that 12 hours is allowed for performing the first Surveillance after reaching 50% RTP. At lower power levels, the calorimetric data from feedwater flow venturi measurements are less accurate. The calculated ΔT power is normally stable (less likely to need adjustment or to be grossly affected by changes in the core loading pattern than the nuclear instrumentation), and its calibration should not be unnecessarily altered by a possibly inaccurate calorimetric measurement at low power.

The third Note is required because at power levels below 70%, calorimetric uncertainty creates the potential for non-conservative adjustment of the ΔT° conversion factor, in cases where the calculated ΔT power would be reduced to match the calorimetric power. Therefore, if the calorimetric heat measurement is less than 70% RTP, and if the calculated ΔT power is lower than the calorimetric measurement by > 5%, then the ΔT° conversion factor shall be adjusted so that the calculated ΔT power matches the calorimetric measurement. No ΔT° conversion factor adjustment is required if the calculated ΔT power is higher than the calorimetric measurement.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of every 24 hours is based on plant operating experience, considering instrument reliability and the limited effects of fuel burnup and rod position changes on the accuracy of the calculated ΔT power.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.3.1.3

This SR compares the AXIAL FLUX DIFFERENCE determined using the incore system to the nuclear instrument channel AXIAL FLUX DIFFERENCE ~~every 31 effective full power days (EFPD)~~ and adjusts the excore nuclear instrument channel if the absolute difference between the incore and excore AFD is ≥ 1.5% AFD.

Each nuclear instrument channel is calibrated to an average weighted peripheral AFD, which accounts for the fact that neutron leakage from the peripheral fuel assemblies nearest each excore detector will have the largest effect on the channel response. This calibration method reduces the effect of changes in the radial power distribution, caused by either burnup or control rod motion, on the channel AFD calibration. The calibration method is consistent with the development of the f(ΔI) penalty functions for the overpower ΔT and overtemperature ΔT functions, which are made a function of the same average weighted peripheral AFD (i.e., the AFD used in determining the f(ΔI) penalty is calculated using the same radial weighting factors as are used to calibrate the excore detector nuclear instrument channels). The incore AFD used as the basis for comparison when performing this SR is also calculated in the same weighted peripheral manner.

If the absolute difference is ≥ 1.5% AFD the nuclear instrument channel is still OPERABLE, but must be readjusted. If the nuclear instrument channel cannot be properly readjusted, the channel is declared inoperable. This surveillance is performed to verify the f(ΔI) input to the overpower ΔT and overtemperature ΔT functions.

Two Notes modify this SR. The first Note indicates that the excore nuclear instrument channel shall be adjusted if the absolute difference between the incore and excore AFD is ≥ 1.5% AFD. Note 2 clarifies that the Surveillance is required only if THERMAL POWER is ≥ 20% RTP and that 24 hours is allowed for performing the first Surveillance after reaching 20% RTP. Below 20% RTP, the design of the incore detector system, low core power density, and detector accuracy make use of the incore detectors inadequate for use as a reference standard for comparison to the excore channels.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of every 31 EFPD is adequate based on plant operating experience, considering instrument reliability and operating history data for instrument drift. Also, the slow changes in neutron flux during the fuel cycle can be detected during this interval.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.3.1.4

This SR is a calibration of the excore channels to the incore channels. If the measurements do not agree, the excore channels are not declared inoperable but must be adjusted to agree with the incore detector measurements. If the excore channels cannot be adjusted, the channels are declared inoperable. This Surveillance is performed to verify the f(ΔI) input to the overpower ΔT and the overtemperature ΔT Function.

A Note modifies this SR. The Note states that this Surveillance is required only if THERMAL POWER is > 50% RTP and that 24 hours is allowed for performing the first surveillance after reaching 50% RTP.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 92 EFPD is adequate based on industry operating experience, considering instrument reliability and operating history data for instrument drift.~~

SR 3.3.1.5

~~A CHANNEL CALIBRATION is performed every 24 months, or approximately at every refueling.~~ CHANNEL CALIBRATION is a complete check of the instrument loop, including the sensor. The test verifies that the channel responds to a measured parameter within the necessary range and accuracy. The CHANNEL CALIBRATION is performed in a manner that is consistent with the assumptions used in analytically calculating the required channel accuracies. The Surveillance Frequency is controlled under the SFCP.

The test is performed in accordance with the SP. If the actual setting of the channel is found to be outside the as-found tolerance, the channel is considered inoperable. This condition of the channel will be further evaluated during performance of the SR. This evaluation will consist of resetting the channel setpoint to the NTS (within the allowed tolerance), and evaluating the channel response. If the channel is functioning as required and is expected to pass the next surveillance, then the channel is OPERABLE and can be restored to service at the completion of the surveillance. After the surveillance is completed, the channel as-found condition will be entered into the Corrective Action Program for further evaluation. Transmitter calibration must be performed consistent with the assumptions of the setpoint methodology. The differences between the current as-found values and the previous as-left values must be consistent with the transmitter drift allowance used in the setpoint methodology.

BASES

SURVEILLANCE REQUIREMENTS (continued)

The setpoint methodology requires that 30 months drift be used ~~(1.25 times the surveillance calibration interval, 24 months)~~.

Interlocks implicitly required to support the Function's OPERABILITY are also addressed by this CHANNEL CALIBRATION. This portion of the CHANNEL CALIBRATION ensures the associated Function is not bypassed when required to be enabled. This can be accomplished by ensuring the interlocks are calibrated properly in accordance with the SP. If the interlock is not automatically functioning as designed, the condition is entered into the Corrective Action Program and appropriate OPERABILITY evaluations performed for the affected Function. The affected Function's OPERABILITY can be met if the interlock is manually enforced to properly enable the affected Function. When an interlock is not supporting the associated Function's OPERABILITY at the existing plant conditions, the affected Function's channels must be declared inoperable and appropriate ACTIONS taken.

The RCP speed sensor (Function 9) is calibrated using a qualitative assessment of the behavior that is obtained through insulation resistance and sensor resistance testing to complement the normal calibration of the remaining adjustable devices in Function 9.

This SR is modified by a Note stating that this test shall include verification that the time constants are adjusted to within limits where applicable.

SR 3.3.1.6

This SR is the performance of a CHANNEL CALIBRATION ~~every 24 months~~. This SR is modified by a Note stating that neutron detectors are excluded from the CHANNEL CALIBRATION. The CHANNEL CALIBRATION is performed in a manner that is consistent with the assumptions used in analytically calculating the required channel accuracies.

The test is performed in accordance with the SP. If the actual setting of the channel is found to be outside the as-found tolerance, the channel is considered inoperable. This condition of the channel will be further evaluated during performance of the SR. This evaluation will consist of resetting the channel setpoint to the NTS (within the allowed tolerance), and evaluating the channel's response. If the channel is functioning as required and is expected to pass the next surveillance, then the channel is OPERABLE and can be restored to service at the completion of the surveillance. After the surveillance is completed, the channel as-found condition will be entered into the Corrective Action Program for further evaluation.

BASES

SURVEILLANCE REQUIREMENTS (continued)

The CHANNEL CALIBRATION for the power range neutron detectors consists of a normalization of the detectors based on a power calorimetric and flux map performed above 20% RTP. Below 20% RTP, the design of the incore detector system, low core power density, and detector accuracy make use of the incore detectors inadequate for use as a reference standard for comparison to the excore channels.

Interlocks implicitly required to support the Function's OPERABILITY are also addressed by this CHANNEL CALIBRATION. This portion of the CHANNEL CALIBRATION ensures the associated Function is not bypassed when required to be enabled. This can be accomplished by ensuring the interlocks are calibrated properly in accordance with the SP. If the interlock is not automatically functioning as designed, the condition is entered into the Corrective Action Program and appropriate OPERABILITY evaluations performed for the affected Function. The affected Function's OPERABILITY can be met if the interlock is manually enforced to properly enable the affected Function. When an interlock is not supporting the associated Function's OPERABILITY at the existing plant conditions, the affected Function's channels must be declared inoperable and appropriate ACTIONS taken.

The Surveillance Frequency is controlled under the SFCP. ~~The 24 month Frequency is based on the need to perform this Surveillance under the conditions that apply during a plant outage and the potential for an unplanned transient if the Surveillance were performed with the reactor at power. Operating experience has shown these components usually pass the Surveillance when performed on the 24 month Frequency.~~

SR 3.3.1.7

This SR is the performance of a TADOT of the Passive Residual Heat Removal Actuation valve position indicator contact inputs. The Surveillance Frequency is controlled under the SFCP. ~~This TADOT is performed every 24 months.~~

~~The Frequency is based on the known reliability of the Function and the multichannel redundancy available, and has been shown to be acceptable through operating experience.~~

The SR is modified by a Note that excludes verification of setpoints from the TADOT. The Functions affected have no setpoints associated with them.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.3.1.8

This SR verifies that the individual channel/division actuation response times are less than or equal to the maximum values assumed in the accident analysis. Response Time testing criteria are included in Reference 1.

In lieu of measurement, the response time for the protection and safety monitoring system equipment is based on allocated values (Ref. 8). The overall response time may be determined by a series of overlapping tests and allocated values such that the entire response time is verified, with the resulting verified response time compared to the appropriate FSAR Chapter 7 (Ref. 1) response time.

Response time may be verified by actual response time tests in any series of sequential, overlapping or total channel measurements, or by the allocated sensor, signal processing and actuation logic response times, where approved, with actual response time tests on the remainder of the channel. Allocations for signal processing and actuation logic response times may be obtained from the Protection and Safety Monitoring System Functional Requirements document. The response time of the RCP speed sensor (Function 9) and its associated preamplifier is allocated in lieu of measurement, based on the specific design of the component models being used (Ref. 10).

The Passive Residual Heat Removal (PRHR) Actuation Function RTS RESPONSE TIME is the time interval between input of a PRHR discharge valve not-fully-closed position feedback signal and the loss of gripper coil voltage. The RTS RESPONSE TIME for the PRHR actuation does not include testing actuation of the discharge valves by ESFAS instrumentation signals because it cannot be tested if an ESFAS function (e.g., CMT Actuation) has already caused a reactor trip.

The Surveillance Frequency is controlled under the SFCP.~~Each division response must be verified every 24 months on a STAGGERED TEST BASIS (i.e., all four Protection Channel Sets would be tested after 96 months). Response times cannot be determined during plant operation because equipment operation is required to measure response times. Experience has shown that these components usually pass this surveillance when performed on a refueling frequency. Therefore, the Frequency was concluded to be acceptable from a reliability standpoint.~~

BASES

ACTIONS (continued)

E.1 and E.2

Condition E is entered when the Required Action and associated Completion Time of Condition D is not met. If three of the four required source range instrumentation channels are not restored to OPERABLE status within the allowed Completion Time, Required Action E.1 requires that action be initiated to fully insert all rods within 1 hour, and Required Action E.2 requires that the PLS be placed in a condition incapable of rod withdrawal within 1 hour. The allowed Completion Time is reasonable, based on operating experience, to reach the specified condition in an orderly manner and without challenging plant systems.

F.1

Condition F addresses the situation where three or more source range instrumentation channels are inoperable. With three or more channels inoperable, single failure criterion cannot be met and the reactor trip breakers must be opened immediately.

SURVEILLANCE REQUIREMENTS

SR 3.3.2.1

Performance of the CHANNEL CHECK ~~once every 12 hours~~ ensures that gross failure of instrumentation has not occurred. A CHANNEL CHECK is normally a comparison of the parameter indicated on one channel to a similar parameter on other channels. It is based on the assumption that instrument channels monitoring the same parameter should read approximately the same value. Significant deviations between the two instrument channels could be an indication of excessive instrument drift in one of the channels or of even something more serious. A CHANNEL CHECK will detect gross channel failure; thus, it is key to verifying that the instrumentation continues to operate properly between each CHANNEL CALIBRATION. Agreement criteria are determined by the plant staff based on a combination of the channel instrument uncertainties, including indication and readability. If a channel is outside the criteria, it may be an indication that the sensor or the signal processing equipment have drifted outside their corresponding limits.

The Surveillance Frequency is controlled under the SFCP.~~The Frequency is based on operating experience that demonstrates that channel failure is rare.~~ Automated operator aids may be used to facilitate the performance of the CHANNEL CHECK.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.3.2.2

This SR is the performance of a CHANNEL CALIBRATION ~~every 24 months~~. This SR is modified by a Note stating that neutron detectors are excluded from the CHANNEL CALIBRATION. The CHANNEL CALIBRATION is performed in a manner that is consistent with the assumptions used in analytically calculating the required channel accuracies.

The test is performed in accordance with the SP. If the actual setting of the channel is found to be outside the as-found tolerance, the channel is considered inoperable. This condition of the channel will be further evaluated during performance of the SR. This evaluation will consist of resetting the channel setpoint to the NTS (within the allowed tolerance), and evaluating the channel's response. If the channel is functioning as required and is expected to pass the next surveillance, then the channel is OPERABLE and can be restored to service at the completion of the surveillance. After the surveillance is completed, the channel as-found condition will be entered into the Corrective Action Program for further evaluation.

The CHANNEL CALIBRATION for the source range neutron detectors consists of obtaining the preamp discriminator curves, evaluating those curves, and comparing the curves to the manufacturer's data.

Interlocks implicitly required to support the Function's OPERABILITY are also addressed by this CHANNEL CALIBRATION. This portion of the CHANNEL CALIBRATION ensures the associated Function is not bypassed when required to be enabled. This can be accomplished by ensuring the interlocks are calibrated properly in accordance with the SP. If the interlock is not automatically functioning as designed, the condition is entered into the Corrective Action Program and appropriate OPERABILITY evaluations performed for the affected Function. The Function's OPERABILITY can be met if the interlock is manually enforced to properly enable the affected Function. When an interlock is not supporting the associated Function's OPERABILITY at the existing plant conditions, the affected Function's channels must be declared inoperable and appropriate ACTIONS taken.

BASES

SURVEILLANCE REQUIREMENTS (continued)

The Surveillance Frequency is controlled under the SFCP. ~~The 24 month Frequency is based on the need to perform this Surveillance under the conditions that apply during a plant outage and the potential for an unplanned transient if the Surveillance were performed with the reactor at power. Operating experience has shown these components usually pass the Surveillance when performed on the 24 month Frequency.~~

SR 3.3.2.3

This SR verifies that the individual channel actuation response times are less than or equal to the maximum values assumed in the accident analysis. Response Time testing criteria are included in Reference 1.

In lieu of measurement, the response time for the protection and safety monitoring system equipment is based on allocated values (Ref. 3). The overall response time may be determined by a series of overlapping tests and allocated values such that the entire response time is verified, with the resulting verified response time compared to the appropriate FSAR Chapter 7 (Ref. 1) response time.

Response time may be verified by actual response time tests in any series of sequential, overlapping or total channel measurements, or by the allocated sensor, signal processing and actuation logic response times, where approved, with actual response time tests on the remainder of the channel.

The Surveillance Frequency is controlled under the SFCP. ~~Each channel response must be verified every 24 months on a STAGGERED TEST BASIS (i.e., all four Protection Channel Sets would be tested after 96 months). Response times cannot be determined during plant operation because equipment operation is required to measure response times. Experience has shown that these components usually pass this surveillance when performed on a refueling frequency. Therefore, the Frequency was concluded to be acceptable from a reliability standpoint.~~

This SR is modified by a note exempting neutron detectors from RTS RESPONSE TIME testing. This Note is necessary because of the difficulty in generating an appropriate detector input signal. Excluding the detectors is acceptable because the principles of detector operation ensure a virtually instantaneous response.

BASES

ACTIONS (continued)

intermediate range instrumentation channels inoperable, three of the four required channels must be restored to OPERABLE status prior to increasing THERMAL POWER above the P-6 setpoint. With the unit in this condition, below P-6, the Source Range Neutron Flux channels perform the monitoring and protection functions.

D.1, D.2, and D.3

Condition D addresses the situation where three or more intermediate range instrumentation channels are inoperable. With three or more channels inoperable, operations involving positive reactivity addition must be suspended immediately. This will preclude any power level increase since there are insufficient OPERABLE Intermediate Range channels to adequately monitor power escalation. In addition, THERMAL POWER must be reduced below the P-6 interlock setpoint within 2 hours, and the plant must be placed in MODE 3 within 7 hours. The allowed Completion Times for Required Actions D.2 and D.3 are reasonable, based on operating experience, to reach the specified condition from full power conditions in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.3.3.1

Performance of the CHANNEL CHECK ~~once every 12 hours~~ ensures that gross failure of instrumentation has not occurred. A CHANNEL CHECK is normally a comparison of the parameter indicated on one channel to a similar parameter on other channels. It is based on the assumption that instrument channels monitoring the same parameter should read approximately the same value. Significant deviations between the two instrument channels could be an indication of excessive instrument drift in one of the channels or of even something more serious. A CHANNEL CHECK will detect gross channel failure; thus, it is key to verifying that the instrumentation continues to operate properly between each CHANNEL CALIBRATION.

Agreement criteria are determined by the plant staff based on a combination of the channel instrument uncertainties, including indication and readability. If a channel is outside the criteria, it may be an indication that the sensor or the signal processing equipment have drifted outside their corresponding limits.

BASES

SURVEILLANCE REQUIREMENTS (continued)

This SR is modified by a Note. The Note states that the SR is not applicable in MODE 1. In MODE 1, the OPERABILITY of the Intermediate Range Neutron Instrumentation is verified by self-checking features in lieu of performing a CHANNEL CHECK.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is based on operating experience that demonstrates that channel failure is rare.~~ Automated operator aids may be used to facilitate the performance of the CHANNEL CHECK.

SR 3.3.3.2

This SR is the performance of a CHANNEL CALIBRATION ~~every 24 months~~. This SR is modified by a Note stating that neutron detectors are excluded from the CHANNEL CALIBRATION. The CHANNEL CALIBRATION is performed in a manner that is consistent with the assumptions used in analytically calculating the required channel accuracies.

The test is performed in accordance with the SP. If the actual setting of the channel is found to be outside the as-found tolerance, the channel is considered inoperable. This condition of the channel will be further evaluated during performance of the SR. This evaluation will consist of resetting the channel setpoint to the NTS (within the allowed tolerance), and evaluating the channel's response. If the channel is functioning as required and is expected to pass the next surveillance, then the channel is OPERABLE and can be restored to service at the completion of the surveillance. After the surveillance is completed, the channel as-found condition will be entered into the Corrective Action Program for further evaluation.

The CHANNEL CALIBRATION for the intermediate range neutron detectors consists of obtaining the detector plateau curves, evaluating those curves, and comparing the curves to the manufacturer's data.

Interlocks implicitly required to support the Function's OPERABILITY are also addressed by this CHANNEL CALIBRATION. This portion of the CHANNEL CALIBRATION ensures the associated Function is not bypassed when required to be enabled. This can be accomplished by ensuring the interlocks are calibrated properly in accordance with the SP. If the interlock is not automatically functioning as designed, the condition is entered into the Corrective Action Program and appropriate OPERABILITY evaluations performed for the affected Function. The

BASES

SURVEILLANCE REQUIREMENTS (continued)

affected Function's OPERABILITY can be met if the interlock is manually enforced to properly enable the affected Function. When an interlock is not supporting the associated Function's OPERABILITY at the existing plant conditions, the affected Function's channels must be declared inoperable and appropriate ACTIONS taken.

The Surveillance Frequency is controlled under the SFCP. ~~The 24 month Frequency is based on the need to perform this Surveillance under the conditions that apply during a plant outage and the potential for an unplanned transient if the Surveillance were performed with the reactor at power. Operating experience has shown these components usually pass the Surveillance when performed on the 24 month Frequency.~~

SR 3.3.3.3

This SR verifies that the individual channel actuation response times are less than or equal to the maximum values assumed in the accident analysis. Response Time testing criteria are included in Reference 1.

In lieu of measurement, the response time for the protection and safety monitoring system equipment is based on allocated values (Ref. 3). The overall response time may be determined by a series of overlapping tests and allocated values such that the entire response time is verified, with the resulting verified response time compared to the appropriate FSAR Chapter 7 (Ref. 1) response time.

Response time may be verified by actual response time tests in any series of sequential, overlapping or total channel measurements, or by the allocated sensor, signal processing and actuation logic response times, where approved, with actual response time tests on the remainder of the channel.

The Surveillance Frequency is controlled under the SFCP. ~~Each channel response must be verified every 24 months on a STAGGERED TEST BASIS (i.e., all four Protection Channel Sets would be tested after 96 months). Response times cannot be determined during plant operation because equipment operation is required to measure response times. Experience has shown that these components usually pass this surveillance when performed on a refueling frequency. Therefore, the Frequency was concluded to be acceptable from a reliability standpoint.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

The Surveillance Frequency is controlled under the SFCP. ~~Each channel response must be verified every 24 months on a STAGGERED TEST BASIS (i.e., all four Protection Channel Sets would be tested after 96 months). Response times cannot be determined during plant operation because equipment operation is required to measure response times. Experience has shown that these components usually pass this surveillance when performed on a refueling frequency. Therefore, the Frequency was concluded to be acceptable from a reliability standpoint.~~

REFERENCE

1. FSAR Chapter 7, “Instrumentation and Controls.”
2. Safety Evaluation by the Office of Nuclear Reactor Regulation related to Amendment Nos. 168 and 166 to the Combined License Nos. NPF-91 and NPF-92, respectively, Vogtle Electric Generating Plant Units 3 and 4, Docket Nos. 52-025 and 52-026, dated November 21, 2019 [ML19297C988 (Proprietary) and ML19297D159 (non-proprietary)].

BASES

SURVEILLANCE REQUIREMENTS

SR 3.3.5.1

SR 3.3.5.1 is the performance of a TADOT of the RTS inputs for Manual Reactor Trip, and from the ESFAS logic for Safeguards Actuation, ADS Stage 1, 2, and 3 Actuation, and CMT Actuation. ~~This TADOT is performed every 24 months.~~ The test shall independently verify the OPERABILITY of the undervoltage and shunt trip mechanisms for the Manual Reactor Trip Function for the Reactor Trip Breakers.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is based on the known reliability of the Functions and the multichannel redundancy available, and has been shown to be acceptable through operating experience.~~

The SR is modified by a Note that excludes verification of setpoints from the TADOT. The Functions affected have no setpoints associated with them.

REFERENCES None

BASES

ACTIONS (continued)

C.1

Condition C addresses the situation where the Required Action and associated Completion Time of Condition A or B are not met in MODE 1 or 2, or there are one or both RTS Trip Actuation functions within three or more divisions inoperable in MODE 1 or MODE 2. Required Action C.1 directs that the plant must be placed in MODE 3 within 6 hours. The allowed Completion Times for Required Action C.1 is reasonable, based on operating experience, to reach the specified condition from full power conditions in an orderly manner and without challenging plant systems.

D.1 and D.2

Condition D addresses the situation where the Required Action and associated Completion Time of Condition A or B are not met in MODE 3, 4, or 5, or there are one or both RTS Trip Actuation functions within three or more divisions inoperable in MODE 3, 4, or 5. Required Action D.1 requires initiating action to fully insert all control rods within 6 hours, and Required Action D.2 requires that the Plant Control System be placed in a condition incapable of rod withdrawal within 6 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the specified condition in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.3.7.1

SR 3.3.7.1 is the performance of a TADOT on the Reactor Trip Digital Outputs, the Reactor Trip Matrix Termination Units, and on both reactor trip breakers associated with a single division ~~every 92 days on a STAGGERED TEST BASIS for four divisions~~. This test shall verify OPERABILITY by actuation of the end devices.

The Reactor Trip Breaker (RTB) test shall include separate verification of the undervoltage and shunt trip mechanisms. Each RTB in a division shall be tested separately in order to minimize the possibility of an inadvertent trip. Both breakers in a single division are tested during each test~~STAGGERED TEST~~.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of every 92 days on a STAGGERED TEST BASIS is adequate based on industry operating experience, considering instrument reliability and operating history data. In addition, the design provides additional breakers to enhance reliability.~~

REFERENCES

1. FSAR Chapter 15, “Accident Analysis.”

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.3.8.1

Performance of the CHANNEL CHECK ~~once every 12 hours~~ ensures that a gross failure of instrumentation has not occurred. A CHANNEL CHECK is a comparison of the parameter indicated on one channel to a similar parameter on other channels. It is based on the assumption that instrument channels monitoring the same parameter should read approximately the same value. Significant deviations between the two instrument channels could be an indication of excessive instrument drift in one of the channels or even something more serious. A CHANNEL CHECK will detect gross channel failure; thus, it is key to verifying the instrumentation continues to operate properly between each CHANNEL CALIBRATION.

Agreement criteria are determined by the plant staff, based on a combination of the channel instrument uncertainties, including indication and readability. If a channel is outside the match criteria, it may be an indication that the sensor or the signal processing equipment has drifted outside their corresponding limits.

The Surveillance Requirement is modified by a Note. The Note states that this SR is only required for Source Range Neutron Flux Doubling. The OPERABILITY for the other Functions is verified by self-checking features in lieu of performing a CHANNEL CHECK.

The Surveillance Frequency is controlled under the SFCP. ~~The Surveillance Frequency is based on operating experience that demonstrates that channel failure is rare.~~ Automated operator aids may be used to facilitate performance of the CHANNEL CHECK.

SR 3.3.8.2

This SR is the performance of a CHANNEL CALIBRATION ~~every 24 months or approximately at every refueling~~. CHANNEL CALIBRATION is a complete check of the instrument loop, including the sensor and the integrated protection cabinets (IPC). The test is performed in accordance with the SP. If the actual setting of the channel is found to be outside the as-found tolerance, the channel is considered inoperable. This condition of the channel will be further evaluated during performance of the SR. This evaluation will consist of resetting the channel setpoint to the NTS (within the allowed tolerance), and evaluating the channel's response. If the channel is functioning as required and is expected to pass the next surveillance, then the channel is OPERABLE and can be restored to service at the completion of the surveillance. After the surveillance is completed, the channel as-found condition will be entered into the

BASES

SURVEILLANCE REQUIREMENTS (continued)

Corrective Action Program for further evaluation. Transmitter calibration must be performed consistent with the assumptions of the setpoint methodology. The difference between the current as-found values and the previous as-left values must be consistent with the transmitter drift allowance used in the setpoint methodology.

Interlocks implicitly required to support the Function's OPERABILITY are also addressed by this CHANNEL CALIBRATION. This portion of the CHANNEL CALIBRATION ensures the associated Function is not bypassed when required to be enabled. This can be accomplished by ensuring the interlocks are calibrated properly in accordance with the SP. If the interlock is not automatically functioning as designed, the condition is entered into the Corrective Action Program and appropriate OPERABILITY evaluations performed for the affected Function. The affected Function's OPERABILITY can be met if the interlock is manually enforced to properly enable the affected Function. When an interlock is not supporting the associated Function's OPERABILITY at the existing plant conditions, the affected Function's channels must be declared inoperable and appropriate ACTIONS taken. In the case of an interlock that is applied "post-vote," the ESF Actuation Logic Function of LCO 3.3.15 or LCO 3.3.16 (as applicable to the plant condition) is declared inoperable.

~~The setpoint methodology requires that 30 months drift be used (1.25 times the surveillance calibration interval, 24 months).~~

~~The Frequency is based on operating experience and consistency with the refueling cycle.~~ The Surveillance Frequency is controlled under the SFCP.

This Surveillance Requirement is modified by two Notes. Note 1 states that this test should include verification that the time constants are adjusted to within limits where applicable. Note 2 states that neutron detectors are excluded from the CHANNEL CALIBRATION. The CHANNEL CALIBRATION for the source range neutron flux monitors consisting of obtaining the detector plateau or preamp discriminator curves, evaluating those curves, and comparing the curves to the manufacturer's data.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.3.8.3

This SR ensures the individual channel ESF RESPONSE TIME is less than or equal to the maximum value assumed in the accident analysis. Individual component response times are not modeled in the analyses. The analyses model the overall or total elapsed time, from the point at which the parameter exceeds the NTS value at the sensor, to the point at which the equipment reaches the required functional state (e.g., valves in full open or closed position).

In lieu of measurement, the response time for the protection and safety monitoring system equipment is based on allocated values (Ref. 7). The overall response time may be determined by a series of overlapping tests and allocated values such that the entire response time is verified, with the resulting verified response time compared to the appropriate FSAR Chapter 7 (Ref. 1) response time.

Response time may be verified by actual response time tests in any series of sequential, overlapping or total channel measurements, or by the allocated sensor, signal processing and actuation logic response times, where approved, with actual response time tests on the remainder of the channel.

The Surveillance Frequency is controlled under the SFCP. ~~ESF RESPONSE TIME tests are conducted on a 24 month STAGGERED TEST BASIS. Testing of the devices is included in the testing of each channel. The final actuation device in one train is tested with each channel. Therefore, staggered testing results in response time verification of these devices every 24 months. The 24 month Frequency is consistent with the typical refueling cycle and is based on unit operating experience, which shows that random failures of instrumentation components causing serious response time degradation, but not channel failure, are infrequent occurrences.~~

The Surveillance Requirement is modified by two Notes: Note 1 states that it is not applicable to Function 1.a for Containment Pressure – Low. The exception is appropriate because the Containment Pressure – Low signal provides an interlock function for the containment vacuum relief valves manual initiation function and does not directly actuate any ESF. Note 2 exempts neutron detectors from ESF RESPONSE TIME testing. This Note is necessary because of the difficulty in generating an appropriate detector input signal. Excluding the detectors is acceptable because the principles of detector operation ensure a virtually instantaneous response.

BASES

ACTIONS (continued)

Required Action J.3 requires initiation of action within 12 hours to restore the channel(s) to OPERABLE status and to continue this action until restoration is accomplished in order to provide the necessary channels for the Function. The 12 hour Completion Time is consistent with the transition to MODE 5 in accordance with Required Action J.2.

K.1 and K.2 – Not used.

L.1, L.2, and L.3

If the Required Action and associated Completion Time of Condition A are not met or if more than two channels are inoperable, the plant must be placed in a condition in which the likelihood and consequences of an event are minimized. To achieve this status, the plant must be brought to at least MODE 3 within 6 hours and to MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner and without challenging plant systems.

Required Action L.3 requires initiation of action within 36 hours to restore the channel(s) to OPERABLE status and to continue this action until restoration is accomplished in order to provide the necessary channels for the Function. The 36 hour Completion Time is consistent with the transition to MODE 5 in accordance with Required Action L.2.

SURVEILLANCE REQUIREMENTS

SR 3.3.9.1

SR 3.3.9.1 is the performance of a TADOT of the manual initiation for the various ESF Functions. ~~This TADOT is performed every 24 months.~~

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is based on the known reliability of the ESF Functions and the multichannel redundancy available, and has been shown to be acceptable through operating experience.~~

The SR is modified by a Note that states verification of setpoint is not required, since these functions have no setpoint associated with them.

REFERENCES

1. FSAR Chapter 15, “Accident Analysis.”

BASES

SURVEILLANCE REQUIREMENTS

SR 3.3.10.1

This SR is the performance of a CHANNEL CALIBRATION ~~every 24 months or approximately at every refueling~~. CHANNEL CALIBRATION is a complete check of the instrument loop, including the sensor and the integrated protection cabinets (IPC). The test is performed in accordance with the SP. If the actual setting of the channel is found to be outside the as-found tolerance, the channel is considered inoperable. This condition of the channel will be further evaluated during performance of the SR. This evaluation will consist of resetting the channel setpoint to the NTS (within the allowed tolerance), and evaluating the channel's response. If the channel is functioning as required and is expected to pass the next surveillance, then the channel is OPERABLE and can be restored to service at the completion of the surveillance. After the surveillance is completed, the channel as-found condition will be entered into the Corrective Action Program for further evaluation. Transmitter calibration must be performed consistent with the assumptions of the setpoint methodology. The difference between the current as-found values and the previous as-left values must be consistent with the transmitter drift allowance used in the setpoint methodology.

Interlocks implicitly required to support the Function's OPERABILITY are also addressed by this CHANNEL CALIBRATION. This portion of the CHANNEL CALIBRATION ensures the associated Function is not bypassed when required to be enabled. This can be accomplished by ensuring the interlocks are calibrated properly in accordance with the SP. If the interlock is not automatically functioning as designed, the condition is entered into the Corrective Action Program and appropriate OPERABILITY evaluations performed for the affected Function. The affected Function's OPERABILITY can be met if the interlock is manually enforced to properly enable the affected Function. When an interlock is not supporting the associated Function's OPERABILITY at the existing plant conditions, the affected Function's channels must be declared inoperable and appropriate ACTIONS taken.

~~The setpoint methodology requires that 30 months drift be used (1.25 times the surveillance calibration interval, 24 months).~~

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is based on operating experience and consistency with the refueling cycle.~~

This Surveillance Requirement is modified by a Note. The Note states that this test should include verification that the time constants are adjusted to within limits.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.3.10.2

This SR ensures the individual channel ESF RESPONSE TIME is less than or equal to the maximum value assumed in the accident analysis. Individual component response times are not modeled in the analyses. The analyses model the overall or total elapsed time, from the point at which the parameter exceeds the NTS value at the sensor, to the point at which the equipment reaches the required functional state (e.g., valves in full open or closed position).

In lieu of measurement, the response time for the protection and safety monitoring system equipment is based on allocated values (Ref. 6). The overall response time may be determined by a series of overlapping tests and allocated values such that the entire response time is verified, with the resulting verified response time compared to the appropriate FSAR Chapter 7 (Ref. 2) response time.

Response time may be verified by actual response time tests in any series of sequential, overlapping or total channel measurements, or by the allocated sensor, signal processing and actuation logic response times, where approved, with actual response time tests on the remainder of the channel.

The Surveillance Frequency is controlled under the SFCP. ~~ESF RESPONSE TIME tests are conducted on a 24 month STAGGERED TEST BASIS. Testing of the devices is included in the testing of each channel. The final actuation device in one train is tested with each channel. Therefore, staggered testing results in response time verification of these devices every 24 months. The 24 month Frequency is consistent with the typical refueling cycle and is based on unit operating experience, which shows that random failures of instrumentation components causing serious response time degradation, but not channel failure, are infrequent occurrences.~~

BASES

ACTIONS (continued)

In the event a channel's as-found condition is outside the as-found tolerance described in the Setpoint Program, or the channel is not functioning as required, or the transmitter, or the Protection and Safety Monitoring System Division associated with a specific Function is found inoperable, then all affected Functions provided by that channel must be declared inoperable and the LCO Condition(s) entered for the particular protection Function(s) affected.

A.1

With one or more startup feedwater lines with one startup feedwater channel inoperable, the inoperable channel must be placed in a trip condition within 6 hours. If one channel is tripped, the interlock condition is satisfied. The specified Completion Time is reasonable considering the time required to complete this action.

B.1 and B.2

If the Required Action and associated Completion Time of Condition A is not met or if one or more startup feedwater lines has two channels inoperable, the plant must be placed in a MODE in which the LCO does not apply. This is accomplished by placing the plant in MODE 3 within 6 hours and in MODE 4 with the RCS being cooled by the RNS within 24 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.3.11.1

This SR is the performance of a CHANNEL CALIBRATION ~~every 24 months or approximately at every refueling~~. CHANNEL CALIBRATION is a complete check of the instrument loop, including the sensor and the integrated protection cabinets (IPC). The test is performed in accordance with the SP. If the actual setting of the channel is found to be outside the as-found tolerance, the channel is considered inoperable. This condition of the channel will be further evaluated during performance of the SR. This evaluation will consist of resetting the channel setpoint to the NTS (within the allowed tolerance), and evaluating the channel's response. If the channel is functioning as required and is expected to pass the next surveillance, then the channel is OPERABLE and can be restored to service at the completion of the surveillance. After the surveillance is completed, the channel as-found condition will be entered into the

BASES

SURVEILLANCE REQUIREMENTS (continued)

Corrective Action Program for further evaluation. Transmitter calibration must be performed consistent with the assumptions of the setpoint methodology. The difference between the current as-found values and the previous as-left values must be consistent with the transmitter drift allowance used in the setpoint methodology.

~~The setpoint methodology requires that 30 months drift be used (1.25 times the surveillance calibration interval, 24 months).~~

The Surveillance Frequency is controlled under the SFCP.~~The Frequency is based on operating experience and consistency with the refueling cycle.~~

This Surveillance Requirement is modified by a Note. The Note states that this test should include verification that the time constants are adjusted to within limits where applicable.

SR 3.3.11.2

This SR ensures the individual channel ESF RESPONSE TIME is less than or equal to the maximum values assumed in the accident analysis. Individual component response times are not modeled in the analyses. The analyses model the overall or total elapsed time, from the point at which the parameter exceeds the NTS value at the sensor, to the point at which the equipment reaches the required functional state (e.g., valves in full open or closed position).

In lieu of measurement, the response time for the protection and safety monitoring system equipment is based on allocated values (Ref. 4). The overall response time may be determined by a series of overlapping tests and allocated values such that the entire response time is verified, with the resulting verified response time compared to the appropriate FSAR Chapter 7 (Ref. 2) response time.

Response time may be verified by actual response time tests in any series of sequential, overlapping or total channel measurements, or by the allocated sensor, signal processing and actuation logic response times, where approved, with actual response time tests on the remainder of the channel.

BASES

SURVEILLANCE REQUIREMENTS (continued)

The Surveillance Frequency is controlled under the SFCP. ~~ESF RESPONSE TIME tests are conducted on a 24 month STAGGERED TEST BASIS. Testing of the devices is included in the testing of each channel. The final actuation device in one train is tested with each channel. Therefore, staggered testing results in response time verification of these devices every 24 months. The 24 month Frequency is consistent with the typical refueling cycle and is based on unit operating experience, which shows that random failures of instrumentation components causing serious response time degradation, but not channel failure, are infrequent occurrences.~~

REFERENCES

1. FSAR Chapter 15, “Accident Analysis.”
2. FSAR Chapter 7, “Instrumentation and Controls.”
3. APP-GW-GSC-020, “Technical Specification Completion Time and Surveillance Frequency Justification.”
4. Safety Evaluation by the Office of Nuclear Reactor Regulation related to Amendment Nos. 168 and 166 to the Combined License Nos. NPF-91 and NPF-92, respectively, Vogtle Electric Generating Plant Units 3 and 4, Docket Nos. 52-025 and 52-026, dated November 21, 2019 [ML19297C988 (Proprietary) and ML19297D159 (non-proprietary)].

BASES

ACTIONS (continued)

B.1, B.2, and B.3

If the Required Action and associated Completion Time Condition A is not met, or if two or three required divisions are inoperable, the plant must be placed in a MODE in which the likelihood and consequences of an event are minimized.

Under these circumstances, the affected isolation valves(s) must be declared inoperable immediately, per Required Action B.1. Declaring the affected isolation valve inoperable allows the supported system Actions (i.e., for inoperable valves) to dictate the required measures. The respective isolation valve LCO provides appropriate actions for the inoperable components. This actions is in accordance with LCO 3.0.6, which requires that the applicable Conditions and Required Actions for the isolation valves declared inoperable shall be entered in accordance with LCO 3.0.2. Additionally, Required Action B.2 requires that the plant must be brought to MODE 3 within 6 hours and Required Action B.3 requires that the plant be brought to MODE 4 in 12 hours. The allowed time is reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.3.12.1

SR 3.3.12.1 is the performance of a TADOT of the blocks for the reactor trip (P-4) input from the integrated protection cabinets (IPCs). ~~This TADOT is performed every 24 months.~~

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is based on the known reliability of the engineered safety features (ESF) function and the multichannel redundancy available, and has been shown to be acceptable through operating experience.~~

REFERENCES

1. FSAR Chapter 15, “Accident Analysis.”

BASES

SURVEILLANCE REQUIREMENTS

The SRs for each ESFAS Main Control Room Isolation, Air Supply Initiation, and Electrical Load De-energization Function are identified in the SRs column of Table 3.3.13-1 for that Function. A Note has been added to the SR table stating that Table 3.3.13-1 determines which SRs apply to which ESFAS Main Control Room Isolation, Air Supply Initiation, and Electrical Load De-energization Function.

SR 3.3.13.1

This SR is the performance of a CHANNEL CALIBRATION ~~every 24 months or approximately at every refueling~~. CHANNEL CALIBRATION is a complete check of the instrument loop, including the sensor and the integrated protection cabinets (IPC). The test is performed in accordance with the SP. If the actual setting of the channel is found to be outside the as-found tolerance, the channel is considered inoperable. This condition of the channel will be further evaluated during performance of the SR. This evaluation will consist of resetting the channel setpoint to the NTS (within the allowed tolerance), and evaluating the channel's response. If the channel is functioning as required and is expected to pass the next surveillance, then the channel is OPERABLE and can be restored to service at the completion of the surveillance. After the surveillance is completed, the channel as-found condition will be entered into the Corrective Action Program for further evaluation. Transmitter calibration must be performed consistent with the assumptions of the setpoint methodology. The difference between the current as-found values and the previous as-left values must be consistent with the transmitter drift allowance used in the setpoint methodology.

~~The setpoint methodology requires that 30 months drift be used (1.25 times the surveillance calibration interval, 24 months).~~

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is based on operating experience and consistency with the refueling cycle.~~

This Surveillance Requirement is modified by a Note. The Note states that this test should include verification that the time constants are adjusted to within limits where applicable.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.3.13.2

This SR ensures the individual channel ESF RESPONSE TIME is less than or equal to the maximum value assumed in the accident analysis. Individual component response times are not modeled in the analyses. The analyses model the overall or total elapsed time, from the point at which the parameter exceeds the NTS value at the sensor, to the point at which the equipment reaches the required functional state (e.g., valves in full open or closed position).

In lieu of measurement, the response time for the protection and safety monitoring system equipment is based on allocated values (Ref. 4). The overall response time may be determined by a series of overlapping tests and allocated values such that the entire response time is verified, with the resulting verified response time compared to the appropriate FSAR Chapter 7 (Ref. 1) response time.

Response time may be verified by actual response time tests in any series of sequential, overlapping or total channel measurements, or by the allocated sensor, signal processing and actuation logic response times, where approved, with actual response time tests on the remainder of the channel.

The Surveillance Frequency is controlled under the SFCP. ~~ESF RESPONSE TIME tests are conducted on a 24 month STAGGERED TEST BASIS. Testing of the devices is included in the testing of each channel. The final actuation device in one train is tested with each channel. Therefore, staggered testing results in response time verification of these devices every 24 months. The 24 month Frequency is consistent with the typical refueling cycle and is based on unit operating experience, which shows that random failures of instrumentation components causing serious response time degradation, but not channel failure, are infrequent occurrences.~~

BASES

ACTIONS (continued)

Refueling Water Storage Tank (IRWST) - Shutdown, MODE 6") to dictate the required measures. The IRWST LCO(s) provide appropriate Required Actions for the inoperability of the IRWST and Spent Fuel Pool Level Instrumentation. This action is in accordance with LCO 3.0.6, which requires that the applicable Conditions and Required Actions for the IRWST declared inoperable shall be entered in accordance with LCO 3.0.2.

SURVEILLANCE REQUIREMENTS

SR 3.3.14.1

This SR is the performance of a CHANNEL CALIBRATION ~~every 24 months or approximately at every refueling~~. CHANNEL CALIBRATION is a complete check of the instrument loop, including the sensor and the integrated protection cabinets (IPC). The test is performed in accordance with the SP. If the actual setting of the channel is found to be outside the as-found tolerance, the channel is considered inoperable. This condition of the channel will be further evaluated during performance of the SR. This evaluation will consist of resetting the channel setpoint to the NTS (within the allowed tolerance), and evaluating the channel's response. If the channel is functioning as required and is expected to pass the next surveillance, then the channel is OPERABLE and can be restored to service at the completion of the surveillance. After the surveillance is completed, the channel as-found condition will be entered into the Corrective Action Program for further evaluation. Transmitter calibration must be performed consistent with the assumptions of the setpoint methodology. The difference between the current as-found values and the previous as-left values must be consistent with the transmitter drift allowance used in the setpoint methodology.

~~The setpoint methodology requires that 30 months drift be used (1.25 times the surveillance calibration interval, 24 months).~~

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is based on operating experience and consistency with the refueling cycle.~~

This Surveillance Requirement is modified by a Note. The Note states that this test shall include verification that the time constants are adjusted to within limits where applicable.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.3.14.2

This SR ensures the individual channel ESF RESPONSE TIME is less than or equal to the maximum value assumed in the accident analysis. Individual component response times are not modeled in the analyses. The analyses model the overall or total elapsed time, from the point at which the parameter exceeds the NTS value at the sensor, to the point at which the equipment reaches the required functional state (e.g., valves in full open or closed position).

In lieu of measurement, the response time for the protection and safety monitoring system equipment is based on allocated values (Ref. 4). The overall response time may be determined by a series of overlapping tests and allocated values such that the entire response time is verified, with the resulting verified response time compared to the appropriate FSAR Chapter 7 (Ref. 2) response time.

Response time may be verified by actual response time tests in any series of sequential, overlapping or total channel measurements, or by the allocated sensor, signal processing and actuation logic response times, where approved, with actual response time tests on the remainder of the channel.

The Surveillance Frequency is controlled under the SFCP. ~~ESF RESPONSE TIME tests are conducted on a 24 month STAGGERED TEST BASIS. Testing of the devices is included in the testing of each channel. The final actuation device in one train is tested with each channel. Therefore, staggered testing results in response time verification of these devices every 24 months. The 24 month Frequency is consistent with the typical refueling cycle and is based on unit operating experience, which shows that random failures of instrumentation components causing serious response time degradation, but not channel failure, are infrequent occurrences.~~

BASES

ACTIONS (continued)

D.1

This Required Action directs entry into the appropriate Condition referenced in Table 3.3.17-1. The applicable Condition referenced in the Table is Function dependent.

Each time an inoperable channel has not met any Required Action of Condition C, and the associated Completion Time has expired, Condition D is entered for that channel and provides for transfer to the appropriate subsequent Condition.

E.1 and E.2

If the Required Action and associated Completion Time of Condition C are not met for the Functions in Table 3.3.17-1, the plant must be placed in a MODE in which the LCO does not apply. This is done by placing the plant in at least MODE 3 within 6 hours and MODE 4 within 12 hours.

The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

The SRs for each PAM Function are identified in the SRs column of Table 3.3.17-1 for that Function. A Note has been added to the SR table stating that Table 3.3.17-1 determines which SRs apply to each PAM Function.

SR 3.3.17.1

Performance of the CHANNEL CHECK ~~once every 31 days~~ verifies that a gross instrumentation failure has not occurred. A CHANNEL CHECK is a comparison of the parameter indicated on one channel to a similar parameter on other channels. It is based on the assumption that instrument channels monitoring the same parameter should read approximately the same value. Significant deviations between the two instrument channels could be an indication of excessive instrument drift in one of the channels or of something even more serious. CHANNEL CHECK will detect gross channel failure; thus, it is key to verifying that the instrumentation continues to operate properly between each CHANNEL CALIBRATION. The high radiation instrumentation should be compared to similar plant instruments located throughout the plant.

BASES

SURVEILLANCE REQUIREMENTS (continued)

Agreement criteria are determined by the unit staff, based on a combination of the channel instrument uncertainties, including indication and readability. If a channel is outside the match criteria, it may be an indication that the sensor or the signal-processing equipment has drifted outside its limit. If the channels are within the match criteria, it is an indication that the channels are OPERABLE.

As specified in the SR, a CHANNEL CHECK is only required for those channels that are normally energized.

The Surveillance Requirement is modified by a Note. The Note states that this SR is not required for Neutron Flux (Intermediate Range) in MODE 1. In MODE 1, the OPERABILITY of the Intermediate Range Neutron Instrumentation is verified by self-checking features in lieu of performing a CHANNEL CHECK.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 31 days is based on operating experience with regard to channel OPERABILITY and drift, which demonstrates that failure of more than one channel of a given function in any 31 day interval is rare. The CHANNEL CHECK supplements less formal, but more frequent, checks of channels during normal operational use of those displays associated with the required channels of this LCO.~~

SR 3.3.17.2

~~A CHANNEL CALIBRATION is performed every 24 months, or approximately at every refueling.~~ CHANNEL CALIBRATION is a complete check of the instrument loop including the sensor. The test verifies that the channel responds to the measured parameter with the necessary range and accuracy. This SR is modified by a Note that excludes neutron detectors. The calibration method for neutron detectors is specified in the Bases of LCO 3.3.3, "Reactor Trip System (RTS) Intermediate Range Instrumentation." RTD and Thermocouple channels are to be calibrated in place using cross-calibration techniques. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is based on operating experience and consistency with the typical industry refueling cycle.~~

REFERENCES

1. Regulatory Guide 1.97, Rev. 3, “Instrumentation for Light-Water Cooled Nuclear Power Plants to Assess Plant and Environs Conditions During and Following an Accident,” U.S. Nuclear Regulatory Commission.

BASES

SURVEILLANCE REQUIREMENTS

SR 3.3.18.1

SR 3.3.18.1 verifies that each required RSW transfer switch performs the required functions. This ensures that if the control room becomes inaccessible, the unit can be placed and maintained in MODE 4 with T_{avg} < 350°F from the RSW. This Surveillance also demonstrates that the ADS and IRWST injection block is unblocked by the RSW transfer switch.

The Surveillance Frequency is controlled under the SFCP. ~~The Surveillance Frequency is controlled under the SFCP. The 24 month Frequency was developed considering it is prudent that these types of surveillances be performed during a unit outage. However, this surveillance is not required to be performed only during a unit outage. This is due to the unit conditions needed to perform the surveillance and the potential for unplanned transients if the surveillance is performed with the reactor at power. Operating experience demonstrates that RSW transfer switches usually pass the surveillance test when performed on the 24 month Frequency.~~

SR 3.3.18.2

This Surveillance verifies that the RSW communicates controls and indications with Divisions A, B, C, and D of the PMS. The operator can select the controls and indications available through each PMS division.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is based on the known reliability of the Functions and the redundancy available, and has been shown to be acceptable through operating experience.~~

SR 3.3.18.3

SR 3.3.18.3 verifies the OPERABILITY of the RSW hardware and software by performing diagnostics to show that operator displays are capable of being called up and displayed to an operator at the RSW. The RSW has several video display units which can be used by the operator. The video display units are identical to that provided in the control room and the operator can display information on the video display units in a manner which is identical to the way the information is displayed in the control room. The operator normally selects an appropriate set of displays based on the particular operational goals being controlled by the operator at the time. Each display consists of static graphical and legend information which is contained within the display processor associated with each video display unit and dynamic data which is updated by the data display system.

BASES

SURVEILLANCE REQUIREMENTS (continued)

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months is based on the use of the data display capability in the control room as part of the normal unit operation and the availability of multiple video display units at the RSW. The Frequency of 24 months is based upon operating experience and consistency with control room hardware and software.~~

SR 3.3.18.4

SR 3.3.18.4 is the performance of a TRIP ACTUATING DEVICE OPERATIONAL TEST (TADOT) ~~every 24 months~~. This test should verify the OPERABILITY of the reactor trip breakers (RTBs) open and closed indication on the RSW by actuating the RTBs. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months was chosen because the RTBs may not be exercised while the facility is at power and is based on operating experience and consistency with the refueling outage.~~

REFERENCES

1. 10 CFR 50, Appendix A, GDC 19.

2. FSAR Section 7.4.1, “Safe Shutdown.”

BASES

ACTIONS (continued)

C.1

Condition C applies when Required Action A cannot be completed for any DAS manual actuation control (other than reactor trip) within the required completion time of 30 days.

Action C.1 requires that the inoperable DAS manual actuation control(s) be restored to OPERABLE status prior to entering MODE 2 following any plant shutdown to MODE 5 while the control is inoperable. This ACTION is provided to ensure that all DAS manual controls are restored to OPERABLE status following the next plant shutdown.

D.1 and D.2

Condition D is entered if the Required Action associated with Condition B or C is not met within the required Completion Time.

Required Actions D.1 and D.2 ensure that the plant is placed in a condition where the probability and consequences of an event are minimized. The allowed Completion Times are reasonable based on plant operating experience, for reaching the required plant conditions from full power conditions in an orderly manner, without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.3.19.1

SR 3.3.19.1 is the performance of a TADOT of the DAS manual trip and actuation controls for the specified safety-related equipment. ~~This TADOT is performed every 24 months.~~

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is based on the known reliability of the DAS functions and has been shown to be acceptable through operating experience.~~

The SR is modified by a Note that excludes verification of the setpoints from the TADOT. The functions have no setpoints associated with them.

REFERENCES

1. WCAP-15985, “AP1000 Implementation of the Regulatory Treatment of Nonsafety-Related Systems Process,” Revision 2, dated August 2003.

2. FSAR Section 7.7.1.11.

BASES

ACTIONS (continued)

A.1

Condition A addresses the situation where one or more divisions of ADS and IRWST Injection Blocking Device(s) is inoperable (e.g., one or both CMT level channels in one or more divisions inoperable when required, or ADS and IRWST Injection Block in one or more divisions not unblocked when required). In this condition, the component interface module (CIM) in the affected division is required to be unblocked in the affected division within 8 hours. The ADS and IRWST Injection Block manual switches may be utilized to implement the unblock. The 8 hours is reasonable based on the low probability of an event occurring during this interval.

B.1

If the Required Action and associated Completion Time of Condition A is not met the affected ADS and IRWST injection valves must be declared inoperable immediately. Declaring the affected valves inoperable allows the supported system Actions (i.e., for ADS and IRWST inoperable valves) to dictate the required measures. The ADS and/or IRWST LCO(s) provide appropriate actions for the inoperable components. This action is in accordance with LCO 3.0.6, which requires that the applicable Conditions and Required Actions for valves declared inoperable shall be entered in accordance with LCO 3.0.2.

SURVEILLANCE REQUIREMENTS

The SRs for each ADS and IRWST Injection Blocking Device Function are identified in the SRs column of Table 3.3.20-1 for that Function.

A Note has been added to the SR table stating that Table 3.3.20-1 determines which SRs apply to which ADS and IRWST Injection Blocking Device Function.

SR 3.3.20.1

Verification that the position of each ADS and IRWST Injection Block switch is in the “unblock” position is required when less than two CMTs are required to be OPERABLE. This assures the actuation of ADS and IRWST injection is not blocked when there may be reduced or no capability for automatic unblocking from CMT level. The Surveillance Frequency is controlled under the SFCP. ~~The 7 day Frequency is adequate considering the availability of main control room status monitoring of the block signal.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.3.20.2

This SR is the performance of a CHANNEL CALIBRATION ~~every 24 months or approximately at every refueling~~. CHANNEL CALIBRATION is a complete check of the instrument loop. The test is performed in accordance with the SP. If the actual setting of the channel is found to be outside the as-found tolerance, the channel is considered inoperable. This condition of the channel will be further evaluated during performance of the SR. This evaluation will consist of resetting the channel setpoint to the NTS (within the allowed tolerance), and evaluating the channel's response. If the channel is functioning as required and is expected to pass the next surveillance, then the channel is OPERABLE and can be restored to service at the completion of the surveillance. After the surveillance is completed, the channel as-found condition will be entered into the Corrective Action Program for further evaluation. Transmitter calibration must be performed consistent with the assumptions of the setpoint methodology. The difference between the current as-found values and the previous as-left values must be consistent with the transmitter drift allowance used in the setpoint methodology.

The setpoint methodology requires that 30 months drift be used ~~(1.25 times the surveillance calibration interval, 24 months)~~.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is based on operating experience and consistency with the refueling cycle~~.

SR 3.3.20.3

This SR is the performance of an ACTUATION LOGIC TEST for unblocking. This test overlaps the ADS and IRWST injection functional tests (i.e., SR 3.4.11.4, SR 3.4.11.5, and SR 3.5.6.9) that verify actuation on an actual or simulated actuation signal.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months is based on the need to perform this SR during periods in which the plant is shut down for refueling to prevent any additional risks associated with inadvertent operation of the ADS and IRWST injection valves.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.3.20.4

This SR is the performance of a TADOT of the of required ADS and IRWST Injection Block manual switch. ~~This TADOT is performed every 24 months.~~

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is based on the known reliability of manual switch Functions and has been shown to be acceptable through operating experience.~~

The SR is modified by a Note that states verification of setpoint is not required, since these functions have no setpoint associated with them.

SR 3.3.20.5

This SR requires performance of LCO 3.5.2 Surveillances associated with ensuring CMTs are capable of injecting to the RCS. CMT injection supports OPERABILITY of the ADS and IRWST Injection Blocking Devices for automatic unblocking. If one or both CMTs are inoperable for injection, all four divisions of ADS and IRWST Injection Blocking Devices are inoperable. Therefore, SRs 3.5.2.2, 3.5.2.5, and 3.5.2.6 are required to be met. See the corresponding Bases for LCO 3.5.2 for a discussion of each Surveillance and its Frequency.

REFERENCES

1. FSAR Chapter 15, “Accident Analysis.”
2. FSAR Chapter 7, “Instrumentation and Controls.”
3. WCAP-10271, “Evaluation of Surveillance Frequencies and Out of Service Times for the Reactor Protection Instrumentation System, June 1996” Supplement 2.

BASES

ACTIONS

A.1

RCS pressure and RCS average temperature are controllable and measurable parameters. With one or both of these parameters not within LCO limits, action must be taken to restore parameter(s).

RCS total flow rate is not a controllable parameter and is not expected to vary during steady state operation. If the indicated RCS total flow rate is below the LCO limit, power must be reduced, as required by Required Action B.1, to restore DNB margin and eliminate the potential for violation of the accident analysis bounds.

The 2 hour Completion Time for restoration of the parameters provides sufficient time to adjust plant parameters, to determine the cause for the off normal condition, and to restore the readings within limits, and is based on plant operating experience.

B.1

If Required Action A.1 is not met within the associated Completion Time, the plant must be brought to a MODE in which the LCO does not apply. To achieve this status, the plant must be brought to at least MODE 2 within 6 hours. In MODE 2, the reduced power condition eliminates the potential for violation of the accident analysis bounds. The Completion Time of 6 hours is reasonable to reach the required plant conditions in an orderly manner.

SURVEILLANCE REQUIREMENTS

SR 3.4.1.1

This surveillance demonstrates that the pressurizer pressure remains greater than or equal to the limit specified in the COLR. Pressurizer pressure indications are averaged to come up with a value for comparison to the limit. The Surveillance Frequency is controlled under the SFCP. ~~Since Required Action A.1 allows a Completion Time of 2 hours to restore parameters that are not within limits, the 12 hour Surveillance Frequency of pressurizer pressure is sufficient to ensure the pressure can be restored to a normal operation, steady state condition following load changes and other expected transient operations. The 12 hour interval has been shown by operating practice to be sufficient to regularly assess for potential degradation and to verify operation is within safety analysis assumptions.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.4.1.2

This surveillance demonstrates that the average RCS temperature remains less than or equal to the limit specified in the COLR. Indications of temperature are averaged to determine a value for comparison to the limit. The Surveillance Frequency is controlled under the SFCP. ~~Since Required Action A.1 allows a Completion Time of 2 hours to restore parameters that are not within limits, the 12 hour Surveillance Frequency for RCS average temperature is sufficient to ensure the temperature can be restored to a normal operation, steady state condition following load changes and other expected transient operations. The 12 hour Frequency has been shown by operating practice to be sufficient to regularly assess for potential degradation and to verify operation is within safety analysis assumptions.~~

SR 3.4.1.3

This surveillance demonstrates that the RCS total flow rate remains ≥ 301,670 gpm and greater than or equal to the limit specified in the COLR. The surveillance ~~12 hour Surveillance Frequency~~ for RCS total flow rate is performed using the installed differential pressure flow instrumentation. The Surveillance Frequency is controlled under the SFCP. ~~The 12 hour interval has been shown by operating practice to be sufficient to regularly assess potential degradation and to verify operation within safety analysis assumptions.~~

SR 3.4.1.4

A CHANNEL CALIBRATION of the differential pressure RCS total flow rate indication channels is performed ~~every 24 months,~~ at the beginning of each fuel cycle.

CHANNEL CALIBRATION is a complete check of the instrument loop, including the sensor. The test verifies that the channel responds to a measured parameter with the necessary range and accuracy.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is based on consistency with the refueling cycle.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.4.1.5

Measurement of RCS total flow rate by performance of precision test measurements ~~once every 24 months,~~ at the beginning of each fuel cycle, allows the installed RCS flow instrumentation to be normalized and verifies the actual RCS flow is greater than or equal to the minimum required RCS flow rate. These test measurements may be based on a precision heat balance, or by differential pressure measurements of static elements in the RCS piping (such as elbows) that have been calibrated by previous precision tests, or by a combination of those two methods. In all cases, the measured flow, less allowance for error, must exceed the corresponding value used in the safety analysis and specified in the COLR.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months reflects the importance of verifying flow after a refueling outage when the core has been altered, which may have caused an alteration of flow resistance.~~

This SR is modified by a Note that allows entry into MODE 1, without having performed the SR, and placement of the unit in the best condition for performing the SR. The Note states that the SR is not required to be performed until after 24 hours after ≥ 90% RTP. This exception is appropriate since the heat balance requires the plant to be at a minimum of 90% RTP to obtain the stated RCS flow accuracies. The Surveillance shall be performed within 24 hours after reaching 90% RTP.

REFERENCES

1. FSAR Chapter 15, “Accident Analyses.”

BASES

ACTIONS

A.1

If the parameters that are outside the limit cannot be restored, the plant must be brought to a MODE in which the LCO does not apply. To achieve this status, the plant must be brought to MODE 2 with $k_{eff} < 1.0$ within 30 minutes. Rapid reactor shutdown can be readily and practically achieved within a 30 minute period. The allowed time is reasonable, based on operating experience, to reach MODE 2 with $k_{eff} < 1.0$ in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.4.2.1

RCS loop average temperature is required to be verified at or above 551°F ~~every 12 hours~~. The SR to verify RCS loop average temperatures ~~every 12 hours~~ takes into account indications and alarms that are continuously available to the operator in the control room and is consistent with other routine Surveillances which are typically performed once per shift. In addition, operators are trained to be sensitive to RCS temperature during approach to criticality and will ensure that the minimum temperature for criticality is met as criticality is approached.

The Surveillance Frequency is controlled under the SFCP.

REFERENCES

1. FSAR, Chapter 15, “Accident Analyses.”

BASES

ACTIONS (continued)

Besides restoring operation within limits, an evaluation is required to determine if RCS operation can continue. The evaluation must verify that the RCPB integrity remains acceptable and must be completed prior to entry into MODE 4. Several methods may be used, including comparison with pre-analyzed transients in the stress analyses, or inspection of the components.

ASME Code, Section XI, Appendix E (Ref. 6), may be used to support the evaluation. However, its use is restricted to evaluation of the vessel beltline.

Condition C is modified by a Note requiring Required Action C.2 to be completed whenever the Condition is entered. The Note emphasizes the need to perform the evaluation of the effects of the excursion outside the allowable limits. Restoration alone per Required Action C.1 is insufficient because higher than analyzed stresses may have occurred and may have affected the RCPB integrity.

SURVEILLANCE REQUIREMENTS

SR 3.4.3.1

Verification that operation is within PTLR limits is required ~~every 30 minutes~~ when RCS P/T conditions are undergoing planned changes. The Surveillance Frequency is controlled under the SFCP. ~~This Frequency is considered reasonable in view of the control room indication available to monitor RCS status. Also, since temperature rate of change limits are specified in hourly increments, 30 minutes permits assessment and correction for minor deviations within a reasonable time.~~

Surveillance for heatup, cooldown, or RCS ISLH testing may be discontinued when the definition given in the relevant plant procedure for ending the activity is satisfied.

This SR is modified by a Note that only requires this surveillance to be performed during system heatup, cooldown, and RCS ISLH testing. No SR is given for criticality operations because LCO 3.4.2, “RCS Minimum Temperature for Criticality,” contains a more restrictive requirement.

BASES

ACTIONS (continued)

requires these Required Actions to be completed (i.e., to exit the Applicability prior to attempting to restart a RCP) when Condition A is entered. This prevents startup of a RCP and the resulting circulation of cold and/or unborated water from an inactive loop into the core, precluding reactivity excursion events which are unanalyzed and lowers power level; thus reducing the core heat removal needs and minimizing the possibility of violating DNB limits.

When all four reactor coolant pumps are operating, a loss of a single reactor coolant pump above power level P-10 will result in an automatic reactor trip.

The Completion Time of 6 hours is reasonable to allow for an orderly transition to MODE 3. The applicable safety analyses described above bound Design Basis Accidents (DBA) initiated with three reactor coolant pumps operating at power levels below P-10.

B.1, B.2, and B.3

If the requirements of the LCO are not met while in MODE 3, 4 or 5, the Required Action is to suspend the start of any RCP, initiate action to fully insert all rods, and place the PLS in a condition incapable of rod withdrawal. The Condition B Note requires these Required Actions to be completed (i.e., to exit the Applicability prior to attempting to restart a RCP) when Condition B is entered. The actions prevent startup of a RCP and the resulting circulation of cold and/or unborated water from an inactive loop into the core, precluding reactivity excursion events which are unanalyzed and eliminate the possibility of a rod withdrawal event with one or more pumps not operating and thus minimizing the possibility of violating DNB limits.

The Completion Time of 1 hour is reasonable to allow for making PLS incapable of rod withdrawal and fully inserting all control rods, since plant cool-down is not required.

SURVEILLANCE REQUIREMENTS

SR 3.4.4.1

This SR requires verification ~~every 12 hours~~ that each RCS loop is in operation with the pump variable speed control bypassed. Verification helps ensure that forced flow is maintaining the margin to DNB and proper boron mixing. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 12 hours is sufficient considering other indications and alarms available to the operator in the main control room to monitor RCS loop performance.~~

BASES

ACTIONS

A.1, A.2, A.3, and A.4 (continued)

Pressurizer water level control malfunctions or other plant evolutions may result in a pressurizer water level above the nominal upper limit, even with the plant at steady state conditions.

The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.4.5.1

This SR requires that during steady state operation, pressurizer level is maintained below the nominal upper limit to provide a minimum space for a steam bubble. The Surveillance is performed by observing the indicated level. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 12 hours corresponds to verifying the parameter each shift. The 12 hour interval has been shown by operating practice to be sufficient to regularly assess the level for any deviation and verify that operation is within safety analyses assumptions. Alarms are also available for early detection of abnormal level indications.~~

REFERENCES

1. FSAR Chapter 15, “Accident Analysis.”

BASES

SURVEILLANCE REQUIREMENTS

SR 3.4.6.1

~~SRs are specified in the Inservice Testing Program.~~ Pressurizer safety valves are to be tested one at a time and in accordance with the requirements of ~~ASME OM Code~~10 CFR 50.55a(f) (Ref. 4), which provides the activities ~~and Frequency~~ necessary to satisfy the SRs. No additional requirements are specified. The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 4).

The pressurizer safety valve setpoint is ± 1% for OPERABILITY, and the values are reset to remain within ± 1% during the Surveillance to allow for drift.

REFERENCES

1. ASME Boiler and Pressure Vessel Code, Section III, NB 7500.
2. WCAP-16779, “AP1000 Overpressure Protection Report, April 2007.”
3. FSAR Chapter 15, “Accident Analyses.”
4. ~~ASME OM Code, “Code for Operation and Maintenance of Nuclear Power Plants.”~~10 CFR 50.55a(f).

BASES

SURVEILLANCE REQUIREMENTS (continued)

An early warning of pressure boundary LEAKAGE or unidentified LEAKAGE is provided by the automatic systems that monitor the containment atmosphere F18 particulate radioactivity and the containment sump level. These LEAKAGE detection systems are specified in LCO 3.4.9, “RCS LEAKAGE Detection Instrumentation.”

Note 2 states that this SR is not applicable to primary to secondary LEAKAGE because LEAKAGE of 150 gallons per day cannot be measured accurately by an RCS water inventory balance.

The Surveillance Frequency is controlled under the SFCP. ~~The 72-hour Frequency is a reasonable interval to trend LEAKAGE and recognizes the importance of early leakage detection in the prevention of accidents.~~

SR 3.4.7.2

This SR verifies that primary to secondary LEAKAGE is less or equal to 150 gallons per day through any one SG. Satisfying the primary to secondary LEAKAGE limit ensures that the operational LEAKAGE performance criterion in the Steam Generator Program is met. If this SR is not met, compliance with LCO 3.4.17, "Steam Generator Tube Integrity," should be evaluated. The 150 gallons per day limit is measured at room temperature as described in Reference 5. The operational LEAKAGE rate limit applies to LEAKAGE through any one SG. If it is not practical to assign the LEAKAGE to an individual SG, all the primary to secondary LEAKAGE should be conservatively assumed to be from one SG.

The Surveillance is modified by a Note which states that the Surveillance is not required to be performed until 12 hours after establishment of steady state operation. For RCS primary to secondary LEAKAGE determination, steady state is defined as stable RCS pressure, temperature, power level, pressurizer and reactor coolant drain tank and in-containment refueling water storage tank levels, and makeup and letdown flows.

The Surveillance Frequency is controlled under the SFCP. ~~The Surveillance Frequency of 72 hours is a reasonable interval to trend primary to secondary LEAKAGE and recognizes the importance of early leakage detection in the prevention of accidents.~~ The primary to secondary LEAKAGE is determined using continuous process radiation monitors or radiochemical grab sampling in accordance with the EPRI guidelines (Ref. 5).

BASES

ACTIONS (continued)

Condition A is modified by a Note that requires Required Action A.2 to be performed prior to starting any RCP whenever the Condition is entered. This ensures that SR 3.1.1.1 will be performed prior to starting an RCP, even when Condition A is exited prior to performing Required Action A.2. Performance of SR 3.1.1.1 is necessary to assure SDM is properly evaluated prior to starting an RCP.

SURVEILLANCE REQUIREMENTS

SR 3.4.8.1

This Surveillance requires verification ~~every 12 hours~~ that a minimum mixing flow is present in the RCS. The Surveillance Frequency is controlled under the SFCP. ~~A Frequency of 12 hours is adequate considering the low probability of an inadvertent BDE during this time, and the ease of verifying the required RCS flow.~~

REFERENCES None.

BASES

ACTIONS (continued)

D.1 and D.2

If a Required Action of Condition A, B or C cannot be met within the required Completion Time, the reactor must be brought to a MODE in which the LCO does not apply. To achieve this status, the plant must be brought to at least MODE 3 within 6 hours and to MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner without challenging plant systems.

E.1

With all required monitors inoperable, no required means of monitoring leakage is available and plant shutdown in accordance with LCO 3.0.3 is required.

SURVEILLANCE REQUIREMENTS

SR 3.4.9.1

SR 3.4.9.1 requires the performance of a CHANNEL CHECK of the containment atmosphere F18 particulate monitor and the required containment sump level channels. The check gives reasonable confidence that the required instrumentation and associated channels are operating properly. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 12 hours is based on instrument reliability and risk and is reasonable for detecting off normal conditions.~~

SR 3.4.9.2

SR 3.4.9.2 requires the performance of a CHANNEL OPERATIONAL TEST (COT) on the containment atmosphere F18 particulate monitor. The test ensures that the monitor can perform its function in the desired manner. The test verifies the alarm setpoint and relative accuracy of the instrument string. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 92 days considers risks and instrument reliability, and operating experience has shown that it is proper for detecting degradation.~~

SR 3.4.9.3

This SR requires the performance of a CHANNEL CALIBRATION for each of the required RCS Leakage detection instrumentation channels. The calibration verifies the accuracy of the instrument strings, including the instruments located inside containment. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months is a typical~~

~~refueling cycle and considers channel reliability. Operating experience has proven that this Frequency is acceptable.~~

BASES

ACTIONS (continued)

The DOSE EQUIVALENT I-131 must be restored to normal within 48 hours. If the concentration cannot be restored to within the LCO limit in 48 hours, it is assumed that the LCO violation is not the result of normal iodine spiking.

A Note permits the use of the provisions of LCO 3.0.4.c. This allowance permits entry into the applicable MODE(S), relying on Required Actions A.1 and A.2 while the DOSE EQUIVALENT I-131 LCO limit is not met. This allowance is acceptable due to the significant conservatism incorporated into the specific activity limit, the low probability of an event which is limiting due to exceeding this limit, and the ability to restore transient-specific activity excursions while the plant remains at, or proceeds to, power operation.

B.1

With DOSE EQUIVALENT XE-133 in excess of the allowed limit, the plant must be placed in a MODE or condition in which the LCO requirements are not applicable. This is done by placing the plant in at least MODE 3 with RCS average temperature < 500˚F within 6 hours.

The change to MODE 3 and RCS average temperature < 500°F lowers the saturation pressure of the reactor coolant below the set points of the main steam safety valves, and prevents venting the SG to the environment in a SGTR event. The allowed Completion Time of 6 hours is reasonable, based on operating experience to reach MODE 3 from full power conditions in an orderly manner, without challenging plant systems.

C.1

If a Required Action and the associated Completion Time of Condition A is not met or if the DOSE EQUIVALENT I-131 is > 60 μCi/gm, the reactor must be brought to MODE 3 with RCS average temperature < 500°F within 6 hours. The Completion Time of 6 hours is reasonable, based on operation experience, to reach MODE 3 below 500°F from full power conditions in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.4.10.1

SR 3.4.10.1 requires performing a measure of the noble gas specific activity of the reactor coolant ~~at least once every 7 days~~. This is a quantitative measure of radionuclides with half lives longer than

BASES

SURVEILLANCE REQUIREMENTS (continued)

15 minutes. This Surveillance provides an indication of any increase in the release of noble gas activity from fuel rods containing cladding defects.

Trending the results of this Surveillance allows proper remedial action to be taken before reaching the LCO limit under normal operating conditions. The Surveillance Frequency is controlled under the SFCP. ~~The 7 day Frequency considers the unlikelihood of a significant increase in fuel defect level during the time.~~

SR 3.4.10.2

This Surveillance is performed in MODE 1 only to ensure iodine remains within limit during normal operation and following fast power changes when increased releases of iodine from the fuel (iodine spiking) is apt to occur. The Surveillance Frequency is controlled under the SFCP. ~~The 14 day Frequency is adequate to trend changes in the iodine activity level.~~ The Frequency, between 2 and 6 hours after a power change of ≥ 15% RTP within a 1 hour period, is established because the iodine levels peak during this time following fuel failures; samples at other times would provide inaccurate results.

REFERENCES

1. FSAR Section 15.1.5, “Steam System Piping Failure.”
2. FSAR Section 15.6.3, “Steam Generator Tube Rupture.”

BASES

SURVEILLANCE REQUIREMENTS

SR 3.4.11.1

This Surveillance verifies that each ~~Each~~ ADS stage 4 isolation motor operated valve is open~~must be verified to be open every 12 hours~~. Note that these valves receive confirmatory open signals. The Surveillance Frequency is controlled under the SFCP. ~~The Surveillance Frequency is acceptable considering valve position is manually monitored in the control room.~~

SR 3.4.11.2

This Surveillance ~~requires verification~~verifies that each ADS stage 1, 2, 3 valve strokes to its fully open position. Note that this surveillance is performed during shutdown conditions.

The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 5).~~for demonstrating valve OPERABILITY references the Inservice Testing Program~~.

SR 3.4.11.3

This Surveillance requires verification that each ADS stage 4 squib valve is OPERABLE in accordance with 10 CFR 50.55a(f) ~~the Inservice Testing Program~~. The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 5).~~for verifying valve OPERABILITY references the Inservice Testing Program~~.

The squib valves will be tested in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 5).~~ASME OM Code (Ref. 5). The applicable ASME OM Code squib valve requirements are specified in paragraph ISTC-5260.~~

SR 3.4.11.4

This ~~SR~~ Surveillance verifies that each Stage 1, 2, and 3 ADS valve actuate to the correct position on an actual or simulated actuation signal. The ADS and IRWST injection blocking device ACTUATION LOGIC TEST provides overlap with this Surveillance.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months is based on the need to perform this surveillance during periods in which the plant is shutdown for refueling to prevent any upsets of plant operation.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.4.11.5

This ~~SR~~ Surveillance verifies that each Stage 4 ADS valve can actuate to the correct position on an actual or simulated actuation signal. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valves. The ADS and IRWST injection blocking device ACTUATION LOGIC TEST provides overlap with this Surveillance. The OPERABILITY of the squib valves is checked by performing a continuity check of the circuit from the Protection Logic Cabinets to the squib valve.

This Surveillance is modified by a Note that excludes squib valve actuation as a requirement for this Surveillance to be met. This is acceptable because the design of the squib actuated valve was selected for this application because of its very high reliability. The OPERABILITY of squib actuated valves is verified by ~~the Inservice Test Program~~10 CFR 50.55a(f) (Ref. 5) for squib actuated valves.

The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 5). ~~The Frequency of 24 months is based on the need to perform this surveillance during periods in which the plant is shutdown for refueling to prevent any upsets of plant operation.~~

REFERENCES

1. FSAR Section 6.3, “Passive Core Cooling System.”
2. FSAR Section 15.6, “Decrease in Reactor Coolant Inventory.”
3. AP1000 Probabilistic Risk Assessment, Appendix A.
4. FSAR Section 3.9.6, “Inservice Testing of Pumps and Valves.”
5. 10 CFR 50.55a(f). ~~ASME OM Code, “Code for Operation and Maintenance of Nuclear Power Plants.”~~

BASES

ACTIONS (continued)

immediately suspending positive reactivity additions to ensure that the SDM is maintained. Sources of positive reactivity addition include boron dilution, withdrawal of control rods, and excessive cooling of the RCS.

SURVEILLANCE REQUIREMENTS

SR 3.4.13.1

Each flow path in ADS stage 1, 2, and 3 required to be open is verified to be open by verifying that each required ADS stage 1, 2, and 3 valve is in its open position~~ every 12 hours~~, as indicated in the control room. The Surveillance Frequency is controlled under the SFCP. ~~This Surveillance Frequency is acceptable based on administrative controls which preclude repositioning the valves.~~

SR 3.4.13.2

SR 3.4.11.1, SR 3.4.11.3, and SR 3.4.11.5 are applicable to the ADS stage 4 valves required to be OPERABLE. The Frequencies associated with each specified SR are applicable. Refer to the corresponding Bases for LCO 3.4.11 for a discussion of each SR.

REFERENCES

1. FSAR Section 19E.4.8.3, " Loss of Normal Residual Heat Removal System Cooling in Mode 5 with Reactor Coolant System Open."

BASES

ACTIONS (continued)

D.1 and D.2

If the RNS suction relief valves are inoperable and the RCS is not depressurized, there is a potential to overpressurize the RCS and exceed the limits allowed in LCO 3.4.3. The suction relief valves are considered inoperable if the RNS isolation valves have isolated the RNS from the RCS in such a way that the suction relief valves cannot perform their intended safety function, or if the valves themselves will not operate to perform their intended safety function. If the RCS is depressurized but the RCS vent path does not provide a flow area sufficient to mitigate any of the design low temperature overpressure events and the RNS suction relief valves are inoperable, there is a potential to overpressurize the RCS and exceed the limits allowed in LCO 3.4.3. The RCS vent path is considered inoperable if the area of the vent is not equivalent to the area of the inlet pipe to the larger RNS suction relief valve.

Under these conditions, Required Action D.1 or D.2 provide two options, either of which must be accomplished in 12 hours. If the RNS suction relief valves cannot be restored to OPERABLE status, the RCS must be depressurized and vented with a RCS vent which provides a flow area sufficient to mitigate any of the design low temperature overpressure events.

The 12 hour Completion Time represents a reasonable time to repair the relief valves, open the RNS isolation valves or otherwise restore the LTOP to OPERABLE status, or depressurize and vent the RCS, without imposing a lengthy period when the LTOP is not able to mitigate a low temperature overpressure event.

SURVEILLANCE REQUIREMENTS

SR 3.4.14.1

To minimize the potential for a low temperature overpressure event by limiting the mass input capability, the accumulator discharge isolation valves are verified closed and locked out. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 12 hours is sufficient considering other indications and alarms available to the operator in the main control room to verify the required status of the equipment.~~

SR 3.4.14.1 is modified by a Note stating that accumulator isolation is only required when the accumulator pressure is more than or at the maximum RCS pressure for the existing temperature, as allowed by the P/T limit curves. This Note requires the accumulator discharge isolation valve Surveillance to be met only under these pressure and temperature conditions.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.4.14.2

The RNS suction relief valves shall be demonstrated OPERABLE by verifying two RNS suction isolation valves in one flow path are open. This Surveillance is only performed if the RNS suction relief valves are being used to satisfy this LCO.

The Surveillance Frequency is controlled under the SFCP. ~~The RNS suction isolation valves are verified to be opened every 12 hours. The Frequency is considered adequate in view of other administrative controls such as valve status indications available to the operator in the control room that verify the RNS suction isolation valves remain open.~~ This Surveillance is required to be met if the RNS suction relief valves are being used to satisfy the pressure relief requirements of LCO 3.4.14.a.

SR 3.4.14.3

To minimize the potential for a low temperature overpressure event when the RNS suction relief valves are OPERABLE, the flow from the CVS is limited to the capacity of the smaller RNS suction relief valve. The CVS makeup line containment isolation valve, CVS-PL-V091, is closed to limit CVS mass and heat injection flow to the CVS flow-restricting orifice, CVS-PY-R10. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 12 hours is sufficient considering other indications and alarms available to the operator in the main control room to verify the required status of the equipment.~~ This Surveillance is required to be met if the RNS suction relief valves are being used to satisfy the pressure relief requirements of LCO 3.4.14.a.

SR 3.4.14.4

The RCS vent of ≥ 4.15 square inches is verified open ~~either:~~at a Frequency controlled under the SFCP.

~~a. Once every 12 hours for a valve that is not locked (valves that are sealed or secured in the open position are considered "locked" in this context) or~~

~~b. Once every 31 days for other vent path(s) (e.g., a vent valve that is locked, sealed, or secured in position or a removed pressurizer safety valve or open manway also fits this category).~~

This Surveillance is modified by a Note that states it is only required to be met if the vent is being used to satisfy the pressure relief requirements of the LCO 3.4.14.b.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.4.14.5

The RNS suction relief valves shall be demonstrated OPERABLE by verifying that two RNS suction isolation valves in one flow path are open and by testing them in accordance with 10 CFR 50.55a(f) (Ref. 5) ~~the Inservice Testing Program~~. (Refer to SR 3.4.14.2 for the RNS suction isolation valve Surveillance.) The ~~ASME OM Code (Ref. 5)~~surveillance test per ~~Inservice Testing Program~~10 CFR 50.55a(f) verifies OPERABILITY by proving proper relief valve mechanical motion and by measuring and, if required, adjusting the lift setpoint. This Surveillance is modified by a Note specifying that it is only required to be met if the RNS suction relief valves are being used to satisfy the pressure relief requirements of LCO 3.4.14.a.

The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f).

REFERENCES

1. Title 10, Code of Federal Regulations, Part 50, Appendix G, “Fracture Toughness Requirements.”

2. Generic Letter 88-11, “NRC Position on Radiation Embrittlement of Reactor Vessel Materials and Its Impact on Plant Operation.”

3. ASME Boiler and Pressure Vessel Code, Section III.

4. FSAR Section 5.2.2, “Overpressure Protection.”

5. ~~ASME OM Code, “Code for Operation and Maintenance of Nuclear Power Plants.”~~10 CFR 50.55a(f).

BASES

SURVEILLANCE REQUIREMENTS (continued)

valve. The leakage limit of 0.5 gpm per inch nominal valve size up to a maximum of 5 gpm applies to each valve. Leakage testing requires a stable pressure condition.

For the two PIVs in series, the leakage requirement applies to each valve individually and not to the combined leakage across both valves. If the PIVs are not individually leakage tested, one valve may have failed completely and not be detected if the other valve in series meets the leakage requirement. In this situation, the protection provided by redundant valves would be lost.

The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 4). ~~Testing shall be performed every 24 months, a typical refueling cycle. The 24 month Frequency is consistent with 10 CFR 50.55a(g) (Ref. 4) as contained in the Inservice Testing Program and is within frequency allowed by the American Society of Mechanical Engineers (ASME) OM Code (Ref. 5).~~

REFERENCES

1. 10 CFR 50.2.
2. 10 CFR 50.55a(c).
3. 10 CFR 50, Appendix A, Section V, GDC 55.
4. 10 CFR 50.55a(~~g~~f).
5. ~~ASME OM Code, "Code for Operation and Maintenance of Nuclear Power Plants."~~

BASES

SURVEILLANCE REQUIREMENTS

SR 3.4.16.1

The dedicated component level remote manual valve switches in the main control room shall be used to stroke each RVHV valve to demonstrate OPERABILITY of the controls.

This Surveillance requires verification that each RVHV valve strokes to its fully open position. The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 2). ~~The Surveillance Frequency for demonstrating valve OPERABILITY references the Inservice Testing Program.~~

REFERENCES

1. FSAR Section 15.5, “Increase in Reactor Coolant System Inventory.”

~~1.~~2. 10 CFR 50.55a(f).

BASES

ACTIONS (continued)

B.1 and B.2

If the Required Action and associated Completion Time of Condition A is not met or if one or more pressurizer heaters with two circuit breakers are determined to be inoperable, the plant must be brought to a MODE in which the LCO does not apply. To achieve this status, the plant must be brought to at least MODE 3 within 6 hours and to MODE 4 with at least one cold leg temperature ≤ 275°F within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner, without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.4.18.1

This SR demonstrates that the pressurizer heater circuit breakers trip open in response to an actual or simulated actuation signal. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the circuit breakers. The OPERABILITY of the motor control center breakers is checked by opening these breakers using the Plant Control System.

The SR excludes circuit breakers that are maintained in the tripped position since a tripped open circuit breaker is already performing the safety function. Restoration of a circuit breaker to the non-tripped position requires verification that the SR has been met within its required Frequency.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months is based on the need to perform this surveillance during periods in which the plant is shutdown for refueling to prevent any upsets of plant operation. This Frequency is adequate based on the use of multiple circuit breakers to prevent the failure of any single circuit breaker from disabling the function and that all circuit breakers are tested.~~

BASES

SURVEILLANCE REQUIREMENTS

SR 3.4.19.1

This SR demonstrates that the auxiliary spray and purification line isolation valves actuate to the isolation position in response to an actual or simulated actuation signal. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valves.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months is based on the need to perform this surveillance during periods in which the plant is shutdown for refueling to prevent any upsets of plant operation.~~

REFERENCES

1. FSAR Section 7.3.1.2.18, “Auxiliary Spray and Purification Line and Zinc/Hydrogen Addition Isolation.”

BASES

ACTIONS (continued)

A.1

If one CVS letdown isolation vlaave is inoperable, action must be taken to restore the valve within 72 hours. The 72 hours allowed to restore the inoperable valve is reasonable based on the low probability of an event occurring during this interval and OPERABILITY of the redundant letdown isolation valve.

B.1

If the Required Action or associated Completion Time of Condition A is not met or two CVS letdown isolation valves are inoperable, action must be initiated immediately to isolate the CVS letdown flow. The method of isolation must include the use of at least one isolation barrier that cannot be adversely affected by a single active failure. Isolation barriers that meet this criterion are a closed and de-activated automatic valve.

SURVEILLANCE REQUIREMENTS

SR 3.4.20.1

This SR demonstrates that the CVS letdown isolation valves actuate to the isolation position in response to an actual or simulated actuation signal. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valves.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months is based on the need to perform this surveillance during periods in which the plant is shutdown to prevent any upsets of plant operation.~~

REFERENCES

1. FSAR Section 19E.2.1.2.4, “Improved RCS Draindown Method.”

BASES

ACTIONS (continued)

restore either the accumulator or the CMT (per LCO 3.5.2 Condition C or E) to OPERABLE status. This Completion Time is considered reasonable because of the low probability of simultaneously entering these multiple PXS Conditions and the very small likelihood of a LOCA occurring at the same time.

C.1 and C.2

If the Required Action and associated Completion Time of Conditions A or B are not met, the plant must be placed in a MODE or condition in which the LCO does not apply. This is done by placing the plant in MODE 3 within 6 hours and with pressurizer pressure to ≤ 1000 psig within 12 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner and without challenging plant systems.

D.1

If more than one accumulator is inoperable, the plant is in a condition outside the accident analyses; therefore, LCO 3.0.3 must be entered immediately.

SURVEILLANCE REQUIREMENTS

SR 3.5.1.1

Each accumulator valve should be verified to be fully open ~~every 12 hours~~. This verification ensures each accumulator isolation valve is fully open, as indicated in the control room, and timely discovery if a valve should be less than fully open. If an isolation valve is not fully open, the rate of injection to the RCS would be reduced. Although a motor operated valve position should not change with power removed, a partially closed valve could result in not meeting DBA analyses assumptions (Ref. 3). The Surveillance Frequency is controlled under the SFCP. ~~A 12 hour Frequency is considered reasonable in view of the other administrative controls which ensure that a mispositioned isolation valve is unlikely.~~

SR 3.5.1.2 and 3.5.1.3

Verification ~~every 12 hours~~ of the borated water volume and nitrogen cover pressure in each accumulator is sufficient to ensure adequate injection during a LOCA. The Surveillance Frequency is controlled under the SFCP. ~~Because of the static design of the accumulator, a 12 hour Frequency usually allows the operator to identify changes before limits~~

BASES

~~are reached. Considering that control room alarms are provided for both parameters these limits are effectively subject to~~

SURVEILLANCE REQUIREMENTS (continued)

~~continuous monitoring. The 12 hour Frequency is considered reasonable considering the availability of the control room alarms and the likelihood that, with any deviation which may occur, the accumulators will perform their safety function with slight deviations in these parameters.~~

SR 3.5.1.4

The boron concentration should be verified to be within required limits for each accumulator ~~every 31 days~~, since the static design of the accumulators limits the ways in which the concentration can be changed. ~~The 31 day Frequency is adequate to identify changes that could occur from mechanisms such as in-leakage.~~ Sampling the affected accumulator within 6 hours after a 51 cu. ft. (i.e., 3% of nominal required borated water volume of 1700 cu. ft.) volume increase will promptly identify whether the volume change has caused a reduction of boron concentration to below the required limit. It is not necessary to verify boron concentration if the added water inventory is from the in-containment refueling water storage tank (IRWST), because the water contained in the IRWST is within the accumulator boron concentration requirements. This is consistent with the recommendation of NUREG-1366 (Ref. 6).

The Surveillance Frequency is controlled under the SFCP.

SR 3.5.1.5

Verification ~~every 31 days~~ that power is removed from each accumulator isolation valve operator when the pressurizer pressure is ≥ 2000 psig ensures that an active failure could not result in the undetected closure of an accumulator motor operated isolation valve. If this were to occur, reduced accumulator capacity might be available for injection following a DBA that required operation of the accumulators. ~~Since power is removed under administrative control, the 31 day Frequency will provide adequate assurance that power is removed.~~

The Surveillance Frequency is controlled under the SFCP.

This SR allows power to be supplied to the motor operated isolation valves when pressurizer pressure is < 2000 psig, thus allowing operational flexibility by avoiding unnecessary delays to manipulate the breakers during unit startup or shutdowns.

Should closure of a valve occur, the safeguard actuation signal provided to the valve would open a closed valve, if required.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.5.1.6

This SR requires performance of a system performance test of each accumulator to verify flow capabilities. The system performance test demonstrates that the accumulator injection line resistance assumed in accident analyses is maintained. Although the likelihood that system performance would degrade with time is low, it is considered prudent to periodically verify system performance. The System Level Operability Testing Program provides specific test requirements and acceptance criteria.

The Surveillance Frequency is controlled under the SFCP.

REFERENCES

1. IEEE Standard 603-1991, “Criteria for Safety Systems for Nuclear Power Generating Stations.”
2. FSAR Section 6.3 “Passive Core Cooling System.”
3. FSAR Section 15.6 “Decrease in Reactor Coolant Inventory.”
4. AP1000 PRA.
5. 10 CFR 50.46.
6. NUREG-1366, December 1992, “Improvements to Technical Specifications Surveillance Requirements.”

BASES

ACTIONS (continued)

F.1 and F.2

If the Required Action or associated Completion Time of Condition A, B, C, D, or E are not met or two CMTs are inoperable for reasons other than Condition C, the plant must be brought to a MODE in which the LCO does not apply. To achieve this status, the plant must be brought to at least MODE 3 within 6 hours and to MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.5.2.1

Verification ~~every 24 hours~~ that the temperature of the borated water in each CMT is within limits ensures that when a CMT is needed to inject water into the RCS, the injected water temperature will be within the limits assumed in the accident analysis. The Surveillance Frequency is controlled under the SFCP. ~~The 24 hour Frequency is adequate, based on the fact that no mechanism exists to rapidly change the temperature of a large tank of water such as a CMT.~~ This parameter is normally monitored in the control room by indication and alarms. Also, there are provisions for monitoring the temperature of the inlet and outlet lines to detect in-leakage which may affect the CMT water temperature.

SR 3.5.2.2

This SR verifies each ~~Each~~ CMT inlet isolation valve ~~must be verified~~ to be fully open ~~each 12 hours~~. Frequent verification is considered to be important, since a CMT cannot perform its safety function, if the valve is closed. Control room instrumentation is normally available for this verification. The Surveillance Frequency is controlled under the SFCP.

SR 3.5.2.3

This SR verifies ~~Verification~~ that excessive amounts of noncondensible gases have not caused the water level to drop below the sensor in the inlet line ~~is required every 24 hours~~. The inlet line of each CMT has a vertical section of pipe which serves as a high point collection point for noncondensible gases. Control room indication of the water level in the high point collection point is available to verify that noncondensible gases have collected to the extent that the water level is depressed below the allowable level. The thermal dispersion sensor locations on the vertical pipe sections have been selected to permit additional gas accumulation before injection flow is significantly affected so that adequate time may be provided to permit containment entry for venting the gas.

BASES

SURVEILLANCE REQUIREMENTS (continued)

This level is well above the CMT level corresponding to that assumed in the safety analysis for CMT injected volume. Verifying the noncondensible gas volume is within the noncondensible gas limit confirms ≥ 2487 cu. ft. of borated water volume in the CMT.

The Surveillance Frequency is controlled under the SFCP. ~~The 24 hour Frequency is based on the expected low rate of gas accumulation and the availability of control room indication.~~

SR 3.5.2.4

The minimum boron concentration is 3400 ppm to meet the CMT safety analysis minimum reactivity control requirements. The maximum boron concentration allowed within each CMT at any time is 4500 ppm to prevent overboration.

Verification ~~every 31 days~~ that the boron concentration in each CMT is within the required limits ensures that the reactivity control from each CMT, assumed in the safety analysis, will be available as required. This surveillance identifies ~~The 31 day Frequency is adequate to identify~~ changes which could occur from mechanisms such as in-leakage considering the provisions for monitoring temperature of the inlet line and top of the CMT. Additionally, the top of the CMT has control room indication and an alarm on increased temperature, which would be indicative of in-leakage.

The Surveillance Frequency is controlled under the SFCP.

SR 3.5.2.5

Verification that the redundant outlet isolation valves are OPERABLE by stroking the valves open ensures that each CMT will function as designed when these valves are actuated. Prior to opening the outlet isolation valves, the inlet isolation valve should be closed temporarily. Closing the inlet isolation valve ensures that the CMT contents will not be diluted or heated by flow from the RCS. Upon completion of the test, the inlet isolation valves must be opened. ~~The Surveillance Frequency references the inservice testing requirements.~~

The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 4).

SR 3.5.2.6

This SR verifies that CMT outlet isolation valve actuates to the correct position on an actual or simulated actuation signal. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valve. This surveillance is performed ~~The Frequency of 24 months is based on the need to perform this surveillance~~ during periods in which the plant is shutdown for refueling to prevent any upsets of plant operation.

The Surveillance Frequency is controlled under the SFCP.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.5.2.7

This SR requires performance of a system performance test of each CMT to verify flow capabilities. The system performance test demonstrates that the CMT injection line resistance assumed in DBA analyses is maintained. Although the likelihood that system performance would degrade with time is low, it is considered prudent to periodically verify system performance. The System Level Operability Testing Program provides specific test requirements and acceptance criteria.

The Surveillance Frequency is controlled under the SFCP.

REFERENCES

1. FSAR Section 6.3, “Passive Core Cooling System.”
2. FSAR Chapter 15, “Accident Analysis.”
3. AP1000 PRA.
4. 10 CFR 50.55a(f).

BASES

SURVEILLANCE REQUIREMENTS

SR 3.5.3.1

The LCO 3.5.2 Surveillance Requirements (SRs 3.5.2.1 through 3.5.2.7, except SR 3.5.2.3) are applicable to the CMT required to be OPERABLE. The Surveillance Frequencies are controlled under the SFCP. ~~The Frequencies associated with each specified SR are applicable.~~ Refer to the corresponding Bases for LCO 3.5.2 for a discussion of each SR.

SR 3.5.3.2

The LCO 3.5.2 SR 3.5.2.3 is applicable to the CMT required to be OPERABLE in MODE 4, with RCS cooling provided by the RNS. It is not required to be met in MODE 5, with the RCS not VENTED, because a noncondensible gas volume in the CMT inlet line does not affect its draindown capability. This level is well above the CMT level corresponding to that assumed in the analysis for loss of RNS CMT volume. Verifying the noncondensible gas volume is within the noncondensible gas limit also confirms the required borated water volume in the CMT. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency associated with the specified SR is applicable.~~ Refer to the corresponding Bases for LCO 3.5.2 for a discussion of this SR.

SR 3.5.3.3

Verification ~~every 7 days~~ that the volume of the borated water is ≥ 2450 cu. ft. for the CMT required to be OPERABLE in MODE 5 with the RCS not VENTED verifies that when the OPERABLE CMT is needed to inject water into the RCS, the injected water volume will be within the limits assumed in the analyzed event. This parameter is normally monitored in the control room by indication and alarm. The Surveillance Frequency is controlled under the SFCP. ~~The 7-day Frequency is adequate, based on the fact that no mechanism exists which will drain water below the top nozzle of the CMT.~~

REFERENCES

1. FSAR Section 6.3, “Passive Core Cooling System.”

2. AP1000 PRA.

BASES

ACTIONS (continued)

inoperable and redundant means of feeding the SGs not available, the unit is in a seriously degraded condition with no means for conducting a controlled cooldown. In such a condition, the unit should not be perturbed by any action, including a power change, that might result in a trip. If redundant means of feeding the SGs are not available, the plant should be maintained in the current MODE until redundant means are restored. LCO 3.0.3 and all other Required Actions shall be suspended until the redundant means are restored, because they could force the unit into a less safe condition.

The Completion Time for Required Action F.1 is intended to allow the operator time to evaluate availability of redundant means. In this Required Action, the Completion Time only begins on discovery that redundant means of feeding the SGs is available.

Action F.2 is modified by a Note which requires that prior to stopping SG feedwater, redundant means of cooling the RCS to cold shutdown conditions must be verified as available. One redundant means of cooling the RCS to cold shutdown includes both subsystems of the normal residual heat removal system (RNS). Without availability of these redundant cooling means, the unit is in a seriously degraded condition with no means for continuing the controlled cooldown. Until the redundant cooling means are restored, heat removal using SG feedwater should be maintained. LCO 3.0.3 and all other Required Actions shall be suspended until the systems and equipment required for further cooldown are restored, because they could force the unit into a less safe condition.

The Completion Time for Required Action F.2 is intended to allow the operator time to evaluate availability of redundant means. In this Required Action, the Completion Time only begins on discovery that redundant means of cooling the RCS to cold shutdown conditions is available

Support equipment should be evaluated when assessing whether redundant means are available.

SURVEILLANCE REQUIREMENTS

SR 3.5.4.1

Verification, using remote indication, that the common PRHR HX outlet manual isolation valve is fully open ensures that the flow path from the heat exchangers to the RCS is available. Misalignment of this valve could render the heat exchanger inoperable. The Surveillance Frequency is controlled under the SFCP. ~~A 12 hour Frequency is reasonable considering that the valve is manually positioned and has control room position indication and alarm.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.5.4.2

Verification that the motor operated PRHR HX inlet valve is fully open, as indicated in the main control room, ensures timely discovery if the valve is not fully open. The Surveillance Frequency is controlled under the SFCP. ~~The 12 hour Frequency is consistent with the ease of verification, confirmatory open signals, and redundant series valve controls that prevent spurious closure.~~

SR 3.5.4.3

This surveillance provides verification ~~Verification~~ that excessive amounts of noncondensible gases have not caused the water level to drop below the sensor in the inlet line ~~is required every 24 hours~~. The inlet line of the PRHR HX has a vertical section of pipe which serves as a high point collection point for noncondensible gases. The thermal dispersion sensor location on the vertical pipe section has have been selected to permit additional gas accumulation before natural circulation flow is significantly affected so that sufficient time may be provided to permit containment entry for venting the gas.

Control room indication of the water level in this high point collection point is available to verify that noncondensible gases have not collected to the extent that the water level is depressed below the allowable level. The Surveillance Frequency is controlled under the SFCP. ~~The 24 hour Frequency is based on the expected low rate of gas accumulation and the availability of control room indication.~~

SR 3.5.4.4

Verification is required to confirm that one Loop 1 RCP is in operation, if any RCPs are operating. If RCPs are only operating in Loop 2 and no RCPs are operating in Loop 1, there is a possibility there may be reverse flow in the PRHR HX. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 12 hours is sufficient considering other indications and alarms available to the operator in the main control room to monitor RCS loop performance.~~

The SR is modified by a Note which only requires that the Surveillance be met if one or more RCPs are in operation. If no RCPs are in operation, there is no need to verify one Loop 1 RCP is in operation.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.5.4.5

Verification is required to confirm that power is removed from the motor operated PRHR HX inlet isolation valve ~~every 31 days~~. Removal of power from this valve reduces the likelihood that the valve will be inadvertently closed as a result of a fire. The Surveillance Frequency is controlled under the SFCP. ~~The 31 day Frequency is acceptable considering the frequent surveillance of valve position and that the valve has a confirmatory open signal.~~

SR 3.5.4.6

Verification that both air operated PRHR HX outlet valves stroke open and both IRWST gutter isolation valves stroke closed ensures that the PRHR HX will actuate on command, with return flow from the gutter to the IRWST. Since these valves are redundant, if one valve is inoperable, the system can function at 100% capacity. Verification requires the actual operation of each valve to move it to its safe position. The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 4). ~~The Surveillance Frequency is provided in the Inservice Testing Program.~~

SR 3.5.4.7

This surveillance requires visual inspection of the IRWST gutter and downspout screens to verify that the return flow to the IRWST will not be restricted by debris. The Surveillance Frequency is controlled under the SFCP. ~~A Frequency of 24 months is adequate since there are no known sources of debris with which the gutter or downspout screens could become restricted.~~

SR 3.5.4.8

This SR verifies that both PRHR HX air operated outlet isolation valves and both IRWST gutter isolation valves actuate to the correct position on an actual or simulated actuation signal. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valves. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months is based on the need to perform this surveillance during periods in which the plant is shutdown for refueling to prevent any upsets of plant operation.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.5.4.9

This SR requires performance of a system performance test of the PRHR HX to verify system heat transfer capabilities. The system performance test demonstrates that the PRHR HX heat transfer assumed in accident analyses is maintained. Although the likelihood that system performance would degrade with time is low, it is considered prudent to periodically verify system performance. The System Level Operability Testing Program provides specific test requirements and acceptance criteria.

The Surveillance Frequency is controlled under the SFCP.

REFERENCES

1. FSAR Section 6.3, “Passive Core Cooling System.”
2. FSAR Chapter 15, “Safety Analysis.”
3. AP1000 PRA.
4. 10 CFR 50.55a(f)

BASES

ACTIONS (continued)

F.1 and F.2

If the IRWST cannot be returned to OPERABLE status within the associated Completion Times of Condition A, B, C, D, or E, or the LCO is not met for reasons other than Conditions A, B, C, D, or E, the plant must be brought to a MODE in which the LCO does not apply. To achieve this status, the plant must be brought to at least MODE 3 within 6 hours and to MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.5.6.1

The IRWST borated water temperature must be verified ~~every 24 hours~~ to ensure that the temperature is within the limit assumed in the accident analysis. The Surveillance Frequency is controlled under the SFCP. ~~This Frequency is sufficient to identify a temperature change that would approach the limit and has been shown to be acceptable through operating experience.~~

SR 3.5.6.2

Verification ~~every 24 hours~~ that the IRWST borated water volume is above the required minimum level will ensure that a sufficient initial supply is available for safety injection and floodup volume for recirculation and as the heat sink for PRHR. During shutdown with the refueling cavity flooded with water from the IRWST, this Surveillance requires that the combined volume of borated water in the IRWST and refueling cavity meet the specified limit. The Surveillance Frequency is controlled under the SFCP. ~~Since the IRWST volume is normally stable, and is monitored by redundant main control indication and alarm, a 24 hour Frequency is appropriate.~~

SR 3.5.6.3

This Surveillance provides verification ~~Verification~~ that excessive amounts of noncondensible gases have not caused the water level to drop below the sensor in the four IRWST injection line squib valve lines ~~is required every 24 hours~~. The 8x8x8 inch tee after the outlet of the IRWST injection line squib valve lines has a vertical section of pipe which serves as a high point collection point for noncondensible gases. The thermal dispersion sensor locations on the vertical pipe sections have been selected to permit additional gas accumulation prior to significantly affecting the injection flow so that adequate time may be provided to permit containment entry for venting the gas. The Surveillance Frequency is controlled under the SFCP.

BASES

SURVEILLANCE REQUIREMENTS (continued)

Control room indication of the water level in this high point collection point is available to verify that noncondensible gases have not collected to the extent that the water level is depressed below the allowable level. The Surveillance Frequency is controlled under the SFCP. ~~The 24 hour Frequency is based on the expected low rate of gas accumulation and the availability of control room indication.~~

SR 3.5.6.4

Verification ~~every 31 days~~ that the boron concentration of the IRWST is greater than the required limit, ensures that the reactor will remain subcritical following a LOCA. The Surveillance Frequency is controlled under the SFCP. ~~Since the IRWST volume is large and normally stable, the 31 day Frequency is acceptable, considering additional verifications are required within 6 hours after each solution volume increase of ≥ 15,000 gal. In addition, the relatively frequent surveillance of the IRWST water volume provides assurance that the IRWST boron concentration is not changed.~~

SR 3.5.6.5

This surveillance requires verification that each motor operated injection and sump recirculation isolation valve is fully open. This surveillance may be performed with available remote position indication instrumentation. The Surveillance Frequency is controlled under the SFCP. ~~The 12 hour Frequency is acceptable, considering the redundant remote indication and alarms and that power is removed from the valve operator.~~

SR 3.5.6.6

Verification is required to confirm that power is removed from each motor operated IRWST isolation valve ~~each 31 days~~. Removal of power from these valves reduces the likelihood that the valves will be inadvertently closed. The Surveillance Frequency is controlled under the SFCP. ~~The 31 day Frequency is acceptable considering frequent surveillance of valve position and that the valve has a confirmatory open signal.~~

SR 3.5.6.7

Each motor operated containment recirculation isolation valve must be verified to be fully open. This valve is required to be open to improve containment recirculation reliability. ~~The 31 day Frequency is acceptable considering the valve has a confirmatory open signal.~~ This surveillance may be performed with available remote position indication instrumentation. The Surveillance Frequency is controlled under the SFCP.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.5.6.8

This Surveillance verifies ~~requires verification~~ that each IRWST injection and each containment recirculation squib valve is OPERABLE in accordance with 10 CFR 50.55a(f). The Surveillance Frequency is controlled under the SFCP in accordance with 10 CFR 50.55a(f) (Ref. 4). ~~the Inservice Testing Program. The Surveillance Frequency for verifying valve OPERABILITY references the Inservice Testing Program.~~

The squib valves will be tested in accordance with the ~~ASME OM Code (Ref. 4). The applicable ASME OM Code squib valve requirements are specified in paragraph ISTC-5260~~requirements of 10 CFR 50.55a(f) (Ref. 4).

SR 3.5.6.9

This SR ensures that each IRWST injection and containment recirculation squib valve can actuate to the correct position on an actual or simulated actuation signal. The ADS and IRWST injection blocking device ACTUATION LOGIC TEST, provides overlap with this Surveillance. The OPERABILITY of the squib valves is checked by performing a continuity check of the circuit from the Protection Logic Cabinets to the squib valve. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months is based on the need to perform this surveillance during periods in which the plant is shutdown for refueling to prevent any upsets of plant operation.~~

SR 3.5.6.10

Visual inspection is required ~~each 24 months~~ to verify that the IRWST screens and the containment recirculation screens are not restricted by debris. The Surveillance Frequency is controlled under the SFCP. ~~A Frequency of 24 months is adequate, since there are no known sources of debris with which these screens could become restricted.~~

SR 3.5.6.11

This SR requires performance of a system inspection and performance test of the IRWST injection and recirculation flow paths to verify system flow capabilities. The system inspection and performance test demonstrates that the IRWST injection and recirculation capabilities assumed in accident analyses is maintained. Although the likelihood that system performance would degrade with time is low, it is considered prudent to periodically verify system performance. The System Level Operability Testing Program provides specific test requirements and

acceptance criteria. The Surveillance Frequency is controlled under the SFCP.

BASES

REFERENCES

1. FSAR Section 6.3, “Passive Core Cooling.”
2. FSAR Section 15.6, “Decrease in Reactor Coolant Inventory.”
3. AP1000 PRA.
4. ~~ASME OM Code, “Code for Operation and Maintenance of Nuclear Power Plants.”~~10 CFR 50.55a(f).

BASES

SURVEILLANCE REQUIREMENTS

SR 3.5.7.1

This SR ensures that the Spent Fuel Pool Cooling System (SFS) containment isolation valves are closed. With the nonsafety-related SFS isolated from the IRWST, a leak in the SFS does not cause draining of the IRWST.

The Surveillance Frequency is controlled under the SFCP.

The Surveillance Requirement is modified by a Note allowing the affected penetration flow path(s) to be unisolated intermittently under administrative controls. These administrative controls consist of stationing a dedicated operator at the valve controls, who is in continuous communication with the control room. In this way, the penetration flow path(s) can be rapidly isolated when a need for isolation is indicated.

SR 3.5.7.2

The LCO 3.5.6 Surveillance Requirements and Frequencies (SR 3.5.6.1 through SR 3.5.6.11, except SR 3.5.6.3) are applicable to the IRWST and the flow paths required to be OPERABLE. Refer to the corresponding Bases for LCO 3.5.6 for a discussion of each SR.

The LCO 3.5.6 Surveillance Requirement 3.5.6.3 is not applicable to the IRWST and the flow paths required to be OPERABLE. During RCS vacuum fill operations, a vapor void may form in the high point vent line, causing the water level to drop below the high point water level sensor. Gas accumulation will not increase to a volume that could potentially challenge the OPERABILITY of the passive safety injection flow.

SR 3.5.7.3

The LCO 3.5.6 Surveillance Requirement 3.5.6.3 is applicable to the IRWST and the flow paths required to be OPERABLE. SR 3.5.7.2 is not required to be met during RCS vacuum fill operations, as a vapor void may form in the high point vent line, causing the water level to drop below the high point water level sensor. Gas accumulation will not increase to a volume that could potentially challenge the OPERABILITY of the passive safety injection flow.

REFERENCES

None.

BASES

ACTIONS (continued)

This is done by immediately suspending positive reactivity additions to ensure that the SDM is maintained. Sources of positive reactivity addition include boron dilution, withdrawal of control rods, and excessive cooling of the RCS.

SURVEILLANCE REQUIREMENTS

SR 3.5.8.1

The IRWST and refueling cavity borated water temperature must be verified ~~every 24 hours~~ to ensure that the temperature is within the limit assumed in accident analysis. The Surveillance Frequency is controlled under the SFCP. ~~This Frequency is sufficient to identify a temperature change that would approach the limit and has been shown to be acceptable through operating experience.~~

SR 3.5.8.2

Verification ~~every 24 hours~~ that the IRWST and refueling cavity borated water volume is above the required minimum level will ensure that a sufficient intitial supply is available for safety injection and floodup volume for recirculation and as the heat sink for PRHR. During shutdown with the refueling cavity flooded with water from the IRWST, the Surveillance requires that the combined volume of borated water in the IRWST and refueling cavity meets the specified limit. The Surveillance Frequency is controlled under the SFCP. ~~Since the IRWST volume is normally stable, and is monitored by redundant main control indication and alarm, a 24 hour Frequency is appropriate.~~

SR 3.5.8.3

This SR ensures that the Spent Fuel Pool Cooling System (SFS) containment isolation valves are closed. With the nonsafety-related SFS isolated from the IRWST and refueling cavity, a leak in the SFS does not cause draining of the IRWST and the refueling cavity. The Surveillance Frequency is controlled under the SFCP.

The Surveillance Requirement is modified by two Notes. Note 1 allows the affected penetration flow path(s) to be unisolated intermittently under administrative controls. These administrative controls consist of stationing a dedicated operator at the valve controls, who is in continuous communication with the control room. In this way, the penetration flow path(s) can be rapidly isolated when a need for isolation is indicated.

BASES

SURVEILLANCE REQUIREMENTS (continued)

The Surveillance Requirement is modified by Note 2 stating that it is only required to be met with refueling cavity and spent fuel pool volumes not in communication. In MODE 6 with refueling cavity and spent fuel pool volumes in communication, Refueling Cavity and SFS Isolation is actuated on a Spent Fuel Pool Level - Low 2 isolation signal as required by LCO 3.3.14, "ESFAS IRWST and Spent Fuel Pool Level Instrumentation."

SR 3.5.8.4

Verification ~~every 31 days~~ that the boron concentration of the IRWST and refueling cavity is greater than the required limit ensures that the reactor will remain subcritical following shutdown events. The Surveillance Frequency is controlled under the SFCP. ~~Since the IRWST volume is large and normally stable, the 31-day Frequency is acceptable, considering additional verifications are required within 6 hours after each combined volume increase of ≥ 15,000 gal.~~

SR 3.5.8.5

LCO 3.5.6 Surveillance Requirements and Frequencies SR 3.5.6.3 and 3.5.6.5 through 3.5.6.11 are applicable to the IRWST and the flow paths required to be OPERABLE. Refer to the corresponding Bases for LCO 3.5.6 for a discussion of each SR.

REFERENCES None.

BASES

ACTIONS (continued)

B.1 and B.2

If the Required Action or associated Completion Time of Condition A is not met or one or more RCPs with two breakers are inoperable in MODE 1, 2, 3, or 4, the plant must be brought to a MODE in which the LCO does not apply. To achieve this status, the plant must be brought to at least MODE 3 within 6 hours and to MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner and without challenging plant systems.

C.1

With one or more RCPs with one breaker inoperable in MODE 5, action must be taken to restore the breaker(s) within 72 hours. The 72 hours is reasonable considering the remaining OPERABLE breaker is capable of performing the associated safety function and the low probability of an event occurring during this interval.

D.1

If the Required Action or associated Completion Time of Condition C is not met or one or more RCPs with two breakers are inoperable in MODE 5, action must be initiated immediately to trip the inoperable RCP breaker(s). Tripping the affected RCP(s) accomplishes the necessary safety function to support CMT OPERABILITY.

SURVEILLANCE REQUIREMENTS

SR 3.5.9.1

This SR verifies that RCP breakers trip open on an actual or simulated actuation signal. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valve.

The SR excludes circuit breakers that are maintained in the tripped position since a tripped open circuit breaker is already performing the safety function. Restoration of a circuit breaker to the non-tripped position requires verification that the SR has been met within its required Frequency.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months is based on the need to perform this surveillance during periods in which the plant is shutdown for refueling to prevent any upsets of plant operation.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

criteria were established during initial air lock and containment OPERABILITY testing. The periodic testing requirements verify that the air lock leakage does not exceed the allowed fraction of the overall containment leakage rate. The Frequency is as required by the Containment Leakage Rate Testing Program.

The SR has been modified by two Notes. Note 1 states that an inoperable air lock door does not invalidate the previous successful performance of the overall air lock leakage test. This is considered reasonable since either air lock door is capable of providing a fission product barrier in the event of a DBA. Note 2 has been added to this SR requiring the results to be evaluated against the acceptance criteria applicable to SR 3.6.1.1. This ensures that air lock leakage is properly accounted for in determining the combined Type B and C containment leakage rate.

SR 3.6.2.2

The air lock door interlock is designed to prevent simultaneous opening of both doors in a single air lock. Since both the inner and outer doors of an air lock are designed to withstand the maximum expected post accident containment pressure, closure of either door will support containment OPERABILITY. Thus, the door interlock feature supports containment OPERABILITY while the air lock is being used for personnel transit in and out of the containment. Periodic testing of this interlock demonstrates that the interlock will function as designed and that simultaneous inner and outer door opening will not inadvertently occur. The Surveillance Frequency is controlled under the SFCP. ~~Due to the purely mechanical nature of this interlock, and given that the interlock mechanism is not normally challenged when the containment air lock door is used for entry and exit (procedures require strict adherence to single door opening), this test is only required to be performed every 24 months. The 24 month Frequency is based on the need to perform this Surveillance under the conditions that apply during a plant outage, and the potential for loss of containment OPERABILITY if the Surveillance were performed with the reactor at power. Operating experience has shown these components usually pass the Surveillance when performed at 24 month Frequency. The 24 month Frequency is based on engineering judgment and is considered adequate given that the interlock is not challenged during the use of the airlock.~~

BASES

ACTIONS (continued)

C.1 and C.2

If the Required Actions and associated Completion Times are not met, the plant must be brought to a MODE in which the LCO does not apply. To achieve this status, the plant must be brought to at least MODE 3 within 6 hours and to MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.6.3.1

This SR ensures that the 16 inch purge valves are closed as required or, if open, open for an allowable reason. If a purge valve is open in violation of this SR, the valve is considered inoperable. If the inoperable valve is not otherwise known to have excessive leakage when closed, it is not considered to have leakage outside of limits. The SR is not required to be met when the 16 inch purge valves are open for the reasons stated. The valves may be opened for pressure control, ALARA or air quality considerations for personnel entry, or for Surveillances that require the valves to be open. The 16 inch purge valves are capable of closing in the environment following a LOCA. Therefore, these valves are allowed to be open for limited periods of time. The Surveillance Frequency is controlled under the SFCP. ~~The 31 day Frequency is consistent with other containment isolation valve requirements discussed in SR 3.6.3.2.~~

SR 3.6.3.2

This SR requires verification that each containment isolation manual valve and blind flange located outside containment and not locked, sealed, or otherwise secured and required to be closed during accident conditions is closed. The SR helps to ensure that post accident leakage of radioactive fluids or gases outside of the containment boundary is within design limits. This SR does not require any testing or valve manipulation. Rather, it involves verification that those containment isolation valves outside containment and capable of being mispositioned are in the correct position. ~~Since verification of valve position for containment isolation valves outside containment is relatively easy, the 31 day Frequency is based on engineering judgment and was chosen to provide added assurance of the correct positions.~~ The SR specifies that containment isolation valves that are open under administrative controls are not required to meet the SR during the time the valves are open. This SR does not apply to valves that are locked, sealed, or otherwise secured in the closed position, since these were verified to be in the correct

BASES

position upon locking, sealing, or securing. The Surveillance Frequency is controlled under the SFCP.

SURVEILLANCE REQUIREMENTS (continued)

The Note applies to valves and blind flanges located in high radiation areas and allows these devices to be verified closed by administrative means. Allowing verification by administrative means is considered acceptable, since access to these areas is typically restricted during MODES 1, 2, 3, and 4 for ALARA reasons. Therefore, the probability of misalignment of these containment isolation valves, once they have been verified to be in the proper position, is small.

SR 3.6.3.3

This SR requires verification that each containment isolation manual valve and blind flange located inside containment and not locked, sealed, or otherwise secured and required to be closed during accident conditions is closed. The SR helps to ensure that post accident leakage of radioactive fluids or gases outside of the containment boundary is within design limits. For containment isolation valves inside containment, the Frequency specified as "prior to entering MODE 4 from MODE 5 if not performed within the previous 92 days" is appropriate since these containment isolation valves are operated under administrative control and the probability of their misalignment is low. The SR specifies that containment isolation valves that are open under administrative controls are not required to meet the SR during the time they are open. This SR does not apply to valves that are locked, sealed, or otherwise secured in the closed position, since these were verified to be in the correct position upon locking, sealing, or securing.

The Note allows valves and blind flanges located in high radiation areas to be verified closed by use of administrative controls. Allowing verification by administrative controls is considered acceptable, since access to these areas is typically restricted during MODES 1, 2, 3, and 4 for ALARA reasons. Therefore, the probability of misalignment of these containment isolation valves, once they have been verified to be in the proper position, is small.

SR 3.6.3.4

Verifying that the isolation time of each automatic power operated containment isolation valve is within limits is required to demonstrate OPERABILITY. The isolation time test ensures that the valve will isolate in a time period less than or equal to that assumed in the safety analysis. The isolation times are specified in FSAR Section 6.2.3 (Ref. 1)~~ and Frequency of this SR is in accordance with the Inservice Testing Program~~. The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f). (Ref. 4)

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.6.3.5

Automatic containment isolation valves close on their respective ESF signal to prevent leakage of radioactive material from containment following a DBA. The actual or simulated actuation signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valve. This SR ensures that each automatic containment isolation valve will actuate to its isolation position on its respective ESF signal. This surveillance is not required for valves that are locked, sealed, or otherwise secured in the required position under administrative controls. The Surveillance Frequency is controlled under the SFCP. ~~The 24 month Frequency is based on the need to perform this Surveillance under the conditions that apply during a plant outage and the potential for an unplanned transient if the Surveillance were performed with the reactor at power. Operating experience has shown that these components usually pass this Surveillance when performed at the 24 month Frequency. Therefore, the Frequency was concluded to be acceptable from a reliability standpoint.~~

REFERENCES

1. FSAR Section 6.2, “Containment Systems.”
2. FSAR Chapter 15, “Accident Analysis.”
3. NUREG-1449, “Shutdown and Low Power Operation at Commercial Nuclear Power Plants in the United States.”
4. 10 CFR 50.55a(f).

BASES

SURVEILLANCE REQUIREMENTS

SR 3.6.4.1

Verifying that containment pressure is within limits ensures that unit operation remains within the limits assumed in the containment analysis. The Surveillance Frequency is controlled under the SFCP. ~~The 12 hour Frequency of this SR was developed based on operating experience related to trending of containment pressure variations during the applicable MODES. Furthermore, the 12 hour Frequency is considered adequate in view of other indications available in the main control room, including alarms, to alert the operator to an abnormal containment pressure condition.~~

REFERENCES

1. FSAR, Section 6.2, “Containment Systems.”

BASES

ACTIONS (continued)

B.1 and B.2

If the containment average air temperature is not restored to within its limit within the required Completion Time, the plant must be placed in a condition in which probability and consequences of an event are minimized. To achieve this status, the plant must be brought to at least MODE 3 within 6 hours and to MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner and without challenging plant systems.

C.1

If the containment average air temperature is not restored to within its limits within the required Completion Time in MODE 5 or 6, Required Action C.1 requires that action be immediately initiated to restore containment average air temperature to within limits and to continue this action until restoration is accomplished.

SURVEILLANCE REQUIREMENTS

SR 3.6.5.1

Verifying that the containment average air temperature is within the LCO limit ensures that containment operation remains within the limits assumed for the containment analyses. In order to determine the containment average air temperature, a weighted average is calculated using measurements taken at locations within the containment selected to provide a representative sample of the associated containment atmosphere. The Surveillance Frequency is controlled under the SFCP. ~~The 12 hour Frequency of this Surveillance Requirement is considered acceptable based on observed slow rates of temperature increase within containment as a result of environmental heat sources (due to the large volume of containment). Furthermore, the 12 hour Frequency is considered adequate in view of other indications available in the main control room, including alarms, to alert the operator to an abnormal containment temperature condition.~~

REFERENCES

1. FSAR Section 6.2, “Containment Systems.”

2. 10 CFR 50.49, “Environmental Qualification of Electric Equipment Important to Safety for Nuclear Power Plants.”

BASES

ACTIONS (continued)

E.1 and E.2

If any of the Required Actions and associated Completion Times of Condition A, B, or C are not met in MODE 5 or 6, or if the LCO is not met for reasons other than Condition A, B, or C when in MODE 5 or 6, action must be initiated, immediately, to restore compliance with the LCO and to continue this action until restoration is accomplished in order to provide the PCS for required heat removal needs. In addition, action be immediately initiated to suspend positive reactivity additions to ensure that the SDM is maintained. Sources of positive reactivity addition include boron dilution, withdrawal of reactivity control assemblies, and excessive cooling of the RCS.

SURVEILLANCE REQUIREMENTS

SR 3.6.6.1

This surveillance requires verification that the PCCWST water temperature is within the limits assumed in the accident analyses. The Surveillance Frequency is controlled under the SFCP. ~~The 24 hour Frequency is adequate to identify a temperature change that would approach the temperature limits since the PCCWST is large and temperature variations are slow.~~

SR 3.6.6.2

Verification that the cooling water volume is above the required minimum ensures that a sufficient supply is available for containment cooling. The Surveillance Frequency is controlled under the SFCP. ~~Since the cooling water volume is normally stable and low level is indicated by a main control room alarm, a 7 day Frequency is appropriate and has been shown to be acceptable in similar applications.~~

SR 3.6.6.3

Verifying the correct alignment of manual, power operated, and automatic valves, excluding check valves, in the PCS flow path provides assurance that the proper flow paths exist for system operation. This SR does not apply to valves that are locked, sealed, or otherwise secured in position since these were verified to be in the correct position prior to being secured. This SR does not require any testing or valve manipulation. Rather, it involves verification, through control room instrumentation or a system walkdown, that valves capable of potentially being mispositioned are in the correct position. The Surveillance Frequency is controlled under the SFCP. ~~The 31 day Frequency is appropriate because the valves are operated under administrative control, and an improper valve position would only affect a single flow path. This Frequency has been~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

~~shown to be acceptable through operating experience.~~

SR 3.6.6.4

This SR requires verification that each automatic isolation valve actuates to its correct position upon receipt of an actual or simulated actuation signal. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valves. This Surveillance is not required for valves that are locked, sealed, or otherwise secured in the required position under administrative controls. The Surveillance Frequency is controlled under the SFCP. ~~The 24 month Frequency is based on the need to perform this Surveillance under the conditions that apply during a plant outage and the potential for an unplanned transient if the Surveillances were performed with the reactor at power. The 24 month Frequency is also acceptable based on consideration of the design reliability of the equipment.~~

~~Operating experience has shown that these components usually pass the Surveillances when performed at the 24 month Frequency. Therefore, the Frequency was concluded to be acceptable from a reliability standpoint.~~

SR 3.6.6.5

Periodic inspections of the PCS air flow path from the shield building annulus inlet to the exit ensure that it is unobstructed, the baffle plates are properly installed, and the upper annulus safety-related drains are unobstructed. The Surveillance Frequency is controlled under the SFCP. ~~Although there are no anticipated mechanisms which would cause air flow path or annulus drain obstruction and the effect of a missing air baffle section is small, it is considered prudent to verify this capability every 24 months. Additionally, the 24 month Frequency is based on the desire to perform this Surveillance under conditions that apply during a plant outage, the need to have access to the locations, and the potential for an unplanned transient if the Surveillance were performed with the reactor at power. This Frequency has been found to be sufficient to detect abnormal degradation in similar situations.~~

SR 3.6.6.6

This SR requires performance of a Passive Containment Cooling System test to verify system flow and water coverage capabilities. The system performance test demonstrates that the containment cooling capability assumed in accident analyses is maintained by verifying the flow rates via

BASES

SURVEILLANCE REQUIREMENTS (continued)

each standpipe and measurement of containment wetting coverage. The System Level OPERABILITY Testing Program provides specific test requirements and acceptance criteria. Although the likelihood that system performance would degrade with time is low, it is considered prudent to periodically verify system performance. The Surveillance Frequency is controlled under the SFCP. ~~The first refueling and 10 year Frequency is based on the ability of the more frequent surveillances to verify the OPERABILITY of the active components and features which could degrade with time.~~

REFERENCES

1. 10 CFR 50, Appendix A, “General Design Criteria for Nuclear Power Plants.”

2. 10 CFR 50, Appendix K, “ECCS Evaluation Models.”

3. FSAR Chapter 6.2, “Containment Systems.”

BASES

ACTIONS

A.1

If the containment equipment hatches, air locks, or any containment penetration that provides direct access from the containment atmosphere to the outside atmosphere is not in the required status, including the containment isolation capability when automatic isolation valves are open, the penetration(s) must be restored to the required status within 1 hour.

B.1 and B.2

If the Required Action and associated Completion Time of Condition A is not met or the LCO is not met for reasons other than Condition A, action must be initiated, immediately, to restore compliance with the LCO and to continue this action until restoration is accomplished. In addition, action be immediately initiated to suspend positive reactivity additions to ensure that the SDM is maintained. Sources of positive reactivity addition include boron dilution, withdrawal of control rods, and excessive cooling of the RCS.

SURVEILLANCE REQUIREMENTS

SR 3.6.7.1

This Surveillance demonstrates that each of the containment penetrations required to be in its closed position is in that position. For any open purge and exhaust valves, this SR will demonstrate that the valves are not blocked from closing. Also if the purge and/or exhaust valves are open and relying on automatic closure, this Surveillance will demonstrate that each valve operator has motive power, which will ensure that each valve is capable of being closed by an automatic containment purge and exhaust isolation signal prior to steaming into containment. Alternately, purge and exhaust penetrations may be verified closed or capable of being closed prior to steaming into the containment, which will limit loss of the cooling water inventory from containment. Open containment spare penetrations shall be verified capable of being closed prior to steaming into the containment by removal of obstructions and installation of the flange or by other closure means which will limit loss of the cooling water inventory from containment.

The Surveillance Frequency is controlled under the SFCP. ~~The Surveillance is performed every 7 days. The Surveillance interval is selected to ensure that the required penetration status is maintained during shutdown inspections, testing, and maintenance.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.6.7.2

Each of the two equipment hatches is provided with a set of hardware, tools, equipment, and self-contained power source for moving the hatch from its storage location and installing it in the opening. The required set of hardware and tools shall be visually inspected to ensure that they can perform the required functions. The equipment and power source shall be inspected and/or operated as necessary to verify that the hatch can be closed properly. The power source shall be verified as containing sufficient energy to install the hatch from the storage location.

The Surveillance Frequency is controlled under the SFCP. ~~The 7 day Frequency is adequate considering that the hardware, tools, equipment, and power sources are dedicated to the associated equipment hatch and not used for any other functions.~~

The SR is modified by a Note which only requires that the surveillance be met for an open equipment hatch. If the equipment hatch is installed in position, then the availability of the means to install the hatch is not required.

REFERENCES

1. FSAR subsection 19E.2.6.2.
2. AP1000 Probabilistic Risk Assessment.

BASES

SURVEILLANCE REQUIREMENTS (continued)

While a weight is specified, the normal manner to confirm the weight limit is met is by measuring the volume of the TSP contained in the pH adjustment baskets. The minimum measured volume of TSP is 540 ft^3 to meet the minimum required mass of TSP (26,460 lbs) and assumes the minimum density of TSP. The minimum TSP density is based on the manufactured density (49 lbm/ft^3), since the density may increase and the volume decrease, during plant operation, due to agglomeration from humidity inside the containment. A TSP volume of 560 ft^3 at the initial loading (i.e., prior to compaction and agglomeration which may occur over time) provides about 4% margin above the minimum required TSP mass of 26,460 lbs.

The Surveillance Frequency is controlled under the SFCP. ~~The periodic verification is required every 24 months, since access to the TSP baskets is only feasible during outages, and normal fuel cycles are scheduled for 24 months. Operating experience has shown this Surveillance Frequency acceptable due to the margin in the volume of TSP placed in the containment building.~~

SR 3.6.8.2

Testing must be performed to ensure the solubility and buffering ability of the TSP after exposure to the containment environment. A representative sample of TSP from one of the baskets in containment is submerged in 1.0 ± 0.01 liter of water at a boron concentration of 3050 ppm that has been heated to a temperature of 71 ± 5°C (160 ± 9°F). The solution is allowed to stand at this temperature for 4 hours without agitation. The solution is then cooled to 25 ± 5°C (77 ± 9°F) and the pH is measured. The solution pH should rise to ≥ 7.0. A TSP sample size of 2.19 grams is used if the TSP volume is verified to be at least 540 ft^3 (minimum required). However, a larger TSP sample size can be used if the measured volume is verified to be greater than the minimum. For example, if the TSP volume is verified to be 560 ft^3, then a representative sample of 2.27 grams can be used.

Agitation of the test solution is prohibited, since an adequate standard for the agitation intensity cannot be specified. The time of 4 hours without agitation is necessary to allow time for the dissolved TSP to naturally diffuse through the sample solution. In the post LOCA sump area, rapid mixing would occur due to liquid flow, significantly decreasing the actual amount of time before the required pH is achieved.

The Surveillance Frequency is controlled under the SFCP.

BASES

ACTIONS (continued)

based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner and without challenging plant systems.

G.1

If the Required Action and associated Completion Time of Condition A, B, or E is not met, or if two vacuum relief check valves are inoperable for opening, or if two vacuum relief isolation valves are inoperable for opening in MODE 5 or 6, action must be immediately initiated to restore compliance with the LCO and to continue this action until restoration is accomplished.

SURVEILLANCE REQUIREMENTS

SR 3.6.9.1

This SR verifies ~~Verification~~ that the containment inside to outside differential air temperature is ≤ 90°F ~~is required every 12 hours~~. The containment inside to outside differential air temperature is the difference between the outside ambient air temperature (measured by the site meteorological instrumentation or equivalent) and the inside containment average air temperature (measured using the same instrumentation as used for SR 3.6.5.1).

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is based on the normally stable containment average air temperature and the relatively small outside ambient air temperature changes within this time.~~

SR 3.6.9.2

~~Each vacuum relief isolation valve must be verified to be closed every 31 days.~~ This SR assures that the vacuum relief isolation valves are closed as required, or if open, the valves are open for an allowable reason. The vacuum relief isolation valves are normally closed. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 31 days is appropriate considering the valves should only be opened to relieve vacuum or manually opened to perform Surveillances and, therefore, the probability of the valve misalignment is low.~~

SR 3.6.9.2 is modified by two Notes. Note 1 states that SR 3.6.9.2 is not required to be met for vacuum relief valves that are open during Surveillances. Surveillances which open the vacuum relief isolation valves are performed infrequently and the valves will only be open for a limited period of time. Note 2 states that SR 3.6.9.2 is not required to be met for vacuum relief valves performing their vacuum relief function. If

BASES

SURVEILLANCE REQUIREMENTS (continued)

vacuum relief is required, the valves open to perform their safety function to maintain the integrity of the containment vessel. Air will flow into containment from the atmosphere due to the differential pressure, which will prevent radiological release from containment during vacuum relief.

SR 3.6.9.3

Each vacuum relief check valve and each vacuum relief isolation valve is tested to be OPERABLE for opening and OPERABLE for closing. This SR satisfies ~~cites the Inservice Testing Program, which establishes~~ the requirement that inservice testing of the ASME Code Class 1, 2, and 3 valves shall be performed in accordance with 10 CFR 50.55a(f) ~~the ASME OM Code~~ (Ref. 2). The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f). ~~Therefore, SR Frequency is governed by the Inservice Testing Program.~~

SR 3.6.9.4

This SR ensures that each vacuum relief isolation valve will actuate to the open position on an actual or simulated actuation signal and will actuate to the closed position on an actual or simulated actuation signal. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valves. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months is based on the need to perform this surveillance during periods in which the plant is shutdown for refueling to prevent any upsets of plant operations.~~

REFERENCES

1. FSAR subsection 6.2.1.1.4, “External Pressure Analysis.”

2. 10 CFR 50.55a(f). ~~ASME OM Code, “Code for Operation and Maintenance of Nuclear Power Plants.”~~

3. FSAR subsection 9.4.7, “Containment Air Filtration System.”

BASES

ACTIONS (continued)

C.1 and C.2

If any Required Action and associated Completion Time of Condition A is not met, or if one or both steam generators have less than two MSSVs OPERABLE, the unit must be placed in a MODE in which the LCO does not apply. To achieve this status, the plant must be placed in at least MODE 3 within 6 hours, and in MODE 4, with RCS cooling provided by the RNS, within 24 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required unit conditions from full power conditions in an orderly manner and without challenging plant systems.

D.1 and D.2

If the Required Action and associated Completion Time of Condition B is not met, the plant must be brought to a MODE in which the LCO does not apply. To achieve this status, the plant must be brought to at least MODE 3 within 6 hours and to MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.7.1.1

This SR verifies the OPERABILITY of the MSSVs by the verification of each MSSV lift setpoint in accordance with the requirements of 10 CFR 50.55a(f). (Ref. 5)~~Inservice Testing Program. The safety and relief valve tests are required to be performed in accordance with ASME OM Code (Ref. 5). According to Reference 5, the following tests are required:~~

~~a. Visual examination;~~

~~b. Seat tightness determination;~~

~~c. Set pressure determination (lift setting);~~

~~d. Compliance with owner's seat tightness criteria; and~~

~~e. Verification of the balancing device integrity on balanced valves.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

~~The ANSI/ASME standard requires that all valves be tested every 5 years and a minimum of 20% of the valves be tested every 24 months. The ASME Code specifies the activities and frequencies necessary to satisfy the requirements.~~ Table 3.7.1-2 allows a ±1% setpoint tolerance for OPERABILITY, and the valves are reset to remain within ±1% during the Surveillance to allow for drift. The lift settings, according to Table 3.7.1-2, correspond to ambient conditions of the valve at nominal operating temperature and pressure.

The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 5).

This SR is modified by a Note that allows entry into and operation in MODE 3 prior to performing the SR. The MSSVs may be either bench tested or tested in situ at hot conditions using an assist device to simulate lift pressure. If the MSSVs are not tested at hot conditions, the lift setting pressure shall be corrected to ambient conditions of the valve at operating temperature and pressure.

REFERENCES

1. FSAR Chapter 10, “Steam and Power Conversion Systems Description.”

2. ASME Boiler and Pressure Vessel Code, Section III, Article NC-7000, “Overpressure Protection,” Class 2 Components.

3. FSAR Section 15.2, “Decreased Heat Removal by Secondary System.”

4. ASME OM Code, Appendix I, “Inservice Testing of Pressure Relief Devices in Light-Water Reactor Nuclear Power Plants.”

5. 10 CFR 50.55a(f).~~ASME OM Code, “Code for Operation and Maintenance of Nuclear Power Plants.”~~

6. NRC Information Notice 94-60, “Potential Overpressurization of the Main Steam System,” August 22, 1994.

BASES

ACTIONS (continued)

is to ensure that these devices are not inadvertently repositioned. Therefore, the probability of misalignment of these valves, once they have been verified to be in the proper position, is small.

F.1, F.2, and F.3

If the main steam line flow path isolation valves cannot be restored to OPERABLE status or affected flow paths isolated within the associated Completion Times of Condition D or E, the unit must be placed in a MODE in which the LCO does not apply. To achieve this status, the plant must be placed in at least MODE 3 within 6 hours, in MODE 4 with normal residual heat removal system in service within 24 hours, and in MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required unit conditions from MODE 2 conditions in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.7.2.1

This SR verifies that MSIV closure time is ≤ 5.0 seconds, on an actual or simulated actuation signal. The MSIV isolation time is assumed in the accident and containment analyses. This Surveillance is normally performed upon returning the unit to operation following a refueling outage. The MSIVs should not be tested at power, since even a part stroke exercise increases the risk of a valve closure when the unit is generating power.

The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 7). ~~The Frequency is in accordance with the Inservice Testing Program.~~

The SR is modified by a note which states that the SR is only required to be performed prior to entry into MODE 2. This allows the option to perform testing in MODES 3 or 4.

SR 3.7.2.2

This SR verifies that the turbine stop, turbine control, turbine bypass, and moisture separator reheater 2nd stage steam isolation valves' closure time is ≤ 5.0 seconds, on an actual or simulated actuation signal. These alternate downstream isolation valves must meet the MSIV isolation time

BASES

SURVEILLANCE REQUIREMENTS (continued)

assumed in the accident and containment analyses. This Surveillance is normally performed upon returning the unit to operation following a refueling outage. The alternate downstream valves should not be tested at power, since even a part stroke exercise increases the risk of a valve closure when the unit is generating power.

The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 7) ~~The Frequency is in accordance with the Inservice Testing Program.~~

The SR is modified by a note which states that the SR is only required to be performed prior to entry into MODE 2. This allows the option to perform testing in MODES 3 or 4.

SR 3.7.2.3

Verifying that the isolation time of each MSIV bypass and steam line drain valve is within limits is required to demonstrate OPERABILITY. The isolation time test ensures that the valve will isolate in a time period less than or equal to that assumed in the safety analysis. The isolation times are specified in FSAR Section 6.2.3 (Ref. 6) ~~and Frequency of this SR is in accordance with the Inservice Testing Program~~. The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref.7).

SR 3.7.2.4

This SR ensures that each MSIV bypass and steam line drain valve will actuate to its isolation position on an actual or simulated actuation signal. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valves. The Surveillance Frequency is controlled under the SFCP. ~~The 24 month Frequency is based on the need to perform this Surveillance during periods in which the plant is shutdown for refueling to prevent any upsets of plant operation.~~

REFERENCES

1. FSAR Section 10.3, “Main Steam System.”
2. FSAR Section 10.4, “Other Features of Steam and Power Conversion Systems.”
3. FSAR Section 6.2.1, “Containment Functional Design.”

BASES

REFERENCES (continued)

4. FSAR Section 15.1, “Increase in Heat Removal by Secondary System.”

5. NUREG-138, Issue 1, “Staff Discussion of Fifteen Technical Issues Listed in Attachment to November 3, 1976 Memorandum from Director NRR to NRR Staff.”

6. FSAR Section 6.2, “Containment Systems.”

7. 10 CFR 50.55a(f).

BASES

ACTIONS (continued)

D.1, D.2, and D.3

If the affected flow path cannot be isolated or the MFIVs, MFCVs, or MFW pump breaker cannot be restored to OPERABLE status within the associated Completion Time, the unit must be placed in a MODE in which the LCO does not apply. To achieve this status, the plant must be placed in at least MODE 3 within 6 hours, in MODE 4 with the normal residual heat removal system in service within 24 hours, and in MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required unit conditions from full power conditions in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.7.3.1

This SR verifies that the closure time of each MFIV and MFCV is ≤ 5.0 seconds, on an actual or simulated actuation signal. The MFIV and MFCV isolation times are assumed in the accident and containment analyses. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valves. This Surveillance is normally performed upon returning the unit to operation following a refueling outage. These valves should not be tested at power, since even a part stroke exercise increases the risk of a valve closure when the unit is generating power.

The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 2) ~~The Frequency is in accordance with the Inservice Testing Program.~~

The SR is modified by a note which states that the SR is only required to be performed prior to entry into MODE 2. This allows the option to perform testing in MODES 3 or 4.

SR 3.7.3.2

This SR verifies that each main feedwater pump breaker trip open in response to an actual or simulated actuation signal. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the pump breaker.

BASES

SURVEILLANCE REQUIREMENTS (continued)

The SR excludes main feedwater pump breakers that are maintained in the tripped position since a tripped open MFW pump breaker is already performing the safety function. Restoration of a breaker to the non-actuated position requires verification that the SR has been met within its required Frequency.

The Surveillance Frequency is controlled under the SFCP. ~~The 24 month Frequency is based on the need to perform this Surveillance during periods in which the plant is shutdown for refueling to prevent any upsets of plant operation.~~

REFERENCES

1. FSAR Section 10.4.7, “Condensate and Feedwater System.”
2. 10 CFR 50.55a(f).

BASES

APPLICABILITY

In MODES 1, 2, 3, and 4 the limits on secondary specific activity apply due to the potential for secondary steam releases to the atmosphere.

In MODES 5 and 6, the steam generators are not being used for heat removal. Both the RCS and steam generators are depressurized, and primary to secondary LEAKAGE is minimal. Therefore, monitoring of secondary specific activity is not required.

ACTIONS

A.1 and A.2

DOSE EQUIVALENT I-131 exceeding the allowable value in the secondary coolant, is an indication of a problem in the RCS and contributes to increased post accident doses. If the secondary specific activity cannot be restored to within limits within the associated Completion Time, the unit must be placed in a MODE in which the LCO does not apply. To achieve this status, the unit must be placed in at least MODE 3 within 6 hours and in MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required unit conditions from full power conditions in an orderly manner and without challenging unit systems.

SURVEILLANCE REQUIREMENTS

SR 3.7.4.1

This SR verifies that the secondary specific activity is within the limits of the accident analysis. A gamma isotopic analysis of the secondary coolant, which determines DOSE EQUIVALENT I-131, confirms the validity of the safety analysis assumptions as to the source terms in post accident releases. It also serves to identify and trend any unusual isotopic concentrations that might indicate changes in reactor coolant activity or leakage. The Surveillance Frequency is controlled under the SFCP. ~~The 31 day Frequency is based on the detection of increasing trends of the level of DOSE EQUIVALENT I-131, and allows for appropriate action to be taken to maintain levels below the LCO limit.~~

REFERENCES

1. FSAR, Chapter 15, “Accident Analyses.”

BASES

SURVEILLANCE REQUIREMENTS

SR 3.7.5.1

This SR verifies sufficient spent fuel pool water is available in the event of a fuel handling accident or loss of spent fuel pool cooling. The water level in the spent fuel pool must be checked periodically. The Surveillance Frequency is controlled under the SFCP. ~~The 7 day Frequency is appropriate because the volume in the pool is normally stable.~~ Water level changes are controlled by plant procedures and are acceptable based on operating experience.

During refueling operations, the level in the spent fuel pool is in equilibrium with the refueling canal, and the level in the refueling canal is checked daily in accordance with SR 3.9.4.1.

REFERENCES

1. FSAR Section 9.1.2, “Spent Fuel Storage.”
2. FSAR Section 9.1.3, “Spent Fuel Pool Cooling System.”
3. FSAR Section 15.7.4, “Fuel Handling Accident.”
4. Regulatory Guide 1.183 Rev. 0, “Alternate Radiological Source Terms for Evaluating Design Basis Accidents at Nuclear Power Reactors.”

BASES

ACTIONS (continued)

of at least 1/8 inch wg when a “High-1” radioactivity concentration (gaseous, particulate, or iodine) is detected in the main control room supply air duct, as well as the availability of the supply air handling unit to provide cooling with recirculation air to maintain the MCR and CSA areas within their design temperatures.

The 7 day Completion Time is based on engineering judgment, considering the low probability of an accident that would result in a significant radiation release, the low probability of radioactivity release, and that the available VBS can provide the required capability. Dose calculations verify that the MCRE dose limits will remain within the requirements of GDC 19 with VBS mitigation in the event of VES inoperability.

G.1 and G.2

In MODE 1, 2, 3, or 4, if the Required Actions and associated Completion Times of Condition D or F are not met, the unit must be placed in a MODE in which the LCO does not apply. To achieve this status, the unit must be placed in at least MODE 3 within 6 hours and in MODE 5 within 36 hours.

H.1

During movement of irradiated fuel assemblies, if the Required Actions and associated Completion Times of Condition F are not met, or the VES is inoperable due to an inoperable MCRE boundary, action must be taken immediately to suspend the movement of fuel. This does not preclude the movement of fuel to a safe position.

SURVEILLANCE REQUIREMENTS

SR 3.7.6.1

Verification is required ~~every 24 hours~~ that compressed air storage tanks contain > 327,574 scf of breathable air.

The standard volume is determined using the compressed air storage tank room temperature (VAS-TE-080A/B) and compressed air storage tanks pressure (VES-PT-001A/B) via the plant control system and chart in the system procedures. Verification that the minimum volume of compressed air is contained in the compressed air storage tanks ensures that there will be an adequate supply of breathable air to maintain MCRE habitability for a period of 72 hours. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 hours is based on the availability of pressure indication in the MCRE.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.7.6.2

SR 3.7.6.2 verifies that the thermal mass of the required heat sinks is within limit(s)~~ every 24 hours~~. One method of satisfying SR 3.7.6.2 is maintaining ambient air temperature below the limit for the MCRE (i.e., ≤ 75°F), the individual required rooms (i.e., ≤ 85°F), and adjacent required room-pairs (i.e., ≤ 85°F) for 4 days. Alternatively, analyses or local measurements can satisfy the verification of the heat sink thermal mass assumed in the main control room heatup calculation.

Satisfying the required heat sink room temperature limits (i.e., ≤ 75°F for the MCRE and ≤ 85°F for the individual required rooms and the adjacent required room-pairs) for sufficient duration (i.e., 4 days) establishes the equilibrium initial heat sink thermal mass assumed in the main control room heatup calculation.

Passive heat sink air temperatures in required dc Equipment Room and required I&C room-pairs (12201/12301, 12203/12302, 12205/12305, and 12207/12304) are verified by temperature elements located in the shared return air ducting (alternatively, local measurement of each room may be utilized). Other required individual rooms (12202, 12204, 12300, 12303, 12313, 12412, and 12501) are verified using indication from the temperature elements in each room.

This is done to verify that the VBS is performing as required to maintain the initial conditions assumed in the safety analyses, and to verify the VES heat sinks provide adequate thermal mass to limit the temperature increase in the MCRE, dc Equipment Rooms, and I&C Rooms from exceeding the allowable limits after VES actuation.

The Surveillance Frequency is controlled under the SFCP. ~~The 24 hour Frequency is acceptable based on the availability of automatic VBS temperature controls, alarms, and indication in the MCRE. Air temperatures may also be verified using local measurement.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.7.6.3

Standby systems should be checked periodically to ensure that they function properly. ~~As the environment and normal operating conditions on this system are not too severe, testing VES once every month provides an adequate check of the system. This monthly~~ VES operation for this surveillance requires operation of one of the two pressure regulating valves. The Surveillance Frequency is controlled under the SFCP. ~~Additionally, the main air delivery line has two parallel air delivery flow paths via automatically actuated isolation valves (VES-V005A/B) that are also required to be exercised quarterly. Monthly testing of the three air delivery flow paths on a STAGGERED TEST BASIS will result in at least quarterly exercising of each air delivery valve. The 31 day on a STAGGERED TEST BASIS Frequency is based on the reliability of the equipment and the availability of system redundancy.~~

SR 3.7.6.4

This SR verifies that the VES air header isolation valves are ~~required to be verified~~ open ~~at 31 day intervals~~. This SR is designed to ensure that the pathways for supplying breathable air to the MCRE are available should loss of VBS occur. These valves should be closed only during required testing or maintenance of downstream components, or to preclude complete depressurization of the system should the VES isolation valves in the air delivery line open inadvertently or begin to leak. The Surveillance Frequency is controlled under the SFCP.

SR 3.7.6.5

~~Verification~~ This SR verifies that the air quality of the compressed air storage tanks meets the requirements of CGA G-7.1, Commodity Specification for Air, Grade E, (Ref. 5) with a pressure dew point of ≤ 40°F at ≥ 3400 psig ~~is required every 92 days~~. If air has not been added to the air storage tanks since the previous verification, verification may be accomplished by confirmation of the acceptability of the previous surveillance results along with examination of the documented record of air makeup. Verification of the initial air quality (in combination with the other surveillances) ensures that breathable air is available for 11 MCRE occupants for at least 72 hours. Confirmation of the pressure dew point verifies that water has not formed in the line, eliminating the potential for freezing at the pressure regulating valve during VES operation. In addition, the dry air allows the MCRE to remain below the maximum relative humidity to support the 90°F WBGT required for human factors performance. The Surveillance Frequency is controlled under the SFCP.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.7.6.6

Verification that the VBS isolation valves and the SDS isolation valves are OPERABLE and will actuate upon demand is required ~~every 24 months~~ to ensure that the MCRE can be isolated upon loss of VBS operation. The SR excludes automatic valves that are locked, sealed, or otherwise secured in the actuated position. The SR does not apply to valves that are locked, sealed, or otherwise secured in the actuated position since the affected valves were verified to be in the actuated position prior to being locked, sealed, or otherwise secured. Placing an automatic valve in a locked, sealed, or otherwise secured position requires an assessment of the OPERABILITY of the system or any supported systems, including whether it is necessary for the valve to be repositioned to the non-actuated position to support the accident analysis. Restoration of an automatic valve to the non-actuated position requires verification that the SR has been met ~~within its required Frequency~~. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valves. The Surveillance Frequency is controlled under the SFCP.

SR 3.7.6.7

This SR verifies ~~Verification~~ that each VES pressure relief isolation valve within the MCRE pressure boundary is OPERABLE ~~is required in accordance with the Inservice Testing Program~~. The valves should actuate on an actual or simulated signal to the correct position. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valves. The SR is used in combination with SR 3.7.6.8 to ensure that adequate vent area is available to mitigate MCRE overpressurization. The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 9).

SR 3.7.6.8

This SR verifies ~~Verification~~ that the VES pressure relief damper ~~is OPERABLE is required at 24 month intervals~~. The SR is used in combination with SR 3.7.6.7 to ensure that adequate vent area is available to mitigate MCRE overpressurization. The Surveillance Frequency is controlled under the SFCP.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.7.6.9

This SR verifies the OPERABILITY of the MCRE boundary by testing for unfiltered air inleakage past the MCRE boundary and into the MCRE. The details of the testing are specified in the Main Control Room Envelope Habitability Program.

The MCRE is considered habitable when the radiological dose to MCRE occupants calculated in the licensing basis analyses of DBA consequences is no more than 5 rem TEDE and the MCRE occupants are protected from hazardous chemicals and smoke. This SR verifies that the unfiltered air inleakage into the MCRE is no greater than the flow rate assumed in the licensing basis analyses of DBA consequences. When unfiltered air inleakage is greater than the assumed flow rate, Condition D must be entered.

SR 3.7.6.10

This SR verifies that the required VES testing is performed in accordance with the Ventilation Filter Testing Program (VFTP). The VES filter tests are in accordance with the UFSAR Appendix 1A position for Regulatory Guide 1.52 (Ref. 7). The VFTP includes testing the performance of the HEPA filter, charcoal adsorber efficiency, minimum flow rate, and physical properties of the activated charcoal. Specific test frequencies and additional information are discussed in detail in the VFTP.

SR 3.7.6.11

Verification that the MCR load shed function actuates on an actual or simulated signal from each PMS Division is required ~~every 24 months~~ to confirm that the non-safety stage 1 and stage 2 MCR heat loads can be de-energized by the VES actuation signal within the required time. The Surveillance Frequency is controlled under the SFCP. ~~The 24 month Frequency is based on the need to perform this Surveillance under the conditions that apply during a plant outage to minimize the potential for adversely affecting MCR operations.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.7.6.12

Verification that the main VES air delivery isolation valves actuate on an actual or simulated signal to the correct position is required ~~every 24 months~~ to confirm that the VES operates as assumed in the safety analysis. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valves. The Surveillance Frequency is controlled under the SFCP. ~~The 24 month Frequency is based on the need to perform this Surveillance under the conditions that apply during a plant outage to minimize adversely affecting MCR operations.~~

REFERENCES

1. FSAR Section 6.4, “Main Control Room Habitability Systems.”
2. FSAR Section 9.5.1, “Fire Protection System.”
3. FSAR Subsection 15.6.5.3, “Radiological Consequences.”
4. Regulatory Guide 1.196, “Control Room Habitability at Light-Water Nuclear Power Reactors.”
5. CGA G-7.1-1997, “Commodity Specification for Air.”
6. NEI 99-03, “Control Room Habitability Assessment,” June 2001.
7. Letter from Eric J. Leeds (NRC) to James W. Davis (NEI) dated January 30, 2004, “NEI Draft White Paper, Use of Generic Letter 91-18 Process and Alternative Source Terms in the Context of Control Room Habitability.” (ADAMS Accession No. ML040300694).
8. Regulatory Guide 1.52, “Design, Inspection, and Testing Criteria for Air Filtration and Adsorption Units of Post-Accident Engineered-Safety-Feature Atmosphere Cleanup Systems in Light-Water-Cooled Nuclear Power Plants,” Revision 3.
9. 10 CFR 50.55a(f).

BASES

ACTIONS (continued)

D.1, D.2, and D.3

If the affected flow paths cannot be isolated or the isolation valves, control valves, or pump breakers cannot be restored to OPERABLE status, closed, or isolated within the associated Completion Times, the unit must be placed in a MODE in which the LCO does not apply. To achieve this status, the plant must be placed in least MODE 3 within 6 hours, in MODE 4 with RCS cooling provided by the normal residual heat removal system within 24 hours, and in MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required unit conditions from full power conditions in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.7.7.1

This surveillance requires verification in accordance with ~~the Inservice Testing Program~~10 CFR 50.55a(f) to assure that each SFW isolation and control valve is OPERABLE. The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 2). ~~The Surveillance Frequency is provided in the Inservice Testing Program.~~

SR 3.7.7.2

This SR ensures that each SFW isolation valve and SFW control valve will actuate to its isolation position on an actual or simulated actuation signal. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valves.

The Surveillance Frequency is controlled under the SFCP. ~~The 24 month Frequency is based on the need to perform this Surveillance during periods in which the plant is shutdown for refueling to prevent any upsets of plant operation.~~

SR 3.7.7.3

This SR verifies that each SFW pump breakers trip open on an actual or simulated actuation signal. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the pump breaker.

BASES

SURVEILLANCE REQUIREMENTS (continued)

The SR excludes SFW pump breakers that are maintained in the tripped position since a tripped open SFW pump breaker is already performing the safety function. Restoration of a breaker to the non-actuated position requires verification that the SR has been met within its required Frequency.

The Surveillance Frequency is controlled under the SFCP. ~~The 24 month Frequency is based on the need to perform this Surveillance during periods in which the plant is shutdown for refueling to prevent any upsets of plant operation.~~

REFERENCES

1. FSAR Section 10.4.9, “Startup Feedwater System.”
2. 10 CFR 50.55a(f).

BASES

ACTIONS (continued)

In the event the affected flow path is isolated in accordance with Required Action C.1, the affected penetration must be verified to be isolated on a periodic basis per Required Action A.2, which remains in effect.

D.1

With two valves in one or more steam generator blowdown flow paths inoperable, action must be taken to isolate the flow path with a closed valve. This action places the flow path in a condition which assures the safety function is performed. The 8 hour Completion Time is reasonable, considering the low probability of an accident occurring during this time period that would require a closure of the SG blowdown flow path isolation valves.

In the event the affected flow path is isolated in accordance with Required Action D.1, the affected penetration must be verified to be isolated on a periodic basis per Required Action B.2, which remains in effect.

E.1, E.2, and E.3

If the SG PORV flow path or blowdown flow path isolation valves cannot be restored to OPERABLE status or are not closed within the associated Completion Time, the unit must be placed in a MODE in which the LCO does not apply. To achieve this status, the unit must be placed at least in MODE 3 within 6 hours, in MODE 4 with the RCS cooling provided by the RNS within 24 hours, and in MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required unit conditions in an orderly manner and without challenging unit systems. Required Action E.3 to be in MODE 5 is modified by a Note stating that it is not applicable to inoperable PORV(s). PORV Applicability is exited on completion of Required Action E.2.

SURVEILLANCE REQUIREMENTS

SR 3.7.10.1

The function of the SG PORV isolation valves (PORV block valves (SGS-PL-V027A & B) and PORVs (SGS-PL-V233A & B)) and blowdown isolation valves (SGS-PL-V074A & B and SGS-PL-V075A & B) is to isolate the steam generators in the event of SGTR, Loss of Feedwater or Feedwater Line Break. Stroking the valves closed demonstrates their capability to perform the isolation function. The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 4). ~~The Frequency for this SR is in accordance with the Inservice Testing Program.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.7.10.2

Verifying that the isolation time of each PORV block valve and SG blowdown isolation valve is within limits is required to demonstrate OPERABILITY. The isolation time test ensures that the valve will isolate in a time period less than or equal to that assumed in the safety analysis. The isolation times are specified in Section 6.2.3 (Ref. 3)~~ and Frequency of this SR is in accordance with the Inservice Testing Program~~. The Surveillance Frequency is controlled under the SFCP in accordance with the requirements of 10 CFR 50.55a(f) (Ref. 4).

SR 3.7.10.3

This Surveillance verifies that each SG PORV, SG PORV block valve, and SG blowdown isolation valve actuates to the isolation position on an actual or simulated actuation signal. The actual or simulated signal is processed through the component interface module to verify the continuity between the output of the component interface module and the valves.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 months is based on the need to perform this Surveillance during periods in which the plant is shutdown for refueling to prevent any upsets of plant operation.~~

REFERENCES

1. FSAR Section 10.3.2.2.3, “Power-Operated Atmospheric Relief Valves.”
2. FSAR Section 10.4.8, “Steam Generator Blowdown System.”
3. FSAR Section 6.2.3, "Containment Isolation System."
4. 10 CFR 50.55a(f).

BASES

SURVEILLANCE REQUIREMENTS

SR 3.7.11.1

This SR verifies that the concentration of boron in the spent fuel pool is within the required limit. As long as this SR is met, the analyzed accidents are fully addressed. The Surveillance Frequency is controlled under the SFCP. ~~The 7 day Frequency is appropriate because no major replenishment of pool water is expected to take place over such a short period of time.~~

REFERENCES

1. FSAR Sections 9.1.2, “Spent Fuel Storage” and 15.7.4, “Fuel Handling Accident.”

2. Double contingency principle of ANSI N16.1-1975, as specified in the April 14, 1978 NRC letter (Section 1.2) and implied in the proposed revision to Regulatory Guide 1.13 (Section 1.4, Appendix A).

3. APP-GW-GLR-029, “AP1000 Spent Fuel Storage Racks Criticality Analysis,” Westinghouse Electric Company LLC (Westinghouse Proprietary).

BASES

SURVEILLANCE REQUIREMENTS (continued)

cells will receive adequate current to optimally charge the battery. The voltage requirements are based on the nominal design voltage of the battery and are consistent with the initial voltages assumed in the battery sizing calculations. This voltage maintains the battery plates in a condition that supports maintaining the grid life (expected to be approximately 20 years). The Surveillance Frequency is controlled under the SFCP. ~~The 7 day Frequency is consistent with manufacturer recommendations and IEEE-450 (Ref. 8).~~

SR 3.8.1.2

This SR verifies the design capacity of the battery chargers. According to Regulatory Guide 1.32 (Ref. 9), the battery charger supply is recommended to be based on the largest combined demands of the various steady state loads and the charging capacity to restore the battery from the design minimum charge state to the fully charged state, irrespective of the status of the unit during these demand occurrences. The minimum required amperes and duration ensure that these requirements can be satisfied.

This SR provides two options. One option requires that each battery charger be capable of supplying 150 amps at the minimum established float voltage for 8 hours. The ampere requirements are based on the design minimum charger size of the chargers. The voltage requirements are based on the charger voltage level after a response to a loss of AC power. The time period is sufficient for the charger temperature to have stabilized and to have been maintained for at least 2 hours.

The other option requires that each battery charger be capable of recharging the battery after a service test coincident with supplying the largest combined demands of the various continuous steady state loads (irrespective of the status of the plant during which these demands occur). This level of loading may not normally be available following the battery service test and will need to be supplemented with additional loads. The duration for this test may be longer than the charger sizing criteria since the battery recharge is affected by float voltage, temperature, an the exponential decay in charging current. The battery is recharged when the measured charging current is ≤ 2 amps.

The Surveillance Frequency is controlled under the SFCP. ~~The Surveillance Frequency is acceptable, given the unit conditions required to perform the test and the other administrative controls existing to ensure adequate charger performance during these 24 month intervals. In addition, this Frequency is intended to be consistent with expected fuel cycle lengths.~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.8.1.3

A battery service test, as defined in IEEE 450-1995 (Ref. 8), is a special test of battery capability, as found, to satisfy the design requirements (battery duty cycle) of the Class 1E DC electrical power system. The discharge rate and test length corresponds to the design duty cycle requirements as specified in Reference 4.

The Surveillance Frequency is controlled under the SFCP. ~~The Surveillance Frequency of 24 months is consistent with accepted industry practice.~~ This Surveillance may be performed during any plant condition with the spare battery and charger providing power to the bus.

This SR is modified by a Note. The Note allows the performance of a modified performance discharge test in lieu of a service test.

The modified performance discharge test is a test, in the "as found" condition, of a battery's capacity with the discharge rate(s) modified to encompass every portion of the battery duty cycle. This allows the performance test to be accomplished in the minimun amount of time while still demonstrating the high rate capability of the battery to meet duty cycle requirements. Thus, a modified performance test satisfies both the requirement to perform periodic performance tests as well as the requirement to perform periodic service tests.

The battery terminal voltage for the modified performance discharge test is required to remain above the minimum battery terminal voltage specified in the battery service test for the ~~the~~ duration of the service test portion of the test. Initial conditions for the modified performance discharge test are identical to those specified for a service test.

SR 3.8.1.4

Verification that the main control room air supply radiation monitor sample pump de-energize on an actual or simulated signal from the PMS is required ~~every 24 months~~ to assure that the non-essential Class 1E 24-hour battery loads are shed to maintain the assumed battery load profile.

The Surveillance Frequency is controlled under the SFCP. ~~The Surveillance Frequency is acceptable, given the unit conditions required to perform the test and other administrative controls existing to ensure adequate charger performance during these 24 month intervals. In addition, this Frequency is intended to be consistent with expected fuel cycle lengths.~~

BASES

ACTIONS (continued)

B.1 and B.2

If the inoperable DC electrical power subsystem cannot be restored to OPERABLE status within the required Completion Time, the unit must be brought to MODE 5 where the probability and consequences on an event are minimized. To achieve this status, the plant must be brought to at least MODE 3 within 6 hours and to MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required unit conditions from full power conditions in an orderly manner and without challenging plant systems.

SURVEILLANCE REQUIREMENTS

SR 3.8.3.1

This Surveillance verifies that the inverters are functioning properly with all required switches and circuit breakers closed and Class 1E AC instrument and control buses energized from the inverter. The verification of proper voltage and frequency output ensures that the required power is readily available for the PMS instrumentation connected to the Class 1E AC instrument and control buses. The Surveillance Frequency is controlled under the SFCP. ~~The 7 day Frequency takes into account the effectiveness of the voltage and frequency instruments, the redundant capability of the inverters, and other indications available in the control room that alert the operator to inverter malfunctions.~~

REFERENCES

1. FSAR Section 8.3.2.1.1.2, “Class 1E Uninterruptible Power Supplies.”
2. FSAR Chapter 6, “Engineered Safety Features.”
3. FSAR Chapter 15, “Accident Analyses.”

BASES

ACTIONS (continued)

could result in failure to meet the minimum SDM or boron concentration limit is required to assure continued safe operation. Introduction of coolant inventory must be from sources that have a boron concentration greater than that what would be required in the RCS for minimum SDM or refueling boron concentration. This may result in an overall reduction in RCS boron concentration, but provides acceptable margin to maintaining subcritical operation. Introduction of temperature changes including temperature increases when operating with a positive MTC must also be evaluated to ensure they do not result in a loss of required SDM.

Suspension of these activities shall not preclude completion of actions to establish a safe conservative condition. These actions minimize the probability of the occurrence of postulated events. It is further required to immediately initiate action to restore the required inverters and to continue this action until restoration is accomplished in order to provide the necessary inverter power to the unit safety systems.

The Completion Time of immediately is consistent with the required times for actions requiring prompt attention. The restoration of the required inverters should be completed as quickly as possible in order to minimize the time the unit safety systems may be without power or powered from a voltage regulating transformer.

SURVEILLANCE REQUIREMENTS

SR 3.8.4.1

This Surveillance verifies that the inverters are functioning properly with all required circuit breakers closed and Class 1E AC instrument and control buses energized from the inverter. The verification of proper voltage and frequency output ensures that the required power is readily available for the instrumentation connected to the Class 1E AC instrument and control buses. The Surveillance Frequency is controlled under the SFCP. ~~The 7 day Frequency takes into account the redundant capability of the inverters, and other indications available in the control room that alert the operator to inverter malfunctions.~~

REFERENCES

1. FSAR Chapter 6, “Engineered Safety Features.”
2. FSAR Chapter 15, “Accident Analysis.”
3. FSAR Appendix 19E, “Shutdown Evaluation.”

BASES

ACTIONS (continued)

The 2 hour Completion Time for one DC electrical power distribution subsystem division is consistent with Regulatory Guide 1.93 (Ref. 4).

E.1 and E.2

If the inoperable division(s) cannot be restored to OPERABLE status within the required Completion Time, the unit must be brought to MODE 5 where the probability and consequences on an event are minimized. To achieve this status, the plant must be brought to at least MODE 3 within 6 hours and to MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required unit conditions from full power conditions in an orderly manner and without challenging plant systems.

F.1

With two inoperable divisions that result in a loss of safety function, adequate core cooling, containment OPERABILITY, and other vital functions for DBA mitigation would be compromised, and immediate plant shutdown in accordance with LCO 3.0.3 is required.

SURVEILLANCE REQUIREMENTS

SR 3.8.5.1

This Surveillance verifies that the Class 1E AC instrument and control and DC electrical power distribution subsystems are functioning properly, with the required circuit breakers and switches properly aligned. The verification of proper voltage availability on the buses ensures that the required voltage is readily available for motive as well as control functions for critical system loads connected to these buses. The Surveillance Frequency is controlled under the SFCP. ~~The 7 day Frequency takes into account the redundant capability of the Class 1E AC instrument and control and DC electrical power distribution subsystems, and other indications available in the control room that alert the operator to subsystem malfunctions.~~

BASES

SURVEILLANCE REQUIREMENTS

SR 3.8.6.1

This Surveillance verifies that the Class 1E AC instrument and control and DC electrical power distribution subsystems are functioning properly, with the required circuit breakers and switches properly aligned. The verification of proper voltage availability on the buses ensures that the required power is readily available for motive as well as control functions for critical system loads connected to these buses. The Surveillance Frequency is controlled under the SFCP. ~~The 7 day Frequency takes into account the redundant capability of the electrical power distribution subsystems and other indications available in the control room that alert the operator to subsystem malfunctions.~~

REFERENCES

1. FSAR Chapter 6, “Engineered Safety Features.”
2. FSAR Chapter 15, “Accident Analysis.”
3. FSAR Section 8.3.2, “DC Power Systems.”
4. FSAR Appendix 19E, “Shutdown Evaluation.”

BASES

ACTIONS (continued)

F.1

With one or more batteries with any battery parameter outside the allowances of the Required Actions for Condition A, B, C, D, or E, sufficient capacity to supply the maximum expected load requirement is not assured and the corresponding battery must be declared inoperable. Additionally, discovering one or more batteries in one Division with one or more battery cells float voltage less than 2.07 V and float current greater than 2 amps indicates that the battery capacity may not be sufficient to perform the intended functions. The battery must therefore be declared inoperable immediately.

SURVEILLANCE REQUIREMENTS

SR 3.8.7.1

Verifying battery float current while on float charge is used to determine the state of charge of the battery. Float charge is the condition in which the charger is supplying the continuous charge required to overcome the internal losses of a battery and maintain the battery in a charged state. The float current requirements are based on the float current indicative of a charged battery. Use of float current to determine the state of charge of the battery is consistent with IEEE-450 (Ref. 3). The Surveillance Frequency is controlled under the SFCP. ~~The 7 day Frequency is consistent with IEEE-450 (Ref. 3).~~

This SR is modified by a Note that states the float current requirement is not required to be met when battery terminal voltage is less than the minimum established float voltage of SR 3.8.1.1. When this float voltage is not maintained the Required Actions of LCO 3.8.1 ACTION A are being taken, which provide the necessary and appropriate verifications of the battery condition. Furthermore, the float current limit of 2 amps is established based on the nominal float voltage value and is not directly applicable when this voltage is not maintained.

SR 3.8.7.2 and SR 3.8.7.5

Optimal long term battery performance is obtained by maintaining a float voltage greater than or equal to the minimum established design limits provided by the battery manufacturer, which corresponds to 264.0 V at the battery terminals, or 2.20 Volts per cell. This provides adequate over-potential, which limits the formation of lead sulfate and self discharge, which could eventually render the battery inoperable. Float voltages in this range or less, but greater than 2.07 Volts per cell, are addressed in Specification 5.5.11. SRs 3.8.7.2 and 3.8.7.5 require verification that the cell float voltages are equal to or greater than the short term absolute

minimum voltage of 2.07 V. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency for cell voltage verification~~

BASES

SURVEILLANCE REQUIREMENTS (continued)

~~every 31 days for pilot cell and 92 days for each connected cell is consistent with IEEE-450 (Ref. 3).~~

SR 3.8.7.3

The limit specified for electrolyte level ensures that the plates suffer no physical damage and maintains adequate electron transfer capability. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is consistent with IEEE-450 (Ref. 3).~~

SR 3.8.7.4

This Surveillance verifies that the pilot cell temperature is greater than or equal to the minimum established design limit (i.e., 60°F). Pilot cell electrolyte temperature is maintained above this temperature to assure the battery can provided the required current and voltage to meet the design requirements. Temperatures lower than assumed in battery sizing calculations act to inhibit or reduce battery capacity. The Surveillance Frequency is controlled under the SFCP. ~~The Frequency is consistent with IEEE-450 (Ref. 3).~~

SR 3.8.7.6

A battery performance discharge test is a test of constant current capacity of a battery, normally done in the as found condition, after having been in service, to detect any change in the capacity determined by the acceptance test. The test is intended to determine overall battery degradation due to age and usage.

Either the battery performance discharge test or the modified performance discharge test is acceptable for satisfying SR 3.8.7.6; however, only the modified performance discharge test may be used to satisfy the battery service test requirements of SR 3.8.1.3. This Surveillance may be performed during any plant condition with the spare battery and charger providing power to the bus.

A modified discharge test is a test of the battery capacity and its ability to provide a high rate, short duration load (usually the highest rate of the duty cycle). This will often confirm the battery's ability to meet the critical period of the load duty cycle, in addition to determining its percentage of rated capacity. Initial conditions for the modified performance discharge test should be identical to those specified for a service test.

The modified performance discharge test is a test, in the "as found" condition, of a battery's capacity with the discharge rate(s) modified to encompass every portion of the battery duty cycle. This allows the

performance test to be accomplished in the minimum amount of time while still demonstrating the high rate capability of the battery to meet

BASES

SURVEILLANCE REQUIREMENTS (continued)

duty cycle requirements. Thus, a modified performance test satisfies both the requirement to perform periodic performance tests as well as the requirement to perform periodic service tests.

The battery terminal voltage for the modified performance discharge test is required to remain above the minimum battery terminal voltage specified in the battery service test for the duration of the service test portion of the test. Initial conditions for the modified performance discharge test are identical to those specified for a service test.

The acceptance criteria for this Surveillance are consistent with IEEE-450 (Ref. 3) and IEEE-485 (Ref. 4). These references recommend that the battery be replaced if its capacity is below 80% of the manufacturer's rating. A capacity of 80% shows that the battery rate of deterioration is increasing, even if there is ample capacity to meet the load requirements. Furthermore, the battery is sized to meet the assumed duty cycle loads when the battery design capacity reaches this 80% limit.

The Surveillance Frequency for this test is ~~normally 60 months~~in accordance with the SFCP, or 25% of expected life, whichever is less. If the battery shows degradation, or if the battery has reached 85% of its expected life and capacity is < 100% of the manufacturer's rating, the Surveillance Frequency is reduced to 12 months. However, if the battery shows no degradation but has reached 85% of its expected life, the Surveillance Frequency is only reduced to 24 months for batteries that retain capacity ≥ 100% of the manufacturer's ratings. Degradation is indicated, according to IEEE-450 (Ref. 3), when the battery capacity drops by more than 10% relative to its capacity on the previous performance test or when it is below 90% of the manufacturer's rating. These Frequencies are consistent with the SFCP and the recommendations in IEEE-450 (Ref. 3) and Information Notice 2013-05 (Ref. 5).

REFERENCES

1. FSAR Chapter 6, "Engineered Safety Features."

2. FSAR Chapter 15, "Accident Analyses."

3. IEEE-450 1995, "IEEE Recommended Practice for Maintenance, Testing, and Replacement of Vented Lead-Acid Batteries for Stationary Applications."

4. IEEE-485 1983, "IEEE Recommended Practice for Sizing Large Lead Storage Batteries for Generating Stations and Substations."

5 NRC Information Notice 2013-05, March 19, 2013.

BASES

ACTIONS

A.1

Continuation of positive reactivity additions (including actions to reduce boron concentration) is contingent upon maintaining the plant in compliance with the LCO. If the boron concentration of any coolant volume in the RCS, the refueling cavity, or the fuel transfer canal is less than its limit, all operations involving positive reactivity additions must be suspended immediately.

Suspension of positive reactivity additions shall not preclude completion of actions to establish a safe condition, including moving a component to a safe position.

A. 2

In addition to immediately suspending positive reactivity additions, boration to restore the concentration must be initiated immediately.

In determining the required combination of boration flow rate and concentration, no unique design basis accident (DBA) must be satisfied. The only requirement is to restore the boron concentration to its required value as soon as possible. In order to raise the boron concentration as soon as possible, the operator shall begin boration with the best source available for plant operations.

Once boration is initiated, it must be continued until the boron concentration is restored. The restoration time depends on the amount of boron that must be injected to reach the required concentration.

SURVEILLANCE REQUIREMENTS

SR 3.9.1.1

This SR verifies that the coolant boron concentration in the RCS, the refueling cavity and the fuel transfer canal is within the COLR limit. The boron concentration of the coolant in each volume is determined periodically by chemical analysis.

The Surveillance Frequency is controlled under the SFCP. ~~A minimum Frequency of once every 72 hours is a reasonable amount of time between verifications of the boron concentration. The surveillance interval is based on operating experience, isolation of unborated water sources in accordance with LCO 3.9.2, and the availability of the source range neutron flux monitors required by LCO 3.9.3.~~

BASES

ACTIONS

The ACTIONS Table has been modified by a Note which allows separate Condition entry for each unborated water source flow path.

A.1

Preventing unplanned dilution of the reactor coolant boron concentration is dependent on maintaining the unborated water isolation valves secured closed. Securing the valves in the closed position verifies that the valves cannot be inadvertently opened. The Completion Time of "Immediately" requires an operator to initiate actions to close an open valve and secure the isolation valve in the closed position immediately. Once actions are initiated, they must be continued until the valves are secured in the closed position.

A.2

Due to the potential of having diluted the boron concentration of the reactor coolant, SR 3.9.1.1 (verification of boron concentration) must be performed whenever Condition A is entered to verify that the required boron concentration exists. The Completion Time of 4 hours is sufficient to obtain and analyze a reactor coolant sample for boron concentration.

SURVEILLANCE REQUIREMENTS

SR 3.9.2.1

These valves are to be secured closed to isolate possible dilution flow paths. The likelihood of a significant reduction in the boron concentration during MODE 6 operations is remote due to the large mass of borated water in the refueling cavity and the fact that all unborated water source flow paths are isolated, precluding a dilution. The boron concentration is checked ~~every 72 hours~~ during MODE 6 under SR 3.9.1.1. This surveillance demonstrates that the valves are closed. The Surveillance Frequency is controlled under the SFCP. ~~The 31 day Frequency is based on engineering judgement and is considered reasonable in view of other administrative controls that will verify that the valve opening is an unlikely possibility.~~

REFERENCES

1. FSAR Chapter 15, "Accident Analyses."

2. NUREG-0800, Standard Review Plan, Section 15.4.6, "Chemical and Volume Control System Malfunction that Results in a Decrease in Boron Concentration in the RCS."

BASES

SURVEILLANCE REQUIREMENTS

SR 3.9.3.1

SR 3.9.3.1 is the performance of a CHANNEL CHECK, which is the comparison of the indicated parameter values monitored by each of these instruments. It is based on the assumption that the two indication channels should be consistent for the existing core conditions. Changes in core geometry due to fuel loading can result in significant differences between the source range channels, however each channel should be consistent with its local conditions.

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 12 hours is consistent with the CHANNEL CHECK Frequency specified for these same instruments in LCO 3.3.2, "Reactor Trip System (RTS) Source Range Instrumentation" and LCO 3.3.8, "Engineered Safety Feature Actuation System (ESFAS) Instrumentation," Function 17.~~

SR 3.9.3.2

This SR is the performance of a CHANNEL CALIBRATION ~~every 24 months~~. This SR is modified by a Note stating that neutron detectors are excluded from the CHANNEL CALIBRATION. The CHANNEL CALIBRATION for the source range neutron flux monitors consists ~~consisting~~ of obtaining the detector plateau or preamp discriminator curves, evaluating those curves, and comparing the curves to the manufacturer's data. The Surveillance Frequency is controlled under the SFCP. ~~The 24 month Frequency is based on the need to perform this Surveillance under the conditions that apply during a plant outage. Operating experience has shown these components usually pass the Surveillance when performed at a 24 month Frequency.~~

REFERENCES

1. FSAR Chapter 15, "Accident Analysis."
2. FSAR Section 14.2.7.1, "Initial Fuel Loading."

BASES

ACTIONS

A.1

With a water level of < 23 ft above the top of the reactor vessel flange, all operations involving movement of irradiated fuel assemblies within containment shall be suspended immediately to ensure that a fuel handling accident cannot occur.

The suspension of fuel movement shall not preclude completion of movement to safe position.

SURVEILLANCE REQUIREMENTS

SR 3.9.4 1

Verification of a minimum water level of 23 ft above the top of the reactor vessel flange ensures that the design basis for the analysis of the postulated fuel handling accident during refueling operations is met. Water at the required level above the top of the reactor vessel flange limits the consequences of damaged fuel rods that are postulated to result from a fuel handling accident inside containment (Ref. 3).

The Surveillance Frequency is controlled under the SFCP. ~~The Frequency of 24 hours is based on engineering judgement and is considered adequate in view of the large volume of water and the normal procedural controls of valve positions which make significant unplanned level changes unlikely.~~

REFERENCES

1. FSAR Chapter 15, “Accident Analysis.”
2. Regulatory Guide 1.183, “Alternate Radiological Source Terms for Evaluating Design Basis Accidents at Nuclear Power Reactors.”
3. FSAR Section 15.7.4, “Fuel Handling Accident.”

Attachment 6 to NL-26-0017
Proposed No Significant Hazards Consideration

License Amendment Request to Revise Technical Specifications to Include a Surveillance Frequency Control Program

Southern Nuclear Operating Company (SNC) has evaluated the proposed change to the TS using the criteria in 10 CFR 50.92 and has determined that the proposed change does not involve a significant hazards consideration.

Vogtle Electric Generating Plant (VEGP) Units 3 and 4 requests adoption of approved changes to the standard technical specifications (STS) and unit-specific technical specifications (TS), to allow relocation of specific TS surveillance frequencies to a licensee-controlled program. The proposed changes establish consistency with NRC-approved Industry / Technical Specifications Task Force (TSTF) travelers TSTF-425, Rev. 3, “Relocate Surveillance Frequencies to Licensee Control – RITSTF Initiative 5b,” and TSTF-596, Rev. 2, “Expand the Applicability of the Surveillance Frequency Control Program (SFCP),” for the NUREG-1431 STS for Westinghouse Plants. The proposed changes are also consistent with the NRC-approved changes to NUREG-2194, “Standard Technical Specifications, Westinghouse Advanced Passive (AP1000) Plants,” presented in TSTF-563, “Revise Instrument Testing Definitions to Incorporate the Surveillance Frequency Control Program.” In addition, VEGP Units 3 and 4 requests administrative changes to a TS definition of the Surveillance Frequency Control Program (SFCP) and modify some TS surveillance requirements that are no longer applicable.

As required by 10 CFR 50.91(a), an analysis of the issue of no significant hazards consideration is presented below:

1. Does the proposed change involve a significant increase in the probability or consequences of an accident previously evaluated?

   Response: No.

   The proposed change relocates the specified frequencies for periodic surveillance requirements to licensee control under a new Surveillance Frequency Control Program (SFCP) and revises the TS definitions of Channel Calibration, Channel Operational Test, and Trip Actuating Device Operational Test to allow the frequency for testing the components or devices in each step to be determined in accordance with the TS SFCP, such that all components in the channel continue to be calibrated. Surveillance frequencies are not an initiator to any accident previously evaluated. As a result, the probability of any accident previously evaluated is not significantly increased. The systems and components required by the technical specifications for which the surveillance frequencies are relocated are still required to be operable, meet the acceptance criteria for the surveillance requirements, and be capable of performing any mitigation function assumed in the accident analysis. As a result, the consequences of any accident previously evaluated are not significantly increased.

   The administrative TS changes do not affect any accident evaluations, as there are no changes to the plant, no changes to analysis for the plant, and no changes to testing or operation of the plant. The proposed changes continue to maintain the initial conditions and operating limits assumed during normal operation, assumed by the accident analysis, and assumed in anticipated operational occurrences. The proposed changes do not involve a change to any mitigation sequence or the predicted radiological releases due to postulated accident conditions.

   Therefore, the proposed change does not involve a significant increase in the probability or consequences of an accident previously evaluated.

2. Does the proposed change create the possibility of a new or different kind of accident from any accident previously evaluated?

   Response: No.

   The proposed change relocates the specified frequencies for periodic surveillance requirements to a licensee-controlled SFCP, revises the TS definitions of instrumentation testing such that all components in the channel continue to be calibrated, and make administrative changes to define the SFCP and modify some TS surveillance requirements that are no longer applicable. No new or different accidents result from utilizing the proposed change. The changes do not involve a physical alteration of the plant (*i.e.,* no new or different type of equipment will be installed) or a change in the methods governing normal plant operation. The proposed changes do not introduce any credible new failure mechanisms, malfunctions, or accident initiators not considered in the design or licensing bases. In addition, the changes do not impose any new or different requirements. The changes do not alter assumptions made in the safety analysis. The proposed changes are consistent with the safety analysis assumptions and current plant operating practice.

   Therefore, the proposed change does not create the possibility of a new or different kind of accident from any accident previously evaluated.

3. Does the proposed change involve a significant reduction in a margin of safety?

   Response: No.

   The design, operation, testing methods, and acceptance criteria for systems, structures, and components (SSCs), specified in applicable codes and standards (or alternatives approved for use by the NRC) will continue to be met as described in the plant licensing basis (including the final safety analysis report and bases to TS), since these are not affected by changes to the surveillance frequencies or the TS definitions. Similarly, there is no impact to safety analysis acceptance criteria as described in the plant licensing basis. The SFCP assures sufficient safety margins are maintained, and that that design, operation, surveillance methods, and acceptance criteria specified in applicable codes and standards (or alternatives approved for use by the NRC) will continue to be met as described in the plants' licensing basis. To evaluate a change in the relocated surveillance frequency, SNC will perform a probabilistic risk evaluation using the guidance contained in NRC approved NEI 04–10, Rev. 1 in accordance with the TS SFCP. NEI 04–10, Rev. 1, methodology provides reasonable acceptance guidelines and methods for evaluating the risk increase of proposed changes to surveillance frequencies consistent with Regulatory Guide 1.177.

   Therefore, the proposed change does not involve a significant reduction in a margin of safety.

Based on the above, SNC concludes that the proposed change presents no significant hazards consideration under the standards set forth in 10 CFR 50.92(c), and, accordingly, a finding of "no significant hazards consideration" is justified.